Controller DUNEW2PLC1
Controller Fault Handler
Power-Up Handler

Tasks

DUNEWINDER2
- **MainProgram**
  - **MainRoutine**
- **UnServo_9**
  - **UnServo_All**
- **EOT_Trip_11**
  - **EOT_TRIP**
- **Error_State_10**
  - **Error**
- **Latch_UnLatch_State_6_7_8**
  - **Latch_Driver**
- **MoveXY_State_2_3**
  - **MoveXY**
  - **xy_speed_regulator**
- **MoveZ_State_4_5**
  - **MoveZ**
- **Ready_State_1**
  - **Ready**
- **Initialize**
  First state to set up machine
  - **Init**
- **PID_Tension_Servo**
  Using code from Sarah Johnson
  - **PID_Tension**
- **pid2**
- **motionQueue**
  - **queue_line_arc**

SafetyTask
- **Safety**
  - **Safety**

Unscheduled
- **Camera**
  Controls to capture data from the Cognex camera.
  - **CameraControl**

Motion Groups

DUNEW2AXIS
- **DUNEW1AMP**
- **X_axis**
- **Y_axis**
- **Z_axis**
- **X_Y**

Ungrouped Axes

Add-On Instructions

- **SCP**
  Scale with Parameters
  - **Logic**

Data Types

- **User-Defined**
  - **MotionSeg**
    motion segments for move queue, lines or arcs
- **Strings**

Add-On-Defined
- **SCP**
  - Scale with Parameters

Module-Defined
- **AB:1734_3SLOT:I:0**
- **AB:1734_3SLOT:O:0**
- **AB:1734_5SLOT:I:0**
- **AB:1734_5SLOT:O:0**
- **AB:1734_IB8S:O:0**
- **AB:1734_IB8S_Safety2:I:0**
- **AB:1734_OB8S:O:0**
- **AB:1734_OB8S_Safety1:I:0**
- **AB:1769_DI32:I:0**
- **AB:1769_DO16:C:0**
- **AB:1769_DO16:I:0**
- **AB:1769_DO16:O:0**
- **AB:1769_IF4:C:0**
- **AB:1769_IF4:I:0**
- **AB:1769_OF4:C:0**
- **AB:1769_OF4:I:0**
- **AB:1769_OF4:O:0**
- **AB:5000_AI8:C:0**
- **AB:5000_AI8:I:0**
- **AB:5000_AI8:O:0**
- **AB:5000_AI_Channel:C:0**
- **AB:5000_AO4:C:0**
- **AB:5000_AO4:I:0**
- **AB:5000_AO4:O:0**
- **AB:5000_AO8:C:0**
- **AB:5000_AO8:I:0**
- **AB:5000_AO8:O:0**
- **AB:5000_AO_Channel:C:0**
- **AB:5000_DI16:C:0**
- **AB:5000_DI16:I:0**
- **AB:5000_DI_Channel:C:0**
- **AB:5000_DO16:O:0**
- **AB:5000_DO16_Diag:C:0**
- **AB:5000_DO16_Diag:I:0**
- **AB:5000_DO_Diag_Channel:C:0**
- **AB:5000_HART_Command_Control_Struct:I:0**
- **AB:5000_HART_Command_Control_Struct:O:0**
- **AB:5000_HART_Static_Struct:I:0**
- **AB:5000_String16_Struct:I:0**
- **AB:5000_String32_Struct:I:0**
- **AB:CIP_Motion_Device_Safety2:SI:0**
- **AB:CIP_Motion_Device_Safety2:SO:0**
- **AB:MotionDevice_Diagnostics:S:0**
- **AB:Motion_Diagnostics:S:1**
- **AB:STRATIX_5700_10PORT_GB_MANAGED:I:0**
- **CC:InSight11_Control:O:0**
- **CC:InSight11_DINT8:I:0**
- **CC:InSight11_DINT8:O:0**
- **CC:InSight11_Status:I:0**
- **CHANNEL_AI:I:0**

- **CHANNEL_AI:O:0**
- **CHANNEL_AI_CAL:O:0**
- **CHANNEL_AI_CJ_DIAG:I:0**
- **CHANNEL_AI_DIAG:I:0**
- **CHANNEL_AI_DIAG_CAL:I:0**
- **CHANNEL_AI_HART:I:0**
- **CHANNEL_AO:O:0**
- **CHANNEL_AO_CAL:O:0**
- **CHANNEL_AO_DIAG:I:0**
- **CHANNEL_AO_DIAG_CAL:I:0**
- **CHANNEL_DI:I:0**
- **CHANNEL_DI_COUNTER:I:0**
- **CHANNEL_DI_COUNTER:O:0**
- **CHANNEL_DI_TIMESTAMP:I:0**
- **CHANNEL_DI_TIMESTAMP:O:0**
- **CHANNEL_DO:I:0**
- **CHANNEL_DO:O:0**
- **CHANNEL_DO_DIAG:I:0**
- **CHANNEL_DO_OVERRIDE:O:0**
- **CHANNEL_DO_SCHEDULED:O:0**
- **CHANNEL_HSC:I:0**
- **CHANNEL_HSC:O:0**
- **PAX_HART_DEVICE:I:0**
- **_02A6:is8xxx_78F5E13D:O:0**
- **_02A6:is8xxx_F1075143:I:0**

Trends

- **Latch**
  Things associated with latching
- **MORE_STATS_S_0**
- **servos**
- **STEPPER**
- **Tension**

I/O Configuration

- **5069 Backplane**
  - **[0] 5069-L310ERMS2 DUNEW2PLC1**
  - **[1] 5069-IB16/A**
  - **[2] 5069-IB16/A**
  - **[3] 5069-OB16/A**
  - **[4] 5069-IF8/A**
  - **[5] 5069-OF8/A**
  - **[6] 5069-IB16/A**
  - **[7] 5069-OB16/A**
- **A1/A2, Ethernet**
  - **5069-L310ERMS2 DUNEW2PLC1**
  - **1783-BMS10CGP DUNEW2NETSW**
  - **1734-AENT/C DUNEW2PLC2**
    - **PointIO 3 Slot Chassis**
      - **[0] 1734-AENT/C DUNEW2PLC2**
      - **[1] 1734-IB8S/B**
      - **[2] 1734-OB8S/B**
  - **2198-P031 DUNEW2AMP**
  - **2198-D006-ERS4 DUNEW2SERVO1**
  - **2198-D006-ERS4 DUNEW2SERVO2**
  - **In-Sight 8000 Series DUNE2CAM1**

General

Vendor:	Rockwell Automation/Allen-Bradley		
Revision:	33.011	Mode:	Remote Run
Chassis Type:	<none>	Key Switch Position:	REM
Slot:	0	Created:	8/21/2023 3:14:37 PM
Safety Partner Slot:	<internal>	Edited:	3/13/2026 11:14:07 AM
EtherNet/IP Mode:	A1/A2: Linear/DLR		

Date/Time

Date and Time:	8/3/1972 8:51:46 PM
Time Zone:	(UTC-06:00) Central Time (US & Canada)
Daylight Saving (+01:00):	Yes
Enable Time Synchronization:	Yes
Is the system time master:	On
Is a synchronized time slave:	Off
Duplicate CST Master Detected:	Off
CST Mastership disabled:	Off
No CST Master:	Off

Advanced Time Sync

CIP Sync Time Synchronization:	Enabled
UTC System Time:	8/4/1972 01:51:51 AM

Grandmaster Clock

User Name:	DUNEW2PLC1
User Location:	
Protocol Address:	00.00.00.00
Physical Address:	00-00-00-00
Clock Type:	Boundary, End-To-end Transparent
Manufacturer Name:	Rockwell Automation
Model:	5069-L310ERMS2/B
Serial Number:	D02827B7
Hardware Revision:	1.1
Firmware Revision:	33.11.77
Software Revision:	
Profile Identity:	00-21-6C-00-01-00
Physical Protocol:	Backplane
Network Protocol:	Local
Port Number:	1
Identity:	5C8816FFFEABE140
Class:	248
Accuracy:	254
Variance:	65535
Source:	Oscillator
Priority 1:	128
Priority 2:	128

Local Clock

Synchronization Status:	Master
Offset from Master:	0 ns
Backplane State:	Master (Port 1)
A1/A2, Ethernet State:	Master (Port 2)
Identity:	5C8816FFFEABE140
Class:	248
Accuracy:	254
Variance:	65535
Source:	Oscillator
Priority 1:	128 (Master Override)
Priority 2:	128 (Tie Breaker)

Advanced

Controller Fault Handler:	<none>	Allow Consumed Tags to Use RPI Provided by Producer:	No
Power-Up Handler:	<none>	Report Overflow Faults:	No
Match Project To Controller:	No	Enable Automatic Diagnostics:	No
Serial Number:	D02827B7		

SFC Execution

Execution Control:	Execute current active steps only	Last Scan of Active Step:	Don't scan
Restart Position:	Restart at most recently executed step		

Nonvolatile Memory

<Empty>

Safety

Safety Application:	Unlocked
Safety Status:	Safety Task OK
Safety Signature:	<none>
Protect Signature in Run Mode:	No
When replacing Safety I/O:	Configure Only When No Safety Signature Exists.
Safety Network Numbers:	
5069 Backplane	4555_0322_E9A5 8/5/2020 9:36:59.685 AM
A1/A2, Ethernet	448F_0298_455C 1/20/2020 6:05:33.660 AM

Memory

Standard Capacity:

Total:	1,048,576 blocks
Available:	740,625 blocks
Used:	307,951 blocks

Safety Capacity:

Total:	524,288 blocks
Available:	505,038 blocks
Used:	19,250 blocks

Security

Primary Security Authority:	No Protection	Restrict Communications Except Through Selected Slots:	No
Use only the selected Security Authority for Authentication and Authorization:	No	Selected Slots:	
Secondary Security Authority:	No Protection	Changes To Detect:	16#ffff_ffff_ffff_ffff
Use only the selected Security Authority for Authentication and Authorization:	No	Audit Value:	16#3997_60d3_95a6_ca81
Permission Set:		Enable Controller Web Pages:	No

Port Configuration

Port A1

Enable:	Yes

Port A2

Enable:	Yes

Advanced Network

Network Topology:	Linear/Star
Supervisor Mode:	Disabled

Name	Value	Data Type	Scope
AbortQueue	0	BOOL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
AbortQueue - motionQueue/queue_line_arc - 119(XIC), 120(XIC), 121(XIC), 122(XIC), 123(XIC), 124(XIC), 125(XIC), 126(XIC), 127(XIC), 128(XIC), 129(XIC), 130(XIC), 131(XIC), 132(XIC), 133(XIC), 134(XIC), 135(XIC), 136(XIC), 137(XIC), 138(XIC), 139(XIC), 140(XIC), 141(XIC), 142(XIC), 143(XIC), 144(XIC), 145(XIC)			
accel_xy	0.0	REAL	DUNEW2PLC1
Tangential acceleration of the winding head in xy-plane			
External Access:	Read/Write		
*accel_xy - MainProgram/MainRoutine - *57(CPT)*			
accel_xy - MoveXY_State_2_3/MoveXY - 9(CPT), 9(CPT)			
ActiveSeq	0	DINT	DUNEW2PLC1
External Access:	Read/Write		
*ActiveSeq - motionQueue/queue_line_arc - *112(MOV), *143(MOV), *46(MOV), *58(MOV)*			
ACTUATOR_POS	2	DINT	DUNEW2PLC1
External Access:	Read Only		
*ACTUATOR_POS - Latch_UnLatch_State_6_7_8/Latch_Driver - *2(CPT), *3(CPT), *4(CPT), *5(CPT), 12(CPT), 13(CMP), 13(CMP), 13(CMP), 13(NEQ)*			
ACTUATOR_POS - MainProgram/MainRoutine - 59(EQU), 59(EQU)			
ACTUATOR_POS - MoveZ_State_4_5/MoveZ - 0(EQU), 7(CMP)			
AFI	0	BOOL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
AFI - Latch_UnLatch_State_6_7_8/Latch_Driver - 22(XIC), 35(XIC)			
ALL_EOT_GOOD	0	BOOL	DUNEW2PLC1
External Access:	Read Only		
ALL_EOT_GOOD - EOT_Trip_11/EOT_TRIP - 0(XIO), 22(XIC)			
*ALL_EOT_GOOD - MainProgram/MainRoutine - *15(OTE)*			
ALL_EOT_GOOD - MoveXY_State_2_3/MoveXY - 17(XIO)			
ALL_EOT_GOOD - MoveZ_State_4_5/MoveZ - 16(XIO)			
APA_IS_VERTICAL	1	BOOL	DUNEW2PLC1
External Access:	Read Only		
*APA_IS_VERTICAL - MainProgram/MainRoutine - *23(OTE)*			
APA_IS_VERTICAL - MoveXY_State_2_3/MoveXY - 1(XIC), 1(XIO)			
AXIS_STAT		MOTION_INSTRUCTION[32]	EOT_Trip_11
External Access:	Read Only		
AXIS_STAT[0]		MOTION_INSTRUCTION	
*AXIS_STAT[0] - EOT_Trip_11/EOT_TRIP - *7(MAM)*			
AXIS_STAT[0].PC	1	BOOL	
AXIS_STAT[0].PC - EOT_Trip_11/EOT_TRIP - 8(XIC)			
AXIS_STAT[1]		MOTION_INSTRUCTION	
*AXIS_STAT[1] - EOT_Trip_11/EOT_TRIP - *10(MAM)*			
AXIS_STAT[1].PC	0	BOOL	
AXIS_STAT[1].PC - EOT_Trip_11/EOT_TRIP - 11(XIC)			
AXIS_STAT[2]		MOTION_INSTRUCTION	
*AXIS_STAT[2] - EOT_Trip_11/EOT_TRIP - *13(MAM)*			
AXIS_STAT[2].PC	0	BOOL	
AXIS_STAT[2].PC - EOT_Trip_11/EOT_TRIP - 14(XIC)			
AXIS_STAT[3]		MOTION_INSTRUCTION	
*AXIS_STAT[3] - EOT_Trip_11/EOT_TRIP - *16(MAM)*			
AXIS_STAT[3].PC	0	BOOL	
AXIS_STAT[3].PC - EOT_Trip_11/EOT_TRIP - 17(XIC)			
AXIS_STAT[4]		MOTION_INSTRUCTION	
*AXIS_STAT[4] - EOT_Trip_11/EOT_TRIP - *19(MAM)*			
AXIS_STAT[4].PC	0	BOOL	
AXIS_STAT[4].PC - EOT_Trip_11/EOT_TRIP - 20(XIC)			

AXIS_STAT (Continued)

AXIS_STAT[28] MOTION_INSTRUCTION
*AXIS_STAT[28] - EOT_Trip_11/EOT_TRIP - *0(MCS)*

AXIS_STAT[29] MOTION_INSTRUCTION
*AXIS_STAT[29] - EOT_Trip_11/EOT_TRIP - *5(MSO)*

AXIS_STAT[30] MOTION_INSTRUCTION
*AXIS_STAT[30] - EOT_Trip_11/EOT_TRIP - *5(MSO)*

AXIS_STAT[31] MOTION_INSTRUCTION
*AXIS_STAT[31] - EOT_Trip_11/EOT_TRIP - *5(MSO)*

blink_off TIMER DUNEW2PLC1
External Access: Read/Write
*blink_off - MainProgram/MainRoutine - *40(TON)*

blink_off.DN 0 BOOL
blink_off.DN - MainProgram/MainRoutine - 41(XIC)

blink_on TIMER DUNEW2PLC1
External Access: Read/Write
*blink_on - MainProgram/MainRoutine - *39(TON), *41(RES)*

blink_on.TT 1 BOOL
blink_on.TT - MainProgram/MainRoutine - 36(XIC), 37(XIC)

blink_on.DN 0 BOOL
blink_on.DN - MainProgram/MainRoutine - 40(XIC)

CAM_F_EN 1 BOOL DUNEW2PLC1
Constant No
External Access: Read/Write
CAM_F_EN - Camera/CameraControl - 0(XIC)

Cam_F_InspectionRes_0 REAL[32] Camera
External Access: Read Only
*Cam_F_InspectionRes_0 - Camera/CameraControl - *12(FFL), *19(FFU)*

Cam_F_InspectionRes_1 REAL[10] Camera
External Access: Read Only
*Cam_F_InspectionRes_1 - Camera/CameraControl - *13(FFL), *20(FFU)*

Cam_F_InspectionRes_2 REAL[10] Camera
External Access: Read Only
*Cam_F_InspectionRes_2 - Camera/CameraControl - *14(FFL), *21(FFU)*

Cam_F_InspectionRes_3 REAL[10] Camera
External Access: Read Only
*Cam_F_InspectionRes_3 - Camera/CameraControl - *15(FFL), *22(FFU)*

CAM_F_TRIGGER 1 BOOL DUNEW2PLC1
Constant No
External Access: Read/Write
CAM_F_TRIGGER - Camera/CameraControl - 1(XIC), 2(XIC), 3(XIC)

check_tension_stable 1 BOOL DUNEW2PLC1
Check if the tension is stable at the setpoint before moving the winding head
Constant No
External Access: Read/Write
check_tension_stable - MoveXY_State_2_3/MoveXY - 1(XIO)
check_tension_stable - MoveZ_State_4_5/MoveZ - 6(XIO)

CheckCurSeg 0 BOOL motionQueue
External Access: Read/Write
*CheckCurSeg - motionQueue/queue_line_arc - *136(OTU), *21(OTU), *33(OTL), 14(XIC), 15(XIC), 16(XIC), 17(XIC), 18(XIC), 19(XIC), 20(XIC), 21(XIC)*

CheckNextSeg 0 BOOL motionQueue
External Access: Read/Write

CheckNextSeg (Continued)
*CheckNextSeg - motionQueue/queue_line_arc - *137(OTU), *29(OTU), *70(OTL), 22(XIC), 23(XIC), 24(XIC), 25(XIC), 26(XIC), 27(XIC), 28(XIC), 29(XIC)*

CmdA_Accel 2000.0 REAL motionQueue
External Access: Read/Write
*CmdA_Accel - motionQueue/queue_line_arc - *37(MOV), *86(MOV), 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM)*

CmdA_CircleType 1 DINT motionQueue
External Access: Read/Write
*CmdA_CircleType - motionQueue/queue_line_arc - *43(MOV), *92(MOV), 101(MCCM), 62(MCCM)*

CmdA_Decel 2000.0 REAL motionQueue
External Access: Read/Write
*CmdA_Decel - motionQueue/queue_line_arc - *38(MOV), *87(MOV), 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM)*

CmdA_Direction 1 DINT motionQueue
External Access: Read/Write
*CmdA_Direction - motionQueue/queue_line_arc - *45(MOV), *94(MOV), 101(MCCM), 62(MCCM)*

CmdA_JerkAccel 20000.0 REAL motionQueue
External Access: Read/Write
*CmdA_JerkAccel - motionQueue/queue_line_arc - *39(MOV), *88(MOV), 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM)*

CmdA_JerkDecel 20000.0 REAL motionQueue
External Access: Read/Write
*CmdA_JerkDecel - motionQueue/queue_line_arc - *40(MOV), *89(MOV), 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM)*

CmdA_SegType 1 DINT motionQueue
External Access: Read/Write
*CmdA_SegType - motionQueue/queue_line_arc - *42(MOV), *91(MOV), 100(EQU), 101(EQU), 61(EQU), 62(EQU)*

CmdA_Speed 437.4951 REAL motionQueue
External Access: Read/Write
*CmdA_Speed - motionQueue/queue_line_arc - *36(MOV), *85(MOV), 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM)*

CmdA_TermType 4 SINT motionQueue
External Access: Read/Write
*CmdA_TermType - motionQueue/queue_line_arc - *41(MOV), *90(MOV), 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM)*

CmdA_ViaCenter REAL[2] motionQueue
External Access: Read/Write
*CmdA_ViaCenter - motionQueue/queue_line_arc - *44(COP), *93(COP)*
CmdA_ViaCenter[0] 0.0 REAL
CmdA_ViaCenter[0] - motionQueue/queue_line_arc - 101(MCCM), 62(MCCM)

CmdA_XY REAL[2] motionQueue
External Access: Read/Write
*CmdA_XY - motionQueue/queue_line_arc - *35(COP), *84(COP), 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM)*

CmdB_Accel 2000.0 REAL motionQueue
External Access: Read/Write
*CmdB_Accel - motionQueue/queue_line_arc - *49(MOV), *74(MOV), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)*

CmdB_CircleType 1 DINT motionQueue
External Access: Read/Write
*CmdB_CircleType - motionQueue/queue_line_arc - *55(MOV), *80(MOV), 64(MCCM), 99(MCCM)*

CmdB_Decel 2000.0 REAL motionQueue
External Access: Read/Write
*CmdB_Decel - motionQueue/queue_line_arc - *50(MOV), *75(MOV), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)*

CmdB_Direction 1 DINT motionQueue
External Access: Read/Write

CmdB_Direction (Continued)
*CmdB_Direction - motionQueue/queue_line_arc - *57(MOV), *82(MOV), 64(MCCM), 99(MCCM)*

CmdB_JerkAccel 20000.0 REAL motionQueue
External Access: Read/Write
*CmdB_JerkAccel - motionQueue/queue_line_arc - *51(MOV), *76(MOV), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)*

CmdB_JerkDecel 20000.0 REAL motionQueue
External Access: Read/Write
*CmdB_JerkDecel - motionQueue/queue_line_arc - *52(MOV), *77(MOV), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)*

CmdB_SegType 1 DINT motionQueue
External Access: Read/Write
*CmdB_SegType - motionQueue/queue_line_arc - *54(MOV), *79(MOV), 63(EQU), 64(EQU), 98(EQU), 99(EQU)*

CmdB_Speed 437.4951 REAL motionQueue
External Access: Read/Write
*CmdB_Speed - motionQueue/queue_line_arc - *48(MOV), *73(MOV), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)*

CmdB_TermType 4 SINT motionQueue
External Access: Read/Write
*CmdB_TermType - motionQueue/queue_line_arc - *53(MOV), *78(MOV), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)*

CmdB_ViaCenter REAL[2] motionQueue
External Access: Read/Write
*CmdB_ViaCenter - motionQueue/queue_line_arc - *56(COP), *81(COP)*
CmdB_ViaCenter[0] 0.0 REAL
CmdB_ViaCenter[0] - motionQueue/queue_line_arc - 64(MCCM), 99(MCCM)

CmdB_XY REAL[2] motionQueue
External Access: Read/Write
*CmdB_XY - motionQueue/queue_line_arc - *47(COP), *72(COP), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)*

CmdTolerance 10 DINT motionQueue
Constant No
External Access: Read/Write
CmdTolerance - motionQueue/queue_line_arc - 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)

constant_cv_out 0 BOOL PID_Tension_Servo
Constant No
External Access: Read/Write
constant_cv_out - PID_Tension_Servo/PID_Tension - 7(XIC)

CurIssued 0 BOOL DUNEW2PLC1
External Access: Read/Write
*CurIssued - motionQueue/queue_line_arc - *117(OTU), *118(OTU), *120(OTU), *65(OTL), 104(XIC), 105(XIC), 117(XIC), 118(XIC), 30(XIO), 67(XIC), 68(XIC)*

current_command 0.0 REAL DUNEW2PLC1
External Access: Read/Write
*current_command - PID_Tension_Servo/PID_Tension - *9(CPT)*

current_command_high 1.5 REAL DUNEW2PLC1
Constant Yes
External Access: Read/Write
current_command_high - PID_Tension_Servo/PID_Tension - 10(CPT), 9(CPT)

current_command_low -0.5 REAL DUNEW2PLC1
Constant Yes
External Access: Read/Write
current_command_low - PID_Tension_Servo/PID_Tension - 10(CPT), 9(CPT)

CurSeg MotionSeg motionQueue

CurSeg (Continued)
motion segments for move queue, lines or arcs
External Access: Read/Write
*CurSeg - motionQueue/queue_line_arc - *106(COP), *32(FFU)*

CurSeg.Valid	0	BOOL	
motion segments for move queue, lines or arcs segment valid flag			
*CurSeg.Valid - motionQueue/queue_line_arc - *138(OTU), 14(XIC), 15(XIO), 16(XIO), 17(XIC), 18(XIC), 19(XIC), 20(XIC)*			
CurSeg.SegType	1	SINT	
motion segments for move queue, lines or arcs 1=line, 2=arc			
CurSeg.SegType - motionQueue/queue_line_arc - 14(EQU), 14(EQU), 19(GRT), 19(LES), 20(GRT), 20(LES), 42(MOV), 54(MOV)			
CurSeg.XY		REAL	
motion segments for move queue, lines or arcs end xy			
CurSeg.XY - motionQueue/queue_line_arc - 35(COP), 47(COP)			
CurSeg.XY[0]	5990.0	REAL	
motion segments for move queue, lines or arcs end xy			
CurSeg.XY[1]	1250.0	REAL	
motion segments for move queue, lines or arcs end xy			
CurSeg.Speed	437.4951	REAL	
motion segments for move queue, lines or arcs >0			
CurSeg.Speed - motionQueue/queue_line_arc - 36(MOV), 48(MOV)			
CurSeg.Accel	2000.0	REAL	
motion segments for move queue, lines or arcs >0			
CurSeg.Accel - motionQueue/queue_line_arc - 37(MOV), 49(MOV)			
CurSeg.Decel	2000.0	REAL	
motion segments for move queue, lines or arcs >0			
CurSeg.Decel - motionQueue/queue_line_arc - 38(MOV), 50(MOV)			
CurSeg.JerkAccel	20000.0	REAL	
motion segments for move queue, lines or arcs			
CurSeg.JerkAccel - motionQueue/queue_line_arc - 39(MOV), 51(MOV)			
CurSeg.JerkDecel	20000.0	REAL	
motion segments for move queue, lines or arcs			
CurSeg.JerkDecel - motionQueue/queue_line_arc - 40(MOV), 52(MOV)			
CurSeg.TermType	4	SINT	
motion segments for move queue, lines or arcs 0 to 6			
CurSeg.TermType - motionQueue/queue_line_arc - 41(MOV), 53(MOV)			
CurSeg.Seq	0	DINT	
motion segments for move queue, lines or arcs Sequence/ack id			
*CurSeg.Seq - motionQueue/queue_line_arc - *139(MOV), 14(GRT), 17(LEQ), 18(LEQ), 19(GRT), 20(GRT), 46(MOV), 58(MOV)*			
CurSeg.CircleType	1	DINT	
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum			
CurSeg.CircleType - motionQueue/queue_line_arc - 43(MOV), 55(MOV)			
CurSeg.ViaCenter		REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
CurSeg.ViaCenter - motionQueue/queue_line_arc - 44(COP), 56(COP)			
CurSeg.ViaCenter[0]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
CurSeg.ViaCenter[1]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
CurSeg.Direction	1	DINT	
motion segments for move queue, lines or arcs 0..3 per mmcm enum			
CurSeg.Direction - motionQueue/queue_line_arc - 45(MOV), 57(MOV)			

cv_to_electrocraft	2.5	REAL	PID_Tension_Servo
Tension_Servo_CMD			
AliasFor:	Local:5:O.Ch00.Data(C)		
Base Tag:	Local:5:O.Ch00.Data(C)		
Constant	No		
External Access:	Read/Write		
*cv_to_electrocraft - PID_Tension_Servo/PID_Tension - *7(MOV), 9(CPT)*			

d_raw	0.0	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*d_raw - MoveXY_State_2_3/xy_speed_regulator - *4(CPT), 4(CPT)*			

Delay_Fixed_Latched		TIMER	Latch_UnLatch_State_6_7_8
External Access:	Read/Write		
*Delay_Fixed_Latched - Latch_UnLatch_State_6_7_8/Latch_Driver - *6(TON)*			
Delay_Fixed_Latched.DN	0	BOOL	
Delay_Fixed_Latched.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 6(XIC)			
delay_mid_position		TIMER	DUNEW2PLC1
External Access:	Read/Write		
*delay_mid_position - Latch_UnLatch_State_6_7_8/Latch_Driver - *3(TON)*			
delay_mid_position.DN	1	BOOL	
delay_mid_position.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 3(XIC)			
Delay_Z_Latched		TIMER	Latch_UnLatch_State_6_7_8
External Access:	Read/Write		
*Delay_Z_Latched - Latch_UnLatch_State_6_7_8/Latch_Driver - *5(TON)*			
Delay_Z_Latched.DN	0	BOOL	
Delay_Z_Latched.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 5(XIC)			
DoneONS_A	0	BOOL	motionQueue
External Access:	Read/Write		
*DoneONS_A - motionQueue/queue_line_arc - *117(ONS)*			
DoneONS_B	0	BOOL	motionQueue
External Access:	Read/Write		
*DoneONS_B - motionQueue/queue_line_arc - *118(ONS)*			
dont_auto_trigger_axes_in_state_1	1	BOOL	Ready_State_1
Constant	No		
External Access:	Read/Write		
dont_auto_trigger_axes_in_state_1 - Ready_State_1/Ready - 2(XIO)			
DUNE2CAM1:I		CC:InSight11_DINT8:I:0	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
DUNE2CAM1:I.Status.InspectionCompleted	1	BOOL	
DUNE2CAM1:I.Status.InspectionCompleted - Camera/CameraControl - 11(XIC)			
DUNE2CAM1:I.InspectionResults[0]	0	DINT	
DUNE2CAM1:I.InspectionResults[0] - Camera/CameraControl - 12(FFL)			
DUNE2CAM1:I.InspectionResults[1]	0	DINT	
DUNE2CAM1:I.InspectionResults[1] - Camera/CameraControl - 13(FFL)			
DUNE2CAM1:I.InspectionResults[2]	0	DINT	
DUNE2CAM1:I.InspectionResults[2] - Camera/CameraControl - 14(FFL)			
DUNE2CAM1:I.InspectionResults[3]	0	DINT	
DUNE2CAM1:I.InspectionResults[3] - Camera/CameraControl - 15(FFL)			
DUNE2CAM1:O		CC:InSight11_DINT8:O:0	DUNEW2PLC1
External Access:	Read/Write		
DUNE2CAM1:O.Control.TriggerEnable	1	BOOL	
*DUNE2CAM1:O.Control.TriggerEnable - Camera/CameraControl - *0(OTE)*			
DUNE2CAM1:O.Control.Trigger	1	BOOL	
*DUNE2CAM1:O.Control.Trigger - Camera/CameraControl - *4(OTE)*			
DUNEW2PLC2:1:I		AB:1734_IB8S_Safety2:I:0	DUNEW2PLC1
Constant	No		

DUNEW2PLC2:1:I (Continued)
External Access: Read/Write
DUNEW2PLC2:1:I.Pt00Data 1 BOOL
LC OSS1
DUNEW2PLC2:1:I.Pt00Data - MainProgram/MainRoutine - 24(XIC)
DUNEW2PLC2:1:I.Pt00Data - Safety /Safety - 1(XIO)
DUNEW2PLC2:1:I.Pt01Data 1 BOOL
LC OSS2
DUNEW2PLC2:1:I.Pt01Data - MainProgram/MainRoutine - 24(XIC)
DUNEW2PLC2:1:I.Pt01Data - Safety /Safety - 1(XIO)
DUNEW2PLC2:1:I.Pt02Data 1 BOOL
EStop Rack
DUNEW2PLC2:1:I.Pt02Data - MainProgram/MainRoutine - 1(XIO)
DUNEW2PLC2:1:I.Pt02Data - Safety /Safety - 0(XIC)
DUNEW2PLC2:1:I.Pt03Data 1 BOOL
EStop Pullcord 1
DUNEW2PLC2:1:I.Pt03Data - MainProgram/MainRoutine - 1(XIO)
DUNEW2PLC2:1:I.Pt03Data - Safety /Safety - 0(XIC)
DUNEW2PLC2:1:I.Pt04Data 1 BOOL
EStop Pullcord 2
DUNEW2PLC2:1:I.Pt04Data - MainProgram/MainRoutine - 1(XIO)
DUNEW2PLC2:1:I.Pt04Data - Safety /Safety - 0(XIC)
DUNEW2PLC2:1:I.Pt05Data 1 BOOL
Run Key
DUNEW2PLC2:1:I.Pt05Data - Safety /Safety - 1(XIO), 2(XIC)
DUNEW2PLC2:1:I.Pt06Data 0 BOOL
DUNEW2PLC2:1:I.Pt06Data - MainProgram/MainRoutine - 58(XIO)
DUNEW2PLC2:1:I.Pt06Data - Safety /Safety - 6(XIO), 7(XIO)

DUNEW2SERVO1:SI AB:CIP_Motion_Device_Safety2:SI:0 DUNEW2PLC1
Constant No
External Access: Read/Write
DUNEW2SERVO1:SI.ConnectionFaulted
0 BOOL
DUNEW2SERVO1:SI.ConnectionFaulted - Safety /Safety - 6(SFX)
DUNEW2SERVO1:SI.FeedbackPosition1
-13 DINT
DUNEW2SERVO1:SI.FeedbackPosition1 - Safety /Safety - 6(SFX)
DUNEW2SERVO1:SI.FeedbackVelocity1
0.0 REAL
DUNEW2SERVO1:SI.FeedbackVelocity1 - Safety /Safety - 6(SFX)
DUNEW2SERVO1:SI.PrimaryFeedbackValid1
1 BOOL
DUNEW2SERVO1:SI.PrimaryFeedbackValid1 - Safety /Safety - 6(SFX), 6(XIC)

DUNEW2SERVO1:SO AB:CIP_Motion_Device_Safety2:SO:0 DUNEW2PLC1
External Access: Read/Write
DUNEW2SERVO1:SO.SLSActive1 0 BOOL
*DUNEW2SERVO1:SO.SLSActive1 - Safety /Safety - *7(SLS), 8(XIC)*
DUNEW2SERVO1:SO.SLSLimit1 0 BOOL
*DUNEW2SERVO1:SO.SLSLimit1 - Safety /Safety - *7(SLS)*
DUNEW2SERVO1:SO.SFHomed1 0 BOOL
*DUNEW2SERVO1:SO.SFHomed1 - Safety /Safety - *6(SFX)*
DUNEW2SERVO1:SO.SFXFault1 0 BOOL
*DUNEW2SERVO1:SO.SFXFault1 - Safety /Safety - *6(SFX)*
DUNEW2SERVO1:SO.SLSFault1 0 BOOL
*DUNEW2SERVO1:SO.SLSFault1 - Safety /Safety - *7(SLS)*
DUNEW2SERVO1:SO.STOOutput1
1 BOOL
*DUNEW2SERVO1:SO.STOOutput1 - Safety /Safety - *4(OTE)*
DUNEW2SERVO1:SO.STOOutput2
1 BOOL

DUNEW2SERVO1:SO (Continued)
*DUNEW2SERVO1:SO.STOOutput2 - Safety /Safety - *4(OTE)*

DUNEW2SERVO2:SO		AB:CIP_Motion_Device_Safety2:SO:0	DUNEW2PLC1
External Access:	Read/Write		
DUNEW2SERVO2:SO.STOOutput1	1	BOOL	
*DUNEW2SERVO2:SO.STOOutput1 - Safety /Safety - *4(OTE)*			
dx	1.22070312e-004	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*dx - MoveXY_State_2_3/MoveXY - *7(CPT), 7(CPT), 7(CPT)*			
dy	4.8001003	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*dy - MoveXY_State_2_3/MoveXY - *7(CPT), 7(CPT), 7(CPT)*			
EN_POS_TRIGGERS	0	BOOL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
EN_POS_TRIGGERS - Camera/CameraControl - 24(XIC), 28(XIC), 29(XIC), 30(XIC), 31(XIC), 32(XIC), 33(XIC), 34(XIC)			
ENABLE_ACTUATOR	0	BOOL	Latch_UnLatch_State_6_7_8
External Access:	Read Only		
*ENABLE_ACTUATOR - Latch_UnLatch_State_6_7_8/Latch_Driver - *7(OTE), 11(XIC), 9(XIO)*			
Enable_tension_motor	0	BOOL	PID_Tension_Servo
AliasFor:	Local:3:O.Pt03.Data(C)		
Base Tag:	Local:3:O.Pt03.Data(C)		
Constant	No		
External Access:	Read/Write		
*Enable_tension_motor - PID_Tension_Servo/PID_Tension - *2(OTE)*			
EnqueueReq	0	BOOL	motionQueue
External Access:	Read/Write		
*EnqueueReq - motionQueue/queue_line_arc - *10(OTU), *128(OTU), *3(OTL), 10(XIC), 4(XIC), 5(XIC), 7(XIC), 8(XIC), 9(XIC)*			
eot_inc		TIMER	DUNEW2PLC1
External Access:	Read/Write		
*eot_inc - EOT_Trip_11/EOT_TRIP - *2(TON), *3(RES)*			
eot_inc.DN	1	BOOL	
eot_inc.DN - EOT_Trip_11/EOT_TRIP - 1(XIC), 12(XIC), 15(XIC), 18(XIC), 6(XIC), 9(XIO)			
eot_pause		TIMER	DUNEW2PLC1
External Access:	Read/Write		
*eot_pause - EOT_Trip_11/EOT_TRIP - *1(TON)*			
eot_pause.DN	0	BOOL	
eot_pause.DN - EOT_Trip_11/EOT_TRIP - 3(XIC)			
eot_triggered	0	BOOL	MoveXY_State_2_3
External Access:	Read/Write		
*eot_triggered - MoveXY_State_2_3/MoveXY - *17(OSR), 18(XIC)*			
ERROR_CODE	0	DINT	DUNEW2PLC1
External Access:	Read Only		

*ERROR_CODE - EOT_Trip_11/EOT_TRIP - *22(CPT)*
*ERROR_CODE - Error_State_10/Error - *12(CPT)*
*ERROR_CODE - Initialize/Init - *4(CPT)*
*ERROR_CODE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *25(CPT), *33(CPT), *36(CPT), *51(CPT)*
ERROR_CODE - MainProgram/MainRoutine - 36(NEQ)
*ERROR_CODE - MoveXY_State_2_3/MoveXY - *1(CPT), *1(CPT), *15(CPT), *2(CPT), *2(CPT)*

ERROR_CODE (Continued)
*ERROR_CODE - MoveZ_State_4_5/MoveZ - *10(CPT), *13(CPT), *20(CPT), *4(CPT), *4(CPT), *7(CPT)*
*ERROR_CODE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT), *16(CPT), 14(NEQ), 16(EQU), 16(EQU)*
*ERROR_CODE - Ready_State_1/Ready - *3(MOV)*

error_servo_off	0	BOOL	Error_State_10
Constant	No		
External Access:	Read/Write		

error_servo_off - Error_State_10/Error - 5(XIC), 6(XIC), 7(XIC)

ESTOP	1	BOOL	DUNEW2PLC1
EStop Rack			
AliasFor:	DUNEW2PLC2:1:I.Pt02Data		
Base Tag:	DUNEW2PLC2:1:I.Pt02Data		
Constant	No		
External Access:	Read/Write		

DUNEW2PLC2:1:I.Pt02Data - MainProgram/MainRoutine - 1(XIO)
DUNEW2PLC2:1:I.Pt02Data - Safety /Safety - 0(XIC)

FaultCode	0	DINT	motionQueue
External Access:	Read/Write		

*FaultCode - motionQueue/queue_line_arc - *11(MOV), *12(MOV), *134(MOV), *15(MOV), *17(MOV), *19(MOV), *23(MOV), *25(MOV), *27(MOV)*

FIFO_CTRL		CONTROL[6]	Camera
External Access:	Read Only		
FIFO_CTRL[0]		CONTROL	
*FIFO_CTRL[0] - Camera/CameraControl - *17(FFU), *9(FFL)*			
FIFO_CTRL[0].POS	0	DINT	
*FIFO_CTRL[0].POS - Camera/CameraControl - *35(MOV)*			
FIFO_CTRL[0].EU	0	BOOL	
FIFO_CTRL[0].EU - Camera/CameraControl - 18(XIC)			
FIFO_CTRL[1]		CONTROL	
*FIFO_CTRL[1] - Camera/CameraControl - *10(FFL), *18(FFU)*			
FIFO_CTRL[1].POS	0	DINT	
*FIFO_CTRL[1].POS - Camera/CameraControl - *35(MOV)*			
FIFO_CTRL[1].EU	0	BOOL	
FIFO_CTRL[1].EU - Camera/CameraControl - 19(XIC)			
FIFO_CTRL[2]		CONTROL	
*FIFO_CTRL[2] - Camera/CameraControl - *12(FFL), *19(FFU)*			
FIFO_CTRL[2].POS	0	DINT	
*FIFO_CTRL[2].POS - Camera/CameraControl - *35(MOV)*			
FIFO_CTRL[2].EU	0	BOOL	
FIFO_CTRL[2].EU - Camera/CameraControl - 20(XIC)			
FIFO_CTRL[3]		CONTROL	
*FIFO_CTRL[3] - Camera/CameraControl - *13(FFL), *20(FFU)*			
FIFO_CTRL[3].POS	0	DINT	
*FIFO_CTRL[3].POS - Camera/CameraControl - *35(MOV)*			
FIFO_CTRL[3].EU	0	BOOL	
FIFO_CTRL[3].EU - Camera/CameraControl - 21(XIC)			
FIFO_CTRL[4]		CONTROL	
*FIFO_CTRL[4] - Camera/CameraControl - *14(FFL), *21(FFU)*			
FIFO_CTRL[4].POS	0	DINT	
*FIFO_CTRL[4].POS - Camera/CameraControl - *35(MOV)*			
FIFO_CTRL[4].EU	0	BOOL	
FIFO_CTRL[4].EU - Camera/CameraControl - 22(XIC)			
FIFO_CTRL[5]		CONTROL	
*FIFO_CTRL[5] - Camera/CameraControl - *15(FFL), *22(FFU)*			
FIFO_CTRL[5].POS	0	DINT	
*FIFO_CTRL[5].POS - Camera/CameraControl - *35(MOV)*			
FIFO_CTRL[5].EU	0	BOOL	
FIFO_CTRL[5].EU - Camera/CameraControl - 23(XIC)			
FIFO_CTRL[5].DN	0	BOOL	
FIFO_CTRL[5].DN - Camera/CameraControl - 16(XIC)			

FIFO_Data REAL[6] DUNEW2PLC1
External Access: Read Only
FIFO_Data[0] 0.0 REAL
*FIFO_Data[0] - Camera/CameraControl - *17(FFU)*
FIFO_Data[1] 0.0 REAL
*FIFO_Data[1] - Camera/CameraControl - *18(FFU)*
FIFO_Data[2] 0.0 REAL
*FIFO_Data[2] - Camera/CameraControl - *19(FFU)*
FIFO_Data[3] 0.0 REAL
*FIFO_Data[3] - Camera/CameraControl - *20(FFU)*
FIFO_Data[4] 0.0 REAL
*FIFO_Data[4] - Camera/CameraControl - *21(FFU)*
FIFO_Data[5] 0.0 REAL
*FIFO_Data[5] - Camera/CameraControl - *22(FFU)*

FIFO_FLUSH 0 BOOL DUNEW2PLC1
Flush the camera FIFOs.
External Access: Read/Write
*FIFO_FLUSH - Camera/CameraControl - *35(OTU), 35(XIC)*

FlipToA 0 BOOL motionQueue
External Access: Read/Write
*FlipToA - motionQueue/queue_line_arc - *108(OTL), *115(OTU), *130(OTU), 110(XIC), 115(XIC)*

FlipToB 0 BOOL motionQueue
External Access: Read/Write
*FlipToB - motionQueue/queue_line_arc - *107(OTL), *116(OTU), *131(OTU), 109(XIC), 116(XIC)*

foundIt 0 BOOL Camera
External Access: Read Only
*foundIt - Camera/CameraControl - *28(OTL), *33(OTU), 29(XIC)*

FRAME_LOC_FT_BTM 1 BOOL DUNEW2PLC1
External Access: Read/Write
*FRAME_LOC_FT_BTM - MainProgram/MainRoutine - *30(OTE)*
FRAME_LOC_FT_BTM - MoveZ_State_4_5/MoveZ - 2(XIO)

FRAME_LOC_FT_MID 0 BOOL DUNEW2PLC1
External Access: Read/Write
*FRAME_LOC_FT_MID - MainProgram/MainRoutine - *29(OTE)*
FRAME_LOC_FT_MID - MoveZ_State_4_5/MoveZ - 2(XIO)

FRAME_LOC_FT_TOP 1 BOOL DUNEW2PLC1
External Access: Read/Write
*FRAME_LOC_FT_TOP - MainProgram/MainRoutine - *28(OTE)*
FRAME_LOC_FT_TOP - MoveZ_State_4_5/MoveZ - 2(XIO)

FRAME_LOC_HD_BTM 1 BOOL DUNEW2PLC1
External Access: Read/Write
*FRAME_LOC_HD_BTM - MainProgram/MainRoutine - *27(OTE)*
FRAME_LOC_HD_BTM - MoveZ_State_4_5/MoveZ - 2(XIO)

FRAME_LOC_HD_MID 1 BOOL DUNEW2PLC1
External Access: Read/Write
*FRAME_LOC_HD_MID - MainProgram/MainRoutine - *26(OTE)*
FRAME_LOC_HD_MID - MoveZ_State_4_5/MoveZ - 2(XIO)

FRAME_LOC_HD_TOP 1 BOOL DUNEW2PLC1
External Access: Read/Write
*FRAME_LOC_HD_TOP - MainProgram/MainRoutine - *25(OTE)*
FRAME_LOC_HD_TOP - MoveZ_State_4_5/MoveZ - 2(XIO)

gamma 24.494896 REAL DUNEW2PLC1
External Access: Read/Write

gamma (Continued)
*gamma - MoveXY_State_2_3/MoveXY - *9(CPT), 9(CPT)*

HEAD_POS	-1	DINT	DUNEW2PLC1
External Access:	Read/Write		
*HEAD_POS - MainProgram/MainRoutine - *59(CPT), *59(CPT), *59(CPT)*			
HomeCounter		COUNTER	Latch_UnLatch_State_6_7_8
External Access:	Read Only		
*HomeCounter - Latch_UnLatch_State_6_7_8/Latch_Driver - *18(RES), *21(CTU)*			
HomeCounter.DN	0	BOOL	
HomeCounter.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 19(XIO), 20(XIO), 21(XIO), 22(XIC), 23(XIC)			
HomeSignal	0	BOOL	Latch_UnLatch_State_6_7_8
External Access:	Read Only		
*HomeSignal - Latch_UnLatch_State_6_7_8/Latch_Driver - *33(OTU)*			
HomeTimer1		TIMER	Latch_UnLatch_State_6_7_8
External Access:	Read Only		
*HomeTimer1 - Latch_UnLatch_State_6_7_8/Latch_Driver - *19(TON)*			
HomeTimer1.TT	0	BOOL	
HomeTimer1.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 20(XIO), 21(XIC)			
HomeTimer2		TIMER	Latch_UnLatch_State_6_7_8
External Access:	Read Only		
*HomeTimer2 - Latch_UnLatch_State_6_7_8/Latch_Driver - *20(TON)*			
HomeTimer2.TT	0	BOOL	
HomeTimer2.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 19(XIO), 44(XIC)			
IncomingSeg		MotionSeg	DUNEW2PLC1
motion segments for move queue, lines or arcs			
Constant	No		
External Access:	Read/Write		
IncomingSeg - motionQueue/queue_line_arc - 7(FFL)			
IncomingSeg.Valid	1	BOOL	
motion segments for move queue, lines or arcs segment valid flag			
IncomingSeg.Valid - motionQueue/queue_line_arc - 4(XIC), 5(XIC)			
IncomingSeg.SegType	1	SINT	
motion segments for move queue, lines or arcs 1=line, 2=arc			
IncomingSeg.SegType - motionQueue/queue_line_arc - 4(EQU), 5(EQU)			
IncomingSeg.XY		REAL	
motion segments for move queue, lines or arcs end xy			
IncomingSeg.XY[0]	5672.515	REAL	
motion segments for move queue, lines or arcs end xy			
IncomingSeg.XY[1]	444.68442	REAL	
motion segments for move queue, lines or arcs end xy			
IncomingSeg.Speed	437.4951	REAL	
motion segments for move queue, lines or arcs >0			
IncomingSeg.Speed - motionQueue/queue_line_arc - 4(GRT), 5(GRT)			
IncomingSeg.Accel	2000.0	REAL	
motion segments for move queue, lines or arcs >0			
IncomingSeg.Accel - motionQueue/queue_line_arc - 4(GRT), 5(GRT)			
IncomingSeg.Decel	2000.0	REAL	
motion segments for move queue, lines or arcs >0			
IncomingSeg.Decel - motionQueue/queue_line_arc - 4(GRT), 5(GRT)			
IncomingSeg.JerkAccel	20000.0	REAL	

IncomingSeg (Continued)
motion segments for move queue, lines or arcs

IncomingSeg.JerkDecel	20000.0	REAL	
motion segments for move queue, lines or arcs			
IncomingSeg.TermType	4	SINT	
motion segments for move queue, lines or arcs 0 to 6			
IncomingSeg.TermType - motionQueue/queue_line_arc - 4(GEQ), 4(LEQ), 5(GEQ), 5(LEQ)			
IncomingSeg.Seq	143	DINT	
motion segments for move queue, lines or arcs Sequence/ack id			
IncomingSeg.Seq - motionQueue/queue_line_arc - 8(MOV)			
IncomingSeg.CircleType	1	DINT	
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum			
IncomingSeg.CircleType - motionQueue/queue_line_arc - 5(GEQ), 5(LEQ)			
IncomingSeg.ViaCenter		REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
IncomingSeg.ViaCenter[0]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
IncomingSeg.ViaCenter[1]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
IncomingSeg.Direction	1	DINT	
motion segments for move queue, lines or arcs 0..3 per mmcm enum			
IncomingSeg.Direction - motionQueue/queue_line_arc - 5(GEQ), 5(LEQ)			
IncomingSegAck	142	DINT	DUNEW2PLC1
External Access:	Read/Write		
*IncomingSegAck - motionQueue/queue_line_arc - *8(MOV)*			
IncomingSegReqID	44	DINT	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
IncomingSegReqID - motionQueue/queue_line_arc - 142(MOV), 3(NEQ), 9(MOV)			
INIT_DONE	1	BOOL	DUNEW2PLC1
External Access:	Read Only		
INIT_DONE - Error_State_10/Error - 0(XIC)			
*INIT_DONE - Initialize/Init - *6(OTE), 7(XIC)*			
INIT_DONE - MainProgram/MainRoutine - 42(XIC), 43(XIC), 44(XIC), 45(XIC), 46(XIC), 47(XIC), 48(XIC), 49(XIC), 50(XIC), 51(XIC), 52(XIC), 53(XIC)			
INIT_DONE - Ready_State_1/Ready - 0(XIC)			
INIT_DONE - UnServo_9/UnServo_All - 0(XIC)			
INIT_SW	1	BOOL	Initialize
External Access:	Read/Write		
*INIT_SW - Initialize/Init - *0(OTU), *2(OTL), 0(XIC), 1(XIC), 2(XIO)*			
INPUTS2		AB:5000_DI16:I:0	DUNEW2PLC1
AliasFor:	Local:2:I		
Base Tag:	Local:2:I		
Constant	No		
External Access:	Read/Write		
INPUTS2.Pt06.Data	0	BOOL	
TENSION ON/OFF			
INPUTS2.Pt07.Data	1	BOOL	
O/T SWITCH OVERTRAVEL SWITCH			
INPUTS2.Pt13.Data	0	BOOL	
Cabinet lighted push button			
IssueCurPulse	0	BOOL	motionQueue
External Access:	Read/Write		
*IssueCurPulse - motionQueue/queue_line_arc - *126(OTU), *59(OTL), *66(OTU), 61(XIC), 62(XIC), 63(XIC), 64(XIC), 65(XIC), 66(XIC)*			
IssueNextPulse	0	BOOL	motionQueue
External Access:	Read/Write		
*IssueNextPulse - motionQueue/queue_line_arc - *103(OTU), *127(OTU), *96(OTL), 100(XIC), 101(XIC), 102(XIC), 103(XIC), 98(XIC), 99(XIC)*			

J 3000.0 REAL DUNEW2PLC1
External Access: Read/Write
*J - MoveXY_State_2_3/MoveXY - *9(MOV), 9(CPT), 9(CPT)*

k 124.99739 REAL MoveXY_State_2_3
External Access: Read/Write
*k - MoveXY_State_2_3/MoveXY - *7(CPT), *7(CPT), 7(CPT)*

LastIncomingSegReqID 44 DINT DUNEW2PLC1
External Access: Read/Write
*LastIncomingSegReqID - motionQueue/queue_line_arc - *142(MOV), *9(MOV), 3(NEQ)*

LATCH_ACTUATOR_HOMED 0 BOOL DUNEW2PLC1
External Access: Read Only
*LATCH_ACTUATOR_HOMED - Initialize/Init - *5(OTU)*
*LATCH_ACTUATOR_HOMED - Latch_UnLatch_State_6_7_8/Latch_Driver - *24(OTL), *25(OTL), *38(OTU), 26(XIC), 27(XIC)*
LATCH_ACTUATOR_HOMED - MoveZ_State_4_5/MoveZ - 7(XIC)

LATCH_ACTUATOR_MID 1 BOOL DUNEW2PLC1
External Access: Read/Write
*LATCH_ACTUATOR_MID - Latch_UnLatch_State_6_7_8/Latch_Driver - *1(OTE), 2(XIO), 3(XIC), 4(XIO)*

LATCH_ACTUATOR_TOP 1 BOOL DUNEW2PLC1
External Access: Read/Write
*LATCH_ACTUATOR_TOP - Latch_UnLatch_State_6_7_8/Latch_Driver - *0(OTE), 2(XIC), 3(XIC), 4(XIO)*

latch_bypass 1 BOOL MoveZ_State_4_5
Constant No
External Access: Read/Write
latch_bypass - MoveZ_State_4_5/MoveZ - 0(XIC)

latch_ouput 0 BOOL DUNEW2PLC1
Latching signal output
AliasFor: Local:3:O.Pt01.Data
Base Tag: Local:3:O.Pt01.Data
Constant No
External Access: Read/Write
*Local:3:O.Pt01.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - *44(OTE)*

LatchCounter2 COUNTER Latch_UnLatch_State_6_7_8
External Access: Read Only
*LatchCounter2 - Latch_UnLatch_State_6_7_8/Latch_Driver - *39(RES), *42(CTU)*
LatchCounter2.DN 1 BOOL
LatchCounter2.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 40(XIO), 41(XIO), 42(XIO), 43(XIC)

LatchCounter3 COUNTER Latch_UnLatch_State_6_7_8
External Access: Read Only
*LatchCounter3 - Latch_UnLatch_State_6_7_8/Latch_Driver - *28(RES), *31(CTU)*
LatchCounter3.DN 1 BOOL
LatchCounter3.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 29(XIO), 30(XIO), 31(XIO), 32(XIC)

Latching_pulse_duration TIMER Latch_UnLatch_State_6_7_8
External Access: Read Only
*Latching_pulse_duration - Latch_UnLatch_State_6_7_8/Latch_Driver - *14(TON)*
Latching_pulse_duration.TT 0 BOOL
Latching_pulse_duration.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 44(XIC)
Latching_pulse_duration.DN 0 BOOL
Latching_pulse_duration.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 15(XIC)

Latching_pulse_interval		TIMER	Latch_UnLatch_State_6_7_8
External Access:	Read/Write		
*Latching_pulse_interval - Latch_UnLatch_State_6_7_8/Latch_Driver - *15(TON)*			
Latching_pulse_interval.DN	0	BOOL	
Latching_pulse_interval.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 14(XIO)			
latching_signal	0	BOOL	Latch_UnLatch_State_6_7_8
External Access:	Read/Write		
*latching_signal - Latch_UnLatch_State_6_7_8/Latch_Driver - *44(OTE)*			
LatchingTimeoutCounter		COUNTER	Latch_UnLatch_State_6_7_8
When you enter state 6 or 7, start counting seconds			
External Access:	Read/Write		
*LatchingTimeoutCounter - Latch_UnLatch_State_6_7_8/Latch_Driver - *46(RES), *49(CTU)*			
LatchingTimeoutCounter.PRE	9999	DINT	
When you enter state 6 or 7, start counting seconds			
LatchingTimeoutCounter.ACC	1	DINT	
When you enter state 6 or 7, start counting seconds			
LatchingTimeoutCounter.CU	0	BOOL	
When you enter state 6 or 7, start counting seconds			
LatchingTimeoutCounter.CD	0	BOOL	
When you enter state 6 or 7, start counting seconds			
LatchingTimeoutCounter.DN	0	BOOL	
When you enter state 6 or 7, start counting seconds			
LatchingTimeoutCounter.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 47(XIO), 48(XIO), 49(XIO), 50(XIC)			
LatchingTimeoutCounter.OV	0	BOOL	
When you enter state 6 or 7, start counting seconds			
LatchingTimeoutCounter.UN	0	BOOL	
When you enter state 6 or 7, start counting seconds			
LatchTimer3		TIMER	Latch_UnLatch_State_6_7_8
External Access:	Read Only		
*LatchTimer3 - Latch_UnLatch_State_6_7_8/Latch_Driver - *40(TON)*			
LatchTimer3.TT	0	BOOL	
LatchTimer3.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 41(XIO), 42(XIC)			
LatchTimer4		TIMER	Latch_UnLatch_State_6_7_8
External Access:	Read Only		
*LatchTimer4 - Latch_UnLatch_State_6_7_8/Latch_Driver - *41(TON)*			
LatchTimer4.TT	0	BOOL	
LatchTimer4.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 40(XIO), 44(XIC)			
LatchTimer5		TIMER	Latch_UnLatch_State_6_7_8
External Access:	Read Only		
*LatchTimer5 - Latch_UnLatch_State_6_7_8/Latch_Driver - *29(TON)*			
LatchTimer5.TT	0	BOOL	
LatchTimer5.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 30(XIO), 31(XIC)			
LatchTimer6		TIMER	Latch_UnLatch_State_6_7_8
External Access:	Read Only		

LatchTimer6 (Continued)
*LatchTimer6 - Latch_UnLatch_State_6_7_8/Latch_Driver - *30(TON)*
LatchTimer6.TT 0 BOOL
LatchTimer6.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 29(XIO), 44(XIC)

LightRing TIMER Camera
External Access: Read Only
*LightRing - Camera/CameraControl - *32(TON)*
LightRing.TT 0 BOOL
LightRing.TT - Camera/CameraControl - 4(XIC), 5(XIC), 6(XIC)
LightRing.DN 0 BOOL
LightRing.DN - Camera/CameraControl - 33(XIC), 34(XIC)

LoadCurReq 0 BOOL motionQueue
External Access: Read/Write
*LoadCurReq - motionQueue/queue_line_arc - *122(OTU), *30(OTL), *34(OTU), 31(XIC), 32(XIC), 33(XIC), 34(XIC)*

LoadNextReq 0 BOOL motionQueue
External Access: Read/Write
*LoadNextReq - motionQueue/queue_line_arc - *123(OTU), *67(OTL), *68(OTL), *71(OTU), 69(XIC), 70(XIC), 71(XIC)*

Local:1:I AB:5000_DI16:I:0 DUNEW2PLC1
Constant No
External Access: Read/Write
Local:1:I.Pt00.Data 1 BOOL
Local:1:I.Pt00.Data - MainProgram/MainRoutine - 0(XIC)
Local:1:I.Pt01.Data 0 BOOL
Local:1:I.Pt01.Data - MainProgram/MainRoutine - 2(XIO)
Local:1:I.Pt02.Data 1 BOOL
Local:1:I.Pt02.Data - MainProgram/MainRoutine - 3(XIC)
Local:1:I.Pt03.Data 0 BOOL
Local:1:I.Pt03.Data - MainProgram/MainRoutine - 4(XIO)
Local:1:I.Pt04.Data 0 BOOL
Local:1:I.Pt04.Data - MainProgram/MainRoutine - 6(XIC)
Local:1:I.Pt06.Data 1 BOOL
Local:1:I.Pt06.Data - MainProgram/MainRoutine - 10(XIC)
Local:1:I.Pt07.Data 1 BOOL
Local:1:I.Pt07.Data - MainProgram/MainRoutine - 9(XIC)
Local:1:I.Pt08.Data 0 BOOL
Latch top
Local:1:I.Pt08.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - 0(XIO)
Local:1:I.Pt09.Data 0 BOOL
Latch middle
Local:1:I.Pt09.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - 1(XIO)
Local:1:I.Pt10.Data 1 BOOL
Local:1:I.Pt10.Data - MainProgram/MainRoutine - 16(XIO)
Local:1:I.Pt11.Data 0 BOOL
Local:1:I.Pt11.Data - MainProgram/MainRoutine - 7(XIC)
Local:1:I.Pt12.Data 0 BOOL
Local:1:I.Pt12.Data - MainProgram/MainRoutine - 18(XIC)
Local:1:I.Pt13.Data 0 BOOL
Local:1:I.Pt13.Data - MainProgram/MainRoutine - 19(XIC)
Local:1:I.Pt15.Data 0 BOOL
The wire break switch was causing electrical problems, so we use the load-cell as a proxy for wire break
Local:1:I.Pt15.Data - MainProgram/MainRoutine - 20(XIC)

Local:2:I AB:5000_DI16:I:0 DUNEW2PLC1
Constant No
External Access: Read/Write
Local:2:I.Pt00.Data 0 BOOL
Local:2:I.Pt00.Data - MainProgram/MainRoutine - 21(XIC)
Local:2:I.Pt01.Data 0 BOOL
Local:2:I.Pt01.Data - MainProgram/MainRoutine - 8(XIC)
Local:2:I.Pt02.Data 1 BOOL

Local:2:I (Continued)
Local:2:I.Pt02.Data - MainProgram/MainRoutine - 17(XIO)
Local:2:I.Pt04.Data 0 BOOL
Local:2:I.Pt04.Data - MainProgram/MainRoutine - 22(XIC)
Local:2:I.Pt06.Data 0 BOOL
TENSION ON/OFF
Local:2:I.Pt06.Data - MainProgram/MainRoutine - 12(XIC)
Local:2:I.Pt07.Data 1 BOOL
O/T SWITCH OVERTRAVEL SWITCH
Local:2:I.Pt08.Data 1 BOOL
Local:2:I.Pt08.Data - MainProgram/MainRoutine - 13(XIC)
Local:2:I.Pt10.Data 1 BOOL
Local:2:I.Pt10.Data - MainProgram/MainRoutine - 14(XIC)
Local:2:I.Pt12.Data 0 BOOL
Local:2:I.Pt12.Data - MainProgram/MainRoutine - 11(XIC)
Local:2:I.Pt13.Data 0 BOOL
Cabinet lighted push button
Local:2:I.Pt13.Data - MainProgram/MainRoutine - 54(XIC)
Local:2:I.Pt14.Data 1 BOOL
Local:2:I.Pt14.Data - MainProgram/MainRoutine - 23(XIC)

Local:3:O AB:5000_DO16:O:0 DUNEW2PLC1
External Access: Read/Write
Local:3:O.Pt01.Data 0 BOOL
Latching signal output
*Local:3:O.Pt01.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - *44(OTE)*
Local:3:O.Pt02.Data 0 BOOL
*Local:3:O.Pt02.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - *22(OTE), *35(OTE), 36(XIC)*
Local:3:O.Pt03.Data 0 BOOL
*Enable_tension_motor - PID_Tension_Servo/PID_Tension - *2(OTE)*
Local:3:O.Pt11.Data 0 BOOL
Reset PB - Light
*Local:3:O.Pt11.Data - MainProgram/MainRoutine - *35(OTE)*
Local:3:O.Pt12.Data 0 BOOL
Beacon - Green
*Local:3:O.Pt12.Data - MainProgram/MainRoutine - *35(OTE)*
Local:3:O.Pt13.Data 0 BOOL
Beacon - Amber
*Local:3:O.Pt13.Data - MainProgram/MainRoutine - *36(OTE)*
Local:3:O.Pt14.Data 1 BOOL
Beacon - Red
*Local:3:O.Pt14.Data - MainProgram/MainRoutine - *38(OTE)*
Local:3:O.Pt15.Data 0 BOOL
Beacon - Sounder
*Local:3:O.Pt15.Data - MainProgram/MainRoutine - *37(OTE)*
*Local:3:O.Pt15.Data - PID_Tension_Servo/PID_Tension - *12(OTE)*

Local:4:I AB:5000_AI8:I:0 DUNEW2PLC1
Wire tension (N)
Constant No
External Access: Read/Write
Local:4:I.RunMode 1 BOOL
Wire tension (N)
Local:4:I.ConnectionFaulted 0 BOOL
Wire tension (N)
Local:4:I.DiagnosticActive 0 BOOL
Wire tension (N)
Local:4:I.DiagnosticSequenceCount
0 SINT
Wire tension (N)
Local:4:I.Ch00 CHANNEL_AI_DIAG:I:0
Wire tension (N)
Local:4:I.Ch00.Fault 0 BOOL
Wire tension (N)

Local:4:I (Continued)		
Local:4:I.Ch00.Uncertain	0	BOOL
Wire tension (N)		
Local:4:I.Ch00.OpenWire	0	BOOL
Wire tension (N)		
Local:4:I.Ch00.OverTemperature	0	BOOL
Wire tension (N)		
Local:4:I.Ch00.FieldPowerOff	0	BOOL
Wire tension (N)		
Local:4:I.Ch00.NotANumber	0	BOOL
Wire tension (N)		
Local:4:I.Ch00.Underrange	0	BOOL
Wire tension (N)		
Local:4:I.Ch00.Overrange	0	BOOL
Wire tension (N)		
Local:4:I.Ch00.LLAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch00.LAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch00.HAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch00.HHAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch00.RateAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch00.CalFault	0	BOOL
Wire tension (N)		
Local:4:I.Ch00.Calibrating	0	BOOL
Wire tension (N)		
Local:4:I.Ch00.Data	-0.01852743	REAL
Wire tension (N)		
Local:4:I.Ch00.RollingTimestamp	3169	INT
Wire tension (N)		
Local:4:I.Ch01		CHANNEL_AI_DIAG:I:0
Wire tension (N)		
Local:4:I.Ch01.Fault	0	BOOL
Wire tension (N)		
Local:4:I.Ch01.Uncertain	0	BOOL
Wire tension (N)		
Local:4:I.Ch01.OpenWire	0	BOOL
Wire tension (N)		
Local:4:I.Ch01.OverTemperature	0	BOOL
Wire tension (N)		
Local:4:I.Ch01.FieldPowerOff	0	BOOL
Wire tension (N)		
Local:4:I.Ch01.NotANumber	0	BOOL
Wire tension (N)		
Local:4:I.Ch01.Underrange	0	BOOL
Wire tension (N)		
Local:4:I.Ch01.Overrange	0	BOOL
Wire tension (N)		
Local:4:I.Ch01.LLAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch01.LAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch01.HAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch01.HHAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch01.RateAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch01.CalFault	0	BOOL
Wire tension (N)		
Local:4:I.Ch01.Calibrating	0	BOOL

Local:4:I (Continued)		
Wire tension (N)		
Local:4:I.Ch01.Data	2.09691888e-003	REAL
Wire tension (N)		
tension_tag - PID_Tension_Servo/PID_Tension - 5(CPT), 6(CMP), 6(CPT)		
Local:4:I.Ch01.RollingTimestamp	3229	INT
Wire tension (N)		
Local:4:I.Ch02		CHANNEL_AI_DIAG:I:0
Wire tension (N)		
Local:4:I.Ch02.Fault	1	BOOL
Wire tension (N)		
Local:4:I.Ch02.Uncertain	0	BOOL
Wire tension (N)		
Local:4:I.Ch02.OpenWire	0	BOOL
Wire tension (N)		
Local:4:I.Ch02.OverTemperature	0	BOOL
Wire tension (N)		
Local:4:I.Ch02.FieldPowerOff	0	BOOL
Wire tension (N)		
Local:4:I.Ch02.NotANumber	0	BOOL
Wire tension (N)		
Local:4:I.Ch02.Underrange	0	BOOL
Wire tension (N)		
Local:4:I.Ch02.Overrange	0	BOOL
Wire tension (N)		
Local:4:I.Ch02.LLAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch02.LAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch02.HAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch02.HHAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch02.RateAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch02.CalFault	0	BOOL
Wire tension (N)		
Local:4:I.Ch02.Calibrating	0	BOOL
Wire tension (N)		
Local:4:I.Ch02.Data	1.#QNAN	REAL
Wire tension (N)		
Local:4:I.Ch02.RollingTimestamp	0	INT
Wire tension (N)		
Local:4:I.Ch03		CHANNEL_AI_DIAG:I:0
Wire tension (N)		
Local:4:I.Ch03.Fault	1	BOOL
Wire tension (N)		
Local:4:I.Ch03.Uncertain	0	BOOL
Wire tension (N)		
Local:4:I.Ch03.OpenWire	0	BOOL
Wire tension (N)		
Local:4:I.Ch03.OverTemperature	0	BOOL
Wire tension (N)		
Local:4:I.Ch03.FieldPowerOff	0	BOOL
Wire tension (N)		
Local:4:I.Ch03.NotANumber	0	BOOL
Wire tension (N)		
Local:4:I.Ch03.Underrange	0	BOOL
Wire tension (N)		
Local:4:I.Ch03.Overrange	0	BOOL
Wire tension (N)		
Local:4:I.Ch03.LLAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch03.LAlarm	0	BOOL

Local:4:I (Continued)		
Wire tension (N)		
Local:4:I.Ch03.HAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch03.HHAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch03.RateAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch03.CalFault	0	BOOL
Wire tension (N)		
Local:4:I.Ch03.Calibrating	0	BOOL
Wire tension (N)		
Local:4:I.Ch03.Data	1.#QNAN	REAL
Wire tension (N)		
Local:4:I.Ch03.RollingTimestamp	0	INT
Wire tension (N)		
Local:4:I.Ch04		CHANNEL_AI_DIAG:I:0
Wire tension (N)		
Local:4:I.Ch04.Fault	1	BOOL
Wire tension (N)		
Local:4:I.Ch04.Uncertain	0	BOOL
Wire tension (N)		
Local:4:I.Ch04.OpenWire	0	BOOL
Wire tension (N)		
Local:4:I.Ch04.OverTemperature	0	BOOL
Wire tension (N)		
Local:4:I.Ch04.FieldPowerOff	0	BOOL
Wire tension (N)		
Local:4:I.Ch04.NotANumber	0	BOOL
Wire tension (N)		
Local:4:I.Ch04.Underrange	0	BOOL
Wire tension (N)		
Local:4:I.Ch04.Overrange	0	BOOL
Wire tension (N)		
Local:4:I.Ch04.LLAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch04.LAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch04.HAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch04.HHAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch04.RateAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch04.CalFault	0	BOOL
Wire tension (N)		
Local:4:I.Ch04.Calibrating	0	BOOL
Wire tension (N)		
Local:4:I.Ch04.Data	1.#QNAN	REAL
Wire tension (N)		
Local:4:I.Ch04.RollingTimestamp	0	INT
Wire tension (N)		
Local:4:I.Ch05		CHANNEL_AI_DIAG:I:0
Wire tension (N)		
Local:4:I.Ch05.Fault	1	BOOL
Wire tension (N)		
Local:4:I.Ch05.Uncertain	0	BOOL
Wire tension (N)		
Local:4:I.Ch05.OpenWire	0	BOOL
Wire tension (N)		
Local:4:I.Ch05.OverTemperature	0	BOOL
Wire tension (N)		
Local:4:I.Ch05.FieldPowerOff	0	BOOL
Wire tension (N)		

Local:4:I (Continued)		
Local:4:I.Ch05.NotANumber	0	BOOL
Wire tension (N)		
Local:4:I.Ch05.Underrange	0	BOOL
Wire tension (N)		
Local:4:I.Ch05.Overrange	0	BOOL
Wire tension (N)		
Local:4:I.Ch05.LLAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch05.LAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch05.HAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch05.HHAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch05.RateAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch05.CalFault	0	BOOL
Wire tension (N)		
Local:4:I.Ch05.Calibrating	0	BOOL
Wire tension (N)		
Local:4:I.Ch05.Data	1.#QNAN	REAL
Wire tension (N)		
Local:4:I.Ch05.RollingTimestamp	0	INT
Wire tension (N)		
Local:4:I.Ch06		CHANNEL_AI_DIAG:I:0
Wire tension (N)		
Local:4:I.Ch06.Fault	1	BOOL
Wire tension (N)		
Local:4:I.Ch06.Uncertain	0	BOOL
Wire tension (N)		
Local:4:I.Ch06.OpenWire	0	BOOL
Wire tension (N)		
Local:4:I.Ch06.OverTemperature	0	BOOL
Wire tension (N)		
Local:4:I.Ch06.FieldPowerOff	0	BOOL
Wire tension (N)		
Local:4:I.Ch06.NotANumber	0	BOOL
Wire tension (N)		
Local:4:I.Ch06.Underrange	0	BOOL
Wire tension (N)		
Local:4:I.Ch06.Overrange	0	BOOL
Wire tension (N)		
Local:4:I.Ch06.LLAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch06.LAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch06.HAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch06.HHAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch06.RateAlarm	0	BOOL
Wire tension (N)		
Local:4:I.Ch06.CalFault	0	BOOL
Wire tension (N)		
Local:4:I.Ch06.Calibrating	0	BOOL
Wire tension (N)		
Local:4:I.Ch06.Data	1.#QNAN	REAL
Wire tension (N)		
Local:4:I.Ch06.RollingTimestamp	0	INT
Wire tension (N)		
Local:4:I.Ch07		CHANNEL_AI_DIAG:I:0
Wire tension (N)		
Local:4:I.Ch07.Fault	1	BOOL

Local:4:I (Continued)

Wire tension (N)			
Local:4:I.Ch07.Uncertain	0	BOOL	
Wire tension (N)			
Local:4:I.Ch07.OpenWire	0	BOOL	
Wire tension (N)			
Local:4:I.Ch07.OverTemperature	0	BOOL	
Wire tension (N)			
Local:4:I.Ch07.FieldPowerOff	0	BOOL	
Wire tension (N)			
Local:4:I.Ch07.NotANumber	0	BOOL	
Wire tension (N)			
Local:4:I.Ch07.Underrange	0	BOOL	
Wire tension (N)			
Local:4:I.Ch07.Overrange	0	BOOL	
Wire tension (N)			
Local:4:I.Ch07.LLAlarm	0	BOOL	
Wire tension (N)			
Local:4:I.Ch07.LAlarm	0	BOOL	
Wire tension (N)			
Local:4:I.Ch07.HAlarm	0	BOOL	
Wire tension (N)			
Local:4:I.Ch07.HHAlarm	0	BOOL	
Wire tension (N)			
Local:4:I.Ch07.RateAlarm	0	BOOL	
Wire tension (N)			
Local:4:I.Ch07.CalFault	0	BOOL	
Wire tension (N)			
Local:4:I.Ch07.Calibrating	0	BOOL	
Wire tension (N)			
Local:4:I.Ch07.Data	1.#QNAN	REAL	
Wire tension (N)			
Local:4:I.Ch07.RollingTimestamp	0	INT	
Wire tension (N)			
Local:5:O		AB:5000_AO8:O:0	DUNEW2PLC1
External Access:	Read/Write		
Local:5:O.Ch00.Data	2.5	REAL	
Tension_Servo_CMD			
*cv_to_electrocraft - PID_Tension_Servo/PID_Tension - *7(MOV), 9(CPT)*			
Local:6:I		AB:5000_DI16:I:0	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
Local:6:I.Pt00.Data	1	BOOL	
Local:6:I.Pt00.Data - MainProgram/MainRoutine - 25(XIC)			
Local:6:I.Pt01.Data	1	BOOL	
Local:6:I.Pt01.Data - MainProgram/MainRoutine - 26(XIC)			
Local:6:I.Pt02.Data	1	BOOL	
Local:6:I.Pt02.Data - MainProgram/MainRoutine - 27(XIC)			
Local:6:I.Pt03.Data	1	BOOL	
Local:6:I.Pt03.Data - MainProgram/MainRoutine - 28(XIC)			
Local:6:I.Pt04.Data	0	BOOL	
Local:6:I.Pt04.Data - MainProgram/MainRoutine - 29(XIC)			
Local:6:I.Pt05.Data	1	BOOL	
Local:6:I.Pt05.Data - MainProgram/MainRoutine - 30(XIC)			
MACHINE_SW_STAT		BOOL[32]	DUNEW2PLC1
External Access:	Read Only		
MACHINE_SW_STAT[0]	0	BOOL	
*MACHINE_SW_STAT[0] - Latch_UnLatch_State_6_7_8/Latch_Driver - *26(OTE)*			
MACHINE_SW_STAT[1]	1	BOOL	
*MACHINE_SW_STAT[1] - MainProgram/MainRoutine - *0(OTE)*			
MACHINE_SW_STAT[2]	1	BOOL	

MACHINE_SW_STAT (Continued)
*MACHINE_SW_STAT[2] - MainProgram/MainRoutine - *2(OTE)*
MACHINE_SW_STAT[3] 1 BOOL
*MACHINE_SW_STAT[3] - MainProgram/MainRoutine - *3(OTE)*
MACHINE_SW_STAT[4] 1 BOOL
*MACHINE_SW_STAT[4] - MainProgram/MainRoutine - *4(OTE)*
MACHINE_SW_STAT[5] 0 BOOL
*MACHINE_SW_STAT[5] - MainProgram/MainRoutine - *6(OTE)*
MACHINE_SW_STAT[6] 0 BOOL
*MACHINE_SW_STAT[6] - MainProgram/MainRoutine - *7(OTE)*
MACHINE_SW_STAT[7] 0 BOOL
*MACHINE_SW_STAT[7] - MainProgram/MainRoutine - *8(OTE)*
MACHINE_SW_STAT[8] 1 BOOL
*MACHINE_SW_STAT[8] - MainProgram/MainRoutine - *9(OTE)*
MACHINE_SW_STAT[9] 0 BOOL
*MACHINE_SW_STAT[9] - MainProgram/MainRoutine - *16(OTE)*
MACHINE_SW_STAT[10] 0 BOOL
*MACHINE_SW_STAT[10] - MainProgram/MainRoutine - *17(OTE)*
MACHINE_SW_STAT[12] 1 BOOL
*MACHINE_SW_STAT[12] - Latch_UnLatch_State_6_7_8/Latch_Driver - *0(OTE)*
MACHINE_SW_STAT[13] 1 BOOL
*MACHINE_SW_STAT[13] - Latch_UnLatch_State_6_7_8/Latch_Driver - *1(OTE)*
MACHINE_SW_STAT[14] 0 BOOL
*MACHINE_SW_STAT[14] - MainProgram/MainRoutine - *22(OTE)*
MACHINE_SW_STAT[15] 0 BOOL
*MACHINE_SW_STAT[15] - MainProgram/MainRoutine - *21(OTE)*
MACHINE_SW_STAT[16] 0 BOOL
*MACHINE_SW_STAT[16] - MainProgram/MainRoutine - *19(OTE)*
MACHINE_SW_STAT[17] 0 BOOL
*MACHINE_SW_STAT[17] - MainProgram/MainRoutine - *18(OTE)*
MACHINE_SW_STAT[18] 1 BOOL
*MACHINE_SW_STAT[18] - MainProgram/MainRoutine - *10(OTE)*
MACHINE_SW_STAT[19] 0 BOOL
*MACHINE_SW_STAT[19] - MainProgram/MainRoutine - *11(OTE)*
MACHINE_SW_STAT[20] 1 BOOL
*MACHINE_SW_STAT[20] - MainProgram/MainRoutine - *13(OTE)*
MACHINE_SW_STAT[21] 1 BOOL
*MACHINE_SW_STAT[21] - MainProgram/MainRoutine - *14(OTE)*
MACHINE_SW_STAT[22] 1 BOOL
*MACHINE_SW_STAT[22] - MainProgram/MainRoutine - *23(OTE)*
MACHINE_SW_STAT[23] 0 BOOL
*MACHINE_SW_STAT[23] - MainProgram/MainRoutine - *1(OTE)*
MACHINE_SW_STAT[25] 1 BOOL
*MACHINE_SW_STAT[25] - MainProgram/MainRoutine - *24(OTE)*
MACHINE_SW_STAT[26] 1 BOOL
*MACHINE_SW_STAT[26] - MainProgram/MainRoutine - *25(OTE)*
MACHINE_SW_STAT[27] 1 BOOL
*MACHINE_SW_STAT[27] - MainProgram/MainRoutine - *26(OTE)*
MACHINE_SW_STAT[28] 1 BOOL
*MACHINE_SW_STAT[28] - MainProgram/MainRoutine - *27(OTE)*
MACHINE_SW_STAT[29] 1 BOOL
*MACHINE_SW_STAT[29] - MainProgram/MainRoutine - *28(OTE)*
MACHINE_SW_STAT[30] 0 BOOL
*MACHINE_SW_STAT[30] - MainProgram/MainRoutine - *29(OTE)*
MACHINE_SW_STAT[31] 1 BOOL
*MACHINE_SW_STAT[31] - MainProgram/MainRoutine - *30(OTE)*

Manual_Trig 0 BOOL Camera
Constant No
External Access: Read Only
Manual_Trig - Camera/CameraControl - 29(XIC)

MAS 1 BOOL DUNEW2PLC1
Motion Axis Stop

MAS (Continued)
Constant No
External Access: Read/Write
MAS - MainProgram/MainRoutine - 15(XIC)

MASTER_Z_GO 0 BOOL DUNEW2PLC1
External Access: Read Only
*MASTER_Z_GO - MoveZ_State_4_5/MoveZ - *3(OTE), 4(XIO), 5(XIC), 6(XIC)*

max_integral 1000 DINT MoveXY_State_2_3
Constant No
External Access: Read/Write
max_integral - MoveXY_State_2_3/xy_speed_regulator - 7(GRT), 7(MOV)

max_tolerable_tension 10.0 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
max_tolerable_tension - PID_Tension_Servo/PID_Tension - 12(GRT)

MAX_TOLERABLE_Z 5.0 REAL DUNEW2PLC1
Constant Yes
External Access: Read/Write
MAX_TOLERABLE_Z - MoveXY_State_2_3/MoveXY - 1(GEQ)

MCCD_X_Y_Axis1 MOTION_INSTRUCTION DUNEW2PLC1
External Access: Read/Write
*MCCD_X_Y_Axis1 - MoveXY_State_2_3/MoveXY - *14(MCCD)*

min_integral -1000 DINT MoveXY_State_2_3
Constant No
External Access: Read/Write
min_integral - MoveXY_State_2_3/xy_speed_regulator - 7(LES), 7(MOV)

min_regulated_speed 100 DINT MoveXY_State_2_3
Constant No
External Access: Read/Write
min_regulated_speed - MoveXY_State_2_3/xy_speed_regulator - 7(GRT), 8(LES), 8(MOV)

MINUS_X_EOT 1 BOOL DUNEW2PLC1
External Access: Read Only
MINUS_X_EOT - EOT_Trip_11/EOT_TRIP - 9(XIO)
*MINUS_X_EOT - MainProgram/MainRoutine - *14(OTE), 15(XIC)*

MINUS_Y_EOT 0 BOOL DUNEW2PLC1
External Access: Read Only
MINUS_Y_EOT - EOT_Trip_11/EOT_TRIP - 15(XIO)
*MINUS_Y_EOT - MainProgram/MainRoutine - *11(OTE), 15(XIC)*

MORE_STATS BOOL[32] DUNEW2PLC1
External Access: Read Only
MORE_STATS[2] 0 BOOL
*MORE_STATS[2] - PID_Tension_Servo/PID_Tension - *13(OTE)*

MORE_STATS_S BOOL[32] DUNEW2PLC1
Mapped With: MORE_STATS
External Access: Read/Write
MORE_STATS_S[0] 0 BOOL
*MORE_STATS_S[0] - Safety /Safety - *2(OTE)*
MORE_STATS_S[1] 0 BOOL
*MORE_STATS_S[1] - Safety /Safety - *8(OTE)*

MotionFault 0 BOOL DUNEW2PLC1
External Access: Read/Write
*MotionFault - motionQueue/queue_line_arc - *13(OTE)*

MOVE_TYPE 0 INT DUNEW2PLC1
0 = No Move 1 = XY JOG 2 = XY Plane Move 3 = Z JOG 4 = Z Axis Move 5 = Toggle Latch Actuator 6 = Home the Latch Actuator 7 = Unlock Latch Actuator Motor 8 = Un-Servo Motors 9 = PLC_Init

External Access: Read/Write
*MOVE_TYPE - EOT_Trip_11/EOT_TRIP - *21(CPT), 12(CMP), 15(CMP), 18(CMP), 6(CMP), 9(CMP)*
MOVE_TYPE - Error_State_10/Error - 12(CMP), 8(CMP)
*MOVE_TYPE - Initialize/Init - *2(CPT), *4(CPT), 0(EQU)*
*MOVE_TYPE - Latch_UnLatch_State_6_7_8/Latch_Driver - *13(CPT), *33(CPT), 43(CMP)*
*MOVE_TYPE - MoveXY_State_2_3/MoveXY - *18(CPT), *21(CPT), 15(CMP)*
*MOVE_TYPE - MoveZ_State_4_5/MoveZ - *19(CPT), 13(CMP)*
MOVE_TYPE - Ready_State_1/Ready - 10(CMP), 12(CMP), 14(CMP), 16(CMP), 18(CMP), 4(CMP), 6(CMP), 8(CMP)
MOVE_TYPE - UnServo_9/UnServo_All - 11(CMP)

MoveA MOTION_INSTRUCTION motionQueue
External Access: Read/Write
*MoveA - motionQueue/queue_line_arc - *100(MCLM), *101(MCCM), *61(MCLM), *62(MCCM)*
MoveA.ER 0 BOOL
*MoveA.ER - motionQueue/queue_line_arc - *11(OTU), 11(XIC), 13(XIC)*
MoveA.PC 0 BOOL
MoveA.PC - motionQueue/queue_line_arc - 117(XIC)
MoveA.IP 0 BOOL
MoveA.IP - motionQueue/queue_line_arc - 105(XIC), 67(XIC)

MoveB MOTION_INSTRUCTION motionQueue
External Access: Read/Write
*MoveB - motionQueue/queue_line_arc - *63(MCLM), *64(MCCM), *98(MCLM), *99(MCCM)*
MoveB.ER 0 BOOL
*MoveB.ER - motionQueue/queue_line_arc - *12(OTU), 12(XIC), 13(XIC)*
MoveB.PC 0 BOOL
MoveB.PC - motionQueue/queue_line_arc - 118(XIC)
MoveB.IP 0 BOOL
MoveB.IP - motionQueue/queue_line_arc - 104(XIC), 68(XIC)

near_ending 0 BOOL MoveXY_State_2_3
True if distance to xy-move target is less than the required stopping distance
External Access: Read/Write
*near_ending - MoveXY_State_2_3/MoveXY - *10(OTE), 14(XIO)*
near_ending - MoveXY_State_2_3/xy_speed_regulator - 9(XIC)

near_ending_trigger 0 BOOL MoveXY_State_2_3
External Access: Read/Write
*near_ending_trigger - MoveXY_State_2_3/xy_speed_regulator - *9(OSR)*

neutral_cv 2.5 REAL DUNEW2PLC1
External Access: Read/Write
*neutral_cv - PID_Tension_Servo/PID_Tension - *10(CPT), 7(LES), 7(MOV)*

NextIssued 0 BOOL DUNEW2PLC1
External Access: Read/Write
*NextIssued - motionQueue/queue_line_arc - *102(OTL), *111(OTU), *121(OTU), 104(XIC), 105(XIC), 117(XIO), 118(XIO), 67(XIO), 68(XIO)*

NextSeg MotionSeg motionQueue
motion segments for move queue, lines or arcs
External Access: Read/Write
*NextSeg - motionQueue/queue_line_arc - *69(FFU), 106(COP)*
NextSeg.Valid 0 BOOL
motion segments for move queue, lines or arcs segment valid flag
*NextSeg.Valid - motionQueue/queue_line_arc - *140(OTU), 22(XIC), 23(XIO), 24(XIO), 25(XIC), 26(XIC), 27(XIC), 28(XIC)*
NextSeg.SegType 1 SINT
motion segments for move queue, lines or arcs 1=line, 2=arc
NextSeg.SegType - motionQueue/queue_line_arc - 22(EQU), 22(EQU), 27(GRT), 27(LES), 28(GRT), 28(LES), 79(MOV), 91(MOV)
NextSeg.XY REAL
motion segments for move queue, lines or arcs end xy

NextSeg (Continued)
NextSeg.XY - motionQueue/queue_line_arc - 72(COP), 84(COP)

NextSeg.XY[0]	5972.74	REAL	
motion segments for move queue, lines or arcs end xy			
NextSeg.XY[1]	1347.2893	REAL	
motion segments for move queue, lines or arcs end xy			
NextSeg.Speed	437.4951	REAL	
motion segments for move queue, lines or arcs >0			
NextSeg.Speed - motionQueue/queue_line_arc - 73(MOV), 85(MOV)			
NextSeg.Accel	2000.0	REAL	
motion segments for move queue, lines or arcs >0			
NextSeg.Accel - motionQueue/queue_line_arc - 74(MOV), 86(MOV)			
NextSeg.Decel	2000.0	REAL	
motion segments for move queue, lines or arcs >0			
NextSeg.Decel - motionQueue/queue_line_arc - 75(MOV), 87(MOV)			
NextSeg.JerkAccel	20000.0	REAL	
motion segments for move queue, lines or arcs			
NextSeg.JerkAccel - motionQueue/queue_line_arc - 76(MOV), 88(MOV)			
NextSeg.JerkDecel	20000.0	REAL	
motion segments for move queue, lines or arcs			
NextSeg.JerkDecel - motionQueue/queue_line_arc - 77(MOV), 89(MOV)			
NextSeg.TermType	4	SINT	
motion segments for move queue, lines or arcs 0 to 6			
NextSeg.TermType - motionQueue/queue_line_arc - 78(MOV), 90(MOV)			
NextSeg.Seq	0	DINT	
motion segments for move queue, lines or arcs Sequence/ack id			
*NextSeg.Seq - motionQueue/queue_line_arc - *141(MOV), 22(GRT), 25(LEQ), 26(LEQ), 27(GRT), 28(GRT), 83(MOV), 95(MOV)*			
NextSeg.CircleType	1	DINT	
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum			
NextSeg.CircleType - motionQueue/queue_line_arc - 80(MOV), 92(MOV)			
NextSeg.ViaCenter		REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
NextSeg.ViaCenter - motionQueue/queue_line_arc - 81(COP), 93(COP)			
NextSeg.ViaCenter[0]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
NextSeg.ViaCenter[1]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
NextSeg.Direction	1	DINT	
motion segments for move queue, lines or arcs 0..3 per mmcm enum			
NextSeg.Direction - motionQueue/queue_line_arc - 82(MOV), 94(MOV)			

NEXTSTATE	1	DINT	DUNEW2PLC1
External Access:	Read Only		

*NEXTSTATE - EOT_Trip_11/EOT_TRIP - *22(CPT)*
*NEXTSTATE - Error_State_10/Error - *12(CPT)*
*NEXTSTATE - Initialize/Init - *4(CPT), *9(CPT)*
*NEXTSTATE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *13(CPT), *33(CPT), *43(CPT), *51(CPT)*
NEXTSTATE - MainProgram/MainRoutine - 42(CMP), 43(CMP), 44(CMP), 45(CMP), 46(CMP), 47(CMP), 48(CMP), 49(CMP), 50(CMP), 51(CMP), 52(CMP), 53(CMP)
*NEXTSTATE - MoveXY_State_2_3/MoveXY - *0(CPT), *1(CPT), *1(CPT), *18(CPT), *2(CPT), *21(CPT)*
*NEXTSTATE - MoveZ_State_4_5/MoveZ - *10(CPT), *19(CPT), *20(CPT), *4(CPT), *5(MOV), *7(CPT)*
*NEXTSTATE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT)*
*NEXTSTATE - Ready_State_1/Ready - *11(CPT), *13(CPT), *15(CPT), *17(CPT), *19(CPT), *5(CPT), *7(CPT), *9(CPT)*
*NEXTSTATE - UnServo_9/UnServo_All - *11(CPT)*

no_apa_collision	0	BOOL	DUNEW2PLC1
External Access:	Read/Write		

*no_apa_collision - MoveZ_State_4_5/MoveZ - *1(OTE), 3(XIC)*

no_latch_collision	1	BOOL	DUNEW2PLC1
External Access:	Read/Write		

*no_latch_collision - MoveZ_State_4_5/MoveZ - *0(OTE), 3(XIC)*

no_supports_collision	0	BOOL	DUNEW2PLC1

no_supports_collision (Continued)
External Access: Read/Write
*no_supports_collision - MoveZ_State_4_5/MoveZ - *2(OTE), 3(XIC)*

oneshotob		BOOL[32]	EOT_Trip_11
External Access:	Read Only		
oneshotob[0]	0	BOOL	
*oneshotob[0] - EOT_Trip_11/EOT_TRIP - *4(OSR), 5(XIC)*			
oneshotob[1]	0	BOOL	
*oneshotob[1] - EOT_Trip_11/EOT_TRIP - *6(OSR), 7(XIC)*			
oneshotob[2]	0	BOOL	
*oneshotob[2] - EOT_Trip_11/EOT_TRIP - *8(OSR), 21(XIC)*			
oneshotob[3]	0	BOOL	
*oneshotob[3] - EOT_Trip_11/EOT_TRIP - *9(OSR), 10(XIC)*			
oneshotob[4]	0	BOOL	
*oneshotob[4] - EOT_Trip_11/EOT_TRIP - *11(OSR), 21(XIC)*			
oneshotob[5]	0	BOOL	
*oneshotob[5] - EOT_Trip_11/EOT_TRIP - *12(OSR), 13(XIC)*			
oneshotob[6]	0	BOOL	
*oneshotob[6] - EOT_Trip_11/EOT_TRIP - *14(OSR), 21(XIC)*			
oneshotob[7]	0	BOOL	
*oneshotob[7] - EOT_Trip_11/EOT_TRIP - *15(OSR), 16(XIC)*			
oneshotob[8]	0	BOOL	
*oneshotob[8] - EOT_Trip_11/EOT_TRIP - *17(OSR), 21(XIC)*			
oneshotob[9]	0	BOOL	
*oneshotob[9] - EOT_Trip_11/EOT_TRIP - *18(OSR), 19(XIC)*			
oneshotob[10]	0	BOOL	
*oneshotob[10] - EOT_Trip_11/EOT_TRIP - *20(OSR), 21(XIC)*			
oneshotob[11]	0	BOOL	
*oneshotob[11] - EOT_Trip_11/EOT_TRIP - *21(OSR), 22(XIC)*			

oneshotob		BOOL[32]	MoveZ_State_4_5
External Access:	Read Only		
oneshotob[3]	0	BOOL	
*oneshotob[3] - MoveZ_State_4_5/MoveZ - *8(OSR), 12(XIC), 9(XIC)*			
oneshotob[4]	0	BOOL	
*oneshotob[4] - MoveZ_State_4_5/MoveZ - *11(OSR), 12(XIC)*			
oneshotob[5]	0	BOOL	
*oneshotob[5] - MoveZ_State_4_5/MoveZ - *13(OSR), 19(XIC)*			
oneshotob[6]	0	BOOL	
*oneshotob[6] - MoveZ_State_4_5/MoveZ - *14(OSR), 15(XIC)*			
oneshotob[7]	0	BOOL	
*oneshotob[7] - MoveZ_State_4_5/MoveZ - *18(OSR), 19(XIC)*			
oneshotob[8]	0	BOOL	
*oneshotob[8] - MoveZ_State_4_5/MoveZ - *7(OSR)*			
oneshotob[10]	0	BOOL	
*oneshotob[10] - MoveZ_State_4_5/MoveZ - *17(OSR), 20(XIC)*			

oneshotob		BOOL[32]	MoveXY_State_2_3
External Access:	Read/Write		
oneshotob[0]	0	BOOL	
*oneshotob[0] - MoveXY_State_2_3/MoveXY - *2(OSR), 3(XIC), 5(XIC)*			
oneshotob[1]	0	BOOL	
*oneshotob[1] - MoveXY_State_2_3/MoveXY - *4(OSR), 5(XIC)*			
oneshotob[5]	0	BOOL	
*oneshotob[5] - MoveXY_State_2_3/MoveXY - *15(OSR), 21(XIC)*			
oneshotob[7]	0	BOOL	
*oneshotob[7] - MoveXY_State_2_3/MoveXY - *20(OSR), 21(XIC)*			
oneshotob[13]	0	BOOL	
*oneshotob[13] - MoveXY_State_2_3/MoveXY - *13(OSF), 14(XIC)*			

oneshotob		BOOL[32]	UnServo_9
External Access:	Read Only		
oneshotob[0]	0	BOOL	

oneshotob (Continued)
*oneshotob[0] - UnServo_9/UnServo_All - *1(OSR), 2(XIC), 3(XIC), 4(XIC)*
oneshotob[1] 0 BOOL
*oneshotob[1] - UnServo_9/UnServo_All - *5(OSR), 6(XIC)*
oneshotob[2] 0 BOOL
*oneshotob[2] - UnServo_9/UnServo_All - *7(OSR), 8(XIC)*
oneshotob[3] 0 BOOL
*oneshotob[3] - UnServo_9/UnServo_All - *9(OSR), 10(XIC)*

oneshotob BOOL[32] PID_Tension_Servo
External Access: Read/Write
oneshotob[0] 0 BOOL
*oneshotob[0] - PID_Tension_Servo/PID_Tension - *15(OSF), 16(XIC)*

oneshotob BOOL[32] Latch_UnLatch_State_6_7_8

External Access: Read Only
oneshotob[0] 0 BOOL
*oneshotob[0] - Latch_UnLatch_State_6_7_8/Latch_Driver - *17(OSR), 18(XIC)*
oneshotob[1] 0 BOOL
For one scan, when we enter state 6 and actuator is enabled
*oneshotob[1] - Latch_UnLatch_State_6_7_8/Latch_Driver - *11(OSR), 12(XIC)*
oneshotob[3] 0 BOOL
*oneshotob[3] - Latch_UnLatch_State_6_7_8/Latch_Driver - *50(OSR), 51(XIC)*
oneshotob[4] 0 BOOL
*oneshotob[4] - Latch_UnLatch_State_6_7_8/Latch_Driver - *27(OSR), 28(XIC)*
oneshotob[5] 0 BOOL
*oneshotob[5] - Latch_UnLatch_State_6_7_8/Latch_Driver - *32(OSR), 33(XIC)*
oneshotob[7] 0 BOOL
*oneshotob[7] - Latch_UnLatch_State_6_7_8/Latch_Driver - *23(OSR), 24(XIC), 25(XIC)*
oneshotob[9] 0 BOOL
Entered state 6 but the actuator was not enabled
*oneshotob[9] - Latch_UnLatch_State_6_7_8/Latch_Driver - *9(OSR), 10(XIC)*
oneshotob[10] 0 BOOL
*oneshotob[10] - Latch_UnLatch_State_6_7_8/Latch_Driver - *37(OSR), 38(XIC), 39(XIC)*
oneshotob[12] 0 BOOL
*oneshotob[12] - Latch_UnLatch_State_6_7_8/Latch_Driver - *45(OSR), 46(XIC)*

oneshotob BOOL[32] Error_State_10
External Access: Read Only
oneshotob[0] 0 BOOL
On entry to state 6 or 7
oneshotob[1] 0 BOOL
*oneshotob[1] - Error_State_10/Error - *1(OSR), 2(XIC), 3(XIC)*
oneshotob[2] 0 BOOL
*oneshotob[2] - Error_State_10/Error - *4(OSR), 5(XIC), 6(XIC), 7(XIC)*
oneshotob[4] 0 BOOL
*oneshotob[4] - Error_State_10/Error - *8(OSR), 9(XIC)*

oneshotsb BOOL[32] MoveXY_State_2_3
External Access: Read/Write
oneshotsb[0] 0 BOOL
*oneshotsb[0] - MoveXY_State_2_3/MoveXY - *2(OSR)*
oneshotsb[1] 1 BOOL
*oneshotsb[1] - MoveXY_State_2_3/MoveXY - *4(OSR)*
oneshotsb[3] 0 BOOL
*oneshotsb[3] - MoveXY_State_2_3/MoveXY - *5(OSR)*
oneshotsb[5] 0 BOOL
*oneshotsb[5] - MoveXY_State_2_3/MoveXY - *15(OSR)*
oneshotsb[6] 0 BOOL
*oneshotsb[6] - MoveXY_State_2_3/MoveXY - *16(OSR)*
oneshotsb[7] 1 BOOL
*oneshotsb[7] - MoveXY_State_2_3/MoveXY - *20(OSR)*

oneshotsb (Continued)

oneshotsb[8]	0	BOOL	
*oneshotsb[8] - MoveXY_State_2_3/MoveXY - *17(OSR)*			
oneshotsb[13]	1	BOOL	
*oneshotsb[13] - MoveXY_State_2_3/MoveXY - *13(OSF)*			
oneshotsb[15]	1	BOOL	
*oneshotsb[15] - MoveXY_State_2_3/xy_speed_regulator - *9(OSR)*			
oneshotsb		BOOL[32]	PID_Tension_Servo
External Access:	Read/Write		
oneshotsb[0]	0	BOOL	
*oneshotsb[0] - PID_Tension_Servo/PID_Tension - *0(OSR), *15(OSF)*			
oneshotsb[1]	0	BOOL	
*oneshotsb[1] - PID_Tension_Servo/PID_Tension - *0(OSF)*			
oneshotsb		BOOL[32]	Latch_UnLatch_State_6_7_8
External Access:	Read Only		
oneshotsb[0]	0	BOOL	
*oneshotsb[0] - Latch_UnLatch_State_6_7_8/Latch_Driver - *17(OSR), 19(XIC), 20(XIC), 21(XIC)*			
oneshotsb[1]	0	BOOL	
While in state 6 and operator is enabled			
*oneshotsb[1] - Latch_UnLatch_State_6_7_8/Latch_Driver - *11(OSR), 13(XIC), 14(XIC)*			
oneshotsb[2]	0	BOOL	
Latch successfully moved to the next position			
oneshotsb[2] - Latch_UnLatch_State_6_7_8/Latch_Driver - 15(XIO)			
oneshotsb[3]	0	BOOL	
*oneshotsb[3] - Latch_UnLatch_State_6_7_8/Latch_Driver - *50(OSR)*			
oneshotsb[4]	0	BOOL	
*oneshotsb[4] - Latch_UnLatch_State_6_7_8/Latch_Driver - *27(OSR), 29(XIC), 30(XIC), 31(XIC)*			
oneshotsb[5]	1	BOOL	
*oneshotsb[5] - Latch_UnLatch_State_6_7_8/Latch_Driver - *32(OSR)*			
oneshotsb[6]	0	BOOL	
LatchCounter done			
oneshotsb[7]	0	BOOL	
*oneshotsb[7] - Latch_UnLatch_State_6_7_8/Latch_Driver - *23(OSR)*			
oneshotsb[8]	0	BOOL	
While in state 6			
oneshotsb[9]	0	BOOL	
*oneshotsb[9] - Latch_UnLatch_State_6_7_8/Latch_Driver - *9(OSR)*			
oneshotsb[10]	1	BOOL	
*oneshotsb[10] - Latch_UnLatch_State_6_7_8/Latch_Driver - *37(OSR), 40(XIC), 41(XIC), 42(XIC)*			
oneshotsb[12]	0	BOOL	
*oneshotsb[12] - Latch_UnLatch_State_6_7_8/Latch_Driver - *45(OSR), 47(XIC), 48(XIC), 49(XIC)*			
oneshotsb		BOOL[32]	Error_State_10
External Access:	Read Only		
oneshotsb[0]	0	BOOL	
While in state 6 or 7			
oneshotsb[1]	0	BOOL	
*oneshotsb[1] - Error_State_10/Error - *1(OSR)*			
oneshotsb[2]	1	BOOL	
*oneshotsb[2] - Error_State_10/Error - *4(OSR)*			
oneshotsb[4]	0	BOOL	
*oneshotsb[4] - Error_State_10/Error - *8(OSR)*			
oneshotsb		BOOL[32]	EOT_Trip_11
External Access:	Read Only		
oneshotsb[0]	1	BOOL	
*oneshotsb[0] - EOT_Trip_11/EOT_TRIP - *4(OSR)*			
oneshotsb[1]	0	BOOL	
*oneshotsb[1] - EOT_Trip_11/EOT_TRIP - *6(OSR)*			
oneshotsb[2]	1	BOOL	

oneshotsb (Continued)
*oneshotsb[2] - EOT_Trip_11/EOT_TRIP - *8(OSR)*
oneshotsb[3] 0 BOOL
*oneshotsb[3] - EOT_Trip_11/EOT_TRIP - *9(OSR)*
oneshotsb[4] 0 BOOL
*oneshotsb[4] - EOT_Trip_11/EOT_TRIP - *11(OSR)*
oneshotsb[5] 0 BOOL
*oneshotsb[5] - EOT_Trip_11/EOT_TRIP - *12(OSR)*
oneshotsb[6] 0 BOOL
*oneshotsb[6] - EOT_Trip_11/EOT_TRIP - *14(OSR)*
oneshotsb[7] 0 BOOL
*oneshotsb[7] - EOT_Trip_11/EOT_TRIP - *15(OSR)*
oneshotsb[8] 0 BOOL
*oneshotsb[8] - EOT_Trip_11/EOT_TRIP - *17(OSR)*
oneshotsb[9] 0 BOOL
*oneshotsb[9] - EOT_Trip_11/EOT_TRIP - *18(OSR)*
oneshotsb[10] 0 BOOL
*oneshotsb[10] - EOT_Trip_11/EOT_TRIP - *20(OSR)*
oneshotsb[11] 0 BOOL
*oneshotsb[11] - EOT_Trip_11/EOT_TRIP - *21(OSR)*

oneshotsb BOOL[32] MoveZ_State_4_5
External Access: Read Only
oneshotsb[3] 0 BOOL
*oneshotsb[3] - MoveZ_State_4_5/MoveZ - *8(OSR)*
oneshotsb[4] 0 BOOL
*oneshotsb[4] - MoveZ_State_4_5/MoveZ - *11(OSR)*
oneshotsb[5] 0 BOOL
*oneshotsb[5] - MoveZ_State_4_5/MoveZ - *13(OSR)*
oneshotsb[6] 0 BOOL
*oneshotsb[6] - MoveZ_State_4_5/MoveZ - *14(OSR)*
oneshotsb[7] 0 BOOL
*oneshotsb[7] - MoveZ_State_4_5/MoveZ - *18(OSR)*
oneshotsb[8] 0 BOOL
*oneshotsb[8] - MoveZ_State_4_5/MoveZ - *7(OSR), 8(XIO)*
oneshotsb[10] 0 BOOL
*oneshotsb[10] - MoveZ_State_4_5/MoveZ - *17(OSR)*

oneshotsb BOOL[32] UnServo_9
External Access: Read Only
oneshotsb[0] 0 BOOL
*oneshotsb[0] - UnServo_9/UnServo_All - *1(OSR)*
oneshotsb[1] 0 BOOL
*oneshotsb[1] - UnServo_9/UnServo_All - *5(OSR)*
oneshotsb[2] 0 BOOL
*oneshotsb[2] - UnServo_9/UnServo_All - *7(OSR)*
oneshotsb[3] 0 BOOL
*oneshotsb[3] - UnServo_9/UnServo_All - *9(OSR)*

osob BOOL[32] Camera
External Access: Read Only
osob[0] 0 BOOL
*osob[0] - Camera/CameraControl - *29(OSR), 30(XIC), 31(XIC)*
osob[1] 0 BOOL
*osob[1] - Camera/CameraControl - *24(OSR), 25(XIC), 26(XIC), 27(XIC)*
osob[3] 0 BOOL
*osob[3] - Camera/CameraControl - *11(OSR)*

ossb BOOL[32] Camera
External Access: Read Only
ossb[0] 0 BOOL
*ossb[0] - Camera/CameraControl - *29(OSR), 32(XIC)*
ossb[1] 0 BOOL
*ossb[1] - Camera/CameraControl - *24(OSR)*

ossb (Continued)

ossb[3]	1	BOOL	
*ossb[3] - Camera/CameraControl - *11(OSR)*			
OutBit		BOOL[32]	Initialize
External Access:	Read Only		
OutBit[0]	0	BOOL	
*OutBit[0] - Initialize/Init - *3(OSR), 4(XIC), 5(XIC)*			
OutBit[1]	0	BOOL	
*OutBit[1] - Initialize/Init - *8(OSR), 9(XIC)*			
OutBit		BOOL[32]	Ready_State_1
External Access:	Read/Write		
OutBit[0]	0	BOOL	
*OutBit[0] - Ready_State_1/Ready - *4(OSR), 5(XIC)*			
OutBit[1]	0	BOOL	
*OutBit[1] - Ready_State_1/Ready - *6(OSR), 7(XIC)*			
OutBit[2]	0	BOOL	
*OutBit[2] - Ready_State_1/Ready - *8(OSR), 9(XIC)*			
OutBit[3]	0	BOOL	
*OutBit[3] - Ready_State_1/Ready - *10(OSR), 11(XIC)*			
OutBit[4]	0	BOOL	
*OutBit[4] - Ready_State_1/Ready - *12(OSR), 13(XIC)*			
OutBit[5]	0	BOOL	
*OutBit[5] - Ready_State_1/Ready - *14(OSR), 15(XIC)*			
OutBit[6]	0	BOOL	
*OutBit[6] - Ready_State_1/Ready - *16(OSR), 17(XIC)*			
OutBit[7]	0	BOOL	
*OutBit[7] - Ready_State_1/Ready - *18(OSR), 19(XIC)*			
overtension_timer		TIMER	DUNEW2PLC1
External Access:	Read/Write		
*overtension_timer - PID_Tension_Servo/PID_Tension - *12(TON)*			
overtension_timer.DN	0	BOOL	
overtension_timer.DN - PID_Tension_Servo/PID_Tension - 13(XIC)			
PendingSeq	0	DINT	DUNEW2PLC1
External Access:	Read/Write		
*PendingSeq - motionQueue/queue_line_arc - *113(MOV), *144(MOV), *83(MOV), *95(MOV), 112(MOV)*			
pid_cv_high_limit	10.0	REAL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
pid_cv_high_limit - PID_Tension_Servo/PID_Tension - 10(CPT), 9(CPT)			
pid_cv_low_limit	0.0	REAL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
pid_cv_low_limit - PID_Tension_Servo/PID_Tension - 10(CPT)			
PID_LOOP_TIMER		TIMER	DUNEW2PLC1
External Access:	Read/Write		
*PID_LOOP_TIMER - PID_Tension_Servo/PID_Tension - *4(TON)*			
PID_LOOP_TIMER.DN	0	BOOL	
PID_LOOP_TIMER.DN - PID_Tension_Servo/PID_Tension - 4(XIO), 7(XIC)			
pid_loop_timer_bypass	0	BOOL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
pid_loop_timer_bypass - PID_Tension_Servo/PID_Tension - 7(XIC)			
PLUS_X_EOT	1	BOOL	DUNEW2PLC1
External Access:	Read Only		
PLUS_X_EOT - EOT_Trip_11/EOT_TRIP - 6(XIO)			

PLUS_X_EOT (Continued)
*PLUS_X_EOT - MainProgram/MainRoutine - *13(OTE), 15(XIC)*

PLUS_Y_EOT 1 BOOL DUNEW2PLC1
External Access: Read Only
PLUS_Y_EOT - EOT_Trip_11/EOT_TRIP - 12(XIO)
*PLUS_Y_EOT - MainProgram/MainRoutine - *10(OTE), 15(XIC)*

Pos_trigger 1 BOOL Camera
External Access: Read Only
*Pos_trigger - Camera/CameraControl - *27(OTL), *29(OTU), *34(OTL), 28(XIC)*

POSITION REAL[1,2] DUNEW2PLC1
Constant Yes
External Access: Read/Write
POSITION[0,0] -898.3585 REAL
X_POSITION - MoveXY_State_2_3/MoveXY - 12(MCLM), 15(CMP), 15(CMP), 22(MCLM), 7(CPT), 8(CPT)
X_POSITION - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)
POSITION[0,1] -13.7993 REAL
Y_POSITION - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 7(CPT), 8(CPT)

PrepCurMove 0 BOOL motionQueue
External Access: Read/Write
*PrepCurMove - motionQueue/queue_line_arc - *124(OTU), *14(OTL), *60(OTU), 35(XIC), 36(XIC), 37(XIC), 38(XIC), 39(XIC), 40(XIC), 41(XIC), 42(XIC), 43(XIC), 44(XIC), 45(XIC), 46(XIC), 47(XIC), 48(XIC), 49(XIC), 50(XIC), 51(XIC), 52(XIC), 53(XIC), 54(XIC), 55(XIC), 56(XIC), 57(XIC), 58(XIC), 59(XIC), 60(XIC)*

PrepNextMove 0 BOOL motionQueue
External Access: Read/Write
*PrepNextMove - motionQueue/queue_line_arc - *125(OTU), *22(OTL), *97(OTU), 72(XIC), 73(XIC), 74(XIC), 75(XIC), 76(XIC), 77(XIC), 78(XIC), 79(XIC), 80(XIC), 81(XIC), 82(XIC), 83(XIC), 84(XIC), 85(XIC), 86(XIC), 87(XIC), 88(XIC), 89(XIC), 90(XIC), 91(XIC), 92(XIC), 93(XIC), 94(XIC), 95(XIC), 96(XIC), 97(XIC)*

PREV_ACT_POS 2 DINT Latch_UnLatch_State_6_7_8

Store the previous actuator position
External Access: Read Only
*PREV_ACT_POS - Latch_UnLatch_State_6_7_8/Latch_Driver - *12(CPT), 13(CMP), 13(CMP), 13(CMP), 13(CMP)*

QueueCount 0 DINT DUNEW2PLC1
External Access: Read/Write
*QueueCount - motionQueue/queue_line_arc - *2(MOV)*

QueueCtl CONTROL motionQueue
External Access: Read/Write
*QueueCtl - motionQueue/queue_line_arc - *119(RES), *32(FFU), *69(FFU), *7(FFL)*
QueueCtl.POS 0 DINT
QueueCtl.POS - motionQueue/queue_line_arc - 0(EQU), 1(GEQ), 2(MOV), 30(GEQ)

QueueEmpty 1 BOOL motionQueue
External Access: Read/Write
*QueueEmpty - motionQueue/queue_line_arc - *0(OTE), 67(XIO), 68(XIO)*

QueueFault 0 BOOL DUNEW2PLC1
External Access: Read/Write
*QueueFault - motionQueue/queue_line_arc - *135(OTU), *16(OTL), *18(OTL), *20(OTL), *24(OTL), *26(OTL), *28(OTL), 13(XIC), 30(XIO), 67(XIO), 68(XIO)*

QueueFull 0 BOOL DUNEW2PLC1
External Access: Read/Write
*QueueFull - motionQueue/queue_line_arc - *1(OTE), 7(XIO), 8(XIO), 9(XIO)*

READ_FIFOS 0 BOOL DUNEW2PLC1

READ_FIFOS (Continued)
External Access: Read/Write
*READ_FIFOS - Camera/CameraControl - *16(OTL), *23(OTU), 17(XIC)*

RegistrationData MOTION_INSTRUCTION[8] Camera
External Access: Read Only
RegistrationData[0] MOTION_INSTRUCTION
*RegistrationData[0] - Camera/CameraControl - *6(MAR)*
RegistrationData[0].PC 0 BOOL
RegistrationData[0].PC - Camera/CameraControl - 7(XIC), 9(XIC)
RegistrationData[1] MOTION_INSTRUCTION
*RegistrationData[1] - Camera/CameraControl - *5(MAR)*
RegistrationData[1].PC 0 BOOL
RegistrationData[1].PC - Camera/CameraControl - 10(XIC), 12(XIC), 13(XIC), 14(XIC), 15(XIC), 8(XIC)

regulated_speed 869.62634 REAL MoveXY_State_2_3
External Access: Read/Write
*regulated_speed - MoveXY_State_2_3/xy_speed_regulator - *8(CPT), *8(MOV), *8(MOV), 7(GRT), 7(LES), 8(LES), 8(LES)*

regulator_loop_timer TIMER MoveXY_State_2_3
External Access: Read/Write
*regulator_loop_timer - MoveXY_State_2_3/xy_speed_regulator - *2(TON)*
regulator_loop_timer.PRE 1 DINT
*regulator_loop_timer.PRE - MoveXY_State_2_3/xy_speed_regulator - *1(MOV)*
regulator_loop_timer.DN 0 BOOL
regulator_loop_timer.DN - MoveXY_State_2_3/xy_speed_regulator - 2(XIO), 6(XIC), 7(XIC), 8(XIC)

ResetPB 0 BOOL DUNEW2PLC1
External Access: Read/Write
*ResetPB - MainProgram/MainRoutine - *54(OTE)*
resetPB_S - Safety /Safety - 5(XIC)

resetPB_S 0 BOOL DUNEW2PLC1
Constant No
Mapped With: ResetPB
External Access: Read/Write
resetPB_S - Safety /Safety - 5(XIC)
*ResetPB - MainProgram/MainRoutine - *54(OTE)*

RotateMoves 0 BOOL motionQueue
External Access: Read/Write
*RotateMoves - motionQueue/queue_line_arc - *104(OTL), *105(OTL), *114(OTU), *129(OTU), 106(XIC), 107(XIC), 108(XIC), 111(XIC), 112(XIC), 113(XIC), 114(XIC)*

RotateONS_AtoB 0 BOOL motionQueue
External Access: Read/Write
*RotateONS_AtoB - motionQueue/queue_line_arc - *104(ONS)*

RotateONS_BtoA 0 BOOL motionQueue
External Access: Read/Write
*RotateONS_BtoA - motionQueue/queue_line_arc - *105(ONS)*

Run_Alarm 0 BOOL DUNEW2PLC1
Constant No
External Access: Read/Write
*Safety_System_Healthy - Safety /Safety - *0(OTE), 3(XIO), 5(XIC)*

Safety_System_Healthy 1 BOOL DUNEW2PLC1
Mapped With: Run_Alarm
External Access: Read/Write
*Safety_System_Healthy - Safety /Safety - *0(OTE), 3(XIO), 5(XIC)*

Safety_Tripped_S 0 BOOL DUNEW2PLC1
External Access: Read/Write

Safety_Tripped_S (Continued)
Safety_Tripped_S - MainProgram/MainRoutine - 34(XIO), 35(XIC)
Safety_Tripped_S - PID_Tension_Servo/PID_Tension - 2(XIO)
*Safety_Tripped_S - Safety /Safety - *3(OTL), *5(OTU), 4(XIO)*

SegQueue		MotionSeg[32]	motionQueue
motion segments for move queue, lines or arcs			
External Access:	Read/Write		
*SegQueue - motionQueue/queue_line_arc - *145(FLL), *32(FFU), *69(FFU), *7(FFL)*			
SegQueue[0]		MotionSeg	
motion segments for move queue, lines or arcs			
SegQueue[0].Valid	0	BOOL	
motion segments for move queue, lines or arcs segment valid flag			
SegQueue[0].SegType	0	SINT	
motion segments for move queue, lines or arcs 1=line, 2=arc			
SegQueue[0].XY		REAL	
motion segments for move queue, lines or arcs end xy			
SegQueue[0].XY[0]	0.0	REAL	
motion segments for move queue, lines or arcs end xy			
SegQueue[0].XY[1]	0.0	REAL	
motion segments for move queue, lines or arcs end xy			
SegQueue[0].Speed	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[0].Accel	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[0].Decel	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[0].JerkAccel	0.0	REAL	
motion segments for move queue, lines or arcs			
SegQueue[0].JerkDecel	0.0	REAL	
motion segments for move queue, lines or arcs			
SegQueue[0].TermType	0	SINT	
motion segments for move queue, lines or arcs 0 to 6			
SegQueue[0].Seq	0	DINT	
motion segments for move queue, lines or arcs Sequence/ack id			
SegQueue[0].CircleType	0	DINT	
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum			
SegQueue[0].ViaCenter		REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
SegQueue[0].ViaCenter[0]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
SegQueue[0].ViaCenter[1]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
SegQueue[0].Direction	0	DINT	
motion segments for move queue, lines or arcs 0..3 per mmcm enum			
SegQueue[1]		MotionSeg	
motion segments for move queue, lines or arcs			
SegQueue[1].Valid	0	BOOL	
motion segments for move queue, lines or arcs segment valid flag			
SegQueue[1].SegType	0	SINT	
motion segments for move queue, lines or arcs 1=line, 2=arc			
SegQueue[1].XY		REAL	
motion segments for move queue, lines or arcs end xy			
SegQueue[1].XY[0]	0.0	REAL	
motion segments for move queue, lines or arcs end xy			
SegQueue[1].XY[1]	0.0	REAL	
motion segments for move queue, lines or arcs end xy			
SegQueue[1].Speed	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[1].Accel	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[1].Decel	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[1].JerkAccel	0.0	REAL	

SegQueue (Continued)
motion segments for move queue, lines or arcs

SegQueue[1].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[1].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[1].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[1].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[1].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[1].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[1].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[1].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[2]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[2].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[2].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[2].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[2].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[2].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[2].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[2].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[2].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[2].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[2].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[2].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[2].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[2].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[2].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[2].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[2].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[2].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[3]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[3].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[3].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[3].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[3].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		

SegQueue (Continued)

SegQueue[3].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[3].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[3].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[3].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[3].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[3].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[3].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[3].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[3].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[3].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[3].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[3].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[3].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[4]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[4].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[4].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[4].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[4].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[4].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[4].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[4].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[4].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[4].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[4].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[4].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[4].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[4].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[4].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[4].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[4].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[4].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[5]		MotionSeg

SegQueue (Continued)

motion segments for move queue, lines or arcs		
SegQueue[5].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[5].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[5].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[5].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[5].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[5].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[5].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[5].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[5].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[5].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[5].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[5].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[5].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[5].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[5].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[5].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[5].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[6]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[6].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[6].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[6].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[6].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[6].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[6].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[6].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[6].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[6].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[6].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[6].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[6].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[6].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		

SegQueue (Continued)

SegQueue[6].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[6].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[6].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[6].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[7]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[7].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[7].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[7].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[7].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[7].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[7].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[7].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[7].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[7].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[7].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[7].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[7].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[7].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[7].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[7].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[7].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[7].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[8]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[8].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[8].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[8].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[8].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[8].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[8].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[8].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[8].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[8].JerkAccel	0.0	REAL

SegQueue (Continued)
motion segments for move queue, lines or arcs

SegQueue[8].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[8].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[8].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[8].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[8].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[8].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[8].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[8].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[9]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[9].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[9].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[9].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[9].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[9].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[9].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[9].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[9].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[9].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[9].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[9].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[9].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[9].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[9].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[9].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[9].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[9].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[10]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[10].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[10].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[10].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[10].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		

SegQueue (Continued)

SegQueue[10].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[10].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[10].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[10].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[10].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[10].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[10].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[10].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[10].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[10].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[10].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[10].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[10].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[11]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[11].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[11].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[11].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[11].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[11].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[11].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[11].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[11].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[11].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[11].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[11].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[11].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[11].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[11].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[11].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[11].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[11].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[12]		MotionSeg

SegQueue (Continued)

motion segments for move queue, lines or arcs		
SegQueue[12].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[12].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[12].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[12].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[12].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[12].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[12].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[12].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[12].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[12].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[12].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[12].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[12].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[12].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[12].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[12].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[12].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[13]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[13].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[13].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[13].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[13].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[13].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[13].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[13].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[13].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[13].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[13].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[13].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[13].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[13].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		

SegQueue (Continued)

SegQueue[13].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[13].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[13].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[13].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[14]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[14].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[14].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[14].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[14].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[14].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[14].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[14].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[14].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[14].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[14].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[14].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[14].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[14].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[14].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[14].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[14].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[14].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[15]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[15].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[15].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[15].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[15].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[15].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[15].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[15].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[15].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[15].JerkAccel	0.0	REAL

SegQueue (Continued)

motion segments for move queue, lines or arcs		
SegQueue[15].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[15].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[15].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[15].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[15].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[15].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[15].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[15].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[16]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[16].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[16].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[16].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[16].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[16].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[16].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[16].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[16].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[16].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[16].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[16].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[16].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[16].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[16].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[16].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[16].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[16].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[17]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[17].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[17].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[17].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[17].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		

SegQueue (Continued)

SegQueue[17].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[17].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[17].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[17].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[17].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[17].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[17].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[17].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[17].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[17].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[17].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[17].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[17].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[18]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[18].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[18].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[18].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[18].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[18].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[18].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[18].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[18].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[18].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[18].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[18].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[18].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[18].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[18].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[18].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[18].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[18].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[19]		MotionSeg

SegQueue (Continued)

motion segments for move queue, lines or arcs		
SegQueue[19].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[19].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[19].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[19].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[19].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[19].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[19].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[19].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[19].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[19].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[19].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[19].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[19].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[19].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[19].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[19].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[19].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[20]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[20].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[20].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[20].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[20].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[20].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[20].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[20].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[20].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[20].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[20].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[20].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[20].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[20].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		

SegQueue (Continued)

SegQueue[20].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[20].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[20].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[20].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[21]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[21].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[21].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[21].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[21].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[21].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[21].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[21].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[21].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[21].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[21].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[21].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[21].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[21].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[21].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[21].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[21].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[21].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[22]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[22].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[22].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[22].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[22].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[22].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[22].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[22].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[22].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[22].JerkAccel	0.0	REAL

SegQueue (Continued)
motion segments for move queue, lines or arcs

SegQueue[22].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[22].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[22].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[22].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[22].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[22].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[22].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[22].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[23]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[23].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[23].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[23].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[23].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[23].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[23].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[23].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[23].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[23].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[23].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[23].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[23].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[23].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[23].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[23].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[23].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[23].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[24]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[24].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[24].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[24].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[24].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		

SegQueue (Continued)

SegQueue[24].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[24].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[24].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[24].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[24].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[24].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[24].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[24].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[24].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[24].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[24].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[24].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[24].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[25]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[25].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[25].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[25].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[25].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[25].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[25].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[25].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[25].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[25].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[25].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[25].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[25].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[25].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[25].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[25].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[25].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[25].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[26]		MotionSeg

SegQueue (Continued)

motion segments for move queue, lines or arcs		
SegQueue[26].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[26].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[26].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[26].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[26].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[26].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[26].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[26].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[26].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[26].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[26].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[26].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[26].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[26].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[26].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[26].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[26].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[27]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[27].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[27].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[27].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[27].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[27].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[27].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[27].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[27].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[27].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[27].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[27].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[27].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[27].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		

SegQueue (Continued)

SegQueue[27].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[27].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[27].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[27].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[28]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[28].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[28].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[28].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[28].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[28].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[28].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[28].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[28].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[28].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[28].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[28].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[28].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[28].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[28].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[28].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[28].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[28].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[29]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[29].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[29].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[29].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[29].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[29].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[29].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[29].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[29].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[29].JerkAccel	0.0	REAL

SegQueue (Continued)
motion segments for move queue, lines or arcs

SegQueue[29].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[29].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[29].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[29].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[29].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[29].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[29].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[29].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[30]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[30].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[30].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[30].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[30].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[30].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[30].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[30].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[30].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[30].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[30].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[30].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[30].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[30].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[30].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[30].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[30].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[30].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[31]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[31].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[31].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[31].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[31].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		

SegQueue (Continued)

SegQueue[31].XY[1]	0.0	REAL	
motion segments for move queue, lines or arcs end xy			
SegQueue[31].Speed	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[31].Accel	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[31].Decel	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[31].JerkAccel	0.0	REAL	
motion segments for move queue, lines or arcs			
SegQueue[31].JerkDecel	0.0	REAL	
motion segments for move queue, lines or arcs			
SegQueue[31].TermType	0	SINT	
motion segments for move queue, lines or arcs 0 to 6			
SegQueue[31].Seq	0	DINT	
motion segments for move queue, lines or arcs Sequence/ack id			
SegQueue[31].CircleType	0	DINT	
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum			
SegQueue[31].ViaCenter		REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
SegQueue[31].ViaCenter[0]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
SegQueue[31].ViaCenter[1]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
SegQueue[31].Direction	0	DINT	
motion segments for move queue, lines or arcs 0..3 per mmcm enum			
SegValid	0	BOOL	motionQueue
External Access:	Read/Write		
*SegValid - motionQueue/queue_line_arc - *6(OTE), 7(XIC), 8(XIC), 9(XIC)*			
SegValidArc	0	BOOL	motionQueue
External Access:	Read/Write		
*SegValidArc - motionQueue/queue_line_arc - *5(OTE), 6(XIC)*			
SegValidLine	0	BOOL	motionQueue
External Access:	Read/Write		
*SegValidLine - motionQueue/queue_line_arc - *4(OTE), 6(XIC)*			
SetBit		BOOL[32]	Initialize
External Access:	Read Only		
SetBit[0]	1	BOOL	
*SetBit[0] - Initialize/Init - *3(OSR), 6(XIC)*			
SetBit[1]	1	BOOL	
*SetBit[1] - Initialize/Init - *8(OSR)*			
SetBit		BOOL[32]	Ready_State_1
External Access:	Read/Write		
SetBit[0]	0	BOOL	
*SetBit[0] - Ready_State_1/Ready - *4(OSR)*			
SetBit[1]	0	BOOL	
*SetBit[1] - Ready_State_1/Ready - *6(OSR)*			
SetBit[2]	0	BOOL	
*SetBit[2] - Ready_State_1/Ready - *8(OSR)*			
SetBit[3]	0	BOOL	
*SetBit[3] - Ready_State_1/Ready - *10(OSR)*			
SetBit[4]	0	BOOL	
*SetBit[4] - Ready_State_1/Ready - *12(OSR)*			
SetBit[5]	0	BOOL	
*SetBit[5] - Ready_State_1/Ready - *14(OSR)*			
SetBit[6]	0	BOOL	
*SetBit[6] - Ready_State_1/Ready - *16(OSR)*			
SetBit[7]	0	BOOL	

SetBit (Continued)
*SetBit[7] - Ready_State_1/Ready - *18(OSR)*

SetPoint_Override 6.0 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
SetPoint_Override - PID_Tension_Servo/PID_Tension - 7(MOV)

speed_regulator_switch 1 BOOL DUNEW2PLC1
This is our "enable safe speed" switch, repurposed to switch the speed regulator
External Access: Read/Write
*speed_regulator_switch - MainProgram/MainRoutine - *58(OTE)*
speed_regulator_switch - MoveXY_State_2_3/MoveXY - 12(XIC), 12(XIO), 13(XIC)

speed_set_to_zero 0 BOOL MoveXY_State_2_3
External Access: Read/Write
*speed_set_to_zero - MoveXY_State_2_3/MoveXY - *16(OSR), 19(XIC)*

speed_tension_setpoint 8.5 REAL MoveXY_State_2_3
Constant No
External Access: Read/Write
speed_tension_setpoint - MoveXY_State_2_3/xy_speed_regulator - 5(CPT)

StartCurONS 0 BOOL motionQueue
External Access: Read/Write
*StartCurONS - motionQueue/queue_line_arc - *30(ONS)*

starting_x -898.3584 REAL MoveXY_State_2_3
External Access: Read/Write
*starting_x - MoveXY_State_2_3/MoveXY - *6(MOV), 7(CPT)*

starting_y -8.9992 REAL MoveXY_State_2_3
External Access: Read/Write
*starting_y - MoveXY_State_2_3/MoveXY - *6(MOV), 7(CPT)*

StartNextA_ONS 0 BOOL motionQueue
External Access: Read/Write
*StartNextA_ONS - motionQueue/queue_line_arc - *67(ONS)*

StartNextB_ONS 0 BOOL motionQueue
External Access: Read/Write
*StartNextB_ONS - motionQueue/queue_line_arc - *68(ONS)*

StartQueuedPath 0 BOOL DUNEW2PLC1
External Access: Read/Write
*StartQueuedPath - motionQueue/queue_line_arc - *133(OTU), *31(OTU), 30(XIC)*

STATE 1 DINT DUNEW2PLC1
CURRENT State Machine State
External Access: Read Only
STATE - Error_State_10/Error - 0(CMP), 12(CMP)
*STATE - Initialize/Init - *4(CPT), 7(CMP)*
STATE - Latch_UnLatch_State_6_7_8/Latch_Driver - 16(CMP), 34(CMP), 45(CMP), 45(CMP), 51(CMP), 8(CMP)
*STATE - MainProgram/MainRoutine - *42(CPT), *43(CPT), *44(CPT), *45(CPT), *46(CPT), *47(CPT), *48(CPT), *49(CPT), *50(CPT), *51(CPT), *52(CPT), *53(CPT)*
STATE - MoveXY_State_2_3/MoveXY - 0(CMP), 1(CMP), 11(CMP)
STATE - MoveZ_State_4_5/MoveZ - 4(CMP), 4(CMP), 5(CMP), 6(CMP)
STATE - Ready_State_1/Ready - 0(CMP)
STATE - UnServo_9/UnServo_All - 0(CMP)

STATE1_IND 1 BOOL Ready_State_1
External Access: Read Only
*STATE1_IND - Ready_State_1/Ready - *0(OTE), 1(XIC), 10(XIC), 12(XIC), 14(XIC), 16(XIC), 18(XIC), 3(XIC), 4(XIC), 6(XIC), 8(XIC)*

STATE3_IND 0 BOOL MoveXY_State_2_3
External Access: Read Only
*STATE3_IND - MoveXY_State_2_3/MoveXY - *1(OTE), 15(XIC), 2(XIC)*

STATE5_IND 0 BOOL MoveZ_State_4_5
External Access: Read Only
*STATE5_IND - MoveZ_State_4_5/MoveZ - *6(OTE), 10(XIC), 13(XIC), 13(XIC), 7(XIC), 8(XIC)*

STATE6_IND 0 BOOL DUNEW2PLC1
External Access: Read Only
*STATE6_IND - Latch_UnLatch_State_6_7_8/Latch_Driver - *8(OTE), 11(XIC), 9(XIC)*

STATE7_IND 0 BOOL Latch_UnLatch_State_6_7_8
External Access: Read Only
*STATE7_IND - Latch_UnLatch_State_6_7_8/Latch_Driver - *16(OTE), 17(XIC), 33(XIC)*

STATE8_IND 1 BOOL DUNEW2PLC1
External Access: Read Only
*STATE8_IND - Latch_UnLatch_State_6_7_8/Latch_Driver - *34(OTE), 35(XIC), 37(XIC), 43(XIC)*

STATE9_IND 0 BOOL UnServo_9
External Access: Read Only
*STATE9_IND - UnServo_9/UnServo_All - *0(OTE), 1(XIC), 11(XIC), 2(XIC), 3(XIC), 4(XIC)*

STATE10_IND 0 BOOL Error_State_10
External Access: Read Only
*STATE10_IND - Error_State_10/Error - *0(OTE), 1(XIC)*

STATE11_IND 1 BOOL EOT_Trip_11
External Access: Read Only
*STATE11_IND - EOT_Trip_11/EOT_TRIP - *0(OTE), 12(XIC), 15(XIC), 18(XIC), 4(XIC), 6(XIC), 9(XIC)*

stopping_distance 0.0 REAL MoveXY_State_2_3
Calculate the distance required to perform a jerk-limited stop at the gurrent jerk, acceleration, and speed
External Access: Read/Write
*stopping_distance - MoveXY_State_2_3/MoveXY - *9(CPT), 10(CMP), 11(CPT)*

stopping_ratio 1.1959486 REAL MoveXY_State_2_3
External Access: Read/Write
*stopping_ratio - MoveXY_State_2_3/MoveXY - *11(CPT)*

support_collision_window_bttm 0 BOOL DUNEW2PLC1
External Access: Read/Write
*support_collision_window_bttm - MainProgram/MainRoutine - *31(OTE)*
support_collision_window_bttm - MoveZ_State_4_5/MoveZ - 2(XIC), 2(XIC), 2(XIO)

support_collision_window_mid 0 BOOL DUNEW2PLC1
External Access: Read/Write
*support_collision_window_mid - MainProgram/MainRoutine - *32(OTE)*
support_collision_window_mid - MoveZ_State_4_5/MoveZ - 2(XIC), 2(XIC), 2(XIO)

support_collision_window_top 0 BOOL DUNEW2PLC1
External Access: Read/Write
*support_collision_window_top - MainProgram/MainRoutine - *33(OTE)*
support_collision_window_top - MoveZ_State_4_5/MoveZ - 2(XIC), 2(XIC), 2(XIO)

T01 TIMER DUNEW2PLC1
External Access: Read/Write
*T01 - MainProgram/MainRoutine - *34(TON)*
T01.TT 0 BOOL
T01.TT - MainProgram/MainRoutine - 36(XIC), 37(XIC), 38(XIC)
T01.DN 1 BOOL

T01 (Continued)
T01.DN - MainProgram/MainRoutine - 38(XIC)

tension 2.13597459e-003 REAL DUNEW2PLC1
wire tension from load cell (N)
External Access: Read/Write
tension - MainProgram/MainRoutine - 20(GRT)
tension - MoveXY_State_2_3/xy_speed_regulator - 5(CPT)
*tension - PID_Tension_Servo/PID_Tension - *5(CPT), *6(CPT), *7(MOV), 11(CMP), 12(GRT), 7(PID), 8(CPT)*

TENSION_CONTROL_OK 0 BOOL DUNEW2PLC1
External Access: Read/Write
TENSION_CONTROL_OK - MoveXY_State_2_3/MoveXY - 1(XIO), 12(XIC), 12(XIC), 12(XIO)
TENSION_CONTROL_OK - MoveZ_State_4_5/MoveZ - 6(XIO)
*TENSION_CONTROL_OK - PID_Tension_Servo/PID_Tension - *3(OTE), 3(XIC), 7(XIC), 7(XIO), 7(XIO)*

tension_motor_cv 2.5 REAL DUNEW2PLC1
External Access: Read/Write
*tension_motor_cv - PID_Tension_Servo/PID_Tension - *7(MOV), *7(MOV), *7(PID), 7(LES), 7(MOV), 8(CPT)*

tension_motor_difference -2.497887 REAL PID_Tension_Servo
External Access: Read/Write
*tension_motor_difference - PID_Tension_Servo/PID_Tension - *8(CPT)*

tension_motor_out 2.5 REAL DUNEW2PLC1
Tension_Servo_CMD
AliasFor: Local:5:O.Ch00.Data
Base Tag: Local:5:O.Ch00.Data
Constant No
External Access: Read/Write
*cv_to_electrocraft - PID_Tension_Servo/PID_Tension - *7(MOV), 9(CPT)*

tension_off_oneshot 0 BOOL DUNEW2PLC1
External Access: Read/Write
*tension_off_oneshot - PID_Tension_Servo/PID_Tension - *0(OSF)*

tension_on_oneshot 0 BOOL DUNEW2PLC1
External Access: Read/Write
*tension_on_oneshot - PID_Tension_Servo/PID_Tension - *0(OSR)*

TENSION_ON_SWITCH 0 BOOL DUNEW2PLC1
External Access: Read/Write
*TENSION_ON_SWITCH - MainProgram/MainRoutine - *12(OTE)*
TENSION_ON_SWITCH - PID_Tension_Servo/PID_Tension - 0(XIC), 12(XIC), 15(XIC), 2(XIC), 7(XIO)

tension_on_switch_delay_on_start TIMER DUNEW2PLC1
External Access: Read/Write
*tension_on_switch_delay_on_start - PID_Tension_Servo/PID_Tension - *0(TON)*
tension_on_switch_delay_on_start.TT
0 BOOL
tension_on_switch_delay_on_start.TT - PID_Tension_Servo/PID_Tension - 2(XIC), 3(XIC), 7(XIC)
tension_on_switch_delay_on_start.DN
0 BOOL
tension_on_switch_delay_on_start.DN - PID_Tension_Servo/PID_Tension - 14(XIC)

tension_setpoint 5.5 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
tension_setpoint - PID_Tension_Servo/PID_Tension - 11(CMP), 7(MOV)

tension_stable_time 1 DINT PID_Tension_Servo
Constant No
External Access: Read/Write
tension_stable_time - PID_Tension_Servo/PID_Tension - 11(MOV)

tension_stable_timer		TIMER	DUNEW2PLC1
External Access:	Read/Write		
*tension_stable_timer - PID_Tension_Servo/PID_Tension - *11(TON)*			
tension_stable_timer.PRE	1	DINT	
*tension_stable_timer.PRE - PID_Tension_Servo/PID_Tension - *11(MOV)*			
tension_stable_timer.DN	0	BOOL	
tension_stable_timer.DN - MoveXY_State_2_3/MoveXY - 1(XIC)			
tension_stable_timer.DN - MoveZ_State_4_5/MoveZ - 6(XIC)			
tension_stable_tolerance	0.1	REAL	PID_Tension_Servo
Constant	No		
External Access:	Read/Write		
tension_stable_tolerance - PID_Tension_Servo/PID_Tension - 11(CMP)			
tension_tag	2.09120335e-003	REAL	DUNEW2PLC1
wire tension (N)			
AliasFor:	Local:4:I.Ch01.Data		
Base Tag:	Local:4:I.Ch01.Data		
Constant	No		
External Access:	Read/Write		
tension_tag - PID_Tension_Servo/PID_Tension - 5(CPT), 6(CMP), 6(CPT)			
TimeoutTimer1		TIMER	Latch_UnLatch_State_6_7_8
External Access:	Read/Write		
*TimeoutTimer1 - Latch_UnLatch_State_6_7_8/Latch_Driver - *47(TON)*			
TimeoutTimer1.TT	0	BOOL	
TimeoutTimer1.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 48(XIO), 49(XIC)			
TimeoutTimer2		TIMER	Latch_UnLatch_State_6_7_8
External Access:	Read/Write		
*TimeoutTimer2 - Latch_UnLatch_State_6_7_8/Latch_Driver - *48(TON)*			
TimeoutTimer2.TT	0	BOOL	
TimeoutTimer2.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 47(XIO)			
TIMER		TIMER	Initialize
External Access:	Read/Write		
*TIMER - Initialize/Init - *1(TON)*			
TIMER.DN	1	BOOL	
TIMER.DN - Initialize/Init - 3(XIC)			
trigger_axes_ob	0	BOOL	Ready_State_1
External Access:	Read/Write		
*trigger_axes_ob - Ready_State_1/Ready - *1(OSR), 2(XIC)*			
trigger_axes_sb	1	BOOL	Ready_State_1
External Access:	Read/Write		
*trigger_axes_sb - Ready_State_1/Ready - *1(OSR)*			
trigger_xy_move	0	BOOL	MoveXY_State_2_3
External Access:	Read/Write		
*trigger_xy_move - MoveXY_State_2_3/MoveXY - *5(OSR), 12(XIC), 6(XIC), 7(XIC)*			
trigger_xy_move - MoveXY_State_2_3/xy_speed_regulator - 0(XIC), 1(XIC)			
triggerOneShotOB	0	BOOL	Camera
External Access:	Read Only		
*triggerOneShotOB - Camera/CameraControl - *1(OSR)*			
triggerOneShotSB	1	BOOL	Camera
External Access:	Read Only		
*triggerOneShotSB - Camera/CameraControl - *1(OSR), 2(XIC), 3(XIC)*			

TriggerResetTimer1 TIMER Camera
External Access: Read Only
*TriggerResetTimer1 - Camera/CameraControl - *2(TON)*
TriggerResetTimer1.TT 1 BOOL
TriggerResetTimer1.TT - Camera/CameraControl - 3(XIO), 4(XIC), 5(XIC), 6(XIC)

TriggerResetTimer2 TIMER Camera
External Access: Read Only
*TriggerResetTimer2 - Camera/CameraControl - *3(TON)*
TriggerResetTimer2.TT 0 BOOL
TriggerResetTimer2.TT - Camera/CameraControl - 2(XIO)

UNLOCK_LATCH_MOTOR_SHAFT
1 BOOL Latch_UnLatch_State_6_7_8

Constant No
External Access: Read Only
UNLOCK_LATCH_MOTOR_SHAFT - Latch_UnLatch_State_6_7_8/Latch_Driver - 34(XIC)

UseAasCurrent 1 BOOL motionQueue
1=A current/B next 0=b current/A next
External Access: Read/Write
*UseAasCurrent - motionQueue/queue_line_arc - *109(OTU), *110(OTL), *132(OTL), 100(XIO), 101(XIO), 104(XIC), 105(XIO), 107(XIC), 108(XIO), 117(XIC), 118(XIO), 35(XIC), 36(XIC), 37(XIC), 38(XIC), 39(XIC), 40(XIC), 41(XIC), 42(XIC), 43(XIC), 44(XIC), 45(XIC), 46(XIC), 47(XIO), 48(XIO), 49(XIO), 50(XIO), 51(XIO), 52(XIO), 53(XIO), 54(XIO), 55(XIO), 56(XIO), 57(XIO), 58(XIO), 61(XIC), 62(XIC), 63(XIO), 64(XIO), 67(XIC), 68(XIO), 72(XIC), 73(XIC), 74(XIC), 75(XIC), 76(XIC), 77(XIC), 78(XIC), 79(XIC), 80(XIC), 81(XIC), 82(XIC), 83(XIC), 84(XIO), 85(XIO), 86(XIO), 87(XIO), 88(XIO), 89(XIO), 90(XIO), 91(XIO), 92(XIO), 93(XIO), 94(XIO), 95(XIO), 98(XIC), 99(XIC)*

v_0 0.049999997 REAL DUNEW2PLC1
External Access: Read/Write
*v_0 - MoveXY_State_2_3/MoveXY - *9(MOV), 9(CPT)*

v_max 600.0 REAL MoveXY_State_2_3
External Access: Read/Write
*v_max - MoveXY_State_2_3/MoveXY - *7(CPT), 7(CPT), 7(LEQ), 7(LES)*
v_max - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM), 7(LES), 8(LES), 8(MOV)

v_xy 0.0 REAL DUNEW2PLC1
velocity of winding head in xy plane
External Access: Read/Write
*v_xy - MainProgram/MainRoutine - *56(CPT), 57(CPT), 57(NEQ)*

v_xyz 0.049999997 REAL DUNEW2PLC1
Velocity of winding head in xyz space
External Access: Read/Write
*v_xyz - MainProgram/MainRoutine - *55(CPT)*
v_xyz - MoveXY_State_2_3/MoveXY - 9(MOV)

winding_head_pid PID DUNEW2PLC1
External Access: Read/Write
*winding_head_pid - PID_Tension_Servo/PID_Tension - *7(PID)*
winding_head_pid.SP 0.0 REAL
*winding_head_pid.SP - PID_Tension_Servo/PID_Tension - *7(MOV), *7(MOV)*

wire_break_debounce TIMER PID_Tension_Servo
External Access: Read/Write
*wire_break_debounce - PID_Tension_Servo/PID_Tension - *1(TON)*
wire_break_debounce.DN 0 BOOL
wire_break_debounce.DN - PID_Tension_Servo/PID_Tension - 14(XIC), 2(XIO), 3(XIO)

wire_break_proxy 0 BOOL DUNEW2PLC1
External Access: Read/Write
*wire_break_proxy - MainProgram/MainRoutine - *20(OTE)*

wire_break_proxy (Continued)
wire_break_proxy - PID_Tension_Servo/PID_Tension - 1(XIO), 2(XIC), 3(XIC)

wire_break_switch_delay_on_start TIMER PID_Tension_Servo
External Access: Read/Write
*wire_break_switch_delay_on_start - PID_Tension_Servo/PID_Tension - *3(TON)*
wire_break_switch_delay_on_start.DN
0 BOOL
wire_break_switch_delay_on_start.DN - PID_Tension_Servo/PID_Tension - 3(XIO)

wire_broken_tension 0.2 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
wire_broken_tension - MainProgram/MainRoutine - 20(GRT)

X_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
X_axis - Camera/CameraControl - 6(MAR)
X_axis - EOT_Trip_11/EOT_TRIP - 10(MAM), 5(MSO), 7(MAM)
X_axis - Error_State_10/Error - 11(MAFR), 5(MSF)
X_axis - MoveXY_State_2_3/MoveXY - 3(MSO)
X_axis - Ready_State_1/Ready - 2(MSO)
X_axis - UnServo_9/UnServo_All - 2(MSF), 6(MAFR)
X_axis.ActualPosition -898.1537 REAL
X_axis.ActualPosition - Camera/CameraControl - 25(CPT), 28(CMP), 28(CMP)
X_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)
X_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 2(LIM), 2(LIM)
X_axis.ActualVelocity 0.0 REAL
X_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)
X_axis.Registration2NegativeEdgePosition
0.0 REAL
X_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 7(CPT)
X_axis.CommandAcceleration 0.0 REAL
X_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)
X_axis.SafeTorqueOffInhibit 0 BOOL
X_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)
X_axis.SLSActiveStatus 0 BOOL
X_axis.SLSActiveStatus - MainProgram/MainRoutine - 36(XIC)

X_axis_Cam_F_Pos REAL[10] Camera
External Access: Read Only
*X_axis_Cam_F_Pos - Camera/CameraControl - *17(FFU), *9(FFL)*

X_axis_Cam_F_Position -5.1799655 REAL Camera
External Access: Read Only
*X_axis_Cam_F_Position - Camera/CameraControl - *7(CPT), 9(FFL)*

X_Axis_SFX SAFETY_FEEDBACK_INTERFACE
DUNEW2PLC1
External Access: Read/Write
*X_Axis_SFX - Safety /Safety - *6(SFX), *7(SLS)*
X_Axis_SFX.HomeTrigger 0 BOOL
X_Axis_SFX.HomeTrigger - Safety /Safety - 6(SFX)
X_Axis_SFX.Reset 0 BOOL
*X_Axis_SFX.Reset - Safety /Safety - *5(OTE), 6(SFX)*

X_Axis_SLS SAFELY_LIMITED_SPEED DUNEW2PLC1
External Access: Read/Write
*X_Axis_SLS - Safety /Safety - *7(SLS)*
X_Axis_SLS.Request 0 BOOL
*X_Axis_SLS.Request - Safety /Safety - *1(OTE), 7(SLS)*
X_Axis_SLS.Reset 0 BOOL
*X_Axis_SLS.Reset - Safety /Safety - *5(OTE), 7(SLS)*
X_Axis_SLS.ActiveLimit 10.0 REAL

X_Axis_SLS (Continued)
X_Axis_SLS.ActiveLimit - Safety /Safety - 7(SLS)

X_AXIS_STAT		MOTION_INSTRUCTION[32]	UnServo_9
External Access:	Read Only		
X_AXIS_STAT[0]		MOTION_INSTRUCTION	
*X_AXIS_STAT[0] - UnServo_9/UnServo_All - *2(MSF)*			
X_AXIS_STAT[0].DN	0	BOOL	
X_AXIS_STAT[0].DN - UnServo_9/UnServo_All - 5(XIC)			
X_AXIS_STAT[1]		MOTION_INSTRUCTION	
*X_AXIS_STAT[1] - UnServo_9/UnServo_All - *6(MAFR)*			
X_AXIS_STAT[1].DN	0	BOOL	
X_AXIS_STAT[1].DN - UnServo_9/UnServo_All - 11(XIC)			
X_AXIS_STAT		MOTION_INSTRUCTION[32]	Error_State_10
External Access:	Read Only		
X_AXIS_STAT[0]		MOTION_INSTRUCTION	
*X_AXIS_STAT[0] - Error_State_10/Error - *2(MCS)*			
X_AXIS_STAT[0].PC	1	BOOL	
X_AXIS_STAT[0].PC - Error_State_10/Error - 4(XIC)			
X_AXIS_STAT[1]		MOTION_INSTRUCTION	
*X_AXIS_STAT[1] - Error_State_10/Error - *5(MSF)*			
X_AXIS_STAT[1].DN	0	BOOL	
X_AXIS_STAT[1].DN - Error_State_10/Error - 12(XIC), 8(XIC)			
X_AXIS_STAT[2]		MOTION_INSTRUCTION	
*X_AXIS_STAT[2] - Error_State_10/Error - *11(MAFR)*			
X_DELTA	0.0	REAL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
X_DELTA - Camera/CameraControl - 28(CMP), 28(CMP), 28(CMP), 28(CMP), 30(CPT)			
x_dist_to_target	0.20483398	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*x_dist_to_target - MoveXY_State_2_3/MoveXY - *8(CPT), 8(CPT)*			
X_max_speed	825.0	REAL	MoveXY_State_2_3
Constant	Yes		
External Access:	Read/Write		
X_max_speed - MoveXY_State_2_3/MoveXY - 7(CPT)			
x_on_mso		MOTION_INSTRUCTION	Ready_State_1
External Access:	Read/Write		
*x_on_mso - Ready_State_1/Ready - *2(MSO)*			
X_PARKED	0	BOOL	DUNEW2PLC1
External Access:	Read Only		
*X_PARKED - MainProgram/MainRoutine - *22(OTE)*			
x_pos	564.005	REAL	Camera
External Access:	Read Only		
*x_pos - Camera/CameraControl - *25(CPT), *34(CPT), 28(CMP), 28(CMP), 30(CPT), 34(CMP)*			
x_pos_next	564.005	REAL	Camera
External Access:	Read/Write		
*x_pos_next - Camera/CameraControl - *30(CPT), 34(CMP), 34(CPT)*			
X_POSITION	-898.3585	REAL	DUNEW2PLC1
AliasFor:	POSITION[0,0]		
Base Tag:	POSITION[0,0]		
Constant	Yes		
External Access:	Read/Write		
X_POSITION - MoveXY_State_2_3/MoveXY - 12(MCLM), 15(CMP), 15(CMP), 22(MCLM), 7(CPT), 8(CPT)			
X_POSITION - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)			

x_time 6758400.0 REAL MoveXY_State_2_3
External Access: Read/Write
*x_time - MoveXY_State_2_3/MoveXY - *7(CPT), 7(CPT), 7(LEQ), 7(LES)*

X_XFER_OK 0 BOOL DUNEW2PLC1
If at the head, check head support arms. Same for foot
External Access: Read Only
*X_XFER_OK - MainProgram/MainRoutine - *21(OTE)*
X_XFER_OK - MoveZ_State_4_5/MoveZ - 1(XIC), 2(XIC)

X_Y COORDINATE_SYSTEM DUNEW2PLC1
External Access: Read/Write
X_Y - EOT_Trip_11/EOT_TRIP - 0(MCS)
X_Y - Error_State_10/Error - 2(MCS)
X_Y - motionQueue/queue_line_arc - 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)
X_Y - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 19(MCS), 22(MCLM)
X_Y - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)
X_Y.MovePendingStatus 0 BOOL
X_Y.MovePendingStatus - motionQueue/queue_line_arc - 104(XIO), 105(XIO), 117(XIO), 118(XIO), 67(XIO), 68(XIO)
X_Y.PhysicalAxisFault 0 BOOL
X_Y.PhysicalAxisFault - MoveXY_State_2_3/MoveXY - 2(XIC)

xy_accel_jerk 1500 DINT MoveXY_State_2_3
Constant No
External Access: Read/Write
xy_accel_jerk - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 22(MCLM)
xy_accel_jerk - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)

XY_ACCELERATION 800.0 REAL DUNEW2PLC1
Maximum positive acceleration for position move.
Constant No
External Access: Read/Write
XY_ACCELERATION - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 22(MCLM)

XY_AXIS_STAT MOTION_INSTRUCTION[32] MoveXY_State_2_3
External Access: Read/Write
XY_AXIS_STAT[0] MOTION_INSTRUCTION
*XY_AXIS_STAT[0] - MoveXY_State_2_3/MoveXY - *3(MSO)*
XY_AXIS_STAT[0].DN 1 BOOL
XY_AXIS_STAT[0].DN - MoveXY_State_2_3/MoveXY - 4(XIC), 5(XIC)
XY_AXIS_STAT[1] MOTION_INSTRUCTION
*XY_AXIS_STAT[1] - MoveXY_State_2_3/MoveXY - *3(MSO)*
XY_AXIS_STAT[1].DN 1 BOOL
XY_AXIS_STAT[1].DN - MoveXY_State_2_3/MoveXY - 4(XIC), 5(XIC)
XY_AXIS_STAT[4] MOTION_INSTRUCTION
*XY_AXIS_STAT[4] - MoveXY_State_2_3/MoveXY - *12(MCLM), *22(MCLM)*
*XY_AXIS_STAT[4] - MoveXY_State_2_3/xy_speed_regulator - *0(MCLM)*
XY_AXIS_STAT[4].ER 1 BOOL
XY_AXIS_STAT[4].ER - MoveXY_State_2_3/MoveXY - 15(XIC)
XY_AXIS_STAT[4].PC 0 BOOL
XY_AXIS_STAT[4].PC - MoveXY_State_2_3/MoveXY - 15(XIC)
XY_AXIS_STAT[4].IP 0 BOOL
XY_AXIS_STAT[4].IP - MoveXY_State_2_3/MoveXY - 1(XIO), 16(XIC), 17(XIC)
XY_AXIS_STAT[5] MOTION_INSTRUCTION
*XY_AXIS_STAT[5] - MoveXY_State_2_3/MoveXY - *19(MCS)*
XY_AXIS_STAT[5].PC 1 BOOL
XY_AXIS_STAT[5].PC - MoveXY_State_2_3/MoveXY - 20(XIC)

xy_d_alpha 1.0 REAL MoveXY_State_2_3
Constant No
External Access: Read/Write
xy_d_alpha - MoveXY_State_2_3/xy_speed_regulator - 4(CPT)

xy_d_dt 50 DINT MoveXY_State_2_3

xy_d_dt (Continued)
Constant No
External Access: Read/Write
xy_d_dt - MoveXY_State_2_3/xy_speed_regulator - 1(MOV), 4(CPT)

xy_d_term 0.0 REAL MoveXY_State_2_3
External Access: Read/Write
*xy_d_term - MoveXY_State_2_3/xy_speed_regulator - *4(CPT), 4(MOV), 8(CPT)*

xy_d_term_prev 0.0 REAL MoveXY_State_2_3
External Access: Read/Write
*xy_d_term_prev - MoveXY_State_2_3/xy_speed_regulator - *4(MOV), 4(CPT)*

xy_d_timer TIMER MoveXY_State_2_3
External Access: Read/Write
*xy_d_timer - MoveXY_State_2_3/xy_speed_regulator - *3(TON)*
xy_d_timer.PRE 50 DINT
*xy_d_timer.PRE - MoveXY_State_2_3/xy_speed_regulator - *1(MOV)*
xy_d_timer.DN 0 BOOL
xy_d_timer.DN - MoveXY_State_2_3/xy_speed_regulator - 3(XIO), 4(XIC)

xy_decel_jerk 3000 DINT MoveXY_State_2_3
Constant No
External Access: Read/Write
xy_decel_jerk - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 22(MCLM), 9(MOV)
xy_decel_jerk - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)

XY_DECELERATION 800.0 REAL DUNEW2PLC1
Maximum negative acceleration for position move.
Constant No
External Access: Read/Write
XY_DECELERATION - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 22(MCLM)

xy_default_speed 0 DINT MoveXY_State_2_3
Constant No
External Access: Read/Write
xy_default_speed - MoveXY_State_2_3/xy_speed_regulator - 8(CPT)

xy_dist_to_target 1.3109018 REAL MoveXY_State_2_3
External Access: Read/Write
*xy_dist_to_target - MoveXY_State_2_3/MoveXY - *8(CPT), 10(CMP), 11(CPT)*

xy_dt 1 DINT MoveXY_State_2_3
Constant No
External Access: Read/Write
xy_dt - MoveXY_State_2_3/xy_speed_regulator - 1(MOV), 7(CPT)

xy_error 8.493781 REAL MoveXY_State_2_3
External Access: Read/Write
*xy_error - MoveXY_State_2_3/xy_speed_regulator - *5(CPT), 4(CPT), 4(MOV), 6(CPT), 7(CPT), 7(GRT), 7(LES)*

xy_error_prev 8.493781 REAL MoveXY_State_2_3
External Access: Read/Write
*xy_error_prev - MoveXY_State_2_3/xy_speed_regulator - *4(MOV), 4(CPT)*

xy_i_term 869.64233 REAL MoveXY_State_2_3
External Access: Read/Write
*xy_i_term - MoveXY_State_2_3/xy_speed_regulator - *1(MOV), *7(CPT), *7(MOV), *7(MOV), 7(CPT), 7(GRT), 7(LES), 8(CPT)*

xy_kd 0.0 REAL MoveXY_State_2_3
Constant No
External Access: Read/Write
xy_kd - MoveXY_State_2_3/xy_speed_regulator - 4(CPT)

xy_ki 300.0 REAL MoveXY_State_2_3
Constant No
External Access: Read/Write
xy_ki - MoveXY_State_2_3/xy_speed_regulator - 7(CPT)

xy_kp 0.0 REAL MoveXY_State_2_3
Constant No
External Access: Read/Write
xy_kp - MoveXY_State_2_3/xy_speed_regulator - 6(CPT)

xy_p_term 0.0 REAL MoveXY_State_2_3
External Access: Read/Write
*xy_p_term - MoveXY_State_2_3/xy_speed_regulator - *6(CPT), 8(CPT)*

XY_POS_LOCK_STAT 0 BOOL MoveXY_State_2_3
External Access: Read Only
*XY_POS_LOCK_STAT - MoveXY_State_2_3/MoveXY - *15(OTE)*

xy_regulated_acceleration 2000.0 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
xy_regulated_acceleration - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)

xy_regulated_deceleration 2000.0 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
xy_regulated_deceleration - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)

XY_SPEED 1020.0 REAL DUNEW2PLC1
Maximum velocity for position move.
Constant No
External Access: Read/Write
XY_SPEED - MoveXY_State_2_3/MoveXY - 16(CMP), 7(CPT), 7(LEQ), 7(LES)

XY_SPEED_REQ 600.0 REAL DUNEW2PLC1
XY Requested speed copy of XY_Speed to be used for MCLM and MCCD in MoveXY
External Access: Read/Write
*XY_SPEED_REQ - MoveXY_State_2_3/MoveXY - *7(CPT), *7(CPT), 12(MCLM), 14(MCCD), 22(MCLM)*

Y_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
Y_axis - Camera/CameraControl - 5(MAR)
Y_axis - EOT_Trip_11/EOT_TRIP - 13(MAM), 16(MAM), 5(MSO)
Y_axis - Error_State_10/Error - 10(MAFR), 6(MSF)
Y_axis - MoveXY_State_2_3/MoveXY - 3(MSO)
Y_axis - Ready_State_1/Ready - 2(MSO)
Y_axis - UnServo_9/UnServo_All - 3(MSF), 8(MAFR)

Y_axis.ActualPosition -15.0941 REAL
Y_axis.ActualPosition - Camera/CameraControl - 26(CPT), 28(CMP), 28(CMP)
Y_axis.ActualPosition - MainProgram/MainRoutine - 31(LIM), 32(LIM), 33(LIM)
Y_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)

Y_axis.ActualVelocity 0.0 REAL
Y_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)

Y_axis.Registration2NegativeEdgePosition
0.0 REAL
Y_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 8(CPT)

Y_axis.CommandAcceleration 0.0 REAL
Y_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)

Y_axis.SafeTorqueOffInhibit 0 BOOL
Y_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)

Y_axis_Cam_F_Pos REAL[10] Camera
External Access: Read Only
*Y_axis_Cam_F_Pos - Camera/CameraControl - *10(FFL), *18(FFU)*

Y_axis_Cam_F_Position	12.123628	REAL	Camera
External Access:	Read Only		
*Y_axis_Cam_F_Position - Camera/CameraControl - *8(CPT), 10(FFL)*			
Y_AXIS_STAT		MOTION_INSTRUCTION[32]	UnServo_9
External Access:	Read Only		
Y_AXIS_STAT[0]		MOTION_INSTRUCTION	
*Y_AXIS_STAT[0] - UnServo_9/UnServo_All - *3(MSF)*			
Y_AXIS_STAT[0].DN	0	BOOL	
Y_AXIS_STAT[0].DN - UnServo_9/UnServo_All - 7(XIC)			
Y_AXIS_STAT[1]		MOTION_INSTRUCTION	
*Y_AXIS_STAT[1] - UnServo_9/UnServo_All - *8(MAFR)*			
Y_AXIS_STAT[1].DN	0	BOOL	
Y_AXIS_STAT[1].DN - UnServo_9/UnServo_All - 11(XIC)			
Y_AXIS_STAT		MOTION_INSTRUCTION[32]	Error_State_10
External Access:	Read Only		
Y_AXIS_STAT[1]		MOTION_INSTRUCTION	
*Y_AXIS_STAT[1] - Error_State_10/Error - *6(MSF)*			
Y_AXIS_STAT[1].DN	0	BOOL	
Y_AXIS_STAT[1].DN - Error_State_10/Error - 12(XIC), 8(XIC)			
Y_AXIS_STAT[2]		MOTION_INSTRUCTION	
*Y_AXIS_STAT[2] - Error_State_10/Error - *10(MAFR)*			
Y_AXIS_STAT[2].DN	0	BOOL	
Y_AXIS_STAT[2].DN - Error_State_10/Error - 11(XIC)			
Y_DELTA	5.75	REAL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
Y_DELTA - Camera/CameraControl - 28(CMP), 28(CMP), 28(CMP), 28(CMP), 31(CPT)			
y_dist_to_target	-1.2947998	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*y_dist_to_target - MoveXY_State_2_3/MoveXY - *8(CPT), 8(CPT)*			
Y_max_speed	600.0	REAL	MoveXY_State_2_3
Constant	Yes		
External Access:	Read/Write		
Y_max_speed - MoveXY_State_2_3/MoveXY - 7(CPT)			
Y_MOUNT_XFER_OK	0	BOOL	DUNEW2PLC1
External Access:	Read Only		
*Y_MOUNT_XFER_OK - MainProgram/MainRoutine - *19(OTE)*			
y_on_mso		MOTION_INSTRUCTION	Ready_State_1
External Access:	Read/Write		
*y_on_mso - Ready_State_1/Ready - *2(MSO)*			
y_pos	2465.8398	REAL	Camera
External Access:	Read/Write		
*y_pos - Camera/CameraControl - *26(CPT), *34(CPT), 28(CMP), 28(CMP), 31(CPT), 34(CMP)*			
y_pos_next	2465.8398	REAL	Camera
External Access:	Read/Write		
*y_pos_next - Camera/CameraControl - *31(CPT), 34(CMP), 34(CPT)*			
Y_POSITION	-13.7993	REAL	DUNEW2PLC1
AliasFor:	POSITION[0,1]		
Base Tag:	POSITION[0,1]		
Constant	Yes		
External Access:	Read/Write		
Y_POSITION - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 7(CPT), 8(CPT)			
X_POSITION - MoveXY_State_2_3/MoveXY - 12(MCLM), 22(MCLM)			
X_POSITION - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)			

y_time	124.99739	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*y_time - MoveXY_State_2_3/MoveXY - *7(CPT), 7(CPT), 7(LEQ), 7(LES)*			
Y_XFER_OK	0	BOOL	DUNEW2PLC1
External Access:	Read Only		
*Y_XFER_OK - MainProgram/MainRoutine - *18(OTE)*			
Y_XFER_OK - MoveZ_State_4_5/MoveZ - 1(XIC), 2(XIC)			
z_accel_jerk	1500	DINT	MoveZ_State_4_5
Constant	No		
External Access:	Read/Write		
z_accel_jerk - MoveZ_State_4_5/MoveZ - 12(MAM)			
Z_ACCELERATION	800.0	REAL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
Z_ACCELERATION - MoveZ_State_4_5/MoveZ - 12(MAM)			
Z_axis		AXIS_CIP_DRIVE	DUNEW2PLC1
External Access:	Read Only		
Z_axis - EOT_Trip_11/EOT_TRIP - 19(MAM), 5(MSO)			
Z_axis - Error_State_10/Error - 3(MAS), 7(MSF), 9(MAFR)			
Z_axis - Initialize/Init - 7(MAFR)			
Z_axis - MoveZ_State_4_5/MoveZ - 12(MAM), 12(MAM), 15(MAS), 16(MAS), 9(MSO)			
Z_axis - Ready_State_1/Ready - 2(MSO)			
Z_axis - UnServo_9/UnServo_All - 10(MAFR), 4(MSF)			
Z_axis.PhysicalAxisFault	0	BOOL	
Z_axis.PhysicalAxisFault - MoveZ_State_4_5/MoveZ - 10(XIC)			
Z_axis.ActualPosition	-1.2238	REAL	
Z_axis.ActualPosition - MainProgram/MainRoutine - 6(CMP)			
Z_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 1(GEQ)			
Z_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 13(CMP), 13(CMP)			
Z_axis.ActualVelocity	-0.049999997	REAL	
Z_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT)			
Z_AXIS_STAT		MOTION_INSTRUCTION[32]	MoveZ_State_4_5
External Access:	Read Only		
Z_AXIS_STAT[3]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[3] - MoveZ_State_4_5/MoveZ - *9(MSO)*			
Z_AXIS_STAT[3].DN	0	BOOL	
Z_AXIS_STAT[3].DN - MoveZ_State_4_5/MoveZ - 11(XIC), 12(XIC)			
Z_AXIS_STAT[4]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[4] - MoveZ_State_4_5/MoveZ - *12(MAM), *12(MAM)*			
Z_AXIS_STAT[4].PC	0	BOOL	
Z_AXIS_STAT[4].PC - MoveZ_State_4_5/MoveZ - 13(XIC)			
Z_AXIS_STAT[4].IP	0	BOOL	
Z_AXIS_STAT[4].IP - MoveZ_State_4_5/MoveZ - 14(XIC), 16(XIC), 17(XIO)			
Z_AXIS_STAT[5]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[5] - MoveZ_State_4_5/MoveZ - *15(MAS)*			
Z_AXIS_STAT[5].PC	0	BOOL	
Z_AXIS_STAT[5].PC - MoveZ_State_4_5/MoveZ - 18(XIC)			
Z_AXIS_STAT[7]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[7] - MoveZ_State_4_5/MoveZ - *16(MAS)*			
Z_AXIS_STAT[7].PC	0	BOOL	
Z_AXIS_STAT[7].PC - MoveZ_State_4_5/MoveZ - 17(XIC)			
Z_AXIS_STAT		MOTION_INSTRUCTION[32]	UnServo_9
External Access:	Read Only		
Z_AXIS_STAT[0]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[0] - UnServo_9/UnServo_All - *4(MSF)*			
Z_AXIS_STAT[0].DN	0	BOOL	
Z_AXIS_STAT[0].DN - UnServo_9/UnServo_All - 9(XIC)			
Z_AXIS_STAT[1]		MOTION_INSTRUCTION	

Z_AXIS_STAT (Continued)
*Z_AXIS_STAT[1] - UnServo_9/UnServo_All - *10(MAFR)*
Z_AXIS_STAT[1].DN 0 BOOL
Z_AXIS_STAT[1].DN - UnServo_9/UnServo_All - 11(XIC)

Z_AXIS_STAT MOTION_INSTRUCTION[32] Initialize
External Access: Read Only
Z_AXIS_STAT[0] MOTION_INSTRUCTION
*Z_AXIS_STAT[0] - Initialize/Init - *7(MAFR)*
Z_AXIS_STAT[0].DN 1 BOOL
Z_AXIS_STAT[0].DN - Initialize/Init - 8(XIC)

Z_AXIS_STAT MOTION_INSTRUCTION[32] Error_State_10
External Access: Read Only
Z_AXIS_STAT[0] MOTION_INSTRUCTION
*Z_AXIS_STAT[0] - Error_State_10/Error - *3(MAS)*
Z_AXIS_STAT[0].PC 1 BOOL
Z_AXIS_STAT[0].PC - Error_State_10/Error - 4(XIC)
Z_AXIS_STAT[1] MOTION_INSTRUCTION
*Z_AXIS_STAT[1] - Error_State_10/Error - *7(MSF)*
Z_AXIS_STAT[1].DN 0 BOOL
Z_AXIS_STAT[1].DN - Error_State_10/Error - 12(XIC), 8(XIC)
Z_AXIS_STAT[2] MOTION_INSTRUCTION
*Z_AXIS_STAT[2] - Error_State_10/Error - *9(MAFR)*
Z_AXIS_STAT[2].DN 0 BOOL
Z_AXIS_STAT[2].DN - Error_State_10/Error - 10(XIC)

z_decel_jerk 3500 DINT MoveZ_State_4_5
Constant No
External Access: Read/Write
z_decel_jerk - MoveZ_State_4_5/MoveZ - 12(MAM)

Z_DECELLERATION 800.0 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
Z_DECELLERATION - MoveZ_State_4_5/MoveZ - 12(MAM), 15(MAS)

Z_EOT 1 BOOL DUNEW2PLC1
External Access: Read Only
Z_EOT - EOT_Trip_11/EOT_TRIP - 18(XIO)
*Z_EOT - MainProgram/MainRoutine - *9(OTE), 15(XIC)*

z_eot_bypass 1 BOOL DUNEW2PLC1
Constant No
External Access: Read/Write
z_eot_bypass - MainProgram/MainRoutine - 9(XIC)
z_eot_bypass - MoveZ_State_4_5/MoveZ - 16(XIO), 17(XIO)

Z_EXTENDED 0 BOOL DUNEW2PLC1
External Access: Read Only
Z_EXTENDED - Latch_UnLatch_State_6_7_8/Latch_Driver - 7(XIC)
*Z_EXTENDED - MainProgram/MainRoutine - *6(OTE)*

Z_FIXED_LATCHED 0 BOOL DUNEW2PLC1
External Access: Read Only
Z_FIXED_LATCHED - Latch_UnLatch_State_6_7_8/Latch_Driver - 6(XIC)
*Z_FIXED_LATCHED - MainProgram/MainRoutine - *8(OTE), 59(XIC), 59(XIC), 59(XIO)*
Z_FIXED_LATCHED - MoveZ_State_4_5/MoveZ - 0(XIC), 0(XIO), 12(XIC), 12(XIO), 6(XIC), 7(XIC)

Z_FIXED_PRESENT 0 BOOL DUNEW2PLC1
External Access: Read Only
Z_FIXED_PRESENT - Latch_UnLatch_State_6_7_8/Latch_Driver - 7(XIC)
*Z_FIXED_PRESENT - MainProgram/MainRoutine - *17(OTE)*

Z_OK_TO_ENGAGE 1 BOOL DUNEW2PLC1
External Access: Read Only
*Z_OK_TO_ENGAGE - Latch_UnLatch_State_6_7_8/Latch_Driver - *3(OTE)*

z_on_mso MOTION_INSTRUCTION Ready_State_1
External Access: Read/Write
*z_on_mso - Ready_State_1/Ready - *2(MSO)*

Z_POS_LOCK_STAT 0 BOOL MoveZ_State_4_5
External Access: Read Only
*Z_POS_LOCK_STAT - MoveZ_State_4_5/MoveZ - *13(OTE)*

Z_POSITION 0.0 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
Z_POSITION - MoveZ_State_4_5/MoveZ - 12(MAM), 12(MAM), 13(CMP), 13(CMP)

Z_RETRACTED 1 BOOL DUNEW2PLC1
External Access: Read Only
*Z_RETRACTED - MainProgram/MainRoutine - *5(OTE)*
Z_RETRACTED - MoveXY_State_2_3/MoveXY - 1(XIC), 1(XIO)

Z_RETRACTED_1A 1 BOOL MainProgram
External Access: Read Only
*Z_RETRACTED_1A - MainProgram/MainRoutine - *0(OTE), 5(XIC)*

Z_RETRACTED_1B 1 BOOL MainProgram
External Access: Read Only
*Z_RETRACTED_1B - MainProgram/MainRoutine - *2(OTE), 5(XIC)*

Z_RETRACTED_2A 1 BOOL MainProgram
External Access: Read Only
*Z_RETRACTED_2A - MainProgram/MainRoutine - *3(OTE), 5(XIC)*

Z_RETRACTED_2B 1 BOOL MainProgram
External Access: Read Only
*Z_RETRACTED_2B - MainProgram/MainRoutine - *4(OTE), 5(XIC)*

Z_SAFE_TO_WITHDRAW 0 BOOL DUNEW2PLC1
External Access: Read/Write
*Z_SAFE_TO_WITHDRAW - Latch_UnLatch_State_6_7_8/Latch_Driver - *6(OTE)*

Z_SPEED 300.0 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
Z_SPEED - MoveZ_State_4_5/MoveZ - 12(MAM), 14(CMP)

Z_STAGE_LATCHED 0 BOOL DUNEW2PLC1
External Access: Read Only
Z_STAGE_LATCHED - Latch_UnLatch_State_6_7_8/Latch_Driver - 24(XIC), 25(XIO), 4(XIO), 5(XIC)
*Z_STAGE_LATCHED - MainProgram/MainRoutine - *7(OTE), 59(XIC), 59(XIO)*

Z_STAGE_PRESENT 0 BOOL DUNEW2PLC1
Winding head is on the Z-arm
External Access: Read Only
Z_STAGE_PRESENT - Latch_UnLatch_State_6_7_8/Latch_Driver - 7(XIC)
*Z_STAGE_PRESENT - MainProgram/MainRoutine - *16(OTE)*

Z_STAGE_UNLATCHED 0 BOOL DUNEW2PLC1
External Access: Read Only
*Z_STAGE_UNLATCHED - Latch_UnLatch_State_6_7_8/Latch_Driver - *2(OTE)*

General

Class:	Standard

Configuration

Type:	Continuous	Disable automatic output processing to reduce task overhead:	No
Watchdog:	500.000 ms	Inhibit task:	No

Program Schedule

Scheduled

MainProgram	UnServo_9	EOT_Trip_11	Error_State_10
Latch_UnLatch_State_6_7_8	MoveXY_State_2_3	MoveZ_State_4_5	Ready_State_1
Initialize	PID_Tension_Servo	pid2	motionQueue

Unscheduled

Camera

Monitor

Scan Times(Elapsed Time)

Max:	0.731000 ms	Last:	0.098000 ms

Interval Times(Elapsed Times Between Triggers)

Max:	0.741000 ms	Min:	0.086000 ms

Task overlap count: 0

General

Class:	Standard

Configuration

Main:	MainRoutine	Inhibit program:	No
Fault:	None		

Monitor

Scan Times(Execution Time)

Max:	125 us	Last:	10 us

0
Local:1:I.Pt00.Data — MACHINE_SW_STAT[1] — Z_RETRACTED_1A

1
EStop Rack
DUNEW2PLC2:1:I.Pt02Data — MACHINE_SW_STAT[23]
EStop Pullcord 1
DUNEW2PLC2:1:I.Pt03Data
EStop Pullcord 2
DUNEW2PLC2:1:I.Pt04Data

2
Local:1:I.Pt01.Data — MACHINE_SW_STAT[2] — Z_RETRACTED_1B

3
Local:1:I.Pt02.Data — MACHINE_SW_STAT[3] — Z_RETRACTED_2A

4
Local:1:I.Pt03.Data — MACHINE_SW_STAT[4] — Z_RETRACTED_2B

5
Z_RETRACTED_1A — Z_RETRACTED_1B — Z_RETRACTED_2B — Z_RETRACTED
Z_RETRACTED_2A

6
Local:1:I.Pt04.Data — MACHINE_SW_STAT[5] — Z_EXTENDED
CMP
Expression Z_axis.ActualPosition > 415

7
Local:1:I.Pt11.Data — MACHINE_SW_STAT[6] — Z_STAGE_LATCHED

8
Local:2:I.Pt01.Data — MACHINE_SW_STAT[7] — Z_FIXED_LATCHED

9
Local:1:I.Pt07.Data — MACHINE_SW_STAT[8] — Z_EOT
z_eot_bypass

10
Local:1:I.Pt06.Data — MACHINE_SW_STAT[18] — PLUS_Y_EOT

11
Local:2:I.Pt12.Data — MACHINE_SW_STAT[19] — MINUS_Y_EOT

12
TENSION ON/OFF
Local:2:I.Pt06.Data — TENSION_ON_SWITCH

13
Local:2:I.Pt08.Data — MACHINE_SW_STAT[20] — PLUS_X_EOT

14
Local:2:I.Pt10.Data — MACHINE_SW_STAT[21] — MINUS_X_EOT

15
Z_EOT — PLUS_Y_EOT — MINUS_Y_EOT — PLUS_X_EOT — MINUS_X_EOT — Motion Axis Stop MAS — ALL_EOT_GOOD

16
Local:1:I.Pt10.Data (/) — MACHINE_SW_STAT[9] — Winding head is on the Z-arm Z_STAGE_PRESENT

17
Local:2:I.Pt02.Data (/) — MACHINE_SW_STAT[10] — Z_FIXED_PRESENT

18
Local:1:I.Pt12.Data — MACHINE_SW_STAT[17] — Y_XFER_OK

19
Local:1:I.Pt13.Data — MACHINE_SW_STAT[16] — Y_MOUNT_XFER_OK

20
The wire break switch was causing electrical problems, so we use the load-cell as a proxy for wire break
Local:1:I.Pt15.Data — wire_break_proxy

wire tension from load cell (N)

GRT	
Source A	tension 0.00214169
Source B	wire_broken_tension 0.2

21
Local:2:I.Pt00.Data — MACHINE_SW_STAT[15] — If at the head, check head support arms. Same for foot X_XFER_OK

22
Local:2:I.Pt04.Data — MACHINE_SW_STAT[14] — X_PARKED

23
Local:2:I.Pt14.Data — MACHINE_SW_STAT[22] — APA_IS_VERTICAL

24
LC OSS1 DUNEW2PLC2:1:I.Pt00Data — LC OSS2 DUNEW2PLC2:1:I.Pt01Data — MACHINE_SW_STAT[25]

25
Local:6:I.Pt00.Data — MACHINE_SW_STAT[26] — FRAME_LOC_HD_TOP

26
Local:6:I.Pt01.Data — MACHINE_SW_STAT[27] — FRAME_LOC_HD_MID

27
Local:6:I.Pt02.Data — MACHINE_SW_STAT[28] — FRAME_LOC_HD_BTM

28 Local:6:I.Pt03.Data — MACHINE_SW_STAT[29] — FRAME_LOC_FT_TOP

29 Local:6:I.Pt04.Data — MACHINE_SW_STAT[30] — FRAME_LOC_FT_MID

30 Local:6:I.Pt05.Data — MACHINE_SW_STAT[31] — FRAME_LOC_FT_BTM

31 LIM
Low Limit 80
Test Y_axis.ActualPosition -15.0941
High Limit 450
— support_collision_window_bttm

32 LIM
Low Limit 1050
Test Y_axis.ActualPosition -15.0941
High Limit 1550
— support_collision_window_mid

33 LIM
Low Limit 2200
Test Y_axis.ActualPosition -15.0941
High Limit 2650
— support_collision_window_top

34 Safety_Tripped_S — TON
Timer T01
Preset 5000
Accum 5000
EN
DN

35 Safety_Tripped_S — Reset PB - Light Local:3:O.Pt11.Data
Beacon - Green Local:3:O.Pt12.Data

36 blink_on.TT — T01.TT — Beacon - Amber Local:3:O.Pt13.Data
NEQ
Source A ERROR_CODE 0
Source B 0
X_axis.SLSActiveStatus

37 blink_on.TT — T01.TT — Beacon - Sounder Local:3:O.Pt15.Data

38
T01.TT
T01.DN
Beacon - Red
Local:3:O.Pt14.Data

39
TON
Timer blink_on
Preset 500
Accum 399
EN
DN

40
blink_on.DN
TON
Timer blink_off
Preset 500
Accum 0
EN
DN

41
blink_off.DN
blink_on
RES

42
INIT_DONE
CMP
Expression NEXTSTATE=0
CURRENT State
Machine State
CPT
Dest STATE
1
Expression 0

43
INIT_DONE
CMP
Expression NEXTSTATE=1
CURRENT State
Machine State
CPT
Dest STATE
1
Expression 1

44
INIT_DONE
CMP
Expression NEXTSTATE=2
CURRENT State
Machine State
CPT
Dest STATE
1
Expression 2

45
INIT_DONE
CMP
Expression NEXTSTATE=3
CURRENT State
Machine State
CPT
Dest STATE
1
Expression 3

46
INIT_DONE
CMP
Expression NEXTSTATE=4
CURRENT State
Machine State
CPT
Dest STATE
1
Expression 4

47
INIT_DONE
CMP
Expression NEXTSTATE=5
CURRENT State
Machine State
CPT
Dest STATE
1
Expression 5

48
INIT_DONE
CMP
Expression NEXTSTATE=6
CURRENT State Machine State
CPT
Dest STATE[1]
Expression 6

49
INIT_DONE
CMP
Expression NEXTSTATE=7
CURRENT State Machine State
CPT
Dest STATE[1]
Expression 7

50
INIT_DONE
CMP
Expression NEXTSTATE=8
CURRENT State Machine State
CPT
Dest STATE[1]
Expression 8

51
INIT_DONE
CMP
Expression NEXTSTATE=9
CURRENT State Machine State
CPT
Dest STATE[1]
Expression 9

52
INIT_DONE
CMP
Expression NEXTSTATE=10
CURRENT State Machine State
CPT
Dest STATE[1]
Expression 10

53
INIT_DONE
CMP
Expression NEXTSTATE=11
CURRENT State Machine State
CPT
Dest STATE[1]
Expression 11

54
Cabinet lighted push button
Local:2:I.Pt13.Data
ResetPB

55
Velocity of winding head in xyz space
CPT
Dest
Expression SQR(X_axis.ActualVelocity*X_axis.ActualVelocity+Y_axis.ActualVelocity*Y_axis.ActualVelocity)+SQR(Z_axis.ActualVelocity*Z_a

v_xyz
0.049999997
s.ActualVelocity)

56

velocity of winding head in xy plane

CPT
Dest v_xy 0.0
Expression SQR(X_axis.ActualVelocity*X_axis.ActualVelocity+Y_axis.ActualVelocity*Y_axis.ActualVelocity)

57

velocity of winding head in xy plane

NEQ
Source A v_xy 0.0
Source B 0

Tangential acceleration of the winding head in xy-plane

CPT
Dest
Expression (X_axis.ActualVelocity*X_axis.CommandAcceleration+Y_axis.ActualVelocity*Y_axis.Commar

accel_xy
0.0
Acceleration)/v_xy

58

DUNEW2PLC2:1:I.Pt06Data /

This is our "enable safe speed" switch, repurposed to switch the speed regulator
speed_regulator_switch

59

The HEAD_POS tag is used in our python code for the head transfers.

Z_STAGE_LATCHED
Z_FIXED_LATCHED

EQU
Source A ACTUATOR_POS 2
Source B 3

CPT
Dest HEAD_POS -1
Expression 0

Z_FIXED_LATCHED

EQU
Source A ACTUATOR_POS 2
Source B 2

CPT
Dest HEAD_POS -1
Expression 3

Z_STAGE_LATCHED / Z_FIXED_LATCHED /

CPT
Dest HEAD_POS -1
Expression -1

(End)

Name	Value	Data Type	Scope
accel_xy	0.0	REAL	DUNEW2PLC1
Tangential acceleration of the winding head in xy-plane			
External Access:	Read/Write		
*accel_xy - MainProgram/MainRoutine - *57(CPT)*			
accel_xy - MoveXY_State_2_3/MoveXY - 9(CPT), 9(CPT)			
ACTUATOR_POS	2	DINT	DUNEW2PLC1
External Access:	Read Only		
*ACTUATOR_POS - Latch_UnLatch_State_6_7_8/Latch_Driver - *2(CPT), *3(CPT), *4(CPT), *5(CPT), 12(CPT), 13(CMP), 13(CMP), 13(CMP), 13(NEQ)*			
ACTUATOR_POS - MainProgram/MainRoutine - 59(EQU), 59(EQU)			
ACTUATOR_POS - MoveZ_State_4_5/MoveZ - 0(EQU), 7(CMP)			
ALL_EOT_GOOD	0	BOOL	DUNEW2PLC1
External Access:	Read Only		
ALL_EOT_GOOD - EOT_Trip_11/EOT_TRIP - 0(XIO), 22(XIC)			
*ALL_EOT_GOOD - MainProgram/MainRoutine - *15(OTE)*			
ALL_EOT_GOOD - MoveXY_State_2_3/MoveXY - 17(XIO)			
ALL_EOT_GOOD - MoveZ_State_4_5/MoveZ - 16(XIO)			
APA_IS_VERTICAL	1	BOOL	DUNEW2PLC1
External Access:	Read Only		
*APA_IS_VERTICAL - MainProgram/MainRoutine - *23(OTE)*			
APA_IS_VERTICAL - MoveXY_State_2_3/MoveXY - 1(XIC), 1(XIO)			
blink_off		TIMER	DUNEW2PLC1
External Access:	Read/Write		
*blink_off - MainProgram/MainRoutine - *40(TON)*			
blink_off.DN	0	BOOL	
blink_off.DN - MainProgram/MainRoutine - 41(XIC)			
blink_on		TIMER	DUNEW2PLC1
External Access:	Read/Write		
*blink_on - MainProgram/MainRoutine - *39(TON), *41(RES)*			
blink_on.TT	0	BOOL	
blink_on.TT - MainProgram/MainRoutine - 36(XIC), 37(XIC)			
blink_on.DN	1	BOOL	
blink_on.DN - MainProgram/MainRoutine - 40(XIC)			
DUNEW2PLC2:1:I		AB:1734_IB8S_Safety2:I:0	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
DUNEW2PLC2:1:I.Pt00Data	1	BOOL	
LC OSS1			
DUNEW2PLC2:1:I.Pt00Data - MainProgram/MainRoutine - 24(XIC)			
DUNEW2PLC2:1:I.Pt00Data - Safety /Safety - 1(XIO)			
DUNEW2PLC2:1:I.Pt01Data	1	BOOL	
LC OSS2			
DUNEW2PLC2:1:I.Pt01Data - MainProgram/MainRoutine - 24(XIC)			
DUNEW2PLC2:1:I.Pt01Data - Safety /Safety - 1(XIO)			
DUNEW2PLC2:1:I.Pt02Data	1	BOOL	
EStop Rack			
DUNEW2PLC2:1:I.Pt02Data - MainProgram/MainRoutine - 1(XIO)			
DUNEW2PLC2:1:I.Pt02Data - Safety /Safety - 0(XIC)			
DUNEW2PLC2:1:I.Pt03Data	1	BOOL	
EStop Pullcord 1			
DUNEW2PLC2:1:I.Pt03Data - MainProgram/MainRoutine - 1(XIO)			
DUNEW2PLC2:1:I.Pt03Data - Safety /Safety - 0(XIC)			
DUNEW2PLC2:1:I.Pt04Data	1	BOOL	
EStop Pullcord 2			
DUNEW2PLC2:1:I.Pt04Data - MainProgram/MainRoutine - 1(XIO)			
DUNEW2PLC2:1:I.Pt04Data - Safety /Safety - 0(XIC)			
DUNEW2PLC2:1:I.Pt05Data	1	BOOL	

DUNEW2PLC2:1:I (Continued)
Run Key
DUNEW2PLC2:1:I.Pt05Data - Safety /Safety - 1(XIO), 2(XIC)

DUNEW2PLC2:1:I.Pt06Data 0 BOOL
DUNEW2PLC2:1:I.Pt06Data - MainProgram/MainRoutine - 58(XIO)
DUNEW2PLC2:1:I.Pt06Data - Safety /Safety - 6(XIO), 7(XIO)

ERROR_CODE 0 DINT DUNEW2PLC1
External Access: Read Only
*ERROR_CODE - EOT_Trip_11/EOT_TRIP - *22(CPT)*
*ERROR_CODE - Error_State_10/Error - *12(CPT)*
*ERROR_CODE - Initialize/Init - *4(CPT)*
*ERROR_CODE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *25(CPT), *33(CPT), *36(CPT), *51(CPT)*
ERROR_CODE - MainProgram/MainRoutine - 36(NEQ)
*ERROR_CODE - MoveXY_State_2_3/MoveXY - *1(CPT), *1(CPT), *15(CPT), *2(CPT), *2(CPT)*
*ERROR_CODE - MoveZ_State_4_5/MoveZ - *10(CPT), *13(CPT), *20(CPT), *4(CPT), *4(CPT), *7(CPT)*
*ERROR_CODE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT), *16(CPT), 14(NEQ), 16(EQU), 16(EQU)*
*ERROR_CODE - Ready_State_1/Ready - *3(MOV)*

FRAME_LOC_FT_BTM 1 BOOL DUNEW2PLC1
External Access: Read/Write
*FRAME_LOC_FT_BTM - MainProgram/MainRoutine - *30(OTE)*
FRAME_LOC_FT_BTM - MoveZ_State_4_5/MoveZ - 2(XIO)

FRAME_LOC_FT_MID 0 BOOL DUNEW2PLC1
External Access: Read/Write
*FRAME_LOC_FT_MID - MainProgram/MainRoutine - *29(OTE)*
FRAME_LOC_FT_MID - MoveZ_State_4_5/MoveZ - 2(XIO)

FRAME_LOC_FT_TOP 1 BOOL DUNEW2PLC1
External Access: Read/Write
*FRAME_LOC_FT_TOP - MainProgram/MainRoutine - *28(OTE)*
FRAME_LOC_FT_TOP - MoveZ_State_4_5/MoveZ - 2(XIO)

FRAME_LOC_HD_BTM 1 BOOL DUNEW2PLC1
External Access: Read/Write
*FRAME_LOC_HD_BTM - MainProgram/MainRoutine - *27(OTE)*
FRAME_LOC_HD_BTM - MoveZ_State_4_5/MoveZ - 2(XIO)

FRAME_LOC_HD_MID 1 BOOL DUNEW2PLC1
External Access: Read/Write
*FRAME_LOC_HD_MID - MainProgram/MainRoutine - *26(OTE)*
FRAME_LOC_HD_MID - MoveZ_State_4_5/MoveZ - 2(XIO)

FRAME_LOC_HD_TOP 1 BOOL DUNEW2PLC1
External Access: Read/Write
*FRAME_LOC_HD_TOP - MainProgram/MainRoutine - *25(OTE)*
FRAME_LOC_HD_TOP - MoveZ_State_4_5/MoveZ - 2(XIO)

HEAD_POS -1 DINT DUNEW2PLC1
External Access: Read/Write
*HEAD_POS - MainProgram/MainRoutine - *59(CPT), *59(CPT), *59(CPT)*

INIT_DONE 1 BOOL DUNEW2PLC1
External Access: Read Only
INIT_DONE - Error_State_10/Error - 0(XIC)
*INIT_DONE - Initialize/Init - *6(OTE), 7(XIC)*
INIT_DONE - MainProgram/MainRoutine - 42(XIC), 43(XIC), 44(XIC), 45(XIC), 46(XIC), 47(XIC), 48(XIC), 49(XIC), 50(XIC), 51(XIC), 52(XIC), 53(XIC)
INIT_DONE - Ready_State_1/Ready - 0(XIC)
INIT_DONE - UnServo_9/UnServo_All - 0(XIC)

Local:1:I AB:5000_DI16:I:0 DUNEW2PLC1
Constant No

Local:1:I (Continued)
External Access: Read/Write
Local:1:I.Pt00.Data 1 BOOL
Local:1:I.Pt00.Data - MainProgram/MainRoutine - 0(XIC)
Local:1:I.Pt01.Data 0 BOOL
Local:1:I.Pt01.Data - MainProgram/MainRoutine - 2(XIO)
Local:1:I.Pt02.Data 1 BOOL
Local:1:I.Pt02.Data - MainProgram/MainRoutine - 3(XIC)
Local:1:I.Pt03.Data 0 BOOL
Local:1:I.Pt03.Data - MainProgram/MainRoutine - 4(XIO)
Local:1:I.Pt04.Data 0 BOOL
Local:1:I.Pt04.Data - MainProgram/MainRoutine - 6(XIC)
Local:1:I.Pt06.Data 1 BOOL
Local:1:I.Pt06.Data - MainProgram/MainRoutine - 10(XIC)
Local:1:I.Pt07.Data 1 BOOL
Local:1:I.Pt07.Data - MainProgram/MainRoutine - 9(XIC)
Local:1:I.Pt08.Data 0 BOOL
Latch top
Local:1:I.Pt08.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - 0(XIO)
Local:1:I.Pt09.Data 0 BOOL
Latch middle
Local:1:I.Pt09.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - 1(XIO)
Local:1:I.Pt10.Data 1 BOOL
Local:1:I.Pt10.Data - MainProgram/MainRoutine - 16(XIO)
Local:1:I.Pt11.Data 0 BOOL
Local:1:I.Pt11.Data - MainProgram/MainRoutine - 7(XIC)
Local:1:I.Pt12.Data 0 BOOL
Local:1:I.Pt12.Data - MainProgram/MainRoutine - 18(XIC)
Local:1:I.Pt13.Data 0 BOOL
Local:1:I.Pt13.Data - MainProgram/MainRoutine - 19(XIC)
Local:1:I.Pt15.Data 0 BOOL
The wire break switch was causing electrical problems, so we use the load-cell as a proxy for wire break
Local:1:I.Pt15.Data - MainProgram/MainRoutine - 20(XIC)

Local:2:I AB:5000_DI16:I:0 DUNEW2PLC1
Constant No
External Access: Read/Write
Local:2:I.Pt00.Data 0 BOOL
Local:2:I.Pt00.Data - MainProgram/MainRoutine - 21(XIC)
Local:2:I.Pt01.Data 0 BOOL
Local:2:I.Pt01.Data - MainProgram/MainRoutine - 8(XIC)
Local:2:I.Pt02.Data 1 BOOL
Local:2:I.Pt02.Data - MainProgram/MainRoutine - 17(XIO)
Local:2:I.Pt04.Data 0 BOOL
Local:2:I.Pt04.Data - MainProgram/MainRoutine - 22(XIC)
Local:2:I.Pt06.Data 0 BOOL
TENSION ON/OFF
Local:2:I.Pt06.Data - MainProgram/MainRoutine - 12(XIC)
Local:2:I.Pt07.Data 1 BOOL
O/T SWITCH OVERTRAVEL SWITCH
Local:2:I.Pt08.Data 1 BOOL
Local:2:I.Pt08.Data - MainProgram/MainRoutine - 13(XIC)
Local:2:I.Pt10.Data 1 BOOL
Local:2:I.Pt10.Data - MainProgram/MainRoutine - 14(XIC)
Local:2:I.Pt12.Data 0 BOOL
Local:2:I.Pt12.Data - MainProgram/MainRoutine - 11(XIC)
Local:2:I.Pt13.Data 0 BOOL
Cabinet lighted push button
Local:2:I.Pt13.Data - MainProgram/MainRoutine - 54(XIC)
Local:2:I.Pt14.Data 1 BOOL
Local:2:I.Pt14.Data - MainProgram/MainRoutine - 23(XIC)

Local:3:O AB:5000_DO16:O:0 DUNEW2PLC1
External Access: Read/Write

Local:3:O (Continued)
Local:3:O.Pt01.Data 0 BOOL
Latching signal output
*Local:3:O.Pt01.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - *44(OTE)*
Local:3:O.Pt02.Data 0 BOOL
*Local:3:O.Pt02.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - *22(OTE), *35(OTE), 36(XIC)*
Local:3:O.Pt03.Data 0 BOOL
*Enable_tension_motor - PID_Tension_Servo/PID_Tension - *2(OTE)*
Local:3:O.Pt11.Data 0 BOOL
Reset PB - Light
*Local:3:O.Pt11.Data - MainProgram/MainRoutine - *35(OTE)*
Local:3:O.Pt12.Data 0 BOOL
Beacon - Green
*Local:3:O.Pt12.Data - MainProgram/MainRoutine - *35(OTE)*
Local:3:O.Pt13.Data 0 BOOL
Beacon - Amber
*Local:3:O.Pt13.Data - MainProgram/MainRoutine - *36(OTE)*
Local:3:O.Pt14.Data 1 BOOL
Beacon - Red
*Local:3:O.Pt14.Data - MainProgram/MainRoutine - *38(OTE)*
Local:3:O.Pt15.Data 0 BOOL
Beacon - Sounder
*Local:3:O.Pt15.Data - MainProgram/MainRoutine - *37(OTE)*
*Local:3:O.Pt15.Data - PID_Tension_Servo/PID_Tension - *12(OTE)*

Local:6:I AB:5000_DI16:I:0 DUNEW2PLC1
Constant No
External Access: Read/Write
Local:6:I.Pt00.Data 1 BOOL
Local:6:I.Pt00.Data - MainProgram/MainRoutine - 25(XIC)
Local:6:I.Pt01.Data 1 BOOL
Local:6:I.Pt01.Data - MainProgram/MainRoutine - 26(XIC)
Local:6:I.Pt02.Data 1 BOOL
Local:6:I.Pt02.Data - MainProgram/MainRoutine - 27(XIC)
Local:6:I.Pt03.Data 1 BOOL
Local:6:I.Pt03.Data - MainProgram/MainRoutine - 28(XIC)
Local:6:I.Pt04.Data 0 BOOL
Local:6:I.Pt04.Data - MainProgram/MainRoutine - 29(XIC)
Local:6:I.Pt05.Data 1 BOOL
Local:6:I.Pt05.Data - MainProgram/MainRoutine - 30(XIC)

MACHINE_SW_STAT BOOL[32] DUNEW2PLC1
External Access: Read Only
MACHINE_SW_STAT[0] 0 BOOL
*MACHINE_SW_STAT[0] - Latch_UnLatch_State_6_7_8/Latch_Driver - *26(OTE)*
MACHINE_SW_STAT[1] 1 BOOL
*MACHINE_SW_STAT[1] - MainProgram/MainRoutine - *0(OTE)*
MACHINE_SW_STAT[2] 1 BOOL
*MACHINE_SW_STAT[2] - MainProgram/MainRoutine - *2(OTE)*
MACHINE_SW_STAT[3] 1 BOOL
*MACHINE_SW_STAT[3] - MainProgram/MainRoutine - *3(OTE)*
MACHINE_SW_STAT[4] 1 BOOL
*MACHINE_SW_STAT[4] - MainProgram/MainRoutine - *4(OTE)*
MACHINE_SW_STAT[5] 0 BOOL
*MACHINE_SW_STAT[5] - MainProgram/MainRoutine - *6(OTE)*
MACHINE_SW_STAT[6] 0 BOOL
*MACHINE_SW_STAT[6] - MainProgram/MainRoutine - *7(OTE)*
MACHINE_SW_STAT[7] 0 BOOL
*MACHINE_SW_STAT[7] - MainProgram/MainRoutine - *8(OTE)*
MACHINE_SW_STAT[8] 1 BOOL
*MACHINE_SW_STAT[8] - MainProgram/MainRoutine - *9(OTE)*
MACHINE_SW_STAT[9] 0 BOOL
*MACHINE_SW_STAT[9] - MainProgram/MainRoutine - *16(OTE)*
MACHINE_SW_STAT[10] 0 BOOL

MACHINE_SW_STAT (Continued)
*MACHINE_SW_STAT[10] - MainProgram/MainRoutine - *17(OTE)*
MACHINE_SW_STAT[12] 1 BOOL
*MACHINE_SW_STAT[12] - Latch_UnLatch_State_6_7_8/Latch_Driver - *0(OTE)*
MACHINE_SW_STAT[13] 1 BOOL
*MACHINE_SW_STAT[13] - Latch_UnLatch_State_6_7_8/Latch_Driver - *1(OTE)*
MACHINE_SW_STAT[14] 0 BOOL
*MACHINE_SW_STAT[14] - MainProgram/MainRoutine - *22(OTE)*
MACHINE_SW_STAT[15] 0 BOOL
*MACHINE_SW_STAT[15] - MainProgram/MainRoutine - *21(OTE)*
MACHINE_SW_STAT[16] 0 BOOL
*MACHINE_SW_STAT[16] - MainProgram/MainRoutine - *19(OTE)*
MACHINE_SW_STAT[17] 0 BOOL
*MACHINE_SW_STAT[17] - MainProgram/MainRoutine - *18(OTE)*
MACHINE_SW_STAT[18] 1 BOOL
*MACHINE_SW_STAT[18] - MainProgram/MainRoutine - *10(OTE)*
MACHINE_SW_STAT[19] 0 BOOL
*MACHINE_SW_STAT[19] - MainProgram/MainRoutine - *11(OTE)*
MACHINE_SW_STAT[20] 1 BOOL
*MACHINE_SW_STAT[20] - MainProgram/MainRoutine - *13(OTE)*
MACHINE_SW_STAT[21] 1 BOOL
*MACHINE_SW_STAT[21] - MainProgram/MainRoutine - *14(OTE)*
MACHINE_SW_STAT[22] 1 BOOL
*MACHINE_SW_STAT[22] - MainProgram/MainRoutine - *23(OTE)*
MACHINE_SW_STAT[23] 0 BOOL
*MACHINE_SW_STAT[23] - MainProgram/MainRoutine - *1(OTE)*
MACHINE_SW_STAT[25] 1 BOOL
*MACHINE_SW_STAT[25] - MainProgram/MainRoutine - *24(OTE)*
MACHINE_SW_STAT[26] 1 BOOL
*MACHINE_SW_STAT[26] - MainProgram/MainRoutine - *25(OTE)*
MACHINE_SW_STAT[27] 1 BOOL
*MACHINE_SW_STAT[27] - MainProgram/MainRoutine - *26(OTE)*
MACHINE_SW_STAT[28] 1 BOOL
*MACHINE_SW_STAT[28] - MainProgram/MainRoutine - *27(OTE)*
MACHINE_SW_STAT[29] 1 BOOL
*MACHINE_SW_STAT[29] - MainProgram/MainRoutine - *28(OTE)*
MACHINE_SW_STAT[30] 0 BOOL
*MACHINE_SW_STAT[30] - MainProgram/MainRoutine - *29(OTE)*
MACHINE_SW_STAT[31] 1 BOOL
*MACHINE_SW_STAT[31] - MainProgram/MainRoutine - *30(OTE)*

MAS 1 BOOL DUNEW2PLC1
Motion Axis Stop
Constant No
External Access: Read/Write
MAS - MainProgram/MainRoutine - 15(XIC)

MINUS_X_EOT 1 BOOL DUNEW2PLC1
External Access: Read Only
MINUS_X_EOT - EOT_Trip_11/EOT_TRIP - 9(XIO)
*MINUS_X_EOT - MainProgram/MainRoutine - *14(OTE), 15(XIC)*

MINUS_Y_EOT 0 BOOL DUNEW2PLC1
External Access: Read Only
MINUS_Y_EOT - EOT_Trip_11/EOT_TRIP - 15(XIO)
*MINUS_Y_EOT - MainProgram/MainRoutine - *11(OTE), 15(XIC)*

NEXTSTATE 1 DINT DUNEW2PLC1
External Access: Read Only
*NEXTSTATE - EOT_Trip_11/EOT_TRIP - *22(CPT)*
*NEXTSTATE - Error_State_10/Error - *12(CPT)*
*NEXTSTATE - Initialize/Init - *4(CPT), *9(CPT)*
*NEXTSTATE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *13(CPT), *33(CPT), *43(CPT), *51(CPT)*

NEXTSTATE (Continued)
NEXTSTATE - MainProgram/MainRoutine - 42(CMP), 43(CMP), 44(CMP), 45(CMP), 46(CMP), 47(CMP), 48(CMP), 49(CMP), 50(CMP), 51(CMP), 52(CMP), 53(CMP)
*NEXTSTATE - MoveXY_State_2_3/MoveXY - *0(CPT), *1(CPT), *1(CPT), *18(CPT), *2(CPT), *21(CPT)*
*NEXTSTATE - MoveZ_State_4_5/MoveZ - *10(CPT), *19(CPT), *20(CPT), *4(CPT), *5(MOV), *7(CPT)*
*NEXTSTATE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT)*
*NEXTSTATE - Ready_State_1/Ready - *11(CPT), *13(CPT), *15(CPT), *17(CPT), *19(CPT), *5(CPT), *7(CPT), *9(CPT)*
*NEXTSTATE - UnServo_9/UnServo_All - *11(CPT)*

PLUS_X_EOT 1 BOOL DUNEW2PLC1
External Access: Read Only
PLUS_X_EOT - EOT_Trip_11/EOT_TRIP - 6(XIO)
*PLUS_X_EOT - MainProgram/MainRoutine - *13(OTE), 15(XIC)*

PLUS_Y_EOT 1 BOOL DUNEW2PLC1
External Access: Read Only
PLUS_Y_EOT - EOT_Trip_11/EOT_TRIP - 12(XIO)
*PLUS_Y_EOT - MainProgram/MainRoutine - *10(OTE), 15(XIC)*

ResetPB 0 BOOL DUNEW2PLC1
External Access: Read/Write
*ResetPB - MainProgram/MainRoutine - *54(OTE)*
resetPB_S - Safety /Safety - 5(XIC)

Safety_Tripped_S 0 BOOL DUNEW2PLC1
External Access: Read/Write
Safety_Tripped_S - MainProgram/MainRoutine - 34(XIO), 35(XIC)
Safety_Tripped_S - PID_Tension_Servo/PID_Tension - 2(XIO)
*Safety_Tripped_S - Safety /Safety - *3(OTL), *5(OTU), 4(XIO)*

speed_regulator_switch 1 BOOL DUNEW2PLC1
This is our "enable safe speed" switch, repurposed to switch the speed regulator
External Access: Read/Write
*speed_regulator_switch - MainProgram/MainRoutine - *58(OTE)*
speed_regulator_switch - MoveXY_State_2_3/MoveXY - 12(XIC), 12(XIO), 13(XIC)

STATE 1 DINT DUNEW2PLC1
CURRENT State Machine State
External Access: Read Only
STATE - Error_State_10/Error - 0(CMP), 12(CMP)
*STATE - Initialize/Init - *4(CPT), 7(CMP)*
STATE - Latch_UnLatch_State_6_7_8/Latch_Driver - 16(CMP), 34(CMP), 45(CMP), 45(CMP), 51(CMP), 8(CMP)
*STATE - MainProgram/MainRoutine - *42(CPT), *43(CPT), *44(CPT), *45(CPT), *46(CPT), *47(CPT), *48(CPT), *49(CPT), *50(CPT), *51(CPT), *52(CPT), *53(CPT)*
STATE - MoveXY_State_2_3/MoveXY - 0(CMP), 1(CMP), 11(CMP)
STATE - MoveZ_State_4_5/MoveZ - 4(CMP), 4(CMP), 5(CMP), 6(CMP)
STATE - Ready_State_1/Ready - 0(CMP)
STATE - UnServo_9/UnServo_All - 0(CMP)

support_collision_window_bttm 0 BOOL DUNEW2PLC1
External Access: Read/Write
*support_collision_window_bttm - MainProgram/MainRoutine - *31(OTE)*
support_collision_window_bttm - MoveZ_State_4_5/MoveZ - 2(XIC), 2(XIC), 2(XIO)

support_collision_window_mid 0 BOOL DUNEW2PLC1
External Access: Read/Write
*support_collision_window_mid - MainProgram/MainRoutine - *32(OTE)*
support_collision_window_mid - MoveZ_State_4_5/MoveZ - 2(XIC), 2(XIC), 2(XIO)

support_collision_window_top 0 BOOL DUNEW2PLC1
External Access: Read/Write
*support_collision_window_top - MainProgram/MainRoutine - *33(OTE)*
support_collision_window_top - MoveZ_State_4_5/MoveZ - 2(XIC), 2(XIC), 2(XIO)

T01 TIMER DUNEW2PLC1
External Access: Read/Write
*T01 - MainProgram/MainRoutine - *34(TON)*
T01.TT 0 BOOL
T01.TT - MainProgram/MainRoutine - 36(XIC), 37(XIC), 38(XIC)
T01.DN 1 BOOL
T01.DN - MainProgram/MainRoutine - 38(XIC)

tension 2.15502619e-003 REAL DUNEW2PLC1
wire tension from load cell (N)
External Access: Read/Write
tension - MainProgram/MainRoutine - 20(GRT)
tension - MoveXY_State_2_3/xy_speed_regulator - 5(CPT)
*tension - PID_Tension_Servo/PID_Tension - *5(CPT), *6(CPT), *7(MOV), 11(CMP), 12(GRT), 7(PID), 8(CPT)*

TENSION_ON_SWITCH 0 BOOL DUNEW2PLC1
External Access: Read/Write
*TENSION_ON_SWITCH - MainProgram/MainRoutine - *12(OTE)*
TENSION_ON_SWITCH - PID_Tension_Servo/PID_Tension - 0(XIC), 12(XIC), 15(XIC), 2(XIC), 7(XIO)

v_xy 0.0 REAL DUNEW2PLC1
velocity of winding head in xy plane
External Access: Read/Write
*v_xy - MainProgram/MainRoutine - *56(CPT), 57(CPT), 57(NEQ)*

v_xyz 0.049999997 REAL DUNEW2PLC1
Velocity of winding head in xyz space
External Access: Read/Write
*v_xyz - MainProgram/MainRoutine - *55(CPT)*
v_xyz - MoveXY_State_2_3/MoveXY - 9(MOV)

wire_break_proxy 0 BOOL DUNEW2PLC1
External Access: Read/Write
*wire_break_proxy - MainProgram/MainRoutine - *20(OTE)*
wire_break_proxy - PID_Tension_Servo/PID_Tension - 1(XIO), 2(XIC), 3(XIC)

wire_broken_tension 0.2 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
wire_broken_tension - MainProgram/MainRoutine - 20(GRT)

X_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
X_axis - Camera/CameraControl - 6(MAR)
X_axis - EOT_Trip_11/EOT_TRIP - 10(MAM), 5(MSO), 7(MAM)
X_axis - Error_State_10/Error - 11(MAFR), 5(MSF)
X_axis - MoveXY_State_2_3/MoveXY - 3(MSO)
X_axis - Ready_State_1/Ready - 2(MSO)
X_axis - UnServo_9/UnServo_All - 2(MSF), 6(MAFR)
X_axis.ActualPosition -898.1537 REAL
X_axis.ActualPosition - Camera/CameraControl - 25(CPT), 28(CMP), 28(CMP)
X_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)
X_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 2(LIM), 2(LIM)
X_axis.ActualVelocity 0.0 REAL
X_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)
X_axis.Registration2NegativeEdgePosition
0.0 REAL
X_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 7(CPT)
X_axis.CommandAcceleration 0.0 REAL
X_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)
X_axis.SafeTorqueOffInhibit 0 BOOL
X_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)
X_axis.SLSActiveStatus 0 BOOL
X_axis.SLSActiveStatus - MainProgram/MainRoutine - 36(XIC)

X_PARKED 0 BOOL DUNEW2PLC1
External Access: Read Only
*X_PARKED - MainProgram/MainRoutine - *22(OTE)*

X_XFER_OK 0 BOOL DUNEW2PLC1
If at the head, check head support arms. Same for foot
External Access: Read Only
*X_XFER_OK - MainProgram/MainRoutine - *21(OTE)*
X_XFER_OK - MoveZ_State_4_5/MoveZ - 1(XIC), 2(XIC)

Y_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
Y_axis - Camera/CameraControl - 5(MAR)
Y_axis - EOT_Trip_11/EOT_TRIP - 13(MAM), 16(MAM), 5(MSO)
Y_axis - Error_State_10/Error - 10(MAFR), 6(MSF)
Y_axis - MoveXY_State_2_3/MoveXY - 3(MSO)
Y_axis - Ready_State_1/Ready - 2(MSO)
Y_axis - UnServo_9/UnServo_All - 3(MSF), 8(MAFR)
Y_axis.ActualPosition -15.0941 REAL
Y_axis.ActualPosition - Camera/CameraControl - 26(CPT), 28(CMP), 28(CMP)
Y_axis.ActualPosition - MainProgram/MainRoutine - 31(LIM), 32(LIM), 33(LIM)
Y_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)
Y_axis.ActualVelocity 0.0 REAL
Y_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)
Y_axis.Registration2NegativeEdgePosition
0.0 REAL
Y_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 8(CPT)
Y_axis.CommandAcceleration 0.0 REAL
Y_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)
Y_axis.SafeTorqueOffInhibit 0 BOOL
Y_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)

Y_MOUNT_XFER_OK 0 BOOL DUNEW2PLC1
External Access: Read Only
*Y_MOUNT_XFER_OK - MainProgram/MainRoutine - *19(OTE)*

Y_XFER_OK 0 BOOL DUNEW2PLC1
External Access: Read Only
*Y_XFER_OK - MainProgram/MainRoutine - *18(OTE)*
Y_XFER_OK - MoveZ_State_4_5/MoveZ - 1(XIC), 2(XIC)

Z_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
Z_axis - EOT_Trip_11/EOT_TRIP - 19(MAM), 5(MSO)
Z_axis - Error_State_10/Error - 3(MAS), 7(MSF), 9(MAFR)
Z_axis - Initialize/Init - 7(MAFR)
Z_axis - MoveZ_State_4_5/MoveZ - 12(MAM), 12(MAM), 15(MAS), 16(MAS), 9(MSO)
Z_axis - Ready_State_1/Ready - 2(MSO)
Z_axis - UnServo_9/UnServo_All - 10(MAFR), 4(MSF)
Z_axis.PhysicalAxisFault 0 BOOL
Z_axis.PhysicalAxisFault - MoveZ_State_4_5/MoveZ - 10(XIC)
Z_axis.ActualPosition -1.2238 REAL
Z_axis.ActualPosition - MainProgram/MainRoutine - 6(CMP)
Z_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 1(GEQ)
Z_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 13(CMP), 13(CMP)
Z_axis.ActualVelocity 0.049999997 REAL
Z_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT)

Z_EOT 1 BOOL DUNEW2PLC1
External Access: Read Only
Z_EOT - EOT_Trip_11/EOT_TRIP - 18(XIO)
*Z_EOT - MainProgram/MainRoutine - *9(OTE), 15(XIC)*

z_eot_bypass 1 BOOL DUNEW2PLC1

z_eot_bypass (Continued)
Constant No
External Access: Read/Write
z_eot_bypass - MainProgram/MainRoutine - 9(XIC)
z_eot_bypass - MoveZ_State_4_5/MoveZ - 16(XIO), 17(XIO)

Z_EXTENDED 0 BOOL DUNEW2PLC1
External Access: Read Only
Z_EXTENDED - Latch_UnLatch_State_6_7_8/Latch_Driver - 7(XIC)
*Z_EXTENDED - MainProgram/MainRoutine - *6(OTE)*

Z_FIXED_LATCHED 0 BOOL DUNEW2PLC1
External Access: Read Only
Z_FIXED_LATCHED - Latch_UnLatch_State_6_7_8/Latch_Driver - 6(XIC)
*Z_FIXED_LATCHED - MainProgram/MainRoutine - *8(OTE), 59(XIC), 59(XIC), 59(XIO)*
Z_FIXED_LATCHED - MoveZ_State_4_5/MoveZ - 0(XIC), 0(XIO), 12(XIC), 12(XIO), 6(XIC), 7(XIC)

Z_FIXED_PRESENT 0 BOOL DUNEW2PLC1
External Access: Read Only
Z_FIXED_PRESENT - Latch_UnLatch_State_6_7_8/Latch_Driver - 7(XIC)
*Z_FIXED_PRESENT - MainProgram/MainRoutine - *17(OTE)*

Z_RETRACTED 1 BOOL DUNEW2PLC1
External Access: Read Only
*Z_RETRACTED - MainProgram/MainRoutine - *5(OTE)*
Z_RETRACTED - MoveXY_State_2_3/MoveXY - 1(XIC), 1(XIO)

Z_RETRACTED_1A 1 BOOL MainProgram
External Access: Read Only
*Z_RETRACTED_1A - MainProgram/MainRoutine - *0(OTE), 5(XIC)*

Z_RETRACTED_1B 1 BOOL MainProgram
External Access: Read Only
*Z_RETRACTED_1B - MainProgram/MainRoutine - *2(OTE), 5(XIC)*

Z_RETRACTED_2A 1 BOOL MainProgram
External Access: Read Only
*Z_RETRACTED_2A - MainProgram/MainRoutine - *3(OTE), 5(XIC)*

Z_RETRACTED_2B 1 BOOL MainProgram
External Access: Read Only
*Z_RETRACTED_2B - MainProgram/MainRoutine - *4(OTE), 5(XIC)*

Z_STAGE_LATCHED 0 BOOL DUNEW2PLC1
External Access: Read Only
Z_STAGE_LATCHED - Latch_UnLatch_State_6_7_8/Latch_Driver - 24(XIC), 25(XIO), 4(XIO), 5(XIC)
*Z_STAGE_LATCHED - MainProgram/MainRoutine - *7(OTE), 59(XIC), 59(XIO)*

Z_STAGE_PRESENT 0 BOOL DUNEW2PLC1
Winding head is on the Z-arm
External Access: Read Only
Z_STAGE_PRESENT - Latch_UnLatch_State_6_7_8/Latch_Driver - 7(XIC)
*Z_STAGE_PRESENT - MainProgram/MainRoutine - *16(OTE)*

General

Type:	Ladder Diagram	Class:	Standard
In Program:	MainProgram	Number of Rungs:	60

General

Class:	Standard

Configuration

Main:	UnServo_All	Inhibit program:	No
Fault:	None		

Monitor

Scan Times(Execution Time)

Max:	78 us	Last:	4 us

0
INIT_DONE
CMP
Expression STATE=9
STATE9_IND

1
STATE9_IND
OSR
Storage Bit oneshotsb[0]
Output Bit oneshotob[0]
SB
OB

2
STATE9_IND oneshotob[0]
MSF
Axis X_axis ...
Motion Control X_AXIS_STAT[0]
EN
DN
ER

3
STATE9_IND oneshotob[0]
MSF
Axis Y_axis ...
Motion Control Y_AXIS_STAT[0]
EN
DN
ER

4
STATE9_IND oneshotob[0]
MSF
Axis Z_axis ...
Motion Control Z_AXIS_STAT[0]
EN
DN
ER

5
X_AXIS_STAT[0].DN
OSR
Storage Bit oneshotsb[1]
Output Bit oneshotob[1]
SB
OB

6
oneshotob[1]
MAFR
Axis X_axis ...
Motion Control X_AXIS_STAT[1]
EN
DN
ER

7
Y_AXIS_STAT[0].DN
OSR
Storage Bit oneshotsb[2]
Output Bit oneshotob[2]
SB
OB

8
oneshotob[2]
MAFR
Axis Y_axis ...
Motion Control Y_AXIS_STAT[1]
EN
DN
ER

9
Z_AXIS_STAT[0].DN
OSR
Storage Bit oneshotsb[3]
Output Bit oneshotob[3]
SB
OB

10
oneshotob[3]
MAFR
Axis Z_axis ...
Motion Control Z_AXIS_STAT[1]
EN
DN
ER

11

STATE9_IND X_AXIS_STAT[1].DN Y_AXIS_STAT[1].DN Z_AXIS_STAT[1].DN

CMP
Expression MOVE_TYPE=0

CPT
Dest NEXTSTATE
1
Expression 1

(End)

Name	Value	Data Type	Scope
INIT_DONE	1	BOOL	DUNEW2PLC1
External Access:	Read Only		

INIT_DONE - Error_State_10/Error - 0(XIC)
*INIT_DONE - Initialize/Init - *6(OTE), 7(XIC)*
INIT_DONE - MainProgram/MainRoutine - 42(XIC), 43(XIC), 44(XIC), 45(XIC), 46(XIC), 47(XIC), 48(XIC), 49(XIC), 50(XIC), 51(XIC), 52(XIC), 53(XIC)
INIT_DONE - Ready_State_1/Ready - 0(XIC)
INIT_DONE - UnServo_9/UnServo_All - 0(XIC)

MOVE_TYPE	0	INT	DUNEW2PLC1

0 = No Move 1 = XY JOG 2 = XY Plane Move 3 = Z JOG 4 = Z Axis Move 5 = Toggle Latch Actuator 6 = Home the Latch Actuator 7 = Unlock Latch Actuator Motor 8 = Un-Servo Motors 9 = PLC_Init

External Access:	Read/Write		

*MOVE_TYPE - EOT_Trip_11/EOT_TRIP - *21(CPT), 12(CMP), 15(CMP), 18(CMP), 6(CMP), 9(CMP)*
MOVE_TYPE - Error_State_10/Error - 12(CMP), 8(CMP)
*MOVE_TYPE - Initialize/Init - *2(CPT), *4(CPT), 0(EQU)*
*MOVE_TYPE - Latch_UnLatch_State_6_7_8/Latch_Driver - *13(CPT), *33(CPT), 43(CMP)*
*MOVE_TYPE - MoveXY_State_2_3/MoveXY - *18(CPT), *21(CPT), 15(CMP)*
*MOVE_TYPE - MoveZ_State_4_5/MoveZ - *19(CPT), 13(CMP)*
MOVE_TYPE - Ready_State_1/Ready - 10(CMP), 12(CMP), 14(CMP), 16(CMP), 18(CMP), 4(CMP), 6(CMP), 8(CMP)
MOVE_TYPE - UnServo_9/UnServo_All - 11(CMP)

NEXTSTATE	1	DINT	DUNEW2PLC1
External Access:	Read Only		

*NEXTSTATE - EOT_Trip_11/EOT_TRIP - *22(CPT)*
*NEXTSTATE - Error_State_10/Error - *12(CPT)*
*NEXTSTATE - Initialize/Init - *4(CPT), *9(CPT)*
*NEXTSTATE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *13(CPT), *33(CPT), *43(CPT), *51(CPT)*
NEXTSTATE - MainProgram/MainRoutine - 42(CMP), 43(CMP), 44(CMP), 45(CMP), 46(CMP), 47(CMP), 48(CMP), 49(CMP), 50(CMP), 51(CMP), 52(CMP), 53(CMP)
*NEXTSTATE - MoveXY_State_2_3/MoveXY - *0(CPT), *1(CPT), *1(CPT), *18(CPT), *2(CPT), *21(CPT)*
*NEXTSTATE - MoveZ_State_4_5/MoveZ - *10(CPT), *19(CPT), *20(CPT), *4(CPT), *5(MOV), *7(CPT)*
*NEXTSTATE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT)*
*NEXTSTATE - Ready_State_1/Ready - *11(CPT), *13(CPT), *15(CPT), *17(CPT), *19(CPT), *5(CPT), *7(CPT), *9(CPT)*
*NEXTSTATE - UnServo_9/UnServo_All - *11(CPT)*

oneshotob		BOOL[32]	UnServo_9
External Access:	Read Only		
oneshotob[0]	0	BOOL	

*oneshotob[0] - UnServo_9/UnServo_All - *1(OSR), 2(XIC), 3(XIC), 4(XIC)*

oneshotob[1]	0	BOOL	

*oneshotob[1] - UnServo_9/UnServo_All - *5(OSR), 6(XIC)*

oneshotob[2]	0	BOOL	

*oneshotob[2] - UnServo_9/UnServo_All - *7(OSR), 8(XIC)*

oneshotob[3]	0	BOOL	

*oneshotob[3] - UnServo_9/UnServo_All - *9(OSR), 10(XIC)*

oneshotsb		BOOL[32]	UnServo_9
External Access:	Read Only		
oneshotsb[0]	0	BOOL	

*oneshotsb[0] - UnServo_9/UnServo_All - *1(OSR)*

oneshotsb[1]	0	BOOL	

*oneshotsb[1] - UnServo_9/UnServo_All - *5(OSR)*

oneshotsb[2]	0	BOOL	

*oneshotsb[2] - UnServo_9/UnServo_All - *7(OSR)*

oneshotsb[3]	0	BOOL	

*oneshotsb[3] - UnServo_9/UnServo_All - *9(OSR)*

STATE	1	DINT	DUNEW2PLC1

CURRENT State Machine State

External Access:	Read Only		

STATE - Error_State_10/Error - 0(CMP), 12(CMP)

STATE (Continued)
*STATE - Initialize/Init - *4(CPT), 7(CMP)*
STATE - Latch_UnLatch_State_6_7_8/Latch_Driver - 16(CMP), 34(CMP), 45(CMP), 45(CMP), 51(CMP), 8(CMP)
*STATE - MainProgram/MainRoutine - *42(CPT), *43(CPT), *44(CPT), *45(CPT), *46(CPT), *47(CPT), *48(CPT), *49(CPT), *50(CPT), *51(CPT), *52(CPT), *53(CPT)*
STATE - MoveXY_State_2_3/MoveXY - 0(CMP), 1(CMP), 11(CMP)
STATE - MoveZ_State_4_5/MoveZ - 4(CMP), 4(CMP), 5(CMP), 6(CMP)
STATE - Ready_State_1/Ready - 0(CMP)
STATE - UnServo_9/UnServo_All - 0(CMP)

STATE9_IND	0	BOOL	UnServo_9
External Access:	Read Only		

*STATE9_IND - UnServo_9/UnServo_All - *0(OTE), 1(XIC), 11(XIC), 2(XIC), 3(XIC), 4(XIC)*

X_axis		AXIS_CIP_DRIVE	DUNEW2PLC1
External Access:	Read Only		

X_axis - Camera/CameraControl - 6(MAR)
X_axis - EOT_Trip_11/EOT_TRIP - 10(MAM), 5(MSO), 7(MAM)
X_axis - Error_State_10/Error - 11(MAFR), 5(MSF)
X_axis - MoveXY_State_2_3/MoveXY - 3(MSO)
X_axis - Ready_State_1/Ready - 2(MSO)
X_axis - UnServo_9/UnServo_All - 2(MSF), 6(MAFR)

X_axis.ActualPosition	-898.1537	REAL

X_axis.ActualPosition - Camera/CameraControl - 25(CPT), 28(CMP), 28(CMP)
X_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)
X_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 2(LIM), 2(LIM)

X_axis.ActualVelocity	0.0	REAL

X_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)

X_axis.Registration2NegativeEdgePosition	0.0	REAL

X_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 7(CPT)

X_axis.CommandAcceleration	0.0	REAL

X_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)

X_axis.SafeTorqueOffInhibit	0	BOOL

X_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)

X_axis.SLSActiveStatus	0	BOOL

X_axis.SLSActiveStatus - MainProgram/MainRoutine - 36(XIC)

X_AXIS_STAT		MOTION_INSTRUCTION[32]	UnServo_9
External Access:	Read Only		
X_AXIS_STAT[0]		MOTION_INSTRUCTION	

*X_AXIS_STAT[0] - UnServo_9/UnServo_All - *2(MSF)*

X_AXIS_STAT[0].DN	0	BOOL

X_AXIS_STAT[0].DN - UnServo_9/UnServo_All - 5(XIC)

X_AXIS_STAT[1]		MOTION_INSTRUCTION

*X_AXIS_STAT[1] - UnServo_9/UnServo_All - *6(MAFR)*

X_AXIS_STAT[1].DN	0	BOOL

X_AXIS_STAT[1].DN - UnServo_9/UnServo_All - 11(XIC)

Y_axis		AXIS_CIP_DRIVE	DUNEW2PLC1
External Access:	Read Only		

Y_axis - Camera/CameraControl - 5(MAR)
Y_axis - EOT_Trip_11/EOT_TRIP - 13(MAM), 16(MAM), 5(MSO)
Y_axis - Error_State_10/Error - 10(MAFR), 6(MSF)
Y_axis - MoveXY_State_2_3/MoveXY - 3(MSO)
Y_axis - Ready_State_1/Ready - 2(MSO)
Y_axis - UnServo_9/UnServo_All - 3(MSF), 8(MAFR)

Y_axis.ActualPosition	-15.0941	REAL

Y_axis.ActualPosition - Camera/CameraControl - 26(CPT), 28(CMP), 28(CMP)
Y_axis.ActualPosition - MainProgram/MainRoutine - 31(LIM), 32(LIM), 33(LIM)
Y_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)

Y_axis.ActualVelocity	0.0	REAL

Y_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)

Y_axis.Registration2NegativeEdgePosition

Y_axis (Continued)

	0.0	REAL	
Y_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 8(CPT)			
Y_axis.CommandAcceleration	0.0	REAL	
Y_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)			
Y_axis.SafeTorqueOffInhibit	0	BOOL	
Y_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)			
Y_AXIS_STAT		MOTION_INSTRUCTION[32]	UnServo_9
External Access:	Read Only		
Y_AXIS_STAT[0]		MOTION_INSTRUCTION	
*Y_AXIS_STAT[0] - UnServo_9/UnServo_All - *3(MSF)*			
Y_AXIS_STAT[0].DN	0	BOOL	
Y_AXIS_STAT[0].DN - UnServo_9/UnServo_All - 7(XIC)			
Y_AXIS_STAT[1]		MOTION_INSTRUCTION	
*Y_AXIS_STAT[1] - UnServo_9/UnServo_All - *8(MAFR)*			
Y_AXIS_STAT[1].DN	0	BOOL	
Y_AXIS_STAT[1].DN - UnServo_9/UnServo_All - 11(XIC)			
Z_axis		AXIS_CIP_DRIVE	DUNEW2PLC1
External Access:	Read Only		
Z_axis - EOT_Trip_11/EOT_TRIP - 19(MAM), 5(MSO)			
Z_axis - Error_State_10/Error - 3(MAS), 7(MSF), 9(MAFR)			
Z_axis - Initialize/Init - 7(MAFR)			
Z_axis - MoveZ_State_4_5/MoveZ - 12(MAM), 12(MAM), 15(MAS), 16(MAS), 9(MSO)			
Z_axis - Ready_State_1/Ready - 2(MSO)			
Z_axis - UnServo_9/UnServo_All - 10(MAFR), 4(MSF)			
Z_axis.PhysicalAxisFault	0	BOOL	
Z_axis.PhysicalAxisFault - MoveZ_State_4_5/MoveZ - 10(XIC)			
Z_axis.ActualPosition	-1.2238	REAL	
Z_axis.ActualPosition - MainProgram/MainRoutine - 6(CMP)			
Z_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 1(GEQ)			
Z_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 13(CMP), 13(CMP)			
Z_axis.ActualVelocity	0.0	REAL	
Z_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT)			
Z_AXIS_STAT		MOTION_INSTRUCTION[32]	UnServo_9
External Access:	Read Only		
Z_AXIS_STAT[0]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[0] - UnServo_9/UnServo_All - *4(MSF)*			
Z_AXIS_STAT[0].DN	0	BOOL	
Z_AXIS_STAT[0].DN - UnServo_9/UnServo_All - 9(XIC)			
Z_AXIS_STAT[1]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[1] - UnServo_9/UnServo_All - *10(MAFR)*			
Z_AXIS_STAT[1].DN	0	BOOL	
Z_AXIS_STAT[1].DN - UnServo_9/UnServo_All - 11(XIC)			

General

Type:	Ladder Diagram	Class:	Standard
In Program:	UnServo_9	Number of Rungs:	12

General

Class:	Standard

Configuration

Main:	EOT_TRIP	Inhibit program:	No
Fault:	None		

Monitor

Scan Times(Execution Time)

Max:	94 us	Last:	7 us

0
ALL_EOT_GOOD
MCS
Coordinate System X_Y
Motion Control AXIS_STAT[28]
Stop Type All
Change Decel Yes
Decel Rate 10000
Decel Units Units per sec2
Change Decel Jerk Yes
Decel Jerk 1000
Jerk Units Units per sec3
EN
DN
ER
IP
PC
STATE11_IND

1
eot_inc.DN
TON
Timer eot_pause
Preset 500
Accum 0
EN
DN

2
TON
Timer eot_inc
Preset 500
Accum 49
EN
DN

3
eot_pause.DN
eot_inc
RES

4
STATE11_IND
OSR
Storage Bit oneshotsb[0]
Output Bit oneshotob[0]
SB
OB

5
oneshotob[0]
MSO
Axis X_axis
Motion Control AXIS_STAT[29]
EN
DN
ER
MSO
Axis Y_axis
Motion Control AXIS_STAT[30]
EN
DN
ER
MSO
Axis Z_axis
Motion Control AXIS_STAT[31]
EN
DN
ER

6
STATE11_IND
PLUS_X_EOT
CMP
Expression MOVE_TYPE=99
eot_inc.DN
OSR
Storage Bit oneshotsb[1]
Output Bit oneshotob[1]
SB
OB

7

oneshotob[1]

MAM

Axis	X_axis ...
Motion Control	AXIS_STAT[0]
Move Type	1
Position	-1
Speed	25
Speed Units	Units per sec
Accel Rate	100
Accel Units	Units per sec2
Decel Rate	100
Decel Units	Units per sec2
Profile	Trapezoidal
Accel Jerk	100
Decel Jerk	100
Jerk Units	% of Time
Merge	Disabled
Merge Speed	Programmed
Lock Position	0
Lock Direction	None
Event Distance	0
Calculated Data	0

EN DN ER IP PC

8

AXIS_STAT[0].PC

OSR

Storage Bit	oneshotsb[2]
Output Bit	oneshotob[2]

SB OB

9

STATE11_IND MINUS_X_EOT

CMP

Expression	MOVE_TYPE=99

eot_inc.DN

OSR

Storage Bit	oneshotsb[3]
Output Bit	oneshotob[3]

SB OB

10

oneshotob[3]

MAM

Axis	X_axis ...
Motion Control	AXIS_STAT[1]
Move Type	1
Position	1
Speed	25
Speed Units	Units per sec
Accel Rate	100
Accel Units	Units per sec2
Decel Rate	100
Decel Units	Units per sec2
Profile	Trapezoidal
Accel Jerk	100
Decel Jerk	100
Jerk Units	% of Time
Merge	Disabled
Merge Speed	Programmed
Lock Position	0
Lock Direction	None
Event Distance	0
Calculated Data	0

EN DN ER IP PC

11
AXIS_STAT[1].PC
OSR
Storage Bit oneshotsb[4]
Output Bit oneshotob[4]
SB
OB

12
STATE11_IND
PLUS_Y_EOT
CMP
Expression MOVE_TYPE=99
eot_inc.DN
OSR
Storage Bit oneshotsb[5]
Output Bit oneshotob[5]
SB
OB

13
oneshotob[5]
MAM
Axis Y_axis ...
Motion Control AXIS_STAT[2]
Move Type 1
Position -1
Speed 25
Speed Units Units per sec
Accel Rate 100
Accel Units Units per sec2
Decel Rate 100
Decel Units Units per sec2
Profile Trapezoidal
Accel Jerk 100
Decel Jerk 100
Jerk Units % of Time
Merge Disabled
Merge Speed Programmed
Lock Position 0
Lock Direction None
Event Distance 0
Calculated Data 0
EN
DN
ER
IP
PC

14
AXIS_STAT[2].PC
OSR
Storage Bit oneshotsb[6]
Output Bit oneshotob[6]
SB
OB

15
STATE11_IND
MINUS_Y_EOT
CMP
Expression MOVE_TYPE=99
eot_inc.DN
OSR
Storage Bit oneshotsb[7]
Output Bit oneshotob[7]
SB
OB

16
oneshotob[7]

MAM
Axis: Y_axis
Motion Control: AXIS_STAT[3]
Move Type: 1
Position: 1
Speed: 25
Speed Units: Units per sec
Accel Rate: 100
Accel Units: Units per sec2
Decel Rate: 100
Decel Units: Units per sec2
Profile: Trapezoidal
Accel Jerk: 100
Decel Jerk: 100
Jerk Units: % of Time
Merge: Disabled
Merge Speed: Programmed
Lock Position: 0
Lock Direction: None
Event Distance: 0
Calculated Data: 0
EN DN ER IP PC

17
AXIS_STAT[3].PC

OSR
Storage Bit: oneshotsb[8]
Output Bit: oneshotob[8]
SB OB

18
STATE11_IND Z_EOT

CMP
Expression: MOVE_TYPE=99

eot_inc.DN

OSR
Storage Bit: oneshotsb[9]
Output Bit: oneshotob[9]
SB OB

19
oneshotob[9]

MAM
Axis: Z_axis
Motion Control: AXIS_STAT[4]
Move Type: 0
Position: 0
Speed: 25
Speed Units: Units per sec
Accel Rate: 100
Accel Units: Units per sec2
Decel Rate: 100
Decel Units: Units per sec2
Profile: Trapezoidal
Accel Jerk: 100
Decel Jerk: 100
Jerk Units: % of Time
Merge: Disabled
Merge Speed: Programmed
Lock Position: 0
Lock Direction: None
Event Distance: 0
Calculated Data: 0
EN DN ER IP PC

20

AXIS_STAT[4].PC

OSR
Storage Bit oneshotsb[10]
Output Bit oneshotob[10]
SB
OB

21

0 = No Move
1 = XY JOG
2 = XY Plane Move
3 = Z JOG
4 = Z Axis Move
5 = Toggle Latch Actuator
6 = Home the Latch Actuator
7 = Unlock Latch Actuator Motor
8 = Un-Servo Motors
9 = PLC_Init

oneshotob[2]
oneshotob[4]
oneshotob[6]
oneshotob[8]
oneshotob[10]

CPT
Dest MOVE_TYPE
0
Expression 0

OSR
Storage Bit oneshotsb[11]
Output Bit oneshotob[11]
SB
OB

22

oneshotob[11] ALL_EOT_GOOD

CPT
Dest ERROR_CODE
0
Expression 0

CPT
Dest NEXTSTATE
1
Expression 1

(End)

Name	Value	Data Type	Scope
ALL_EOT_GOOD	0	BOOL	DUNEW2PLC1
External Access:	Read Only		
ALL_EOT_GOOD - EOT_Trip_11/EOT_TRIP - 0(XIO), 22(XIC)			
*ALL_EOT_GOOD - MainProgram/MainRoutine - *15(OTE)*			
ALL_EOT_GOOD - MoveXY_State_2_3/MoveXY - 17(XIO)			
ALL_EOT_GOOD - MoveZ_State_4_5/MoveZ - 16(XIO)			
AXIS_STAT		MOTION_INSTRUCTION[32]	EOT_Trip_11
External Access:	Read Only		
AXIS_STAT[0]		MOTION_INSTRUCTION	
*AXIS_STAT[0] - EOT_Trip_11/EOT_TRIP - *7(MAM)*			
AXIS_STAT[0].PC	1	BOOL	
AXIS_STAT[0].PC - EOT_Trip_11/EOT_TRIP - 8(XIC)			
AXIS_STAT[1]		MOTION_INSTRUCTION	
*AXIS_STAT[1] - EOT_Trip_11/EOT_TRIP - *10(MAM)*			
AXIS_STAT[1].PC	0	BOOL	
AXIS_STAT[1].PC - EOT_Trip_11/EOT_TRIP - 11(XIC)			
AXIS_STAT[2]		MOTION_INSTRUCTION	
*AXIS_STAT[2] - EOT_Trip_11/EOT_TRIP - *13(MAM)*			
AXIS_STAT[2].PC	0	BOOL	
AXIS_STAT[2].PC - EOT_Trip_11/EOT_TRIP - 14(XIC)			
AXIS_STAT[3]		MOTION_INSTRUCTION	
*AXIS_STAT[3] - EOT_Trip_11/EOT_TRIP - *16(MAM)*			
AXIS_STAT[3].PC	0	BOOL	
AXIS_STAT[3].PC - EOT_Trip_11/EOT_TRIP - 17(XIC)			
AXIS_STAT[4]		MOTION_INSTRUCTION	
*AXIS_STAT[4] - EOT_Trip_11/EOT_TRIP - *19(MAM)*			
AXIS_STAT[4].PC	0	BOOL	
AXIS_STAT[4].PC - EOT_Trip_11/EOT_TRIP - 20(XIC)			
AXIS_STAT[28]		MOTION_INSTRUCTION	
*AXIS_STAT[28] - EOT_Trip_11/EOT_TRIP - *0(MCS)*			
AXIS_STAT[29]		MOTION_INSTRUCTION	
*AXIS_STAT[29] - EOT_Trip_11/EOT_TRIP - *5(MSO)*			
AXIS_STAT[30]		MOTION_INSTRUCTION	
*AXIS_STAT[30] - EOT_Trip_11/EOT_TRIP - *5(MSO)*			
AXIS_STAT[31]		MOTION_INSTRUCTION	
*AXIS_STAT[31] - EOT_Trip_11/EOT_TRIP - *5(MSO)*			
eot_inc		TIMER	DUNEW2PLC1
External Access:	Read/Write		
*eot_inc - EOT_Trip_11/EOT_TRIP - *2(TON), *3(RES)*			
eot_inc.DN	0	BOOL	
eot_inc.DN - EOT_Trip_11/EOT_TRIP - 1(XIC), 12(XIC), 15(XIC), 18(XIC), 6(XIC), 9(XIO)			
eot_pause		TIMER	DUNEW2PLC1
External Access:	Read/Write		
*eot_pause - EOT_Trip_11/EOT_TRIP - *1(TON)*			
eot_pause.DN	0	BOOL	
eot_pause.DN - EOT_Trip_11/EOT_TRIP - 3(XIC)			
ERROR_CODE	0	DINT	DUNEW2PLC1
External Access:	Read Only		
*ERROR_CODE - EOT_Trip_11/EOT_TRIP - *22(CPT)*			
*ERROR_CODE - Error_State_10/Error - *12(CPT)*			
*ERROR_CODE - Initialize/Init - *4(CPT)*			
*ERROR_CODE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *25(CPT), *33(CPT), *36(CPT), *51(CPT)*			
ERROR_CODE - MainProgram/MainRoutine - 36(NEQ)			
*ERROR_CODE - MoveXY_State_2_3/MoveXY - *1(CPT), *1(CPT), *15(CPT), *2(CPT), *2(CPT)*			
*ERROR_CODE - MoveZ_State_4_5/MoveZ - *10(CPT), *13(CPT), *20(CPT), *4(CPT), *4(CPT), *7(CPT)*			
*ERROR_CODE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT), *16(CPT), 14(NEQ), 16(EQU), 16(EQU)*			
*ERROR_CODE - Ready_State_1/Ready - *3(MOV)*			
MINUS_X_EOT	1	BOOL	DUNEW2PLC1

MINUS_X_EOT (Continued)
External Access: Read Only
MINUS_X_EOT - EOT_Trip_11/EOT_TRIP - 9(XIO)
*MINUS_X_EOT - MainProgram/MainRoutine - *14(OTE), 15(XIC)*

MINUS_Y_EOT 0 BOOL DUNEW2PLC1
External Access: Read Only
MINUS_Y_EOT - EOT_Trip_11/EOT_TRIP - 15(XIO)
*MINUS_Y_EOT - MainProgram/MainRoutine - *11(OTE), 15(XIC)*

MOVE_TYPE 0 INT DUNEW2PLC1
0 = No Move 1 = XY JOG 2 = XY Plane Move 3 = Z JOG 4 = Z Axis Move 5 = Toggle Latch Actuator 6 = Home the Latch Actuator 7 = Unlock Latch Actuator Motor 8 = Un-Servo Motors 9 = PLC_Init

External Access: Read/Write
*MOVE_TYPE - EOT_Trip_11/EOT_TRIP - *21(CPT), 12(CMP), 15(CMP), 18(CMP), 6(CMP), 9(CMP)*
MOVE_TYPE - Error_State_10/Error - 12(CMP), 8(CMP)
*MOVE_TYPE - Initialize/Init - *2(CPT), *4(CPT), 0(EQU)*
*MOVE_TYPE - Latch_UnLatch_State_6_7_8/Latch_Driver - *13(CPT), *33(CPT), 43(CMP)*
*MOVE_TYPE - MoveXY_State_2_3/MoveXY - *18(CPT), *21(CPT), 15(CMP)*
*MOVE_TYPE - MoveZ_State_4_5/MoveZ - *19(CPT), 13(CMP)*
MOVE_TYPE - Ready_State_1/Ready - 10(CMP), 12(CMP), 14(CMP), 16(CMP), 18(CMP), 4(CMP), 6(CMP), 8(CMP)
MOVE_TYPE - UnServo_9/UnServo_All - 11(CMP)

NEXTSTATE 1 DINT DUNEW2PLC1
External Access: Read Only
*NEXTSTATE - EOT_Trip_11/EOT_TRIP - *22(CPT)*
*NEXTSTATE - Error_State_10/Error - *12(CPT)*
*NEXTSTATE - Initialize/Init - *4(CPT), *9(CPT)*
*NEXTSTATE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *13(CPT), *33(CPT), *43(CPT), *51(CPT)*
NEXTSTATE - MainProgram/MainRoutine - 42(CMP), 43(CMP), 44(CMP), 45(CMP), 46(CMP), 47(CMP), 48(CMP), 49(CMP), 50(CMP), 51(CMP), 52(CMP), 53(CMP)
*NEXTSTATE - MoveXY_State_2_3/MoveXY - *0(CPT), *1(CPT), *1(CPT), *18(CPT), *2(CPT), *21(CPT)*
*NEXTSTATE - MoveZ_State_4_5/MoveZ - *10(CPT), *19(CPT), *20(CPT), *4(CPT), *5(MOV), *7(CPT)*
*NEXTSTATE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT)*
*NEXTSTATE - Ready_State_1/Ready - *11(CPT), *13(CPT), *15(CPT), *17(CPT), *19(CPT), *5(CPT), *7(CPT), *9(CPT)*
*NEXTSTATE - UnServo_9/UnServo_All - *11(CPT)*

oneshotob BOOL[32] EOT_Trip_11
External Access: Read Only
oneshotob[0] 0 BOOL
*oneshotob[0] - EOT_Trip_11/EOT_TRIP - *4(OSR), 5(XIC)*
oneshotob[1] 0 BOOL
*oneshotob[1] - EOT_Trip_11/EOT_TRIP - *6(OSR), 7(XIC)*
oneshotob[2] 0 BOOL
*oneshotob[2] - EOT_Trip_11/EOT_TRIP - *8(OSR), 21(XIC)*
oneshotob[3] 0 BOOL
*oneshotob[3] - EOT_Trip_11/EOT_TRIP - *9(OSR), 10(XIC)*
oneshotob[4] 0 BOOL
*oneshotob[4] - EOT_Trip_11/EOT_TRIP - *11(OSR), 21(XIC)*
oneshotob[5] 0 BOOL
*oneshotob[5] - EOT_Trip_11/EOT_TRIP - *12(OSR), 13(XIC)*
oneshotob[6] 0 BOOL
*oneshotob[6] - EOT_Trip_11/EOT_TRIP - *14(OSR), 21(XIC)*
oneshotob[7] 0 BOOL
*oneshotob[7] - EOT_Trip_11/EOT_TRIP - *15(OSR), 16(XIC)*
oneshotob[8] 0 BOOL
*oneshotob[8] - EOT_Trip_11/EOT_TRIP - *17(OSR), 21(XIC)*
oneshotob[9] 0 BOOL
*oneshotob[9] - EOT_Trip_11/EOT_TRIP - *18(OSR), 19(XIC)*
oneshotob[10] 0 BOOL
*oneshotob[10] - EOT_Trip_11/EOT_TRIP - *20(OSR), 21(XIC)*
oneshotob[11] 0 BOOL
*oneshotob[11] - EOT_Trip_11/EOT_TRIP - *21(OSR), 22(XIC)*

oneshotsb		BOOL[32]	EOT_Trip_11
External Access:	Read Only		
oneshotsb[0]	1	BOOL	
*oneshotsb[0] - EOT_Trip_11/EOT_TRIP - *4(OSR)*			
oneshotsb[1]	0	BOOL	
*oneshotsb[1] - EOT_Trip_11/EOT_TRIP - *6(OSR)*			
oneshotsb[2]	1	BOOL	
*oneshotsb[2] - EOT_Trip_11/EOT_TRIP - *8(OSR)*			
oneshotsb[3]	0	BOOL	
*oneshotsb[3] - EOT_Trip_11/EOT_TRIP - *9(OSR)*			
oneshotsb[4]	0	BOOL	
*oneshotsb[4] - EOT_Trip_11/EOT_TRIP - *11(OSR)*			
oneshotsb[5]	0	BOOL	
*oneshotsb[5] - EOT_Trip_11/EOT_TRIP - *12(OSR)*			
oneshotsb[6]	0	BOOL	
*oneshotsb[6] - EOT_Trip_11/EOT_TRIP - *14(OSR)*			
oneshotsb[7]	1	BOOL	
*oneshotsb[7] - EOT_Trip_11/EOT_TRIP - *15(OSR)*			
oneshotsb[8]	0	BOOL	
*oneshotsb[8] - EOT_Trip_11/EOT_TRIP - *17(OSR)*			
oneshotsb[9]	0	BOOL	
*oneshotsb[9] - EOT_Trip_11/EOT_TRIP - *18(OSR)*			
oneshotsb[10]	0	BOOL	
*oneshotsb[10] - EOT_Trip_11/EOT_TRIP - *20(OSR)*			
oneshotsb[11]	0	BOOL	
*oneshotsb[11] - EOT_Trip_11/EOT_TRIP - *21(OSR)*			
PLUS_X_EOT	1	BOOL	DUNEW2PLC1
External Access:	Read Only		
PLUS_X_EOT - EOT_Trip_11/EOT_TRIP - 6(XIO)			
*PLUS_X_EOT - MainProgram/MainRoutine - *13(OTE), 15(XIC)*			
PLUS_Y_EOT	1	BOOL	DUNEW2PLC1
External Access:	Read Only		
PLUS_Y_EOT - EOT_Trip_11/EOT_TRIP - 12(XIO)			
*PLUS_Y_EOT - MainProgram/MainRoutine - *10(OTE), 15(XIC)*			
STATE11_IND	1	BOOL	EOT_Trip_11
External Access:	Read Only		
*STATE11_IND - EOT_Trip_11/EOT_TRIP - *0(OTE), 12(XIC), 15(XIC), 18(XIC), 4(XIC), 6(XIC), 9(XIC)*			
X_axis		AXIS_CIP_DRIVE	DUNEW2PLC1
External Access:	Read Only		
X_axis - Camera/CameraControl - 6(MAR)			
X_axis - EOT_Trip_11/EOT_TRIP - 10(MAM), 5(MSO), 7(MAM)			
X_axis - Error_State_10/Error - 11(MAFR), 5(MSF)			
X_axis - MoveXY_State_2_3/MoveXY - 3(MSO)			
X_axis - Ready_State_1/Ready - 2(MSO)			
X_axis - UnServo_9/UnServo_All - 2(MSF), 6(MAFR)			
X_axis.ActualPosition	-898.1537	REAL	
X_axis.ActualPosition - Camera/CameraControl - 25(CPT), 28(CMP), 28(CMP)			
X_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)			
X_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 2(LIM), 2(LIM)			
X_axis.ActualVelocity	0.0	REAL	
X_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)			
X_axis.Registration2NegativeEdgePosition	0.0	REAL	
X_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 7(CPT)			
X_axis.CommandAcceleration	0.0	REAL	
X_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)			
X_axis.SafeTorqueOffInhibit	0	BOOL	
X_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)			
X_axis.SLSActiveStatus	0	BOOL	
X_axis.SLSActiveStatus - MainProgram/MainRoutine - 36(XIC)			

X_Y COORDINATE_SYSTEM DUNEW2PLC1
External Access: Read/Write
X_Y - EOT_Trip_11/EOT_TRIP - 0(MCS)
X_Y - Error_State_10/Error - 2(MCS)
X_Y - motionQueue/queue_line_arc - 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)
X_Y - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 19(MCS), 22(MCLM)
X_Y - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)
X_Y.MovePendingStatus 0 BOOL
X_Y.MovePendingStatus - motionQueue/queue_line_arc - 104(XIO), 105(XIO), 117(XIO), 118(XIO), 67(XIO), 68(XIO)
X_Y.PhysicalAxisFault 0 BOOL
X_Y.PhysicalAxisFault - MoveXY_State_2_3/MoveXY - 2(XIC)

Y_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
Y_axis - Camera/CameraControl - 5(MAR)
Y_axis - EOT_Trip_11/EOT_TRIP - 13(MAM), 16(MAM), 5(MSO)
Y_axis - Error_State_10/Error - 10(MAFR), 6(MSF)
Y_axis - MoveXY_State_2_3/MoveXY - 3(MSO)
Y_axis - Ready_State_1/Ready - 2(MSO)
Y_axis - UnServo_9/UnServo_All - 3(MSF), 8(MAFR)
Y_axis.ActualPosition -15.0941 REAL
Y_axis.ActualPosition - Camera/CameraControl - 26(CPT), 28(CMP), 28(CMP)
Y_axis.ActualPosition - MainProgram/MainRoutine - 31(LIM), 32(LIM), 33(LIM)
Y_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)
Y_axis.ActualVelocity 0.0 REAL
Y_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)
Y_axis.Registration2NegativeEdgePosition
0.0 REAL
Y_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 8(CPT)
Y_axis.CommandAcceleration 0.0 REAL
Y_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)
Y_axis.SafeTorqueOffInhibit 0 BOOL
Y_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)

Z_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
Z_axis - EOT_Trip_11/EOT_TRIP - 19(MAM), 5(MSO)
Z_axis - Error_State_10/Error - 3(MAS), 7(MSF), 9(MAFR)
Z_axis - Initialize/Init - 7(MAFR)
Z_axis - MoveZ_State_4_5/MoveZ - 12(MAM), 12(MAM), 15(MAS), 16(MAS), 9(MSO)
Z_axis - Ready_State_1/Ready - 2(MSO)
Z_axis - UnServo_9/UnServo_All - 10(MAFR), 4(MSF)
Z_axis.PhysicalAxisFault 0 BOOL
Z_axis.PhysicalAxisFault - MoveZ_State_4_5/MoveZ - 10(XIC)
Z_axis.ActualPosition -1.2238 REAL
Z_axis.ActualPosition - MainProgram/MainRoutine - 6(CMP)
Z_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 1(GEQ)
Z_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 13(CMP), 13(CMP)
Z_axis.ActualVelocity 0.049999997 REAL
Z_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT)

Z_EOT 1 BOOL DUNEW2PLC1
External Access: Read Only
Z_EOT - EOT_Trip_11/EOT_TRIP - 18(XIO)
*Z_EOT - MainProgram/MainRoutine - *9(OTE), 15(XIC)*

General

Type:	Ladder Diagram	Class:	Standard
In Program:	EOT_Trip_11	Number of Rungs:	23

General

Class:	Standard

Configuration

Main:	Error	Inhibit program:	No
Fault:	None		

Monitor

Scan Times(Execution Time)

Max:	68 us	Last:	3 us

0
INIT_DONE
CMP
Expression STATE=10
STATE10_IND

1
STATE10_IND
OSR
Storage Bit oneshotsb[1]
Output Bit oneshotob[1]
SB
OB

2
oneshotob[1]
MCS
Coordinate System X_Y ...
Motion Control X_AXIS_STAT[0]
Stop Type All
Change Decel Yes
Decel Rate 1000
Decel Units Units per sec2
Change Decel Jerk Yes
Decel Jerk 1000
Jerk Units Units per sec3
EN
DN
ER
IP
PC

3
oneshotob[1]
MAS
Axis Z_axis ...
Motion Control Z_AXIS_STAT[0]
Stop Type All
Change Decel Yes
Decel Rate 1000
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 10000
Jerk Units % of Time
EN
DN
ER
IP
PC

4
X_AXIS_STAT[0].PC
Z_AXIS_STAT[0].PC
OSR
Storage Bit oneshotsb[2]
Output Bit oneshotob[2]
SB
OB

5
oneshotob[2]
error_servo_off
MSF
Axis X_axis ...
Motion Control X_AXIS_STAT[1]
EN
DN
ER

6
oneshotob[2]
error_servo_off
MSF
Axis Y_axis ...
Motion Control Y_AXIS_STAT[1]
EN
DN
ER

7
oneshotob[2]
error_servo_off
MSF
Axis Z_axis ...
Motion Control Z_AXIS_STAT[1]
EN
DN
ER

8

X_AXIS_STAT[1].DN Y_AXIS_STAT[1].DN Z_AXIS_STAT[1].DN

CMP
Expression MOVE_TYPE=0

OSR
Storage Bit oneshotsb[4]
Output Bit oneshotob[4]
SB
OB

9

oneshotob[4]

MAFR
Axis Z_axis ...
Motion Control Z_AXIS_STAT[2]
EN
DN
ER

10

Z_AXIS_STAT[2].DN

MAFR
Axis Y_axis ...
Motion Control Y_AXIS_STAT[2]
EN
DN
ER

11

Y_AXIS_STAT[2].DN

MAFR
Axis X_axis ...
Motion Control X_AXIS_STAT[2]
EN
DN
ER

12

Z_AXIS_STAT[1].DN Y_AXIS_STAT[1].DN X_AXIS_STAT[1].DN

CMP
Expression MOVE_TYPE=0

CMP
Expression STATE=10

CPT
Dest ERROR_CODE
0
Expression 0

CPT
Dest NEXTSTATE
1
Expression 1

(End)

Name	Value	Data Type	Scope
ERROR_CODE	0	DINT	DUNEW2PLC1
External Access:	Read Only		
*ERROR_CODE - EOT_Trip_11/EOT_TRIP - *22(CPT)*			
*ERROR_CODE - Error_State_10/Error - *12(CPT)*			
*ERROR_CODE - Initialize/Init - *4(CPT)*			
*ERROR_CODE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *25(CPT), *33(CPT), *36(CPT), *51(CPT)*			
ERROR_CODE - MainProgram/MainRoutine - 36(NEQ)			
*ERROR_CODE - MoveXY_State_2_3/MoveXY - *1(CPT), *1(CPT), *15(CPT), *2(CPT), *2(CPT)*			
*ERROR_CODE - MoveZ_State_4_5/MoveZ - *10(CPT), *13(CPT), *20(CPT), *4(CPT), *4(CPT), *7(CPT)*			
*ERROR_CODE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT), *16(CPT), 14(NEQ), 16(EQU), 16(EQU)*			
*ERROR_CODE - Ready_State_1/Ready - *3(MOV)*			
error_servo_off	0	BOOL	Error_State_10
Constant	No		
External Access:	Read/Write		
error_servo_off - Error_State_10/Error - 5(XIC), 6(XIC), 7(XIC)			
INIT_DONE	1	BOOL	DUNEW2PLC1
External Access:	Read Only		
INIT_DONE - Error_State_10/Error - 0(XIC)			
*INIT_DONE - Initialize/Init - *6(OTE), 7(XIC)*			
INIT_DONE - MainProgram/MainRoutine - 42(XIC), 43(XIC), 44(XIC), 45(XIC), 46(XIC), 47(XIC), 48(XIC), 49(XIC), 50(XIC), 51(XIC), 52(XIC), 53(XIC)			
INIT_DONE - Ready_State_1/Ready - 0(XIC)			
INIT_DONE - UnServo_9/UnServo_All - 0(XIC)			
MOVE_TYPE	0	INT	DUNEW2PLC1
0 = No Move 1 = XY JOG 2 = XY Plane Move 3 = Z JOG 4 = Z Axis Move 5 = Toggle Latch Actuator 6 = Home the Latch Actuator 7 = Unlock Latch Actuator Motor 8 = Un-Servo Motors 9 = PLC_Init			
External Access:	Read/Write		
*MOVE_TYPE - EOT_Trip_11/EOT_TRIP - *21(CPT), 12(CMP), 15(CMP), 18(CMP), 6(CMP), 9(CMP)*			
MOVE_TYPE - Error_State_10/Error - 12(CMP), 8(CMP)			
*MOVE_TYPE - Initialize/Init - *2(CPT), *4(CPT), 0(EQU)*			
*MOVE_TYPE - Latch_UnLatch_State_6_7_8/Latch_Driver - *13(CPT), *33(CPT), 43(CMP)*			
*MOVE_TYPE - MoveXY_State_2_3/MoveXY - *18(CPT), *21(CPT), 15(CMP)*			
*MOVE_TYPE - MoveZ_State_4_5/MoveZ - *19(CPT), 13(CMP)*			
MOVE_TYPE - Ready_State_1/Ready - 10(CMP), 12(CMP), 14(CMP), 16(CMP), 18(CMP), 4(CMP), 6(CMP), 8(CMP)			
MOVE_TYPE - UnServo_9/UnServo_All - 11(CMP)			
NEXTSTATE	1	DINT	DUNEW2PLC1
External Access:	Read Only		
*NEXTSTATE - EOT_Trip_11/EOT_TRIP - *22(CPT)*			
*NEXTSTATE - Error_State_10/Error - *12(CPT)*			
*NEXTSTATE - Initialize/Init - *4(CPT), *9(CPT)*			
*NEXTSTATE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *13(CPT), *33(CPT), *43(CPT), *51(CPT)*			
NEXTSTATE - MainProgram/MainRoutine - 42(CMP), 43(CMP), 44(CMP), 45(CMP), 46(CMP), 47(CMP), 48(CMP), 49(CMP), 50(CMP), 51(CMP), 52(CMP), 53(CMP)			
*NEXTSTATE - MoveXY_State_2_3/MoveXY - *0(CPT), *1(CPT), *1(CPT), *18(CPT), *2(CPT), *21(CPT)*			
*NEXTSTATE - MoveZ_State_4_5/MoveZ - *10(CPT), *19(CPT), *20(CPT), *4(CPT), *5(MOV), *7(CPT)*			
*NEXTSTATE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT)*			
*NEXTSTATE - Ready_State_1/Ready - *11(CPT), *13(CPT), *15(CPT), *17(CPT), *19(CPT), *5(CPT), *7(CPT), *9(CPT)*			
*NEXTSTATE - UnServo_9/UnServo_All - *11(CPT)*			
oneshotob		BOOL[32]	Error_State_10
External Access:	Read Only		
oneshotob[0]	0	BOOL	
On entry to state 6 or 7			
oneshotob[1]	0	BOOL	
*oneshotob[1] - Error_State_10/Error - *1(OSR), 2(XIC), 3(XIC)*			
oneshotob[2]	0	BOOL	
*oneshotob[2] - Error_State_10/Error - *4(OSR), 5(XIC), 6(XIC), 7(XIC)*			
oneshotob[4]	0	BOOL	

oneshotob (Continued)
*oneshotob[4] - Error_State_10/Error - *8(OSR), 9(XIC)*

oneshotsb		BOOL[32]	Error_State_10
External Access:	Read Only		
oneshotsb[0]	0	BOOL	
While in state 6 or 7			
oneshotsb[1]	0	BOOL	
*oneshotsb[1] - Error_State_10/Error - *1(OSR)*			
oneshotsb[2]	1	BOOL	
*oneshotsb[2] - Error_State_10/Error - *4(OSR)*			
oneshotsb[4]	0	BOOL	
*oneshotsb[4] - Error_State_10/Error - *8(OSR)*			
STATE	1	DINT	DUNEW2PLC1
CURRENT State Machine State			
External Access:	Read Only		
STATE - Error_State_10/Error - 0(CMP), 12(CMP)			
*STATE - Initialize/Init - *4(CPT), 7(CMP)*			
STATE - Latch_UnLatch_State_6_7_8/Latch_Driver - 16(CMP), 34(CMP), 45(CMP), 45(CMP), 51(CMP), 8(CMP)			
*STATE - MainProgram/MainRoutine - *42(CPT), *43(CPT), *44(CPT), *45(CPT), *46(CPT), *47(CPT), *48(CPT), *49(CPT), *50(CPT), *51(CPT), *52(CPT), *53(CPT)*			
STATE - MoveXY_State_2_3/MoveXY - 0(CMP), 1(CMP), 11(CMP)			
STATE - MoveZ_State_4_5/MoveZ - 4(CMP), 4(CMP), 5(CMP), 6(CMP)			
STATE - Ready_State_1/Ready - 0(CMP)			
STATE - UnServo_9/UnServo_All - 0(CMP)			
STATE10_IND	0	BOOL	Error_State_10
External Access:	Read Only		
*STATE10_IND - Error_State_10/Error - *0(OTE), 1(XIC)*			
X_axis		AXIS_CIP_DRIVE	DUNEW2PLC1
External Access:	Read Only		
X_axis - Camera/CameraControl - 6(MAR)			
X_axis - EOT_Trip_11/EOT_TRIP - 10(MAM), 5(MSO), 7(MAM)			
X_axis - Error_State_10/Error - 11(MAFR), 5(MSF)			
X_axis - MoveXY_State_2_3/MoveXY - 3(MSO)			
X_axis - Ready_State_1/Ready - 2(MSO)			
X_axis - UnServo_9/UnServo_All - 2(MSF), 6(MAFR)			
X_axis.ActualPosition	-898.1537	REAL	
X_axis.ActualPosition - Camera/CameraControl - 25(CPT), 28(CMP), 28(CMP)			
X_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)			
X_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 2(LIM), 2(LIM)			
X_axis.ActualVelocity	0.0	REAL	
X_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)			
X_axis.Registration2NegativeEdgePosition	0.0	REAL	
X_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 7(CPT)			
X_axis.CommandAcceleration	0.0	REAL	
X_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)			
X_axis.SafeTorqueOffInhibit	0	BOOL	
X_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)			
X_axis.SLSActiveStatus	0	BOOL	
X_axis.SLSActiveStatus - MainProgram/MainRoutine - 36(XIC)			
X_AXIS_STAT		MOTION_INSTRUCTION[32]	Error_State_10
External Access:	Read Only		
X_AXIS_STAT[0]		MOTION_INSTRUCTION	
*X_AXIS_STAT[0] - Error_State_10/Error - *2(MCS)*			
X_AXIS_STAT[0].PC	1	BOOL	
X_AXIS_STAT[0].PC - Error_State_10/Error - 4(XIC)			
X_AXIS_STAT[1]		MOTION_INSTRUCTION	
*X_AXIS_STAT[1] - Error_State_10/Error - *5(MSF)*			
X_AXIS_STAT[1].DN	0	BOOL	

X_AXIS_STAT (Continued)
X_AXIS_STAT[1].DN - Error_State_10/Error - 12(XIC), 8(XIC)

X_AXIS_STAT[2]		MOTION_INSTRUCTION	
*X_AXIS_STAT[2] - Error_State_10/Error - *11(MAFR)*			
X_Y		COORDINATE_SYSTEM	DUNEW2PLC1
External Access:	Read/Write		
X_Y - EOT_Trip_11/EOT_TRIP - 0(MCS)			
X_Y - Error_State_10/Error - 2(MCS)			
X_Y - motionQueue/queue_line_arc - 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)			
X_Y - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 19(MCS), 22(MCLM)			
X_Y - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)			
X_Y.MovePendingStatus	0	BOOL	
X_Y.MovePendingStatus - motionQueue/queue_line_arc - 104(XIO), 105(XIO), 117(XIO), 118(XIO), 67(XIO), 68(XIO)			
X_Y.PhysicalAxisFault	0	BOOL	
X_Y.PhysicalAxisFault - MoveXY_State_2_3/MoveXY - 2(XIC)			
Y_axis		AXIS_CIP_DRIVE	DUNEW2PLC1
External Access:	Read Only		
Y_axis - Camera/CameraControl - 5(MAR)			
Y_axis - EOT_Trip_11/EOT_TRIP - 13(MAM), 16(MAM), 5(MSO)			
Y_axis - Error_State_10/Error - 10(MAFR), 6(MSF)			
Y_axis - MoveXY_State_2_3/MoveXY - 3(MSO)			
Y_axis - Ready_State_1/Ready - 2(MSO)			
Y_axis - UnServo_9/UnServo_All - 3(MSF), 8(MAFR)			
Y_axis.ActualPosition	-15.0941	REAL	
Y_axis.ActualPosition - Camera/CameraControl - 26(CPT), 28(CMP), 28(CMP)			
Y_axis.ActualPosition - MainProgram/MainRoutine - 31(LIM), 32(LIM), 33(LIM)			
Y_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)			
Y_axis.ActualVelocity	0.0	REAL	
Y_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)			
Y_axis.Registration2NegativeEdgePosition	0.0	REAL	
Y_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 8(CPT)			
Y_axis.CommandAcceleration	0.0	REAL	
Y_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)			
Y_axis.SafeTorqueOffInhibit	0	BOOL	
Y_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)			
Y_AXIS_STAT		MOTION_INSTRUCTION[32]	Error_State_10
External Access:	Read Only		
Y_AXIS_STAT[1]		MOTION_INSTRUCTION	
*Y_AXIS_STAT[1] - Error_State_10/Error - *6(MSF)*			
Y_AXIS_STAT[1].DN	0	BOOL	
Y_AXIS_STAT[1].DN - Error_State_10/Error - 12(XIC), 8(XIC)			
Y_AXIS_STAT[2]		MOTION_INSTRUCTION	
*Y_AXIS_STAT[2] - Error_State_10/Error - *10(MAFR)*			
Y_AXIS_STAT[2].DN	0	BOOL	
Y_AXIS_STAT[2].DN - Error_State_10/Error - 11(XIC)			
Z_axis		AXIS_CIP_DRIVE	DUNEW2PLC1
External Access:	Read Only		
Z_axis - EOT_Trip_11/EOT_TRIP - 19(MAM), 5(MSO)			
Z_axis - Error_State_10/Error - 3(MAS), 7(MSF), 9(MAFR)			
Z_axis - Initialize/Init - 7(MAFR)			
Z_axis - MoveZ_State_4_5/MoveZ - 12(MAM), 12(MAM), 15(MAS), 16(MAS), 9(MSO)			
Z_axis - Ready_State_1/Ready - 2(MSO)			
Z_axis - UnServo_9/UnServo_All - 10(MAFR), 4(MSF)			
Z_axis.PhysicalAxisFault	0	BOOL	
Z_axis.PhysicalAxisFault - MoveZ_State_4_5/MoveZ - 10(XIC)			
Z_axis.ActualPosition	-1.2238	REAL	
Z_axis.ActualPosition - MainProgram/MainRoutine - 6(CMP)			
Z_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 1(GEQ)			
Z_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 13(CMP), 13(CMP)			

Z_axis (Continued)

Z_axis.ActualVelocity	0.0	REAL	
Z_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT)			
Z_AXIS_STAT		MOTION_INSTRUCTION[32]	Error_State_10
External Access:	Read Only		
Z_AXIS_STAT[0]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[0] - Error_State_10/Error - *3(MAS)*			
Z_AXIS_STAT[0].PC	1	BOOL	
Z_AXIS_STAT[0].PC - Error_State_10/Error - 4(XIC)			
Z_AXIS_STAT[1]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[1] - Error_State_10/Error - *7(MSF)*			
Z_AXIS_STAT[1].DN	0	BOOL	
Z_AXIS_STAT[1].DN - Error_State_10/Error - 12(XIC), 8(XIC)			
Z_AXIS_STAT[2]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[2] - Error_State_10/Error - *9(MAFR)*			
Z_AXIS_STAT[2].DN	0	BOOL	
Z_AXIS_STAT[2].DN - Error_State_10/Error - 10(XIC)			

General

Type:	Ladder Diagram	Class:	Standard
In Program:	Error_State_10	Number of Rungs:	13

General

Class:	Standard

Configuration

Main:	Latch_Driver	Inhibit program:	No
Fault:	None		

Monitor

Scan Times(Execution Time)

Max:	109 us	Last:	12 us

0
Latch top
Local:1:I.Pt08.Data
MACHINE_SW_STAT[12]
LATCH_ACTUATOR_TOP

1
Latch middle
Local:1:I.Pt09.Data
MACHINE_SW_STAT[13]
LATCH_ACTUATOR_MID

2
Actuator Pos 3 - Latch has released lock on Z-stage and is locked locked onto Fixed stage
LATCH_ACTUATOR_TOP
LATCH_ACTUATOR_MID
CPT
Dest ACTUATOR_POS
2
Expression 3
Z_STAGE_UNLATCHED

3
Actuator Pos 2 - Head on Fixed stage, Z-stage mechanism in upper position to allow engaging
LATCH_ACTUATOR_TOP
LATCH_ACTUATOR_MID
TON
Timer delay_mid_position
Preset 100
Accum 100
EN
DN
delay_mid_position.DN
CPT
Dest ACTUATOR_POS
2
Expression 2
Z_OK_TO_ENGAGE

4
Actuator Position 0 - Latch in unknown position between sensors
LATCH_ACTUATOR_TOP
LATCH_ACTUATOR_MID
Z_STAGE_LATCHED
CPT
Dest ACTUATOR_POS
2
Expression 0

5
Actuator Position 1 - Z-stage latched proxy high
Z_STAGE_LATCHED
TON
Timer Delay_Z_Latched
Preset 100
Accum 0
EN
DN
Delay_Z_Latched.DN
CPT
Dest ACTUATOR_POS
2
Expression 1

6
Z_FIXED_LATCHED
TON
Timer Delay_Fixed_Latched
Preset 100
Accum 0
EN
DN
Delay_Fixed_Latched.DN
Z_SAFE_TO_WITHDRAW

7
Winding head is on the Z-arm
Z_STAGE_PRESENT
Z_FIXED_PRESENT
Z_EXTENDED
ENABLE_ACTUATOR

8
CMP
Expression STATE=6
STATE6_IND

9
STATE6_IND ENABLE_ACTUATOR
OSR
Storage Bit oneshotsb[9]
Output Bit oneshotob[9]
SB
OB

10
Entered state 6 but the actuator was not enabled
oneshotob[9]
CPT
Dest ERROR_CODE
0
Expression 6001
CPT
Dest NEXTSTATE
1
Expression 10

11
STATE6_IND ENABLE_ACTUATOR
While in state 6 and operator is enabled
OSR
Storage Bit oneshotsb[1]
Output Bit oneshotob[1]
SB
OB

12
For one scan, when we enter state 6 and actuator is enabled
oneshotob[1]
Store the previous actuator position
CPT
Dest PREV_ACT_POS
2
Expression ACTUATOR_POS

13

While in state 6 and operator is enabled
oneshotsb[1]

CMP
Expression PREV_ACT_POS=1

CMP
Expression ACTUATOR_POS=3

CMP
Expression PREV_ACT_POS=3

CMP
Expression ACTUATOR_POS=2

CMP
Expression PREV_ACT_POS=2

CMP
Expression ACTUATOR_POS=1

CMP
Expression PREV_ACT_POS=0

NEQ
Source A ACTUATOR_POS
2
Source B 0

0 = No Move
1 = XY JOG
2 = XY Plane Move
3 = Z JOG
4 = Z Axis Move
5 = Toggle Latch Actuator
6 = Home the Latch Actuator
7 = Unlock Latch Actuator Motor
8 = Un-Servo Motors
9 = PLC_Init

CPT
Dest MOVE_TYPE
0
Expression 0

CPT
Dest NEXTSTATE
1
Expression 1

14

While in state 6 and operator is enabled
oneshotsb[1]

Latching_pulse_interval.DN

TON
Timer Latching_pulse_duration
Preset 25
Accum 0
EN
DN

15

Latch successfully moved to the next position
oneshotsb[2]

Latching_pulse_duration.DN

TON
Timer Latching_pulse_interval
Preset 250
Accum 0
EN
DN

16

CMP
Expression STATE=7

STATE7_IND

17

STATE7_IND

OSR
Storage Bit oneshotsb[0]
Output Bit oneshotob[0]
SB
OB

18

oneshotob[0]

HomeCounter
RES

19 oneshotsb[0] HomeCounter.DN HomeTimer2.TT
TON Timer HomeTimer1 Preset 500 Accum 0 EN DN

20 oneshotsb[0] HomeCounter.DN HomeTimer1.TT
TON Timer HomeTimer2 Preset 500 Accum 0 EN DN

21 oneshotsb[0] HomeCounter.DN HomeTimer1.TT
CTU Counter HomeCounter Preset 99999 Accum 386 CU DN

22 HomeCounter.DN AFI Local:3:O.Pt02.Data

23 HomeCounter.DN
OSR Storage Bit oneshotsb[7] Output Bit oneshotob[7] SB OB

24 oneshotob[7] Z_STAGE_LATCHED LATCH_ACTUATOR_HOMED L

25 oneshotob[7] Z_STAGE_LATCHED
CPT Dest ERROR_CODE 0 Expression 7002
LATCH_ACTUATOR_HOMED L

26 LATCH_ACTUATOR_HOMED MACHINE_SW_STAT[0]

27 LATCH_ACTUATOR_HOMED
OSR Storage Bit oneshotsb[4] Output Bit oneshotob[4] SB OB

28 oneshotob[4] LatchCounter3 RES

29 oneshotsb[4] LatchCounter3.DN LatchTimer6.TT
TON Timer LatchTimer5 Preset 250 Accum 0 EN DN

30 oneshotsb[4] LatchCounter3.DN LatchTimer5.TT
TON Timer LatchTimer6 Preset 250 Accum 0 EN DN

31 oneshotsb[4] LatchCounter3.DN LatchTimer5.TT
CTU Counter LatchCounter3 Preset 2 Accum 2 CU DN

32
LatchCounter3.DN
OSR
Storage Bit oneshotsb[5]
Output Bit oneshotob[5]
SB
OB

0 = No Move
1 = XY JOG
2 = XY Plane Move
3 = Z JOG
4 = Z Axis Move
5 = Toggle Latch Actuator
6 = Home the Latch Actuator
7 = Unlock Latch Actuator Motor
8 = Un-Servo Motors
9 = PLC_Init

33
STATE7_IND
oneshotob[5]
CPT
Dest ERROR_CODE
0
Expression 7000
HomeSignal
U
CPT
Dest MOVE_TYPE
0
Expression 0
CPT
Dest NEXTSTATE
1
Expression 1

34
CMP
Expression STATE=8
UNLOCK_LATCH_MOTOR_SHAFT
STATE8_IND

35
STATE8_IND
AFI
Local:3:O.Pt02.Data

36
Local:3:O.Pt02.Data
CPT
Dest ERROR_CODE
0
Expression 8000

37
STATE8_IND
OSR
Storage Bit oneshotsb[10]
Output Bit oneshotob[10]
SB
OB

38
oneshotob[10]
LATCH_ACTUATOR_HOMED
U

39
oneshotob[10]
LatchCounter2
RES

40
oneshotsb[10]
LatchCounter2.DN
/
LatchTimer4.TT
/
TON
Timer LatchTimer3
Preset 250
Accum 0
EN
DN

41
oneshotsb[10] LatchCounter2.DN LatchTimer3.TT
TON
Timer LatchTimer4
Preset 250
Accum 0
EN
DN

42
oneshotsb[10] LatchCounter2.DN LatchTimer3.TT
CTU
Counter LatchCounter2
Preset 2
Accum 2
CU
DN

43
STATE8_IND LatchCounter2.DN
CMP
Expression MOVE_TYPE=0
CPT
Dest NEXTSTATE
1
Expression 1

44
Latching_pulse_duration.TT
LatchTimer4.TT
HomeTimer2.TT
LatchTimer6.TT
Latching signal output
Local:3:O.Pt01.Data latching_signal

45
CMP
Expression STATE=6
CMP
Expression STATE=7
OSR
Storage Bit oneshotsb[12]
Output Bit oneshotob[12]
SB
OB

46
oneshotob[12]
When you enter state 6 or 7, start counting seconds
LatchingTimeoutCounter
RES

47
oneshotsb[12] LatchingTimeoutCounter.DN TimeoutTimer2.TT
When you enter state 6 or 7, start counting seconds
TON
Timer TimeoutTimer1
Preset 1000
Accum 0
EN
DN

48
oneshotsb[12] LatchingTimeoutCounter.DN TimeoutTimer1.TT
When you enter state 6 or 7, start counting seconds
TON
Timer TimeoutTimer2
Preset 1000
Accum 0
EN
DN

49
oneshotsb[12] LatchingTimeoutCounter.DN TimeoutTimer1.TT
When you enter state 6 or 7, start counting seconds
When you enter state 6 or 7, start counting seconds
CTU
Counter LatchingTimeoutCounter
Preset 9999
Accum 1
CU
DN

50

When you enter state 6 or 7, start counting seconds
LatchingTimeoutCounter.DN

OSR
Storage Bit oneshotsb[3]
Output Bit oneshotob[3]
SB
OB

51

oneshotob[3]

CMP
Expression STATE=6

CPT
Dest ERROR_CODE
0
Expression 6002

CPT
Dest NEXTSTATE
1
Expression 10

(End)

Name	Value	Data Type	Scope
ACTUATOR_POS	2	DINT	DUNEW2PLC1
External Access:	Read Only		
*ACTUATOR_POS - Latch_UnLatch_State_6_7_8/Latch_Driver - *2(CPT), *3(CPT), *4(CPT), *5(CPT), 12(CPT), 13(CMP), 13(CMP), 13(CMP), 13(NEQ)*			
ACTUATOR_POS - MainProgram/MainRoutine - 59(EQU), 59(EQU)			
ACTUATOR_POS - MoveZ_State_4_5/MoveZ - 0(EQU), 7(CMP)			
AFI	0	BOOL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
AFI - Latch_UnLatch_State_6_7_8/Latch_Driver - 22(XIC), 35(XIC)			
Delay_Fixed_Latched		TIMER	Latch_UnLatch_State_6_7_8
External Access:	Read/Write		
*Delay_Fixed_Latched - Latch_UnLatch_State_6_7_8/Latch_Driver - *6(TON)*			
Delay_Fixed_Latched.DN	0	BOOL	
Delay_Fixed_Latched.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 6(XIC)			
delay_mid_position		TIMER	DUNEW2PLC1
External Access:	Read/Write		
*delay_mid_position - Latch_UnLatch_State_6_7_8/Latch_Driver - *3(TON)*			
delay_mid_position.DN	1	BOOL	
delay_mid_position.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 3(XIC)			
Delay_Z_Latched		TIMER	Latch_UnLatch_State_6_7_8
External Access:	Read/Write		
*Delay_Z_Latched - Latch_UnLatch_State_6_7_8/Latch_Driver - *5(TON)*			
Delay_Z_Latched.DN	0	BOOL	
Delay_Z_Latched.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 5(XIC)			
ENABLE_ACTUATOR	0	BOOL	Latch_UnLatch_State_6_7_8
External Access:	Read Only		
*ENABLE_ACTUATOR - Latch_UnLatch_State_6_7_8/Latch_Driver - *7(OTE), 11(XIC), 9(XIO)*			
ERROR_CODE	0	DINT	DUNEW2PLC1
External Access:	Read Only		
*ERROR_CODE - EOT_Trip_11/EOT_TRIP - *22(CPT)*			
*ERROR_CODE - Error_State_10/Error - *12(CPT)*			
*ERROR_CODE - Initialize/Init - *4(CPT)*			
*ERROR_CODE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *25(CPT), *33(CPT), *36(CPT), *51(CPT)*			
ERROR_CODE - MainProgram/MainRoutine - 36(NEQ)			
*ERROR_CODE - MoveXY_State_2_3/MoveXY - *1(CPT), *1(CPT), *15(CPT), *2(CPT), *2(CPT)*			
*ERROR_CODE - MoveZ_State_4_5/MoveZ - *10(CPT), *13(CPT), *20(CPT), *4(CPT), *4(CPT), *7(CPT)*			
*ERROR_CODE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT), *16(CPT), 14(NEQ), 16(EQU), 16(EQU)*			
*ERROR_CODE - Ready_State_1/Ready - *3(MOV)*			
HomeCounter		COUNTER	Latch_UnLatch_State_6_7_8
External Access:	Read Only		
*HomeCounter - Latch_UnLatch_State_6_7_8/Latch_Driver - *18(RES), *21(CTU)*			
HomeCounter.DN	0	BOOL	
HomeCounter.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 19(XIO), 20(XIO), 21(XIO), 22(XIC), 23(XIC)			
HomeSignal	0	BOOL	Latch_UnLatch_State_6_7_8

HomeSignal (Continued)
External Access: Read Only
*HomeSignal - Latch_UnLatch_State_6_7_8/Latch_Driver - *33(OTU)*

HomeTimer1		TIMER	Latch_UnLatch_State_6_7_8

External Access: Read Only
*HomeTimer1 - Latch_UnLatch_State_6_7_8/Latch_Driver - *19(TON)*

HomeTimer1.TT	0	BOOL	

HomeTimer1.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 20(XIO), 21(XIC)

HomeTimer2		TIMER	Latch_UnLatch_State_6_7_8

External Access: Read Only
*HomeTimer2 - Latch_UnLatch_State_6_7_8/Latch_Driver - *20(TON)*

HomeTimer2.TT	0	BOOL	

HomeTimer2.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 19(XIO), 44(XIC)

LATCH_ACTUATOR_HOMED	0	BOOL	DUNEW2PLC1

External Access: Read Only
*LATCH_ACTUATOR_HOMED - Initialize/Init - *5(OTU)*
*LATCH_ACTUATOR_HOMED - Latch_UnLatch_State_6_7_8/Latch_Driver - *24(OTL), *25(OTL), *38(OTU), 26(XIC), 27(XIC)*
LATCH_ACTUATOR_HOMED - MoveZ_State_4_5/MoveZ - 7(XIC)

LATCH_ACTUATOR_MID	1	BOOL	DUNEW2PLC1

External Access: Read/Write
*LATCH_ACTUATOR_MID - Latch_UnLatch_State_6_7_8/Latch_Driver - *1(OTE), 2(XIO), 3(XIC), 4(XIO)*

LATCH_ACTUATOR_TOP	1	BOOL	DUNEW2PLC1

External Access: Read/Write
*LATCH_ACTUATOR_TOP - Latch_UnLatch_State_6_7_8/Latch_Driver - *0(OTE), 2(XIC), 3(XIC), 4(XIO)*

LatchCounter2		COUNTER	Latch_UnLatch_State_6_7_8

External Access: Read Only
*LatchCounter2 - Latch_UnLatch_State_6_7_8/Latch_Driver - *39(RES), *42(CTU)*

LatchCounter2.DN	1	BOOL	

LatchCounter2.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 40(XIO), 41(XIO), 42(XIO), 43(XIC)

LatchCounter3		COUNTER	Latch_UnLatch_State_6_7_8

External Access: Read Only
*LatchCounter3 - Latch_UnLatch_State_6_7_8/Latch_Driver - *28(RES), *31(CTU)*

LatchCounter3.DN	1	BOOL	

LatchCounter3.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 29(XIO), 30(XIO), 31(XIO), 32(XIC)

Latching_pulse_duration		TIMER	Latch_UnLatch_State_6_7_8

External Access: Read Only
*Latching_pulse_duration - Latch_UnLatch_State_6_7_8/Latch_Driver - *14(TON)*

Latching_pulse_duration.TT	0	BOOL	

Latching_pulse_duration.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 44(XIC)

Latching_pulse_duration.DN	0	BOOL	

Latching_pulse_duration.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 15(XIC)

Latching_pulse_interval		TIMER	Latch_UnLatch_State_6_7_8

External Access: Read/Write

Latching_pulse_interval (Continued)
*Latching_pulse_interval - Latch_UnLatch_State_6_7_8/Latch_Driver - *15(TON)*
Latching_pulse_interval.DN 0 BOOL
Latching_pulse_interval.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 14(XIO)

latching_signal 0 BOOL Latch_UnLatch_State_6_7_8

External Access: Read/Write
*latching_signal - Latch_UnLatch_State_6_7_8/Latch_Driver - *44(OTE)*

LatchingTimeoutCounter COUNTER Latch_UnLatch_State_6_7_8

When you enter state 6 or 7, start counting seconds
External Access: Read/Write
*LatchingTimeoutCounter - Latch_UnLatch_State_6_7_8/Latch_Driver - *46(RES), *49(CTU)*
LatchingTimeoutCounter.PRE 9999 DINT
When you enter state 6 or 7, start counting seconds
LatchingTimeoutCounter.ACC 1 DINT
When you enter state 6 or 7, start counting seconds
LatchingTimeoutCounter.CU 0 BOOL
When you enter state 6 or 7, start counting seconds
LatchingTimeoutCounter.CD 0 BOOL
When you enter state 6 or 7, start counting seconds
LatchingTimeoutCounter.DN 0 BOOL
When you enter state 6 or 7, start counting seconds
LatchingTimeoutCounter.DN - Latch_UnLatch_State_6_7_8/Latch_Driver - 47(XIO), 48(XIO), 49(XIO), 50(XIC)
LatchingTimeoutCounter.OV 0 BOOL
When you enter state 6 or 7, start counting seconds
LatchingTimeoutCounter.UN 0 BOOL
When you enter state 6 or 7, start counting seconds

LatchTimer3 TIMER Latch_UnLatch_State_6_7_8

External Access: Read Only
*LatchTimer3 - Latch_UnLatch_State_6_7_8/Latch_Driver - *40(TON)*
LatchTimer3.TT 0 BOOL
LatchTimer3.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 41(XIO), 42(XIC)

LatchTimer4 TIMER Latch_UnLatch_State_6_7_8

External Access: Read Only
*LatchTimer4 - Latch_UnLatch_State_6_7_8/Latch_Driver - *41(TON)*
LatchTimer4.TT 0 BOOL
LatchTimer4.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 40(XIO), 44(XIC)

LatchTimer5 TIMER Latch_UnLatch_State_6_7_8

External Access: Read Only
*LatchTimer5 - Latch_UnLatch_State_6_7_8/Latch_Driver - *29(TON)*
LatchTimer5.TT 0 BOOL
LatchTimer5.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 30(XIO), 31(XIC)

LatchTimer6 TIMER Latch_UnLatch_State_6_7_8

External Access: Read Only
*LatchTimer6 - Latch_UnLatch_State_6_7_8/Latch_Driver - *30(TON)*
LatchTimer6.TT 0 BOOL
LatchTimer6.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 29(XIO), 44(XIC)

Local:1:I AB:5000_DI16:I:0 DUNEW2PLC1
Constant No
External Access: Read/Write
Local:1:I.Pt00.Data 1 BOOL
Local:1:I.Pt00.Data - MainProgram/MainRoutine - 0(XIC)
Local:1:I.Pt01.Data 0 BOOL
Local:1:I.Pt01.Data - MainProgram/MainRoutine - 2(XIO)
Local:1:I.Pt02.Data 1 BOOL
Local:1:I.Pt02.Data - MainProgram/MainRoutine - 3(XIC)
Local:1:I.Pt03.Data 0 BOOL
Local:1:I.Pt03.Data - MainProgram/MainRoutine - 4(XIO)
Local:1:I.Pt04.Data 0 BOOL
Local:1:I.Pt04.Data - MainProgram/MainRoutine - 6(XIC)
Local:1:I.Pt06.Data 1 BOOL
Local:1:I.Pt06.Data - MainProgram/MainRoutine - 10(XIC)
Local:1:I.Pt07.Data 1 BOOL
Local:1:I.Pt07.Data - MainProgram/MainRoutine - 9(XIC)
Local:1:I.Pt08.Data 0 BOOL
Latch top
Local:1:I.Pt08.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - 0(XIO)
Local:1:I.Pt09.Data 0 BOOL
Latch middle
Local:1:I.Pt09.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - 1(XIO)
Local:1:I.Pt10.Data 1 BOOL
Local:1:I.Pt10.Data - MainProgram/MainRoutine - 16(XIO)
Local:1:I.Pt11.Data 0 BOOL
Local:1:I.Pt11.Data - MainProgram/MainRoutine - 7(XIC)
Local:1:I.Pt12.Data 0 BOOL
Local:1:I.Pt12.Data - MainProgram/MainRoutine - 18(XIC)
Local:1:I.Pt13.Data 0 BOOL
Local:1:I.Pt13.Data - MainProgram/MainRoutine - 19(XIC)
Local:1:I.Pt15.Data 0 BOOL
The wire break switch was causing electrical problems, so we use the load-cell as a proxy for wire break
Local:1:I.Pt15.Data - MainProgram/MainRoutine - 20(XIC)

Local:3:O AB:5000_DO16:O:0 DUNEW2PLC1
External Access: Read/Write
Local:3:O.Pt01.Data 0 BOOL
Latching signal output
*Local:3:O.Pt01.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - *44(OTE)*
Local:3:O.Pt02.Data 0 BOOL
*Local:3:O.Pt02.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - *22(OTE), *35(OTE), 36(XIC)*
Local:3:O.Pt03.Data 0 BOOL
*Enable_tension_motor - PID_Tension_Servo/PID_Tension - *2(OTE)*
Local:3:O.Pt11.Data 0 BOOL
Reset PB - Light
*Local:3:O.Pt11.Data - MainProgram/MainRoutine - *35(OTE)*
Local:3:O.Pt12.Data 0 BOOL
Beacon - Green
*Local:3:O.Pt12.Data - MainProgram/MainRoutine - *35(OTE)*
Local:3:O.Pt13.Data 0 BOOL
Beacon - Amber
*Local:3:O.Pt13.Data - MainProgram/MainRoutine - *36(OTE)*
Local:3:O.Pt14.Data 1 BOOL
Beacon - Red
*Local:3:O.Pt14.Data - MainProgram/MainRoutine - *38(OTE)*
Local:3:O.Pt15.Data 0 BOOL
Beacon - Sounder
*Local:3:O.Pt15.Data - MainProgram/MainRoutine - *37(OTE)*
*Local:3:O.Pt15.Data - PID_Tension_Servo/PID_Tension - *12(OTE)*

MACHINE_SW_STAT BOOL[32] DUNEW2PLC1
External Access: Read Only
MACHINE_SW_STAT[0] 0 BOOL

MACHINE_SW_STAT (Continued)
*MACHINE_SW_STAT[0] - Latch_UnLatch_State_6_7_8/Latch_Driver - *26(OTE)*
MACHINE_SW_STAT[1] 1 BOOL
*MACHINE_SW_STAT[1] - MainProgram/MainRoutine - *0(OTE)*
MACHINE_SW_STAT[2] 1 BOOL
*MACHINE_SW_STAT[2] - MainProgram/MainRoutine - *2(OTE)*
MACHINE_SW_STAT[3] 1 BOOL
*MACHINE_SW_STAT[3] - MainProgram/MainRoutine - *3(OTE)*
MACHINE_SW_STAT[4] 1 BOOL
*MACHINE_SW_STAT[4] - MainProgram/MainRoutine - *4(OTE)*
MACHINE_SW_STAT[5] 0 BOOL
*MACHINE_SW_STAT[5] - MainProgram/MainRoutine - *6(OTE)*
MACHINE_SW_STAT[6] 0 BOOL
*MACHINE_SW_STAT[6] - MainProgram/MainRoutine - *7(OTE)*
MACHINE_SW_STAT[7] 0 BOOL
*MACHINE_SW_STAT[7] - MainProgram/MainRoutine - *8(OTE)*
MACHINE_SW_STAT[8] 1 BOOL
*MACHINE_SW_STAT[8] - MainProgram/MainRoutine - *9(OTE)*
MACHINE_SW_STAT[9] 0 BOOL
*MACHINE_SW_STAT[9] - MainProgram/MainRoutine - *16(OTE)*
MACHINE_SW_STAT[10] 0 BOOL
*MACHINE_SW_STAT[10] - MainProgram/MainRoutine - *17(OTE)*
MACHINE_SW_STAT[12] 1 BOOL
*MACHINE_SW_STAT[12] - Latch_UnLatch_State_6_7_8/Latch_Driver - *0(OTE)*
MACHINE_SW_STAT[13] 1 BOOL
*MACHINE_SW_STAT[13] - Latch_UnLatch_State_6_7_8/Latch_Driver - *1(OTE)*
MACHINE_SW_STAT[14] 0 BOOL
*MACHINE_SW_STAT[14] - MainProgram/MainRoutine - *22(OTE)*
MACHINE_SW_STAT[15] 0 BOOL
*MACHINE_SW_STAT[15] - MainProgram/MainRoutine - *21(OTE)*
MACHINE_SW_STAT[16] 0 BOOL
*MACHINE_SW_STAT[16] - MainProgram/MainRoutine - *19(OTE)*
MACHINE_SW_STAT[17] 0 BOOL
*MACHINE_SW_STAT[17] - MainProgram/MainRoutine - *18(OTE)*
MACHINE_SW_STAT[18] 1 BOOL
*MACHINE_SW_STAT[18] - MainProgram/MainRoutine - *10(OTE)*
MACHINE_SW_STAT[19] 0 BOOL
*MACHINE_SW_STAT[19] - MainProgram/MainRoutine - *11(OTE)*
MACHINE_SW_STAT[20] 1 BOOL
*MACHINE_SW_STAT[20] - MainProgram/MainRoutine - *13(OTE)*
MACHINE_SW_STAT[21] 1 BOOL
*MACHINE_SW_STAT[21] - MainProgram/MainRoutine - *14(OTE)*
MACHINE_SW_STAT[22] 1 BOOL
*MACHINE_SW_STAT[22] - MainProgram/MainRoutine - *23(OTE)*
MACHINE_SW_STAT[23] 0 BOOL
*MACHINE_SW_STAT[23] - MainProgram/MainRoutine - *1(OTE)*
MACHINE_SW_STAT[25] 1 BOOL
*MACHINE_SW_STAT[25] - MainProgram/MainRoutine - *24(OTE)*
MACHINE_SW_STAT[26] 1 BOOL
*MACHINE_SW_STAT[26] - MainProgram/MainRoutine - *25(OTE)*
MACHINE_SW_STAT[27] 1 BOOL
*MACHINE_SW_STAT[27] - MainProgram/MainRoutine - *26(OTE)*
MACHINE_SW_STAT[28] 1 BOOL
*MACHINE_SW_STAT[28] - MainProgram/MainRoutine - *27(OTE)*
MACHINE_SW_STAT[29] 1 BOOL
*MACHINE_SW_STAT[29] - MainProgram/MainRoutine - *28(OTE)*
MACHINE_SW_STAT[30] 0 BOOL
*MACHINE_SW_STAT[30] - MainProgram/MainRoutine - *29(OTE)*
MACHINE_SW_STAT[31] 1 BOOL
*MACHINE_SW_STAT[31] - MainProgram/MainRoutine - *30(OTE)*

MOVE_TYPE 0 INT DUNEW2PLC1
0 = No Move 1 = XY JOG 2 = XY Plane Move 3 = Z JOG 4 = Z Axis Move 5 = Toggle Latch Actuator 6 = Home the Latch Actuator 7 = Unlock Latch Actuator Motor 8 = Un-Servo Motors 9 = PLC_Init

MOVE_TYPE (Continued)
External Access: Read/Write
*MOVE_TYPE - EOT_Trip_11/EOT_TRIP - *21(CPT), 12(CMP), 15(CMP), 18(CMP), 6(CMP), 9(CMP)*
MOVE_TYPE - Error_State_10/Error - 12(CMP), 8(CMP)
*MOVE_TYPE - Initialize/Init - *2(CPT), *4(CPT), 0(EQU)*
*MOVE_TYPE - Latch_UnLatch_State_6_7_8/Latch_Driver - *13(CPT), *33(CPT), 43(CMP)*
*MOVE_TYPE - MoveXY_State_2_3/MoveXY - *18(CPT), *21(CPT), 15(CMP)*
*MOVE_TYPE - MoveZ_State_4_5/MoveZ - *19(CPT), 13(CMP)*
MOVE_TYPE - Ready_State_1/Ready - 10(CMP), 12(CMP), 14(CMP), 16(CMP), 18(CMP), 4(CMP), 6(CMP), 8(CMP)
MOVE_TYPE - UnServo_9/UnServo_All - 11(CMP)

NEXTSTATE 1 DINT DUNEW2PLC1
External Access: Read Only
*NEXTSTATE - EOT_Trip_11/EOT_TRIP - *22(CPT)*
*NEXTSTATE - Error_State_10/Error - *12(CPT)*
*NEXTSTATE - Initialize/Init - *4(CPT), *9(CPT)*
*NEXTSTATE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *13(CPT), *33(CPT), *43(CPT), *51(CPT)*
NEXTSTATE - MainProgram/MainRoutine - 42(CMP), 43(CMP), 44(CMP), 45(CMP), 46(CMP), 47(CMP), 48(CMP), 49(CMP), 50(CMP), 51(CMP), 52(CMP), 53(CMP)
*NEXTSTATE - MoveXY_State_2_3/MoveXY - *0(CPT), *1(CPT), *1(CPT), *18(CPT), *2(CPT), *21(CPT)*
*NEXTSTATE - MoveZ_State_4_5/MoveZ - *10(CPT), *19(CPT), *20(CPT), *4(CPT), *5(MOV), *7(CPT)*
*NEXTSTATE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT)*
*NEXTSTATE - Ready_State_1/Ready - *11(CPT), *13(CPT), *15(CPT), *17(CPT), *19(CPT), *5(CPT), *7(CPT), *9(CPT)*
*NEXTSTATE - UnServo_9/UnServo_All - *11(CPT)*

oneshotob BOOL[32] Latch_UnLatch_State_6_7_8

External Access: Read Only
oneshotob[0] 0 BOOL
*oneshotob[0] - Latch_UnLatch_State_6_7_8/Latch_Driver - *17(OSR), 18(XIC)*
oneshotob[1] 0 BOOL
For one scan, when we enter state 6 and actuator is enabled
*oneshotob[1] - Latch_UnLatch_State_6_7_8/Latch_Driver - *11(OSR), 12(XIC)*
oneshotob[3] 0 BOOL
*oneshotob[3] - Latch_UnLatch_State_6_7_8/Latch_Driver - *50(OSR), 51(XIC)*
oneshotob[4] 0 BOOL
*oneshotob[4] - Latch_UnLatch_State_6_7_8/Latch_Driver - *27(OSR), 28(XIC)*
oneshotob[5] 0 BOOL
*oneshotob[5] - Latch_UnLatch_State_6_7_8/Latch_Driver - *32(OSR), 33(XIC)*
oneshotob[7] 0 BOOL
*oneshotob[7] - Latch_UnLatch_State_6_7_8/Latch_Driver - *23(OSR), 24(XIC), 25(XIC)*
oneshotob[9] 0 BOOL
Entered state 6 but the actuator was not enabled
*oneshotob[9] - Latch_UnLatch_State_6_7_8/Latch_Driver - *9(OSR), 10(XIC)*
oneshotob[10] 0 BOOL
*oneshotob[10] - Latch_UnLatch_State_6_7_8/Latch_Driver - *37(OSR), 38(XIC), 39(XIC)*
oneshotob[12] 0 BOOL
*oneshotob[12] - Latch_UnLatch_State_6_7_8/Latch_Driver - *45(OSR), 46(XIC)*

oneshotsb BOOL[32] Latch_UnLatch_State_6_7_8

External Access: Read Only
oneshotsb[0] 0 BOOL
*oneshotsb[0] - Latch_UnLatch_State_6_7_8/Latch_Driver - *17(OSR), 19(XIC), 20(XIC), 21(XIC)*
oneshotsb[1] 0 BOOL
While in state 6 and operator is enabled
*oneshotsb[1] - Latch_UnLatch_State_6_7_8/Latch_Driver - *11(OSR), 13(XIC), 14(XIC)*
oneshotsb[2] 0 BOOL
Latch successfully moved to the next position
oneshotsb[2] - Latch_UnLatch_State_6_7_8/Latch_Driver - 15(XIO)
oneshotsb[3] 0 BOOL
*oneshotsb[3] - Latch_UnLatch_State_6_7_8/Latch_Driver - *50(OSR)*

oneshotsb (Continued)
oneshotsb[4] 0 BOOL
*oneshotsb[4] - Latch_UnLatch_State_6_7_8/Latch_Driver - *27(OSR), 29(XIC), 30(XIC), 31(XIC)*
oneshotsb[5] 1 BOOL
*oneshotsb[5] - Latch_UnLatch_State_6_7_8/Latch_Driver - *32(OSR)*
oneshotsb[6] 0 BOOL
LatchCounter done
oneshotsb[7] 0 BOOL
*oneshotsb[7] - Latch_UnLatch_State_6_7_8/Latch_Driver - *23(OSR)*
oneshotsb[8] 0 BOOL
While in state 6
oneshotsb[9] 0 BOOL
*oneshotsb[9] - Latch_UnLatch_State_6_7_8/Latch_Driver - *9(OSR)*
oneshotsb[10] 1 BOOL
*oneshotsb[10] - Latch_UnLatch_State_6_7_8/Latch_Driver - *37(OSR), 40(XIC), 41(XIC), 42(XIC)*
oneshotsb[12] 0 BOOL
*oneshotsb[12] - Latch_UnLatch_State_6_7_8/Latch_Driver - *45(OSR), 47(XIC), 48(XIC), 49(XIC)*

PREV_ACT_POS 2 DINT Latch_UnLatch_State_6_7_8

Store the previous actuator position
External Access: Read Only
*PREV_ACT_POS - Latch_UnLatch_State_6_7_8/Latch_Driver - *12(CPT), 13(CMP), 13(CMP), 13(CMP), 13(CMP)*

STATE 1 DINT DUNEW2PLC1
CURRENT State Machine State
External Access: Read Only
STATE - Error_State_10/Error - 0(CMP), 12(CMP)
*STATE - Initialize/Init - *4(CPT), 7(CMP)*
STATE - Latch_UnLatch_State_6_7_8/Latch_Driver - 16(CMP), 34(CMP), 45(CMP), 45(CMP), 51(CMP), 8(CMP)
*STATE - MainProgram/MainRoutine - *42(CPT), *43(CPT), *44(CPT), *45(CPT), *46(CPT), *47(CPT), *48(CPT), *49(CPT), *50(CPT), *51(CPT), *52(CPT), *53(CPT)*
STATE - MoveXY_State_2_3/MoveXY - 0(CMP), 1(CMP), 11(CMP)
STATE - MoveZ_State_4_5/MoveZ - 4(CMP), 4(CMP), 5(CMP), 6(CMP)
STATE - Ready_State_1/Ready - 0(CMP)
STATE - UnServo_9/UnServo_All - 0(CMP)

STATE6_IND 0 BOOL DUNEW2PLC1
External Access: Read Only
*STATE6_IND - Latch_UnLatch_State_6_7_8/Latch_Driver - *8(OTE), 11(XIC), 9(XIC)*

STATE7_IND 0 BOOL Latch_UnLatch_State_6_7_8

External Access: Read Only
*STATE7_IND - Latch_UnLatch_State_6_7_8/Latch_Driver - *16(OTE), 17(XIC), 33(XIC)*

STATE8_IND 1 BOOL DUNEW2PLC1
External Access: Read Only
*STATE8_IND - Latch_UnLatch_State_6_7_8/Latch_Driver - *34(OTE), 35(XIC), 37(XIC), 43(XIC)*

TimeoutTimer1 TIMER Latch_UnLatch_State_6_7_8

External Access: Read/Write
*TimeoutTimer1 - Latch_UnLatch_State_6_7_8/Latch_Driver - *47(TON)*
TimeoutTimer1.TT 0 BOOL
TimeoutTimer1.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 48(XIO), 49(XIC)

TimeoutTimer2 TIMER Latch_UnLatch_State_6_7_8

External Access: Read/Write

TimeoutTimer2 (Continued)
*TimeoutTimer2 - Latch_UnLatch_State_6_7_8/Latch_Driver - *48(TON)*

TimeoutTimer2.TT	0	BOOL	

TimeoutTimer2.TT - Latch_UnLatch_State_6_7_8/Latch_Driver - 47(XIO)

UNLOCK_LATCH_MOTOR_SHAFT	1	BOOL	Latch_UnLatch_State_6_7_8

Constant No
External Access: Read Only
UNLOCK_LATCH_MOTOR_SHAFT - Latch_UnLatch_State_6_7_8/Latch_Driver - 34(XIC)

Z_EXTENDED	0	BOOL	DUNEW2PLC1

External Access: Read Only
Z_EXTENDED - Latch_UnLatch_State_6_7_8/Latch_Driver - 7(XIC)
*Z_EXTENDED - MainProgram/MainRoutine - *6(OTE)*

Z_FIXED_LATCHED	0	BOOL	DUNEW2PLC1

External Access: Read Only
Z_FIXED_LATCHED - Latch_UnLatch_State_6_7_8/Latch_Driver - 6(XIC)
*Z_FIXED_LATCHED - MainProgram/MainRoutine - *8(OTE), 59(XIC), 59(XIC), 59(XIO)*
Z_FIXED_LATCHED - MoveZ_State_4_5/MoveZ - 0(XIC), 0(XIO), 12(XIC), 12(XIO), 6(XIC), 7(XIC)

Z_FIXED_PRESENT	0	BOOL	DUNEW2PLC1

External Access: Read Only
Z_FIXED_PRESENT - Latch_UnLatch_State_6_7_8/Latch_Driver - 7(XIC)
*Z_FIXED_PRESENT - MainProgram/MainRoutine - *17(OTE)*

Z_OK_TO_ENGAGE	1	BOOL	DUNEW2PLC1

External Access: Read Only
*Z_OK_TO_ENGAGE - Latch_UnLatch_State_6_7_8/Latch_Driver - *3(OTE)*

Z_SAFE_TO_WITHDRAW	0	BOOL	DUNEW2PLC1

External Access: Read/Write
*Z_SAFE_TO_WITHDRAW - Latch_UnLatch_State_6_7_8/Latch_Driver - *6(OTE)*

Z_STAGE_LATCHED	0	BOOL	DUNEW2PLC1

External Access: Read Only
Z_STAGE_LATCHED - Latch_UnLatch_State_6_7_8/Latch_Driver - 24(XIC), 25(XIO), 4(XIO), 5(XIC)
*Z_STAGE_LATCHED - MainProgram/MainRoutine - *7(OTE), 59(XIC), 59(XIO)*

Z_STAGE_PRESENT	0	BOOL	DUNEW2PLC1

Winding head is on the Z-arm
External Access: Read Only
Z_STAGE_PRESENT - Latch_UnLatch_State_6_7_8/Latch_Driver - 7(XIC)
*Z_STAGE_PRESENT - MainProgram/MainRoutine - *16(OTE)*

Z_STAGE_UNLATCHED	0	BOOL	DUNEW2PLC1

External Access: Read Only
*Z_STAGE_UNLATCHED - Latch_UnLatch_State_6_7_8/Latch_Driver - *2(OTE)*

General

Type:	Ladder Diagram	Class:	Standard
In Program:	Latch_UnLatch_State_6_7_8	Number of Rungs:	52

General

Class:	Standard

Configuration

Main:	MoveXY	Inhibit program:	No
Fault:	None		

Monitor

Scan Times(Execution Time)

Max:	122 us	Last:	9 us

0

CMP
Expression STATE=2

CPT
Dest NEXTSTATE
1
Expression 1

1

XY_AXIS_STAT[4].IP

CMP
Expression STATE=3

tension_stable_timer.DN

Check if the tension is stable at the setpoint before moving the winding head

check_tension_stable

TENSION_CONTROL_OK

Z_RETRACTED

GEQ
Source A Z_axis.ActualPosit
-1.22
Source B MAX_TOLERABLE

Z_RETRACTED

APA_IS_VERT

APA_IS_VERT

CPT
Dest ERROR_CODE
0
Expression 3001

CPT
Dest NEXTSTATE
1
Expression 10

n
7
Z
0

CAL

STATE3_IND

CAL

CPT
Dest ERROR_CODE
0
Expression 3005

CPT
Dest NEXTSTATE
1
Expression 10

2

STATE3_IND

OSR
Storage Bit oneshotsb[0]
Output Bit oneshotob[0]

SB
OB

X_Y.PhysicalAxisFault

CPT
Dest ERROR_CODE
0
Expression 3002

X_axis.SafeTorqueOffInhibit

Y_axis.SafeTorqueOffInhibit

CPT
Dest ERROR_CODE
0
Expression 3004

CPT
Dest NEXTSTATE
1
Expression 10

3

oneshotob[0]

MSO
Axis X_axis ...
Motion Control XY_AXIS_STAT[0]
EN DN ER

MSO
Axis Y_axis ...
Motion Control XY_AXIS_STAT[1]
EN DN ER

4

XY_AXIS_STAT[0].DN XY_AXIS_STAT[1].DN

OSR
Storage Bit oneshotsb[1]
Output Bit oneshotob[1]
SB OB

5

oneshotob[1]

XY_AXIS_STAT[0].DN XY_AXIS_STAT[1].DN oneshotob[0]

OSR
Storage Bit oneshotsb[3]
Output Bit trigger_xy_move
SB OB

6

trigger_xy_move

MOV
Source X_axis.ActualPosition
-898.1537
Dest starting_x
-898.3584

MOV
Source Y_axis.ActualPosition
-15.0941
Dest starting_y
-8.9992

7

trigger_xy_move

CPT
Dest dx
1.22070312e-004
Expression ABS(starting_x-X_POSITION)

CPT
Dest dy
4.8001003
Expression ABS(starting_y-Y_POSITION)

CPT
Dest x_time
6758400.0
Expression X_max_speed/dx

CPT
Dest y_time
124.99739
Expression Y_max_speed/dy

LES
Source A x_time
6758400.0
Source B y_time
124.99739

CPT
Dest k
124.99739
Expression x_time

CPT
Dest v_max
600.0
Expression k*SQR(dx*dx+dy*dy)

LEQ
Source A y_time
124.99739
Source B x_time
6758400.0

CPT
Dest k
124.99739
Expression y_time

XY Requested speed copy of XY_Speed to be used for MCLM and MCCD in MoveXY

LES
Source A v_max
600.0
Source B XY_SPEED
1020.0

CPT
Dest XY_SPEED_REQ
600.0
Expression v_max

Maximum velocity for position move.

XY Requested speed copy of XY_Speed to be used for MCLM and MCCD in MoveXY

LEQ
Source A XY_SPEED
1020.0
Source B v_max
600.0

CPT
Dest XY_SPEED_REQ
600.0
Expression XY_SPEED

8

CPT
Dest x_dist_to_target
0.20483398
Expression X_axis.ActualPosition-X_POSITION

CPT
Dest y_dist_to_target
-1.2947998
Expression Y_axis.ActualPosition-Y_POSITION

CPT
Dest xy_dist_to_target
1.3109018
Expression SQR(x_dist_to_target*x_dist_to_target+y_dist_to_target*y_dist_to_target)

9

MOV
Source xy_decel_jerk
3000
Dest J
3000.0

MOV
Source v_xyz
0.0
Dest v_0
0.0

CPT
Dest gamma
0.0
Expression SQR(accel_xy*accel_xy+4*J*v_0)

Calculate the distance required to perform a jerk-limited stop at the gurrent jerk, acceleration, and speed

CPT
Dest stopping_distance
0.0
Expression (accel_xy+gamma)*(accel_xy+gamma)*(accel_xy+gamma)/(6*J*J)

10

CMP
Expression xy_dist_to_target<stopping_distance*1

True if distance to xy-move target is less than the required stopping distance
near_ending

11

CMP
Expression STATE=3

CPT
Dest stopping_ratio
1.1959486
Expression stopping_distance/xy_dist_to_target

This rung executes the moves. If either the tension is off or the speed regulator switch is off, it moves normally. Otherwise select whether to use the PI controller or not (i.e. the tension limited control)

12

TENSION_CONTROL_OK

This is our "enable safe speed" switch, repurposed to switch the speed regulator

speed_regulator_switch

JSR
Routine Name xy_speed_regulator

TENSION_CONTROL_OK /

trigger_xy_move

This is our "enable safe speed" switch, repurposed to switch the speed regulator

TENSION_CONTROL_OK

speed_regulator_switch /

MCLM

Coordinate System	X_Y	...
Motion Control	XY_AXIS_STAT[4]	
Move Type	0	
Position	X_POSITION <POSITION[0,0]>	...
X_axis	??	
Y_axis	??	
Speed	XY_SPEED_REQ 600.0	
Speed Units	Units per sec	
Accel Rate	XY_ACCELERATION 800.0	
Accel Units	Units per sec2	
Decel Rate	XY_DECELERATION 800.0	
Decel Units	Units per sec2	
Profile	S-Curve	
Accel Jerk	xy_accel_jerk 1500	
Decel Jerk	xy_decel_jerk 3000	
Jerk Units	Units per sec3	
Termination Type	0	
Merge	Disabled	
Merge Speed	Programmed	
Command Tolerance	50	
Lock Position	0	
Lock Direction	None	
Event Distance	0	
Calculated Data	0	

EN
DN
ER
IP
AC
PC

13

This is our "enable safe speed" switch, repurposed to switch the speed regulator

speed_regulator_switch

OSF
Storage Bit oneshotsb[13]
Output Bit oneshotob[13]

SB
OB

14

oneshotob[13]

True if distance to xy-move target is less than the required stopping distance
near_ending

MCCD
Coordinate System X_Y ...
Motion Control MCCD_X_Y_Axis1
Motion Type Coordinated Move
Change Speed Yes
Speed XY_SPEED_REQ
600.0
Speed Units Units per sec
Change Accel Yes
Accel Rate XY_ACCELERATION
800.0
Accel Units Units per sec2
Change Decel Yes
Decel Rate XY_DECELERATION
800.0
Decel Units Units per sec2
Change Accel Jerk No
Accel Jerk xy_accel_jerk
1500
Change Decel Jerk No
Decel Jerk xy_decel_jerk
3000
Jerk Units Units per sec3
Scope Active Motion
EN
DN
ER

15

XY_AXIS_STAT[4].PC

XY_AXIS_STAT[4].ER

CMP
Expression X_axis.ActualPosition<(X_POSITION+0.1)

CMP
Expression X_axis.ActualPositi

STATE3_IND

CMP
Expression MOVE_TYPE=0

>(X_POSITION-0.1)

CMP
Expression Y_axis.ActualPosition<(Y_POSITION+0.1)

CMP
Expression Y_axis.ActualPosition>(Y_P

XY_P

CPT
Dest ER
Expressio

ITION-0.1)

S_LOCK_STAT

OR_CODE
0
3003

OSR
Storage Bit oneshotsb[5]
Output Bit oneshotob[5]
SB
OB

16 XY_AXIS_STAT[4].IP

CMP
Expression XY_SPEED=0

OSR
Storage Bit oneshotsb[6]
Output Bit speed_set_to_zero
SB
OB

17 XY_AXIS_STAT[4].IP ALL_EOT_GOOD

OSR
Storage Bit oneshotsb[8]
Output Bit eot_triggered
SB
OB

0 = No Move
1 = XY JOG
2 = XY Plane Move
3 = Z JOG
4 = Z Axis Move
5 = Toggle Latch Actuator
6 = Home the Latch Actuator
7 = Unlock Latch Actuator Motor
8 = Un-Servo Motors
9 = PLC_Init

18 eot_triggered

CPT
Dest NEXTSTATE
1
Expression 11

CPT
Dest MOVE_TYPE
0
Expression 0

19 speed_set_to_zero

MCS
Coordinate System X_Y ...
Motion Control XY_AXIS_STAT[5]
Stop Type All
Change Decel Yes
Decel Rate 1000
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 1000
Jerk Units Units per sec3
EN
DN
ER
IP
PC

20

XY_AXIS_STAT[5].PC

OSR
Storage Bit oneshotsb[7]
Output Bit oneshotob[7]
SB
OB

21

oneshotob[5]
oneshotob[7]

0 = No Move
1 = XY JOG
2 = XY Plane Move
3 = Z JOG
4 = Z Axis Move
5 = Toggle Latch Actuator
6 = Home the Latch Actuator
7 = Unlock Latch Actuator Motor
8 = Un-Servo Motors
9 = PLC_Init

CPT
Dest MOVE_TYPE
0
Expression 0

CPT
Dest NEXTSTATE
1
Expression 1

22

MCLM
Coordinate System X_Y ...
Motion Control XY_AXIS_STAT[4]
Move Type 0

Position X_POSITION ...
<POSITION[0,0]>
X_axis ??
Y_axis ??
Speed XY_SPEED_REQ
600.0
Speed Units Units per sec
Accel Rate XY_ACCELERATION
800.0
Accel Units Units per sec2
Decel Rate XY_DECELERATION
800.0
Decel Units Units per sec2
Profile S-Curve
Accel Jerk xy_accel_jerk
1500
Decel Jerk xy_decel_jerk
3000
Jerk Units Units per sec3
Termination Type 0

Merge Coordinated Motion
Merge Speed Programmed
Command Tolerance 50

Lock Position 0

Lock Direction None
Event Distance 0
Calculated Data 0

EN
DN
ER
IP
AC
PC

(End)

Name	Value	Data Type	Scope
accel_xy	0.0	REAL	DUNEW2PLC1
Tangential acceleration of the winding head in xy-plane			
External Access:	Read/Write		
*accel_xy - MainProgram/MainRoutine - *57(CPT)*			
accel_xy - MoveXY_State_2_3/MoveXY - 9(CPT), 9(CPT)			
ALL_EOT_GOOD	0	BOOL	DUNEW2PLC1
External Access:	Read Only		
ALL_EOT_GOOD - EOT_Trip_11/EOT_TRIP - 0(XIO), 22(XIC)			
*ALL_EOT_GOOD - MainProgram/MainRoutine - *15(OTE)*			
ALL_EOT_GOOD - MoveXY_State_2_3/MoveXY - 17(XIO)			
ALL_EOT_GOOD - MoveZ_State_4_5/MoveZ - 16(XIO)			
APA_IS_VERTICAL	1	BOOL	DUNEW2PLC1
External Access:	Read Only		
*APA_IS_VERTICAL - MainProgram/MainRoutine - *23(OTE)*			
APA_IS_VERTICAL - MoveXY_State_2_3/MoveXY - 1(XIC), 1(XIO)			
check_tension_stable	1	BOOL	DUNEW2PLC1
Check if the tension is stable at the setpoint before moving the winding head			
Constant	No		
External Access:	Read/Write		
check_tension_stable - MoveXY_State_2_3/MoveXY - 1(XIO)			
check_tension_stable - MoveZ_State_4_5/MoveZ - 6(XIO)			
dx	1.22070312e-004	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*dx - MoveXY_State_2_3/MoveXY - *7(CPT), 7(CPT), 7(CPT)*			
dy	4.8001003	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*dy - MoveXY_State_2_3/MoveXY - *7(CPT), 7(CPT), 7(CPT)*			
eot_triggered	0	BOOL	MoveXY_State_2_3
External Access:	Read/Write		
*eot_triggered - MoveXY_State_2_3/MoveXY - *17(OSR), 18(XIC)*			
ERROR_CODE	0	DINT	DUNEW2PLC1
External Access:	Read Only		
*ERROR_CODE - EOT_Trip_11/EOT_TRIP - *22(CPT)*			
*ERROR_CODE - Error_State_10/Error - *12(CPT)*			
*ERROR_CODE - Initialize/Init - *4(CPT)*			
*ERROR_CODE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *25(CPT), *33(CPT), *36(CPT), *51(CPT)*			
ERROR_CODE - MainProgram/MainRoutine - 36(NEQ)			
*ERROR_CODE - MoveXY_State_2_3/MoveXY - *1(CPT), *1(CPT), *15(CPT), *2(CPT), *2(CPT)*			
*ERROR_CODE - MoveZ_State_4_5/MoveZ - *10(CPT), *13(CPT), *20(CPT), *4(CPT), *4(CPT), *7(CPT)*			
*ERROR_CODE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT), *16(CPT), 14(NEQ), 16(EQU), 16(EQU)*			
*ERROR_CODE - Ready_State_1/Ready - *3(MOV)*			
gamma	0.0	REAL	DUNEW2PLC1
External Access:	Read/Write		
*gamma - MoveXY_State_2_3/MoveXY - *9(CPT), 9(CPT)*			
J	3000.0	REAL	DUNEW2PLC1
External Access:	Read/Write		
*J - MoveXY_State_2_3/MoveXY - *9(MOV), 9(CPT), 9(CPT)*			
k	124.99739	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*k - MoveXY_State_2_3/MoveXY - *7(CPT), *7(CPT), 7(CPT)*			
MAX_TOLERABLE_Z	5.0	REAL	DUNEW2PLC1
Constant	Yes		

MAX_TOLERABLE_Z (Continued)
External Access: Read/Write
MAX_TOLERABLE_Z - MoveXY_State_2_3/MoveXY - 1(GEQ)

MCCD_X_Y_Axis1 MOTION_INSTRUCTION DUNEW2PLC1
External Access: Read/Write
*MCCD_X_Y_Axis1 - MoveXY_State_2_3/MoveXY - *14(MCCD)*

MOVE_TYPE 0 INT DUNEW2PLC1
0 = No Move 1 = XY JOG 2 = XY Plane Move 3 = Z JOG 4 = Z Axis Move 5 = Toggle Latch Actuator 6 = Home the Latch Actuator 7 = Unlock Latch Actuator Motor 8 = Un-Servo Motors 9 = PLC_Init

External Access: Read/Write
*MOVE_TYPE - EOT_Trip_11/EOT_TRIP - *21(CPT), 12(CMP), 15(CMP), 18(CMP), 6(CMP), 9(CMP)*
MOVE_TYPE - Error_State_10/Error - 12(CMP), 8(CMP)
*MOVE_TYPE - Initialize/Init - *2(CPT), *4(CPT), 0(EQU)*
*MOVE_TYPE - Latch_UnLatch_State_6_7_8/Latch_Driver - *13(CPT), *33(CPT), 43(CMP)*
*MOVE_TYPE - MoveXY_State_2_3/MoveXY - *18(CPT), *21(CPT), 15(CMP)*
*MOVE_TYPE - MoveZ_State_4_5/MoveZ - *19(CPT), 13(CMP)*
MOVE_TYPE - Ready_State_1/Ready - 10(CMP), 12(CMP), 14(CMP), 16(CMP), 18(CMP), 4(CMP), 6(CMP), 8(CMP)
MOVE_TYPE - UnServo_9/UnServo_All - 11(CMP)

near_ending 0 BOOL MoveXY_State_2_3
True if distance to xy-move target is less than the required stopping distance
External Access: Read/Write
*near_ending - MoveXY_State_2_3/MoveXY - *10(OTE), 14(XIO)*
near_ending - MoveXY_State_2_3/xy_speed_regulator - 9(XIC)

NEXTSTATE 1 DINT DUNEW2PLC1
External Access: Read Only
*NEXTSTATE - EOT_Trip_11/EOT_TRIP - *22(CPT)*
*NEXTSTATE - Error_State_10/Error - *12(CPT)*
*NEXTSTATE - Initialize/Init - *4(CPT), *9(CPT)*
*NEXTSTATE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *13(CPT), *33(CPT), *43(CPT), *51(CPT)*
NEXTSTATE - MainProgram/MainRoutine - 42(CMP), 43(CMP), 44(CMP), 45(CMP), 46(CMP), 47(CMP), 48(CMP), 49(CMP), 50(CMP), 51(CMP), 52(CMP), 53(CMP)
*NEXTSTATE - MoveXY_State_2_3/MoveXY - *0(CPT), *1(CPT), *1(CPT), *18(CPT), *2(CPT), *21(CPT)*
*NEXTSTATE - MoveZ_State_4_5/MoveZ - *10(CPT), *19(CPT), *20(CPT), *4(CPT), *5(MOV), *7(CPT)*
*NEXTSTATE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT)*
*NEXTSTATE - Ready_State_1/Ready - *11(CPT), *13(CPT), *15(CPT), *17(CPT), *19(CPT), *5(CPT), *7(CPT), *9(CPT)*
*NEXTSTATE - UnServo_9/UnServo_All - *11(CPT)*

oneshotob BOOL[32] MoveXY_State_2_3
External Access: Read/Write
oneshotob[0] 0 BOOL
*oneshotob[0] - MoveXY_State_2_3/MoveXY - *2(OSR), 3(XIC), 5(XIC)*
oneshotob[1] 0 BOOL
*oneshotob[1] - MoveXY_State_2_3/MoveXY - *4(OSR), 5(XIC)*
oneshotob[5] 0 BOOL
*oneshotob[5] - MoveXY_State_2_3/MoveXY - *15(OSR), 21(XIC)*
oneshotob[7] 0 BOOL
*oneshotob[7] - MoveXY_State_2_3/MoveXY - *20(OSR), 21(XIC)*
oneshotob[13] 0 BOOL
*oneshotob[13] - MoveXY_State_2_3/MoveXY - *13(OSF), 14(XIC)*

oneshotsb BOOL[32] MoveXY_State_2_3
External Access: Read/Write
oneshotsb[0] 0 BOOL
*oneshotsb[0] - MoveXY_State_2_3/MoveXY - *2(OSR)*
oneshotsb[1] 1 BOOL
*oneshotsb[1] - MoveXY_State_2_3/MoveXY - *4(OSR)*
oneshotsb[3] 0 BOOL
*oneshotsb[3] - MoveXY_State_2_3/MoveXY - *5(OSR)*
oneshotsb[5] 0 BOOL

oneshotsb (Continued)
*oneshotsb[5] - MoveXY_State_2_3/MoveXY - *15(OSR)*

oneshotsb[6]	0	BOOL	
*oneshotsb[6] - MoveXY_State_2_3/MoveXY - *16(OSR)*			
oneshotsb[7]	1	BOOL	
*oneshotsb[7] - MoveXY_State_2_3/MoveXY - *20(OSR)*			
oneshotsb[8]	0	BOOL	
*oneshotsb[8] - MoveXY_State_2_3/MoveXY - *17(OSR)*			
oneshotsb[13]	1	BOOL	
*oneshotsb[13] - MoveXY_State_2_3/MoveXY - *13(OSF)*			
oneshotsb[15]	1	BOOL	
*oneshotsb[15] - MoveXY_State_2_3/xy_speed_regulator - *9(OSR)*			
speed_regulator_switch	1	BOOL	DUNEW2PLC1
This is our "enable safe speed" switch, repurposed to switch the speed regulator			
External Access:	Read/Write		
*speed_regulator_switch - MainProgram/MainRoutine - *58(OTE)*			
speed_regulator_switch - MoveXY_State_2_3/MoveXY - 12(XIC), 12(XIO), 13(XIC)			
speed_set_to_zero	0	BOOL	MoveXY_State_2_3
External Access:	Read/Write		
*speed_set_to_zero - MoveXY_State_2_3/MoveXY - *16(OSR), 19(XIC)*			
starting_x	-898.3584	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*starting_x - MoveXY_State_2_3/MoveXY - *6(MOV), 7(CPT)*			
starting_y	-8.9992	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*starting_y - MoveXY_State_2_3/MoveXY - *6(MOV), 7(CPT)*			
STATE	1	DINT	DUNEW2PLC1
CURRENT State Machine State			
External Access:	Read Only		
STATE - Error_State_10/Error - 0(CMP), 12(CMP)			
*STATE - Initialize/Init - *4(CPT), 7(CMP)*			
STATE - Latch_UnLatch_State_6_7_8/Latch_Driver - 16(CMP), 34(CMP), 45(CMP), 45(CMP), 51(CMP), 8(CMP)			
*STATE - MainProgram/MainRoutine - *42(CPT), *43(CPT), *44(CPT), *45(CPT), *46(CPT), *47(CPT), *48(CPT), *49(CPT), *50(CPT), *51(CPT), *52(CPT), *53(CPT)*			
STATE - MoveXY_State_2_3/MoveXY - 0(CMP), 1(CMP), 11(CMP)			
STATE - MoveZ_State_4_5/MoveZ - 4(CMP), 4(CMP), 5(CMP), 6(CMP)			
STATE - Ready_State_1/Ready - 0(CMP)			
STATE - UnServo_9/UnServo_All - 0(CMP)			
STATE3_IND	0	BOOL	MoveXY_State_2_3
External Access:	Read Only		
*STATE3_IND - MoveXY_State_2_3/MoveXY - *1(OTE), 15(XIC), 2(XIC)*			
stopping_distance	2.72165489e-004	REAL	MoveXY_State_2_3
Calculate the distance required to perform a jerk-limited stop at the gurrent jerk, acceleration, and speed			
External Access:	Read/Write		
*stopping_distance - MoveXY_State_2_3/MoveXY - *9(CPT), 10(CMP), 11(CPT)*			
stopping_ratio	1.1959486	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*stopping_ratio - MoveXY_State_2_3/MoveXY - *11(CPT)*			
TENSION_CONTROL_OK	0	BOOL	DUNEW2PLC1
External Access:	Read/Write		
TENSION_CONTROL_OK - MoveXY_State_2_3/MoveXY - 1(XIO), 12(XIC), 12(XIC), 12(XIO)			
TENSION_CONTROL_OK - MoveZ_State_4_5/MoveZ - 6(XIO)			
*TENSION_CONTROL_OK - PID_Tension_Servo/PID_Tension - *3(OTE), 3(XIC), 7(XIC), 7(XIO), 7(XIO)*			
tension_stable_timer		TIMER	DUNEW2PLC1

tension_stable_timer (Continued)
External Access: Read/Write
*tension_stable_timer - PID_Tension_Servo/PID_Tension - *11(TON)*

tension_stable_timer.PRE	1	DINT	
*tension_stable_timer.PRE - PID_Tension_Servo/PID_Tension - *11(MOV)*			
tension_stable_timer.DN	0	BOOL	
tension_stable_timer.DN - MoveXY_State_2_3/MoveXY - 1(XIC)			
tension_stable_timer.DN - MoveZ_State_4_5/MoveZ - 6(XIC)			
trigger_xy_move	0	BOOL	MoveXY_State_2_3
External Access:	Read/Write		
*trigger_xy_move - MoveXY_State_2_3/MoveXY - *5(OSR), 12(XIC), 6(XIC), 7(XIC)*			
trigger_xy_move - MoveXY_State_2_3/xy_speed_regulator - 0(XIC), 1(XIC)			
v_0	0.049999997	REAL	DUNEW2PLC1
External Access:	Read/Write		
*v_0 - MoveXY_State_2_3/MoveXY - *9(MOV), 9(CPT)*			
v_max	600.0	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*v_max - MoveXY_State_2_3/MoveXY - *7(CPT), 7(CPT), 7(LEQ), 7(LES)*			
v_max - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM), 7(LES), 8(LES), 8(MOV)			
v_xyz	0.0	REAL	DUNEW2PLC1
Velocity of winding head in xyz space			
External Access:	Read/Write		
*v_xyz - MainProgram/MainRoutine - *55(CPT)*			
v_xyz - MoveXY_State_2_3/MoveXY - 9(MOV)			
X_axis		AXIS_CIP_DRIVE	DUNEW2PLC1
External Access:	Read Only		
X_axis - Camera/CameraControl - 6(MAR)			
X_axis - EOT_Trip_11/EOT_TRIP - 10(MAM), 5(MSO), 7(MAM)			
X_axis - Error_State_10/Error - 11(MAFR), 5(MSF)			
X_axis - MoveXY_State_2_3/MoveXY - 3(MSO)			
X_axis - Ready_State_1/Ready - 2(MSO)			
X_axis - UnServo_9/UnServo_All - 2(MSF), 6(MAFR)			
X_axis.ActualPosition	-898.1537	REAL	
X_axis.ActualPosition - Camera/CameraControl - 25(CPT), 28(CMP), 28(CMP)			
X_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)			
X_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 2(LIM), 2(LIM)			
X_axis.ActualVelocity	0.0	REAL	
X_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)			
X_axis.Registration2NegativeEdgePosition	0.0	REAL	
X_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 7(CPT)			
X_axis.CommandAcceleration	0.0	REAL	
X_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)			
X_axis.SafeTorqueOffInhibit	0	BOOL	
X_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)			
X_axis.SLSActiveStatus	0	BOOL	
X_axis.SLSActiveStatus - MainProgram/MainRoutine - 36(XIC)			
x_dist_to_target	0.20483398	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*x_dist_to_target - MoveXY_State_2_3/MoveXY - *8(CPT), 8(CPT)*			
X_max_speed	825.0	REAL	MoveXY_State_2_3
Constant	Yes		
External Access:	Read/Write		
X_max_speed - MoveXY_State_2_3/MoveXY - 7(CPT)			
X_POSITION	-898.3585	REAL	DUNEW2PLC1
AliasFor:	POSITION[0,0]		

X_POSITION (Continued)
Base Tag: POSITION[0,0]
Constant Yes
External Access: Read/Write
X_POSITION - MoveXY_State_2_3/MoveXY - 12(MCLM), 15(CMP), 15(CMP), 22(MCLM), 7(CPT), 8(CPT)
X_POSITION - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)

x_time	6758400.0	REAL	MoveXY_State_2_3

External Access: Read/Write
*x_time - MoveXY_State_2_3/MoveXY - *7(CPT), 7(CPT), 7(LEQ), 7(LES)*

X_Y		COORDINATE_SYSTEM	DUNEW2PLC1

External Access: Read/Write
X_Y - EOT_Trip_11/EOT_TRIP - 0(MCS)
X_Y - Error_State_10/Error - 2(MCS)
X_Y - motionQueue/queue_line_arc - 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)
X_Y - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 19(MCS), 22(MCLM)
X_Y - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)

X_Y.MovePendingStatus	0	BOOL

X_Y.MovePendingStatus - motionQueue/queue_line_arc - 104(XIO), 105(XIO), 117(XIO), 118(XIO), 67(XIO), 68(XIO)

X_Y.PhysicalAxisFault	0	BOOL

X_Y.PhysicalAxisFault - MoveXY_State_2_3/MoveXY - 2(XIC)

xy_accel_jerk	1500	DINT	MoveXY_State_2_3

Constant No
External Access: Read/Write
xy_accel_jerk - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 22(MCLM)
xy_accel_jerk - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)

XY_ACCELERATION	800.0	REAL	DUNEW2PLC1

Maximum positive acceleration for position move.
Constant No
External Access: Read/Write
XY_ACCELERATION - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 22(MCLM)

XY_AXIS_STAT		MOTION_INSTRUCTION[32]	MoveXY_State_2_3

External Access: Read/Write

XY_AXIS_STAT[0]		MOTION_INSTRUCTION

*XY_AXIS_STAT[0] - MoveXY_State_2_3/MoveXY - *3(MSO)*

XY_AXIS_STAT[0].DN	1	BOOL

XY_AXIS_STAT[0].DN - MoveXY_State_2_3/MoveXY - 4(XIC), 5(XIC)

XY_AXIS_STAT[1]		MOTION_INSTRUCTION

*XY_AXIS_STAT[1] - MoveXY_State_2_3/MoveXY - *3(MSO)*

XY_AXIS_STAT[1].DN	1	BOOL

XY_AXIS_STAT[1].DN - MoveXY_State_2_3/MoveXY - 4(XIC), 5(XIC)

XY_AXIS_STAT[4]		MOTION_INSTRUCTION

*XY_AXIS_STAT[4] - MoveXY_State_2_3/MoveXY - *12(MCLM), *22(MCLM)*
*XY_AXIS_STAT[4] - MoveXY_State_2_3/xy_speed_regulator - *0(MCLM)*

XY_AXIS_STAT[4].ER	1	BOOL

XY_AXIS_STAT[4].ER - MoveXY_State_2_3/MoveXY - 15(XIC)

XY_AXIS_STAT[4].PC	0	BOOL

XY_AXIS_STAT[4].PC - MoveXY_State_2_3/MoveXY - 15(XIC)

XY_AXIS_STAT[4].IP	0	BOOL

XY_AXIS_STAT[4].IP - MoveXY_State_2_3/MoveXY - 1(XIO), 16(XIC), 17(XIC)

XY_AXIS_STAT[5]		MOTION_INSTRUCTION

*XY_AXIS_STAT[5] - MoveXY_State_2_3/MoveXY - *19(MCS)*

XY_AXIS_STAT[5].PC	1	BOOL

XY_AXIS_STAT[5].PC - MoveXY_State_2_3/MoveXY - 20(XIC)

xy_decel_jerk	3000	DINT	MoveXY_State_2_3

Constant No
External Access: Read/Write
xy_decel_jerk - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 22(MCLM), 9(MOV)
xy_decel_jerk - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)

XY_DECELERATION 800.0 REAL DUNEW2PLC1
Maximum negative acceleration for position move.
Constant No
External Access: Read/Write
XY_DECELERATION - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 22(MCLM)

xy_dist_to_target 1.3109018 REAL MoveXY_State_2_3
External Access: Read/Write
*xy_dist_to_target - MoveXY_State_2_3/MoveXY - *8(CPT), 10(CMP), 11(CPT)*

XY_POS_LOCK_STAT 0 BOOL MoveXY_State_2_3
External Access: Read Only
*XY_POS_LOCK_STAT - MoveXY_State_2_3/MoveXY - *15(OTE)*

XY_SPEED 1020.0 REAL DUNEW2PLC1
Maximum velocity for position move.
Constant No
External Access: Read/Write
XY_SPEED - MoveXY_State_2_3/MoveXY - 16(CMP), 7(CPT), 7(LEQ), 7(LES)

XY_SPEED_REQ 600.0 REAL DUNEW2PLC1
XY Requested speed copy of XY_Speed to be used for MCLM and MCCD in MoveXY
External Access: Read/Write
*XY_SPEED_REQ - MoveXY_State_2_3/MoveXY - *7(CPT), *7(CPT), 12(MCLM), 14(MCCD), 22(MCLM)*

Y_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
Y_axis - Camera/CameraControl - 5(MAR)
Y_axis - EOT_Trip_11/EOT_TRIP - 13(MAM), 16(MAM), 5(MSO)
Y_axis - Error_State_10/Error - 10(MAFR), 6(MSF)
Y_axis - MoveXY_State_2_3/MoveXY - 3(MSO)
Y_axis - Ready_State_1/Ready - 2(MSO)
Y_axis - UnServo_9/UnServo_All - 3(MSF), 8(MAFR)
Y_axis.ActualPosition -15.0941 REAL
Y_axis.ActualPosition - Camera/CameraControl - 26(CPT), 28(CMP), 28(CMP)
Y_axis.ActualPosition - MainProgram/MainRoutine - 31(LIM), 32(LIM), 33(LIM)
Y_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)
Y_axis.ActualVelocity 0.0 REAL
Y_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)
Y_axis.Registration2NegativeEdgePosition
0.0 REAL
Y_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 8(CPT)
Y_axis.CommandAcceleration 0.0 REAL
Y_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)
Y_axis.SafeTorqueOffInhibit 0 BOOL
Y_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)

y_dist_to_target -1.2947998 REAL MoveXY_State_2_3
External Access: Read/Write
*y_dist_to_target - MoveXY_State_2_3/MoveXY - *8(CPT), 8(CPT)*

Y_max_speed 600.0 REAL MoveXY_State_2_3
Constant Yes
External Access: Read/Write
Y_max_speed - MoveXY_State_2_3/MoveXY - 7(CPT)

Y_POSITION -13.7993 REAL DUNEW2PLC1
AliasFor: POSITION[0,1]
Base Tag: POSITION[0,1]
Constant Yes
External Access: Read/Write
Y_POSITION - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 7(CPT), 8(CPT)
X_POSITION - MoveXY_State_2_3/MoveXY - 12(MCLM), 22(MCLM)
X_POSITION - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)

y_time	124.99739	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*y_time - MoveXY_State_2_3/MoveXY - *7(CPT), 7(CPT), 7(LEQ), 7(LES)*			
Z_axis		AXIS_CIP_DRIVE	DUNEW2PLC1
External Access:	Read Only		
Z_axis - EOT_Trip_11/EOT_TRIP - 19(MAM), 5(MSO)			
Z_axis - Error_State_10/Error - 3(MAS), 7(MSF), 9(MAFR)			
Z_axis - Initialize/Init - 7(MAFR)			
Z_axis - MoveZ_State_4_5/MoveZ - 12(MAM), 12(MAM), 15(MAS), 16(MAS), 9(MSO)			
Z_axis - Ready_State_1/Ready - 2(MSO)			
Z_axis - UnServo_9/UnServo_All - 10(MAFR), 4(MSF)			
Z_axis.PhysicalAxisFault	0	BOOL	
Z_axis.PhysicalAxisFault - MoveZ_State_4_5/MoveZ - 10(XIC)			
Z_axis.ActualPosition	-1.2237	REAL	
Z_axis.ActualPosition - MainProgram/MainRoutine - 6(CMP)			
Z_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 1(GEQ)			
Z_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 13(CMP), 13(CMP)			
Z_axis.ActualVelocity	0.0	REAL	
Z_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT)			
Z_RETRACTED	1	BOOL	DUNEW2PLC1
External Access:	Read Only		
*Z_RETRACTED - MainProgram/MainRoutine - *5(OTE)*			
Z_RETRACTED - MoveXY_State_2_3/MoveXY - 1(XIC), 1(XIO)			

General

Type:	Ladder Diagram	Class:	Standard
In Program:	MoveXY_State_2_3	Number of Rungs:	23

0

trigger_xy_move

MCLM

Coordinate System	X_Y
Motion Control	XY_AXIS_STAT[4]
Move Type	0
Position	X_POSITION <POSITION[0,0]>
X_axis	??
Y_axis	??
Speed	v_max 600.0
Speed Units	Units per sec
Accel Rate	xy_regulated_acceleration 2000.0
Accel Units	Units per sec2
Decel Rate	xy_regulated_deceleration 2000.0
Decel Units	Units per sec2
Profile	S-Curve
Accel Jerk	xy_accel_jerk 1500
Decel Jerk	xy_decel_jerk 3000
Jerk Units	Units per sec3
Termination Type	0
Merge	Disabled
Merge Speed	Programmed
Command Tolerance	50
Lock Position	0
Lock Direction	None
Event Distance	0
Calculated Data	0

EN DN ER IP AC PC

1

trigger_xy_move

MOV
Source xy_dt 1
Dest regulator_loop_timer.PRE 1

MOV
Source xy_d_dt 50
Dest xy_d_timer.PRE 50

MOV
Source 0
Dest xy_i_term 869.64233

2

regulator_loop_timer.DN

TON
Timer regulator_loop_timer
Preset 1
Accum 0
EN DN

3

xy_d_timer.DN

TON
Timer xy_d_timer
Preset 50
Accum 0
EN DN

4

xy_d_timer.DN

CPT
Dest d_raw 0.0
Expression xy_kd*(xy_error-xy_error_prev)/xy_d_dt*100

CPT
Dest xy_d_term 0.0
Expression xy_d_alpha*d_raw+(1-xy_d_alpha)*xy_d_term_prev

MOV
Source xy_error 8.493781
Dest xy_error_prev 8.493781

MOV
Source xy_d_term 0.0
Dest xy_d_term_prev 0.0

5

CPT
Dest xy_error 8.493781
Expression speed_tension_setpoint-tension

6

regulator_loop_timer.DN

CPT
Dest xy_p_term 0.0
Expression xy_kp*xy_error

7

regulator_loop_timer.DN

LES
Source A regulated_speed 869.62634
Source B v_max 600.0

LES
Source A xy_error 8.493781
Source B 0

GRT
Source A regulated_speed 869.62634
Source B min_regulated_speed 100

GRT
Source A xy_error 8.493781
Source B 0

CPT
Dest xy_i_term 869.64233
Expression xy_i_term+(xy_ki*xy_error*xy_dt/1000)

LES
Source A xy_i_term 869.64233
Source B min_integral -1000

MOV
Source min_integral -1000
Dest xy_i_term 869.64233

GRT
Source A xy_i_term 869.64233
Source B max_integral 1000

MOV
Source max_integral 1000
Dest xy_i_term 869.64233

8

regulator_loop_timer.DN

CPT
Dest regulated_speed
869.62634
Expression xy_default_speed+xy_p_term+xy_i_term+xy_d_term

LES
Source A v_max
600.0
Source B regulated_speed
869.62634

MOV
Source v_max
600.0
Dest regulated_speed
869.62634

LES
Source A regulated_speed
869.62634
Source B min_regulated_speed
100

MOV
Source min_regulated_speed
100
Dest regulated_speed
869.62634

9

True if distance to xy-move target is less than the required stopping distance
near_ending

OSR
Storage Bit oneshotsb[15]
Output Bit near_ending_trigger
SB
OB

(End)

Name	Value	Data Type	Scope
d_raw	0.0	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*d_raw - MoveXY_State_2_3/xy_speed_regulator - *4(CPT), 4(CPT)*			
max_integral	1000	DINT	MoveXY_State_2_3
Constant	No		
External Access:	Read/Write		
max_integral - MoveXY_State_2_3/xy_speed_regulator - 7(GRT), 7(MOV)			
min_integral	-1000	DINT	MoveXY_State_2_3
Constant	No		
External Access:	Read/Write		
min_integral - MoveXY_State_2_3/xy_speed_regulator - 7(LES), 7(MOV)			
min_regulated_speed	100	DINT	MoveXY_State_2_3
Constant	No		
External Access:	Read/Write		
min_regulated_speed - MoveXY_State_2_3/xy_speed_regulator - 7(GRT), 8(LES), 8(MOV)			
near_ending	0	BOOL	MoveXY_State_2_3
True if distance to xy-move target is less than the required stopping distance			
External Access:	Read/Write		
*near_ending - MoveXY_State_2_3/MoveXY - *10(OTE), 14(XIO)*			
near_ending - MoveXY_State_2_3/xy_speed_regulator - 9(XIC)			
near_ending_trigger	0	BOOL	MoveXY_State_2_3
External Access:	Read/Write		
*near_ending_trigger - MoveXY_State_2_3/xy_speed_regulator - *9(OSR)*			
oneshotsb		BOOL[32]	MoveXY_State_2_3
External Access:	Read/Write		
oneshotsb[0]	0	BOOL	
*oneshotsb[0] - MoveXY_State_2_3/MoveXY - *2(OSR)*			
oneshotsb[1]	1	BOOL	
*oneshotsb[1] - MoveXY_State_2_3/MoveXY - *4(OSR)*			
oneshotsb[3]	0	BOOL	
*oneshotsb[3] - MoveXY_State_2_3/MoveXY - *5(OSR)*			
oneshotsb[5]	0	BOOL	
*oneshotsb[5] - MoveXY_State_2_3/MoveXY - *15(OSR)*			
oneshotsb[6]	0	BOOL	
*oneshotsb[6] - MoveXY_State_2_3/MoveXY - *16(OSR)*			
oneshotsb[7]	1	BOOL	
*oneshotsb[7] - MoveXY_State_2_3/MoveXY - *20(OSR)*			
oneshotsb[8]	0	BOOL	
*oneshotsb[8] - MoveXY_State_2_3/MoveXY - *17(OSR)*			
oneshotsb[13]	1	BOOL	
*oneshotsb[13] - MoveXY_State_2_3/MoveXY - *13(OSF)*			
oneshotsb[15]	1	BOOL	
*oneshotsb[15] - MoveXY_State_2_3/xy_speed_regulator - *9(OSR)*			
regulated_speed	869.62634	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*regulated_speed - MoveXY_State_2_3/xy_speed_regulator - *8(CPT), *8(MOV), *8(MOV), 7(GRT), 7(LES), 8(LES), 8(LES)*			
regulator_loop_timer		TIMER	MoveXY_State_2_3
External Access:	Read/Write		
*regulator_loop_timer - MoveXY_State_2_3/xy_speed_regulator - *2(TON)*			
regulator_loop_timer.PRE	1	DINT	
*regulator_loop_timer.PRE - MoveXY_State_2_3/xy_speed_regulator - *1(MOV)*			
regulator_loop_timer.DN	0	BOOL	
regulator_loop_timer.DN - MoveXY_State_2_3/xy_speed_regulator - 2(XIO), 6(XIC), 7(XIC), 8(XIC)			
speed_tension_setpoint	8.5	REAL	MoveXY_State_2_3

speed_tension_setpoint (Continued)
Constant No
External Access: Read/Write
speed_tension_setpoint - MoveXY_State_2_3/xy_speed_regulator - 5(CPT)

tension 2.10263440e-003 REAL DUNEW2PLC1
wire tension from load cell (N)
External Access: Read/Write
tension - MainProgram/MainRoutine - 20(GRT)
tension - MoveXY_State_2_3/xy_speed_regulator - 5(CPT)
*tension - PID_Tension_Servo/PID_Tension - *5(CPT), *6(CPT), *7(MOV), 11(CMP), 12(GRT), 7(PID), 8(CPT)*

trigger_xy_move 0 BOOL MoveXY_State_2_3
External Access: Read/Write
*trigger_xy_move - MoveXY_State_2_3/MoveXY - *5(OSR), 12(XIC), 6(XIC), 7(XIC)*
trigger_xy_move - MoveXY_State_2_3/xy_speed_regulator - 0(XIC), 1(XIC)

v_max 600.0 REAL MoveXY_State_2_3
External Access: Read/Write
*v_max - MoveXY_State_2_3/MoveXY - *7(CPT), 7(CPT), 7(LEQ), 7(LES)*
v_max - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM), 7(LES), 8(LES), 8(MOV)

X_POSITION -898.3585 REAL DUNEW2PLC1
AliasFor: POSITION[0,0]
Base Tag: POSITION[0,0]
Constant Yes
External Access: Read/Write
X_POSITION - MoveXY_State_2_3/MoveXY - 12(MCLM), 15(CMP), 15(CMP), 22(MCLM), 7(CPT), 8(CPT)
X_POSITION - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)

X_Y COORDINATE_SYSTEM DUNEW2PLC1
External Access: Read/Write
X_Y - EOT_Trip_11/EOT_TRIP - 0(MCS)
X_Y - Error_State_10/Error - 2(MCS)
X_Y - motionQueue/queue_line_arc - 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)
X_Y - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 19(MCS), 22(MCLM)
X_Y - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)
X_Y.MovePendingStatus 0 BOOL
X_Y.MovePendingStatus - motionQueue/queue_line_arc - 104(XIO), 105(XIO), 117(XIO), 118(XIO), 67(XIO), 68(XIO)
X_Y.PhysicalAxisFault 0 BOOL
X_Y.PhysicalAxisFault - MoveXY_State_2_3/MoveXY - 2(XIC)

xy_accel_jerk 1500 DINT MoveXY_State_2_3
Constant No
External Access: Read/Write
xy_accel_jerk - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 22(MCLM)
xy_accel_jerk - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)

XY_AXIS_STAT MOTION_INSTRUCTION[32] MoveXY_State_2_3
External Access: Read/Write
XY_AXIS_STAT[0] MOTION_INSTRUCTION
*XY_AXIS_STAT[0] - MoveXY_State_2_3/MoveXY - *3(MSO)*
XY_AXIS_STAT[0].DN 1 BOOL
XY_AXIS_STAT[0].DN - MoveXY_State_2_3/MoveXY - 4(XIC), 5(XIC)
XY_AXIS_STAT[1] MOTION_INSTRUCTION
*XY_AXIS_STAT[1] - MoveXY_State_2_3/MoveXY - *3(MSO)*
XY_AXIS_STAT[1].DN 1 BOOL
XY_AXIS_STAT[1].DN - MoveXY_State_2_3/MoveXY - 4(XIC), 5(XIC)
XY_AXIS_STAT[4] MOTION_INSTRUCTION
*XY_AXIS_STAT[4] - MoveXY_State_2_3/MoveXY - *12(MCLM), *22(MCLM)*
*XY_AXIS_STAT[4] - MoveXY_State_2_3/xy_speed_regulator - *0(MCLM)*
XY_AXIS_STAT[4].ER 1 BOOL
XY_AXIS_STAT[4].ER - MoveXY_State_2_3/MoveXY - 15(XIC)
XY_AXIS_STAT[4].PC 0 BOOL

XY_AXIS_STAT (Continued)
XY_AXIS_STAT[4].PC - MoveXY_State_2_3/MoveXY - 15(XIC)

XY_AXIS_STAT[4].IP	0	BOOL	
XY_AXIS_STAT[4].IP - MoveXY_State_2_3/MoveXY - 1(XIO), 16(XIC), 17(XIC)			
XY_AXIS_STAT[5]		MOTION_INSTRUCTION	
*XY_AXIS_STAT[5] - MoveXY_State_2_3/MoveXY - *19(MCS)*			
XY_AXIS_STAT[5].PC	1	BOOL	
XY_AXIS_STAT[5].PC - MoveXY_State_2_3/MoveXY - 20(XIC)			
xy_d_alpha	1.0	REAL	MoveXY_State_2_3
Constant	No		
External Access:	Read/Write		
xy_d_alpha - MoveXY_State_2_3/xy_speed_regulator - 4(CPT)			
xy_d_dt	50	DINT	MoveXY_State_2_3
Constant	No		
External Access:	Read/Write		
xy_d_dt - MoveXY_State_2_3/xy_speed_regulator - 1(MOV), 4(CPT)			
xy_d_term	0.0	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*xy_d_term - MoveXY_State_2_3/xy_speed_regulator - *4(CPT), 4(MOV), 8(CPT)*			
xy_d_term_prev	0.0	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*xy_d_term_prev - MoveXY_State_2_3/xy_speed_regulator - *4(MOV), 4(CPT)*			
xy_d_timer		TIMER	MoveXY_State_2_3
External Access:	Read/Write		
*xy_d_timer - MoveXY_State_2_3/xy_speed_regulator - *3(TON)*			
xy_d_timer.PRE	50	DINT	
*xy_d_timer.PRE - MoveXY_State_2_3/xy_speed_regulator - *1(MOV)*			
xy_d_timer.DN	0	BOOL	
xy_d_timer.DN - MoveXY_State_2_3/xy_speed_regulator - 3(XIO), 4(XIC)			
xy_decel_jerk	3000	DINT	MoveXY_State_2_3
Constant	No		
External Access:	Read/Write		
xy_decel_jerk - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 22(MCLM), 9(MOV)			
xy_decel_jerk - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)			
xy_default_speed	0	DINT	MoveXY_State_2_3
Constant	No		
External Access:	Read/Write		
xy_default_speed - MoveXY_State_2_3/xy_speed_regulator - 8(CPT)			
xy_dt	1	DINT	MoveXY_State_2_3
Constant	No		
External Access:	Read/Write		
xy_dt - MoveXY_State_2_3/xy_speed_regulator - 1(MOV), 7(CPT)			
xy_error	8.493781	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*xy_error - MoveXY_State_2_3/xy_speed_regulator - *5(CPT), 4(CPT), 4(MOV), 6(CPT), 7(CPT), 7(GRT), 7(LES)*			
xy_error_prev	8.493781	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*xy_error_prev - MoveXY_State_2_3/xy_speed_regulator - *4(MOV), 4(CPT)*			
xy_i_term	869.64233	REAL	MoveXY_State_2_3
External Access:	Read/Write		
*xy_i_term - MoveXY_State_2_3/xy_speed_regulator - *1(MOV), *7(CPT), *7(MOV), *7(MOV), 7(CPT), 7(GRT), 7(LES), 8(CPT)*			
xy_kd	0.0	REAL	MoveXY_State_2_3

xy_kd (Continued)
Constant No
External Access: Read/Write
xy_kd - MoveXY_State_2_3/xy_speed_regulator - 4(CPT)

xy_ki	300.0	REAL	MoveXY_State_2_3

Constant No
External Access: Read/Write
xy_ki - MoveXY_State_2_3/xy_speed_regulator - 7(CPT)

xy_kp	0.0	REAL	MoveXY_State_2_3

Constant No
External Access: Read/Write
xy_kp - MoveXY_State_2_3/xy_speed_regulator - 6(CPT)

xy_p_term	0.0	REAL	MoveXY_State_2_3

External Access: Read/Write
*xy_p_term - MoveXY_State_2_3/xy_speed_regulator - *6(CPT), 8(CPT)*

xy_regulated_acceleration	2000.0	REAL	DUNEW2PLC1

Constant No
External Access: Read/Write
xy_regulated_acceleration - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)

xy_regulated_deceleration	2000.0	REAL	DUNEW2PLC1

Constant No
External Access: Read/Write
xy_regulated_deceleration - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)

General

Type:	Ladder Diagram	Class:	Standard
In Program:	MoveXY_State_2_3	Number of Rungs:	10

General

Class:	Standard

Configuration

Main:	MoveZ	Inhibit program:	No
Fault:	None		

Monitor

Scan Times(Execution Time)

Max:	76 us	Last:	5 us

0

Z_FIXED_LATCHED

EQU
Source A ACTUATOR_POS
2
Source B 2

no_latch_collision

Z_FIXED_LATCHED

latch_bypass

1

If at the head, check head support arms. Same for foot
X_XFER_OK

Y_XFER_OK

no_apa_collision

2

If at the head, check head support arms. Same for foot
X_XFER_OK

LIM
Low Limit 400
Test X_axis.ActualPosition
-898.1537
High Limit 500

support_collision_window_bttm FRAME_LOC_HD_BTM
support_collision_window_mid FRAME_LOC_HD_MID
support_collision_window_top FRAME_LOC_HD_TOP

LIM
Low Limit 7100
Test X_axis.ActualPosition
-898.1537
High Limit 7200

support_collision_window_bttm FRAME_LOC_FT_BTM
support_collision_window_mid FRAME_LOC_FT_MID
support_collision_window_top FRAME_LOC_FT_TOP

support_collision_window_bttm support_collision_window_mid support_collision_window_top

Y_XFER_OK

no_supports_collision

3

no_latch_collision | no_supports_collision | no_apa_collision → MASTER_Z_GO

If Master Z go is not present and machine attempts to move Z axis (state 4/5) then produce appropriate error code and put machine into error state 10

4

MASTER_Z_GO /

CMP Expression STATE=4 → CPT Dest ERROR_CODE 0 Expression 4001

CMP Expression STATE=5 → CPT Dest ERROR_CODE 0 Expression 5001

→ CPT Dest NEXTSTATE 1 Expression 10

5

MASTER_Z_GO | CMP Expression STATE=4 → MOV Source 1 Dest NEXTSTATE 1

6

MASTER_Z_GO | CMP Expression STATE=5 → tension_stable_timer.DN → STATE5_IND

Check if the tension is stable at the setpoint before moving the winding head

check_tension_stable /

TENSION_CONTROL_OK /

Z_FIXED_LATCHED

7
STATE5_IND Z_FIXED_LATCHED LATCH_ACTUATOR_HOMED
CMP
Expression ACTUATOR_POS<>2
CPT
Dest ERROR_CODE
0
Expression 5004
CPT
Dest NEXTSTATE
1
Expression 10
OSR
Storage Bit oneshotsb[8]
Output Bit oneshotob[8]
SB
OB

8
STATE5_IND oneshotsb[8] /
OSR
Storage Bit oneshotsb[3]
Output Bit oneshotob[3]
SB
OB

9
oneshotob[3]
MSO
Axis Z_axis ...
Motion Control Z_AXIS_STAT[3]
EN
DN
ER

10
STATE5_IND Z_axis.PhysicalAxisFault
CPT
Dest ERROR_CODE
0
Expression 5002
CPT
Dest NEXTSTATE
1
Expression 10

11
Z_AXIS_STAT[3].DN
OSR
Storage Bit oneshotsb[4]
Output Bit oneshotob[4]
SB
OB

Move more abruptly when the winding head isn't on the z-arm

12

oneshotob[4]

Z_AXIS_STAT[3].DN

oneshotob[3]

Z_FIXED_LATCHED

MAM

Axis	Z_axis
Motion Control	Z_AXIS_STAT[4]
Move Type	0
Position	Z_POSITION 0.0
Speed	Z_SPEED 300.0
Speed Units	Units per sec
Accel Rate	Z_ACCELERATION 800.0
Accel Units	Units per sec2
Decel Rate	Z_DECELLERATION 800.0
Decel Units	Units per sec2
Profile	S-Curve
Accel Jerk	z_accel_jerk 1500
Decel Jerk	z_decel_jerk 3500
Jerk Units	Units per sec3
Merge	Disabled
Merge Speed	Programmed
Lock Position	0
Lock Direction	None
Event Distance	0
Calculated Data	0

EN DN ER IP PC

Z_FIXED_LATCHED

MAM

Axis	Z_axis
Motion Control	Z_AXIS_STAT[4]
Move Type	0
Position	Z_POSITION 0.0
Speed	1000
Speed Units	Units per sec
Accel Rate	10000
Accel Units	Units per sec2
Decel Rate	10000
Decel Units	Units per sec2
Profile	S-Curve
Accel Jerk	10000
Decel Jerk	10000
Jerk Units	Units per sec3
Merge	Disabled
Merge Speed	Programmed
Lock Position	0
Lock Direction	None
Event Distance	0
Calculated Data	0

EN DN ER IP PC

13

Z_AXIS_STAT[4].PC STATE5_IND

CMP
Expression Z_axis.ActualPosition<(Z_POSITION+0.2)

CMP
Expression Z_axis.Actua

STATE5_IND

CMP
Expression MOVE_TYPE=0

osition>(Z_POSITION-0.2)

Z_POS_LOCK_STAT

CPT
Dest ERROR_CODE
0
Expression 5003

OSR
Storage Bit oneshotsb[5]
Output Bit oneshotob[5]
SB
OB

14

Z_AXIS_STAT[4].IP

CMP
Expression Z_SPEED=0

OSR
Storage Bit oneshotsb[6]
Output Bit oneshotob[6]
SB
OB

15

oneshotob[6]

MAS
Axis Z_axis ...
Motion Control Z_AXIS_STAT[5]
Stop Type All
Change Decel Yes
Decel Rate Z_DECELLERATION
800.0
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
Jerk Units % of Time
EN
DN
ER
IP
PC

16

z_eot_bypass Z_AXIS_STAT[4].IP ALL_EOT_GOOD

MAS
Axis Z_axis ...
Motion Control Z_AXIS_STAT[7]
Stop Type Jog
Change Decel Yes
Decel Rate 400
Decel Units Units per sec2
Change Decel Jerk No
Decel Jerk 100
Jerk Units % of Time
EN
DN
ER
IP
PC

17

z_eot_bypass Z_AXIS_STAT[7].PC Z_AXIS_STAT[4].IP

OSR
Storage Bit oneshotsb[10]
Output Bit oneshotob[10]
SB
OB

18

Z_AXIS_STAT[5].PC

OSR
Storage Bit oneshotsb[7]
Output Bit oneshotob[7]
SB
OB

0 = No Move
1 = XY JOG
2 = XY Plane Move
3 = Z JOG
4 = Z Axis Move
5 = Toggle Latch Actuator
6 = Home the Latch Actuator
7 = Unlock Latch Actuator Motor
8 = Un-Servo Motors
9 = PLC_Init

19

oneshotob[5]
oneshotob[7]

CPT
Dest MOVE_TYPE
0
Expression 0

CPT
Dest NEXTSTATE
1
Expression 1

20

oneshotob[10]

CPT
Dest ERROR_CODE
0
Expression 5005

CPT
Dest NEXTSTATE
1
Expression 11

(End)

Name	Value	Data Type	Scope
ACTUATOR_POS	2	DINT	DUNEW2PLC1
External Access:	Read Only		
*ACTUATOR_POS - Latch_UnLatch_State_6_7_8/Latch_Driver - *2(CPT), *3(CPT), *4(CPT), *5(CPT), 12(CPT), 13(CMP), 13(CMP), 13(CMP), 13(NEQ)*			
ACTUATOR_POS - MainProgram/MainRoutine - 59(EQU), 59(EQU)			
ACTUATOR_POS - MoveZ_State_4_5/MoveZ - 0(EQU), 7(CMP)			
ALL_EOT_GOOD	0	BOOL	DUNEW2PLC1
External Access:	Read Only		
ALL_EOT_GOOD - EOT_Trip_11/EOT_TRIP - 0(XIO), 22(XIC)			
*ALL_EOT_GOOD - MainProgram/MainRoutine - *15(OTE)*			
ALL_EOT_GOOD - MoveXY_State_2_3/MoveXY - 17(XIO)			
ALL_EOT_GOOD - MoveZ_State_4_5/MoveZ - 16(XIO)			
check_tension_stable	1	BOOL	DUNEW2PLC1
Check if the tension is stable at the setpoint before moving the winding head			
Constant	No		
External Access:	Read/Write		
check_tension_stable - MoveXY_State_2_3/MoveXY - 1(XIO)			
check_tension_stable - MoveZ_State_4_5/MoveZ - 6(XIO)			
ERROR_CODE	0	DINT	DUNEW2PLC1
External Access:	Read Only		
*ERROR_CODE - EOT_Trip_11/EOT_TRIP - *22(CPT)*			
*ERROR_CODE - Error_State_10/Error - *12(CPT)*			
*ERROR_CODE - Initialize/Init - *4(CPT)*			
*ERROR_CODE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *25(CPT), *33(CPT), *36(CPT), *51(CPT)*			
ERROR_CODE - MainProgram/MainRoutine - 36(NEQ)			
*ERROR_CODE - MoveXY_State_2_3/MoveXY - *1(CPT), *1(CPT), *15(CPT), *2(CPT), *2(CPT)*			
*ERROR_CODE - MoveZ_State_4_5/MoveZ - *10(CPT), *13(CPT), *20(CPT), *4(CPT), *4(CPT), *7(CPT)*			
*ERROR_CODE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT), *16(CPT), 14(NEQ), 16(EQU), 16(EQU)*			
*ERROR_CODE - Ready_State_1/Ready - *3(MOV)*			
FRAME_LOC_FT_BTM	1	BOOL	DUNEW2PLC1
External Access:	Read/Write		
*FRAME_LOC_FT_BTM - MainProgram/MainRoutine - *30(OTE)*			
FRAME_LOC_FT_BTM - MoveZ_State_4_5/MoveZ - 2(XIO)			
FRAME_LOC_FT_MID	0	BOOL	DUNEW2PLC1
External Access:	Read/Write		
*FRAME_LOC_FT_MID - MainProgram/MainRoutine - *29(OTE)*			
FRAME_LOC_FT_MID - MoveZ_State_4_5/MoveZ - 2(XIO)			
FRAME_LOC_FT_TOP	1	BOOL	DUNEW2PLC1
External Access:	Read/Write		
*FRAME_LOC_FT_TOP - MainProgram/MainRoutine - *28(OTE)*			
FRAME_LOC_FT_TOP - MoveZ_State_4_5/MoveZ - 2(XIO)			
FRAME_LOC_HD_BTM	1	BOOL	DUNEW2PLC1
External Access:	Read/Write		
*FRAME_LOC_HD_BTM - MainProgram/MainRoutine - *27(OTE)*			
FRAME_LOC_HD_BTM - MoveZ_State_4_5/MoveZ - 2(XIO)			
FRAME_LOC_HD_MID	1	BOOL	DUNEW2PLC1
External Access:	Read/Write		
*FRAME_LOC_HD_MID - MainProgram/MainRoutine - *26(OTE)*			
FRAME_LOC_HD_MID - MoveZ_State_4_5/MoveZ - 2(XIO)			
FRAME_LOC_HD_TOP	1	BOOL	DUNEW2PLC1
External Access:	Read/Write		
*FRAME_LOC_HD_TOP - MainProgram/MainRoutine - *25(OTE)*			
FRAME_LOC_HD_TOP - MoveZ_State_4_5/MoveZ - 2(XIO)			

LATCH_ACTUATOR_HOMED 0 BOOL DUNEW2PLC1
External Access: Read Only
*LATCH_ACTUATOR_HOMED - Initialize/Init - *5(OTU)*
*LATCH_ACTUATOR_HOMED - Latch_UnLatch_State_6_7_8/Latch_Driver - *24(OTL), *25(OTL), *38(OTU), 26(XIC), 27(XIC)*
LATCH_ACTUATOR_HOMED - MoveZ_State_4_5/MoveZ - 7(XIC)

latch_bypass 1 BOOL MoveZ_State_4_5
Constant No
External Access: Read/Write
latch_bypass - MoveZ_State_4_5/MoveZ - 0(XIC)

MASTER_Z_GO 0 BOOL DUNEW2PLC1
External Access: Read Only
*MASTER_Z_GO - MoveZ_State_4_5/MoveZ - *3(OTE), 4(XIO), 5(XIC), 6(XIC)*

MOVE_TYPE 0 INT DUNEW2PLC1
0 = No Move 1 = XY JOG 2 = XY Plane Move 3 = Z JOG 4 = Z Axis Move 5 = Toggle Latch Actuator 6 = Home the Latch Actuator 7 = Unlock Latch Actuator Motor 8 = Un-Servo Motors 9 = PLC_Init

External Access: Read/Write
*MOVE_TYPE - EOT_Trip_11/EOT_TRIP - *21(CPT), 12(CMP), 15(CMP), 18(CMP), 6(CMP), 9(CMP)*
MOVE_TYPE - Error_State_10/Error - 12(CMP), 8(CMP)
*MOVE_TYPE - Initialize/Init - *2(CPT), *4(CPT), 0(EQU)*
*MOVE_TYPE - Latch_UnLatch_State_6_7_8/Latch_Driver - *13(CPT), *33(CPT), 43(CMP)*
*MOVE_TYPE - MoveXY_State_2_3/MoveXY - *18(CPT), *21(CPT), 15(CMP)*
*MOVE_TYPE - MoveZ_State_4_5/MoveZ - *19(CPT), 13(CMP)*
MOVE_TYPE - Ready_State_1/Ready - 10(CMP), 12(CMP), 14(CMP), 16(CMP), 18(CMP), 4(CMP), 6(CMP), 8(CMP)
MOVE_TYPE - UnServo_9/UnServo_All - 11(CMP)

NEXTSTATE 1 DINT DUNEW2PLC1
External Access: Read Only
*NEXTSTATE - EOT_Trip_11/EOT_TRIP - *22(CPT)*
*NEXTSTATE - Error_State_10/Error - *12(CPT)*
*NEXTSTATE - Initialize/Init - *4(CPT), *9(CPT)*
*NEXTSTATE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *13(CPT), *33(CPT), *43(CPT), *51(CPT)*
NEXTSTATE - MainProgram/MainRoutine - 42(CMP), 43(CMP), 44(CMP), 45(CMP), 46(CMP), 47(CMP), 48(CMP), 49(CMP), 50(CMP), 51(CMP), 52(CMP), 53(CMP)
*NEXTSTATE - MoveXY_State_2_3/MoveXY - *0(CPT), *1(CPT), *1(CPT), *18(CPT), *2(CPT), *21(CPT)*
*NEXTSTATE - MoveZ_State_4_5/MoveZ - *10(CPT), *19(CPT), *20(CPT), *4(CPT), *5(MOV), *7(CPT)*
*NEXTSTATE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT)*
*NEXTSTATE - Ready_State_1/Ready - *11(CPT), *13(CPT), *15(CPT), *17(CPT), *19(CPT), *5(CPT), *7(CPT), *9(CPT)*
*NEXTSTATE - UnServo_9/UnServo_All - *11(CPT)*

no_apa_collision 0 BOOL DUNEW2PLC1
External Access: Read/Write
*no_apa_collision - MoveZ_State_4_5/MoveZ - *1(OTE), 3(XIC)*

no_latch_collision 1 BOOL DUNEW2PLC1
External Access: Read/Write
*no_latch_collision - MoveZ_State_4_5/MoveZ - *0(OTE), 3(XIC)*

no_supports_collision 0 BOOL DUNEW2PLC1
External Access: Read/Write
*no_supports_collision - MoveZ_State_4_5/MoveZ - *2(OTE), 3(XIC)*

oneshotob BOOL[32] MoveZ_State_4_5
External Access: Read Only
oneshotob[3] 0 BOOL
*oneshotob[3] - MoveZ_State_4_5/MoveZ - *8(OSR), 12(XIC), 9(XIC)*
oneshotob[4] 0 BOOL
*oneshotob[4] - MoveZ_State_4_5/MoveZ - *11(OSR), 12(XIC)*
oneshotob[5] 0 BOOL
*oneshotob[5] - MoveZ_State_4_5/MoveZ - *13(OSR), 19(XIC)*
oneshotob[6] 0 BOOL

oneshotob (Continued)
*oneshotob[6] - MoveZ_State_4_5/MoveZ - *14(OSR), 15(XIC)*
oneshotob[7] 0 BOOL
*oneshotob[7] - MoveZ_State_4_5/MoveZ - *18(OSR), 19(XIC)*
oneshotob[8] 0 BOOL
*oneshotob[8] - MoveZ_State_4_5/MoveZ - *7(OSR)*
oneshotob[10] 0 BOOL
*oneshotob[10] - MoveZ_State_4_5/MoveZ - *17(OSR), 20(XIC)*

oneshotsb BOOL[32] MoveZ_State_4_5
External Access: Read Only
oneshotsb[3] 0 BOOL
*oneshotsb[3] - MoveZ_State_4_5/MoveZ - *8(OSR)*
oneshotsb[4] 0 BOOL
*oneshotsb[4] - MoveZ_State_4_5/MoveZ - *11(OSR)*
oneshotsb[5] 0 BOOL
*oneshotsb[5] - MoveZ_State_4_5/MoveZ - *13(OSR)*
oneshotsb[6] 0 BOOL
*oneshotsb[6] - MoveZ_State_4_5/MoveZ - *14(OSR)*
oneshotsb[7] 0 BOOL
*oneshotsb[7] - MoveZ_State_4_5/MoveZ - *18(OSR)*
oneshotsb[8] 0 BOOL
*oneshotsb[8] - MoveZ_State_4_5/MoveZ - *7(OSR), 8(XIO)*
oneshotsb[10] 0 BOOL
*oneshotsb[10] - MoveZ_State_4_5/MoveZ - *17(OSR)*

STATE 1 DINT DUNEW2PLC1
CURRENT State Machine State
External Access: Read Only
STATE - Error_State_10/Error - 0(CMP), 12(CMP)
*STATE - Initialize/Init - *4(CPT), 7(CMP)*
STATE - Latch_UnLatch_State_6_7_8/Latch_Driver - 16(CMP), 34(CMP), 45(CMP), 45(CMP), 51(CMP), 8(CMP)
*STATE - MainProgram/MainRoutine - *42(CPT), *43(CPT), *44(CPT), *45(CPT), *46(CPT), *47(CPT), *48(CPT), *49(CPT), *50(CPT), *51(CPT), *52(CPT), *53(CPT)*
STATE - MoveXY_State_2_3/MoveXY - 0(CMP), 1(CMP), 11(CMP)
STATE - MoveZ_State_4_5/MoveZ - 4(CMP), 4(CMP), 5(CMP), 6(CMP)
STATE - Ready_State_1/Ready - 0(CMP)
STATE - UnServo_9/UnServo_All - 0(CMP)

STATE5_IND 0 BOOL MoveZ_State_4_5
External Access: Read Only
*STATE5_IND - MoveZ_State_4_5/MoveZ - *6(OTE), 10(XIC), 13(XIC), 13(XIC), 7(XIC), 8(XIC)*

support_collision_window_bttm 0 BOOL DUNEW2PLC1
External Access: Read/Write
*support_collision_window_bttm - MainProgram/MainRoutine - *31(OTE)*
support_collision_window_bttm - MoveZ_State_4_5/MoveZ - 2(XIC), 2(XIC), 2(XIO)

support_collision_window_mid 0 BOOL DUNEW2PLC1
External Access: Read/Write
*support_collision_window_mid - MainProgram/MainRoutine - *32(OTE)*
support_collision_window_mid - MoveZ_State_4_5/MoveZ - 2(XIC), 2(XIC), 2(XIO)

support_collision_window_top 0 BOOL DUNEW2PLC1
External Access: Read/Write
*support_collision_window_top - MainProgram/MainRoutine - *33(OTE)*
support_collision_window_top - MoveZ_State_4_5/MoveZ - 2(XIC), 2(XIC), 2(XIO)

TENSION_CONTROL_OK 0 BOOL DUNEW2PLC1
External Access: Read/Write
TENSION_CONTROL_OK - MoveXY_State_2_3/MoveXY - 1(XIO), 12(XIC), 12(XIC), 12(XIO)
TENSION_CONTROL_OK - MoveZ_State_4_5/MoveZ - 6(XIO)
*TENSION_CONTROL_OK - PID_Tension_Servo/PID_Tension - *3(OTE), 3(XIC), 7(XIC), 7(XIO), 7(XIO)*

tension_stable_timer TIMER DUNEW2PLC1
External Access: Read/Write
*tension_stable_timer - PID_Tension_Servo/PID_Tension - *11(TON)*
tension_stable_timer.PRE 1 DINT
*tension_stable_timer.PRE - PID_Tension_Servo/PID_Tension - *11(MOV)*
tension_stable_timer.DN 0 BOOL
tension_stable_timer.DN - MoveXY_State_2_3/MoveXY - 1(XIC)
tension_stable_timer.DN - MoveZ_State_4_5/MoveZ - 6(XIC)

X_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
X_axis - Camera/CameraControl - 6(MAR)
X_axis - EOT_Trip_11/EOT_TRIP - 10(MAM), 5(MSO), 7(MAM)
X_axis - Error_State_10/Error - 11(MAFR), 5(MSF)
X_axis - MoveXY_State_2_3/MoveXY - 3(MSO)
X_axis - Ready_State_1/Ready - 2(MSO)
X_axis - UnServo_9/UnServo_All - 2(MSF), 6(MAFR)
X_axis.ActualPosition -898.1537 REAL
X_axis.ActualPosition - Camera/CameraControl - 25(CPT), 28(CMP), 28(CMP)
X_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)
X_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 2(LIM), 2(LIM)
X_axis.ActualVelocity 0.0 REAL
X_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)
X_axis.Registration2NegativeEdgePosition
0.0 REAL
X_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 7(CPT)
X_axis.CommandAcceleration 0.0 REAL
X_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)
X_axis.SafeTorqueOffInhibit 0 BOOL
X_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)
X_axis.SLSActiveStatus 0 BOOL
X_axis.SLSActiveStatus - MainProgram/MainRoutine - 36(XIC)

X_XFER_OK 0 BOOL DUNEW2PLC1
If at the head, check head support arms. Same for foot
External Access: Read Only
*X_XFER_OK - MainProgram/MainRoutine - *21(OTE)*
X_XFER_OK - MoveZ_State_4_5/MoveZ - 1(XIC), 2(XIC)

Y_XFER_OK 0 BOOL DUNEW2PLC1
External Access: Read Only
*Y_XFER_OK - MainProgram/MainRoutine - *18(OTE)*
Y_XFER_OK - MoveZ_State_4_5/MoveZ - 1(XIC), 2(XIC)

z_accel_jerk 1500 DINT MoveZ_State_4_5
Constant No
External Access: Read/Write
z_accel_jerk - MoveZ_State_4_5/MoveZ - 12(MAM)

Z_ACCELERATION 800.0 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
Z_ACCELERATION - MoveZ_State_4_5/MoveZ - 12(MAM)

Z_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
Z_axis - EOT_Trip_11/EOT_TRIP - 19(MAM), 5(MSO)
Z_axis - Error_State_10/Error - 3(MAS), 7(MSF), 9(MAFR)
Z_axis - Initialize/Init - 7(MAFR)
Z_axis - MoveZ_State_4_5/MoveZ - 12(MAM), 12(MAM), 15(MAS), 16(MAS), 9(MSO)
Z_axis - Ready_State_1/Ready - 2(MSO)
Z_axis - UnServo_9/UnServo_All - 10(MAFR), 4(MSF)
Z_axis.PhysicalAxisFault 0 BOOL
Z_axis.PhysicalAxisFault - MoveZ_State_4_5/MoveZ - 10(XIC)

Z_axis (Continued)

Z_axis.ActualPosition	-1.2237	REAL	
Z_axis.ActualPosition - MainProgram/MainRoutine - 6(CMP)			
Z_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 1(GEQ)			
Z_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 13(CMP), 13(CMP)			
Z_axis.ActualVelocity	-0.049999997	REAL	
Z_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT)			
Z_AXIS_STAT		MOTION_INSTRUCTION[32]	MoveZ_State_4_5
External Access:	Read Only		
Z_AXIS_STAT[3]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[3] - MoveZ_State_4_5/MoveZ - *9(MSO)*			
Z_AXIS_STAT[3].DN	0	BOOL	
Z_AXIS_STAT[3].DN - MoveZ_State_4_5/MoveZ - 11(XIC), 12(XIC)			
Z_AXIS_STAT[4]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[4] - MoveZ_State_4_5/MoveZ - *12(MAM), *12(MAM)*			
Z_AXIS_STAT[4].PC	0	BOOL	
Z_AXIS_STAT[4].PC - MoveZ_State_4_5/MoveZ - 13(XIC)			
Z_AXIS_STAT[4].IP	0	BOOL	
Z_AXIS_STAT[4].IP - MoveZ_State_4_5/MoveZ - 14(XIC), 16(XIC), 17(XIO)			
Z_AXIS_STAT[5]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[5] - MoveZ_State_4_5/MoveZ - *15(MAS)*			
Z_AXIS_STAT[5].PC	0	BOOL	
Z_AXIS_STAT[5].PC - MoveZ_State_4_5/MoveZ - 18(XIC)			
Z_AXIS_STAT[7]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[7] - MoveZ_State_4_5/MoveZ - *16(MAS)*			
Z_AXIS_STAT[7].PC	0	BOOL	
Z_AXIS_STAT[7].PC - MoveZ_State_4_5/MoveZ - 17(XIC)			
z_decel_jerk	3500	DINT	MoveZ_State_4_5
Constant	No		
External Access:	Read/Write		
z_decel_jerk - MoveZ_State_4_5/MoveZ - 12(MAM)			
Z_DECELLERATION	800.0	REAL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
Z_DECELLERATION - MoveZ_State_4_5/MoveZ - 12(MAM), 15(MAS)			
z_eot_bypass	1	BOOL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
z_eot_bypass - MainProgram/MainRoutine - 9(XIC)			
z_eot_bypass - MoveZ_State_4_5/MoveZ - 16(XIO), 17(XIO)			
Z_FIXED_LATCHED	0	BOOL	DUNEW2PLC1
External Access:	Read Only		
Z_FIXED_LATCHED - Latch_UnLatch_State_6_7_8/Latch_Driver - 6(XIC)			
*Z_FIXED_LATCHED - MainProgram/MainRoutine - *8(OTE), 59(XIC), 59(XIC), 59(XIO)*			
Z_FIXED_LATCHED - MoveZ_State_4_5/MoveZ - 0(XIC), 0(XIO), 12(XIC), 12(XIO), 6(XIC), 7(XIC)			
Z_POS_LOCK_STAT	0	BOOL	MoveZ_State_4_5
External Access:	Read Only		
*Z_POS_LOCK_STAT - MoveZ_State_4_5/MoveZ - *13(OTE)*			
Z_POSITION	0.0	REAL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
Z_POSITION - MoveZ_State_4_5/MoveZ - 12(MAM), 12(MAM), 13(CMP), 13(CMP)			
Z_SPEED	300.0	REAL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
Z_SPEED - MoveZ_State_4_5/MoveZ - 12(MAM), 14(CMP)			

General

Type:	Ladder Diagram	Class:	Standard
In Program:	MoveZ_State_4_5	Number of Rungs:	21

General

Class:	Standard

Configuration

Main:	Ready	Inhibit program:	No
Fault:	None		

Monitor

Scan Times(Execution Time)

Max:	71 us	Last:	4 us

0
INIT_DONE
CMP
Expression STATE=1
STATE1_IND

1
STATE1_IND
OSR
Storage Bit trigger_axes_sb
Output Bit trigger_axes_ob
SB
OB

2
trigger_axes_ob
dont_auto_trigger_axes_in_state_1
MSO
Axis X_axis ...
Motion Control x_on_mso
EN
DN
ER
MSO
Axis Y_axis ...
Motion Control y_on_mso
EN
DN
ER
MSO
Axis Z_axis ...
Motion Control z_on_mso
EN
DN
ER

3
STATE1_IND
MOV
Source 0
Dest ERROR_CODE
0

4
STATE1_IND
CMP
Expression MOVE_TYPE=1
OSR
Storage Bit SetBit[0]
Output Bit OutBit[0]
SB
OB

5
OutBit[0]
CPT
Dest NEXTSTATE
1
Expression 2

6
STATE1_IND
CMP
Expression MOVE_TYPE=2
OSR
Storage Bit SetBit[1]
Output Bit OutBit[1]
SB
OB

7
OutBit[1]
CPT
Dest NEXTSTATE
1
Expression 3

8
STATE1_IND
CMP
Expression MOVE_TYPE=3
OSR
Storage Bit SetBit[2]
Output Bit OutBit[2]
SB
OB

9
OutBit[2]
CPT
Dest NEXTSTATE
1
Expression 4

10
STATE1_IND
CMP
Expression MOVE_TYPE=4
OSR
Storage Bit SetBit[3]
Output Bit OutBit[3]
SB
OB

11
OutBit[3]
CPT
Dest NEXTSTATE
1
Expression 5

12
STATE1_IND
CMP
Expression MOVE_TYPE=5
OSR
Storage Bit SetBit[4]
Output Bit OutBit[4]
SB
OB

13
OutBit[4]
CPT
Dest NEXTSTATE
1
Expression 6

14
STATE1_IND
CMP
Expression MOVE_TYPE=6
OSR
Storage Bit SetBit[5]
Output Bit OutBit[5]
SB
OB

15
OutBit[5]
CPT
Dest NEXTSTATE
1
Expression 7

16
STATE1_IND
CMP
Expression MOVE_TYPE=7
OSR
Storage Bit SetBit[6]
Output Bit OutBit[6]
SB
OB

17
OutBit[6]
CPT
Dest NEXTSTATE
1
Expression 8

18
STATE1_IND
CMP
Expression MOVE_TYPE=8
OSR
Storage Bit SetBit[7]
Output Bit OutBit[7]
SB
OB

19
OutBit[7]
CPT
Dest NEXTSTATE
1
Expression 9

(End)

Name	Value	Data Type	Scope
dont_auto_trigger_axes_in_state_1	1	BOOL	Ready_State_1
Constant	No		
External Access:	Read/Write		
dont_auto_trigger_axes_in_state_1 - Ready_State_1/Ready - 2(XIO)			
ERROR_CODE	0	DINT	DUNEW2PLC1
External Access:	Read Only		
*ERROR_CODE - EOT_Trip_11/EOT_TRIP - *22(CPT)*			
*ERROR_CODE - Error_State_10/Error - *12(CPT)*			
*ERROR_CODE - Initialize/Init - *4(CPT)*			
*ERROR_CODE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *25(CPT), *33(CPT), *36(CPT), *51(CPT)*			
ERROR_CODE - MainProgram/MainRoutine - 36(NEQ)			
*ERROR_CODE - MoveXY_State_2_3/MoveXY - *1(CPT), *1(CPT), *15(CPT), *2(CPT), *2(CPT)*			
*ERROR_CODE - MoveZ_State_4_5/MoveZ - *10(CPT), *13(CPT), *20(CPT), *4(CPT), *4(CPT), *7(CPT)*			
*ERROR_CODE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT), *16(CPT), 14(NEQ), 16(EQU), 16(EQU)*			
*ERROR_CODE - Ready_State_1/Ready - *3(MOV)*			
INIT_DONE	1	BOOL	DUNEW2PLC1
External Access:	Read Only		
INIT_DONE - Error_State_10/Error - 0(XIC)			
*INIT_DONE - Initialize/Init - *6(OTE), 7(XIC)*			
INIT_DONE - MainProgram/MainRoutine - 42(XIC), 43(XIC), 44(XIC), 45(XIC), 46(XIC), 47(XIC), 48(XIC), 49(XIC), 50(XIC), 51(XIC), 52(XIC), 53(XIC)			
INIT_DONE - Ready_State_1/Ready - 0(XIC)			
INIT_DONE - UnServo_9/UnServo_All - 0(XIC)			
MOVE_TYPE	0	INT	DUNEW2PLC1
0 = No Move 1 = XY JOG 2 = XY Plane Move 3 = Z JOG 4 = Z Axis Move 5 = Toggle Latch Actuator 6 = Home the Latch Actuator 7 = Unlock Latch Actuator Motor 8 = Un-Servo Motors 9 = PLC_Init			
External Access:	Read/Write		
*MOVE_TYPE - EOT_Trip_11/EOT_TRIP - *21(CPT), 12(CMP), 15(CMP), 18(CMP), 6(CMP), 9(CMP)*			
MOVE_TYPE - Error_State_10/Error - 12(CMP), 8(CMP)			
*MOVE_TYPE - Initialize/Init - *2(CPT), *4(CPT), 0(EQU)*			
*MOVE_TYPE - Latch_UnLatch_State_6_7_8/Latch_Driver - *13(CPT), *33(CPT), 43(CMP)*			
*MOVE_TYPE - MoveXY_State_2_3/MoveXY - *18(CPT), *21(CPT), 15(CMP)*			
*MOVE_TYPE - MoveZ_State_4_5/MoveZ - *19(CPT), 13(CMP)*			
MOVE_TYPE - Ready_State_1/Ready - 10(CMP), 12(CMP), 14(CMP), 16(CMP), 18(CMP), 4(CMP), 6(CMP), 8(CMP)			
MOVE_TYPE - UnServo_9/UnServo_All - 11(CMP)			
NEXTSTATE	1	DINT	DUNEW2PLC1
External Access:	Read Only		
*NEXTSTATE - EOT_Trip_11/EOT_TRIP - *22(CPT)*			
*NEXTSTATE - Error_State_10/Error - *12(CPT)*			
*NEXTSTATE - Initialize/Init - *4(CPT), *9(CPT)*			
*NEXTSTATE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *13(CPT), *33(CPT), *43(CPT), *51(CPT)*			
NEXTSTATE - MainProgram/MainRoutine - 42(CMP), 43(CMP), 44(CMP), 45(CMP), 46(CMP), 47(CMP), 48(CMP), 49(CMP), 50(CMP), 51(CMP), 52(CMP), 53(CMP)			
*NEXTSTATE - MoveXY_State_2_3/MoveXY - *0(CPT), *1(CPT), *1(CPT), *18(CPT), *2(CPT), *21(CPT)*			
*NEXTSTATE - MoveZ_State_4_5/MoveZ - *10(CPT), *19(CPT), *20(CPT), *4(CPT), *5(MOV), *7(CPT)*			
*NEXTSTATE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT)*			
*NEXTSTATE - Ready_State_1/Ready - *11(CPT), *13(CPT), *15(CPT), *17(CPT), *19(CPT), *5(CPT), *7(CPT), *9(CPT)*			
*NEXTSTATE - UnServo_9/UnServo_All - *11(CPT)*			
OutBit		BOOL[32]	Ready_State_1
External Access:	Read/Write		
OutBit[0]	0	BOOL	
*OutBit[0] - Ready_State_1/Ready - *4(OSR), 5(XIC)*			
OutBit[1]	0	BOOL	
*OutBit[1] - Ready_State_1/Ready - *6(OSR), 7(XIC)*			
OutBit[2]	0	BOOL	
*OutBit[2] - Ready_State_1/Ready - *8(OSR), 9(XIC)*			
OutBit[3]	0	BOOL	

OutBit (Continued)
*OutBit[3] - Ready_State_1/Ready - *10(OSR), 11(XIC)*
OutBit[4] 0 BOOL
*OutBit[4] - Ready_State_1/Ready - *12(OSR), 13(XIC)*
OutBit[5] 0 BOOL
*OutBit[5] - Ready_State_1/Ready - *14(OSR), 15(XIC)*
OutBit[6] 0 BOOL
*OutBit[6] - Ready_State_1/Ready - *16(OSR), 17(XIC)*
OutBit[7] 0 BOOL
*OutBit[7] - Ready_State_1/Ready - *18(OSR), 19(XIC)*

SetBit BOOL[32] Ready_State_1
External Access: Read/Write
SetBit[0] 0 BOOL
*SetBit[0] - Ready_State_1/Ready - *4(OSR)*
SetBit[1] 0 BOOL
*SetBit[1] - Ready_State_1/Ready - *6(OSR)*
SetBit[2] 0 BOOL
*SetBit[2] - Ready_State_1/Ready - *8(OSR)*
SetBit[3] 0 BOOL
*SetBit[3] - Ready_State_1/Ready - *10(OSR)*
SetBit[4] 0 BOOL
*SetBit[4] - Ready_State_1/Ready - *12(OSR)*
SetBit[5] 0 BOOL
*SetBit[5] - Ready_State_1/Ready - *14(OSR)*
SetBit[6] 0 BOOL
*SetBit[6] - Ready_State_1/Ready - *16(OSR)*
SetBit[7] 0 BOOL
*SetBit[7] - Ready_State_1/Ready - *18(OSR)*

STATE 1 DINT DUNEW2PLC1
CURRENT State Machine State
External Access: Read Only
STATE - Error_State_10/Error - 0(CMP), 12(CMP)
*STATE - Initialize/Init - *4(CPT), 7(CMP)*
STATE - Latch_UnLatch_State_6_7_8/Latch_Driver - 16(CMP), 34(CMP), 45(CMP), 45(CMP), 51(CMP), 8(CMP)
*STATE - MainProgram/MainRoutine - *42(CPT), *43(CPT), *44(CPT), *45(CPT), *46(CPT), *47(CPT), *48(CPT), *49(CPT), *50(CPT), *51(CPT), *52(CPT), *53(CPT)*
STATE - MoveXY_State_2_3/MoveXY - 0(CMP), 1(CMP), 11(CMP)
STATE - MoveZ_State_4_5/MoveZ - 4(CMP), 4(CMP), 5(CMP), 6(CMP)
STATE - Ready_State_1/Ready - 0(CMP)
STATE - UnServo_9/UnServo_All - 0(CMP)

STATE1_IND 1 BOOL Ready_State_1
External Access: Read Only
*STATE1_IND - Ready_State_1/Ready - *0(OTE), 1(XIC), 10(XIC), 12(XIC), 14(XIC), 16(XIC), 18(XIC), 3(XIC), 4(XIC), 6(XIC), 8(XIC)*

trigger_axes_ob 0 BOOL Ready_State_1
External Access: Read/Write
*trigger_axes_ob - Ready_State_1/Ready - *1(OSR), 2(XIC)*

trigger_axes_sb 1 BOOL Ready_State_1
External Access: Read/Write
*trigger_axes_sb - Ready_State_1/Ready - *1(OSR)*

X_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
X_axis - Camera/CameraControl - 6(MAR)
X_axis - EOT_Trip_11/EOT_TRIP - 10(MAM), 5(MSO), 7(MAM)
X_axis - Error_State_10/Error - 11(MAFR), 5(MSF)
X_axis - MoveXY_State_2_3/MoveXY - 3(MSO)
X_axis - Ready_State_1/Ready - 2(MSO)
X_axis - UnServo_9/UnServo_All - 2(MSF), 6(MAFR)
X_axis.ActualPosition -898.1537 REAL

X_axis (Continued)
X_axis.ActualPosition - Camera/CameraControl - 25(CPT), 28(CMP), 28(CMP)
X_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)
X_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 2(LIM), 2(LIM)

X_axis.ActualVelocity	0.0	REAL
X_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)		
X_axis.Registration2NegativeEdgePosition	0.0	REAL
X_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 7(CPT)		
X_axis.CommandAcceleration	0.0	REAL
X_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)		
X_axis.SafeTorqueOffInhibit	0	BOOL
X_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)		
X_axis.SLSActiveStatus	0	BOOL
X_axis.SLSActiveStatus - MainProgram/MainRoutine - 36(XIC)		

x_on_mso MOTION_INSTRUCTION Ready_State_1
External Access: Read/Write
*x_on_mso - Ready_State_1/Ready - *2(MSO)*

Y_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
Y_axis - Camera/CameraControl - 5(MAR)
Y_axis - EOT_Trip_11/EOT_TRIP - 13(MAM), 16(MAM), 5(MSO)
Y_axis - Error_State_10/Error - 10(MAFR), 6(MSF)
Y_axis - MoveXY_State_2_3/MoveXY - 3(MSO)
Y_axis - Ready_State_1/Ready - 2(MSO)
Y_axis - UnServo_9/UnServo_All - 3(MSF), 8(MAFR)

Y_axis.ActualPosition	-15.0941	REAL
Y_axis.ActualPosition - Camera/CameraControl - 26(CPT), 28(CMP), 28(CMP)		
Y_axis.ActualPosition - MainProgram/MainRoutine - 31(LIM), 32(LIM), 33(LIM)		
Y_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)		
Y_axis.ActualVelocity	0.0	REAL
Y_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)		
Y_axis.Registration2NegativeEdgePosition	0.0	REAL
Y_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 8(CPT)		
Y_axis.CommandAcceleration	0.0	REAL
Y_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)		
Y_axis.SafeTorqueOffInhibit	0	BOOL
Y_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)		

y_on_mso MOTION_INSTRUCTION Ready_State_1
External Access: Read/Write
*y_on_mso - Ready_State_1/Ready - *2(MSO)*

Z_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
Z_axis - EOT_Trip_11/EOT_TRIP - 19(MAM), 5(MSO)
Z_axis - Error_State_10/Error - 3(MAS), 7(MSF), 9(MAFR)
Z_axis - Initialize/Init - 7(MAFR)
Z_axis - MoveZ_State_4_5/MoveZ - 12(MAM), 12(MAM), 15(MAS), 16(MAS), 9(MSO)
Z_axis - Ready_State_1/Ready - 2(MSO)
Z_axis - UnServo_9/UnServo_All - 10(MAFR), 4(MSF)

Z_axis.PhysicalAxisFault	0	BOOL
Z_axis.PhysicalAxisFault - MoveZ_State_4_5/MoveZ - 10(XIC)		
Z_axis.ActualPosition	-1.2237	REAL
Z_axis.ActualPosition - MainProgram/MainRoutine - 6(CMP)		
Z_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 1(GEQ)		
Z_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 13(CMP), 13(CMP)		
Z_axis.ActualVelocity	0.0	REAL
Z_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT)		

z_on_mso MOTION_INSTRUCTION Ready_State_1

z_on_mso (Continued)

External Access: Read/Write

*z_on_mso - Ready_State_1/Ready - *2(MSO)*

General

Type:	Ladder Diagram	Class:	Standard
In Program:	Ready_State_1	Number of Rungs:	20

General

First state to set up machine

Class: Standard

Configuration

Main:	Init	Inhibit program:	No
Fault:	None		

Monitor

Scan Times(Execution Time)

Max:	86 us	Last:	3 us

0 = No Move
1 = XY JOG
2 = XY Plane Move
3 = Z JOG
4 = Z Axis Move
5 = Toggle Latch Actuator
6 = Home the Latch Actuator
7 = Unlock Latch Actuator Motor
8 = Un-Servo Motors
9 = PLC_Init

0

EQU	
Source A	MOVE_TYPE
	0
Source B	9

INIT_SW

INIT_SW (U)

1

INIT_SW

TON	
Timer	TIMER
Preset	2000
Accum	2000

EN
DN

0 = No Move
1 = XY JOG
2 = XY Plane Move
3 = Z JOG
4 = Z Axis Move
5 = Toggle Latch Actuator
6 = Home the Latch Actuator
7 = Unlock Latch Actuator Motor
8 = Un-Servo Motors
9 = PLC_Init

2

INIT_SW

CPT	
Dest	MOVE_TYPE
	0
Expression	0

INIT_SW (L)

3

TIMER.DN

OSR	
Storage Bit	SetBit[0]
Output Bit	OutBit[0]

SB
OB

4

OutBit[0]

0 = No Move
1 = XY JOG
2 = XY Plane Move
3 = Z JOG
4 = Z Axis Move
5 = Toggle Latch Actuator
6 = Home the Latch Actuator
7 = Unlock Latch Actuator Motor
8 = Un-Servo Motors
9 = PLC_Init

CPT	
Dest	MOVE_TYPE
	0
Expression	0

CPT	
Dest	NEXTSTATE
	1
Expression	1

CURRENT State
Machine State

CPT	
Dest	STATE
	1
Expression	0

CPT	
Dest	ERROR_CODE
	0
Expression	0

5

OutBit[0] — LATCH_ACTUATOR_HOMED (U)

6

SetBit[0] — INIT_DONE

7

INIT_DONE

CMP	
Expression	STATE=0

MAFR	
Axis	Z_axis ...
Motion Control	Z_AXIS_STAT[0]

EN, DN, ER

8

Z_AXIS_STAT[0].DN

OSR	
Storage Bit	SetBit[1]
Output Bit	OutBit[1]

SB, OB

9

OutBit[1]

CPT	
Dest	NEXTSTATE
	1
Expression	1

(End)

Name	Value	Data Type	Scope
ERROR_CODE	0	DINT	DUNEW2PLC1
External Access:	Read Only		

*ERROR_CODE - EOT_Trip_11/EOT_TRIP - *22(CPT)*
*ERROR_CODE - Error_State_10/Error - *12(CPT)*
*ERROR_CODE - Initialize/Init - *4(CPT)*
*ERROR_CODE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *25(CPT), *33(CPT), *36(CPT), *51(CPT)*
ERROR_CODE - MainProgram/MainRoutine - 36(NEQ)
*ERROR_CODE - MoveXY_State_2_3/MoveXY - *1(CPT), *1(CPT), *15(CPT), *2(CPT), *2(CPT)*
*ERROR_CODE - MoveZ_State_4_5/MoveZ - *10(CPT), *13(CPT), *20(CPT), *4(CPT), *4(CPT), *7(CPT)*
*ERROR_CODE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT), *16(CPT), 14(NEQ), 16(EQU), 16(EQU)*
*ERROR_CODE - Ready_State_1/Ready - *3(MOV)*

INIT_DONE	1	BOOL	DUNEW2PLC1
External Access:	Read Only		

INIT_DONE - Error_State_10/Error - 0(XIC)
*INIT_DONE - Initialize/Init - *6(OTE), 7(XIC)*
INIT_DONE - MainProgram/MainRoutine - 42(XIC), 43(XIC), 44(XIC), 45(XIC), 46(XIC), 47(XIC), 48(XIC), 49(XIC), 50(XIC), 51(XIC), 52(XIC), 53(XIC)
INIT_DONE - Ready_State_1/Ready - 0(XIC)
INIT_DONE - UnServo_9/UnServo_All - 0(XIC)

INIT_SW	1	BOOL	Initialize
External Access:	Read/Write		

*INIT_SW - Initialize/Init - *0(OTU), *2(OTL), 0(XIC), 1(XIC), 2(XIO)*

LATCH_ACTUATOR_HOMED	0	BOOL	DUNEW2PLC1
External Access:	Read Only		

*LATCH_ACTUATOR_HOMED - Initialize/Init - *5(OTU)*
*LATCH_ACTUATOR_HOMED - Latch_UnLatch_State_6_7_8/Latch_Driver - *24(OTL), *25(OTL), *38(OTU), 26(XIC), 27(XIC)*
LATCH_ACTUATOR_HOMED - MoveZ_State_4_5/MoveZ - 7(XIC)

MOVE_TYPE	0	INT	DUNEW2PLC1

0 = No Move 1 = XY JOG 2 = XY Plane Move 3 = Z JOG 4 = Z Axis Move 5 = Toggle Latch Actuator 6 = Home the Latch Actuator 7 = Unlock Latch Actuator Motor 8 = Un-Servo Motors 9 = PLC_Init

External Access:	Read/Write		

*MOVE_TYPE - EOT_Trip_11/EOT_TRIP - *21(CPT), 12(CMP), 15(CMP), 18(CMP), 6(CMP), 9(CMP)*
MOVE_TYPE - Error_State_10/Error - 12(CMP), 8(CMP)
*MOVE_TYPE - Initialize/Init - *2(CPT), *4(CPT), 0(EQU)*
*MOVE_TYPE - Latch_UnLatch_State_6_7_8/Latch_Driver - *13(CPT), *33(CPT), 43(CMP)*
*MOVE_TYPE - MoveXY_State_2_3/MoveXY - *18(CPT), *21(CPT), 15(CMP)*
*MOVE_TYPE - MoveZ_State_4_5/MoveZ - *19(CPT), 13(CMP)*
MOVE_TYPE - Ready_State_1/Ready - 10(CMP), 12(CMP), 14(CMP), 16(CMP), 18(CMP), 4(CMP), 6(CMP), 8(CMP)
MOVE_TYPE - UnServo_9/UnServo_All - 11(CMP)

NEXTSTATE	1	DINT	DUNEW2PLC1
External Access:	Read Only		

*NEXTSTATE - EOT_Trip_11/EOT_TRIP - *22(CPT)*
*NEXTSTATE - Error_State_10/Error - *12(CPT)*
*NEXTSTATE - Initialize/Init - *4(CPT), *9(CPT)*
*NEXTSTATE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *13(CPT), *33(CPT), *43(CPT), *51(CPT)*
NEXTSTATE - MainProgram/MainRoutine - 42(CMP), 43(CMP), 44(CMP), 45(CMP), 46(CMP), 47(CMP), 48(CMP), 49(CMP), 50(CMP), 51(CMP), 52(CMP), 53(CMP)
*NEXTSTATE - MoveXY_State_2_3/MoveXY - *0(CPT), *1(CPT), *1(CPT), *18(CPT), *2(CPT), *21(CPT)*
*NEXTSTATE - MoveZ_State_4_5/MoveZ - *10(CPT), *19(CPT), *20(CPT), *4(CPT), *5(MOV), *7(CPT)*
*NEXTSTATE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT)*
*NEXTSTATE - Ready_State_1/Ready - *11(CPT), *13(CPT), *15(CPT), *17(CPT), *19(CPT), *5(CPT), *7(CPT), *9(CPT)*
*NEXTSTATE - UnServo_9/UnServo_All - *11(CPT)*

OutBit		BOOL[32]	Initialize
External Access:	Read Only		
OutBit[0]	0	BOOL	

*OutBit[0] - Initialize/Init - *3(OSR), 4(XIC), 5(XIC)*

OutBit (Continued)

OutBit[1]	0	BOOL	
*OutBit[1] - Initialize/Init - *8(OSR), 9(XIC)*			
SetBit		BOOL[32]	Initialize
External Access:	Read Only		
SetBit[0]	1	BOOL	
*SetBit[0] - Initialize/Init - *3(OSR), 6(XIC)*			
SetBit[1]	1	BOOL	
*SetBit[1] - Initialize/Init - *8(OSR)*			
STATE	1	DINT	DUNEW2PLC1
CURRENT State Machine State			
External Access:	Read Only		
STATE - Error_State_10/Error - 0(CMP), 12(CMP)			
*STATE - Initialize/Init - *4(CPT), 7(CMP)*			
STATE - Latch_UnLatch_State_6_7_8/Latch_Driver - 16(CMP), 34(CMP), 45(CMP), 45(CMP), 51(CMP), 8(CMP)			
*STATE - MainProgram/MainRoutine - *42(CPT), *43(CPT), *44(CPT), *45(CPT), *46(CPT), *47(CPT), *48(CPT), *49(CPT), *50(CPT), *51(CPT), *52(CPT), *53(CPT)*			
STATE - MoveXY_State_2_3/MoveXY - 0(CMP), 1(CMP), 11(CMP)			
STATE - MoveZ_State_4_5/MoveZ - 4(CMP), 4(CMP), 5(CMP), 6(CMP)			
STATE - Ready_State_1/Ready - 0(CMP)			
STATE - UnServo_9/UnServo_All - 0(CMP)			
TIMER		TIMER	Initialize
External Access:	Read/Write		
*TIMER - Initialize/Init - *1(TON)*			
TIMER.DN	1	BOOL	
TIMER.DN - Initialize/Init - 3(XIC)			
Z_axis		AXIS_CIP_DRIVE	DUNEW2PLC1
External Access:	Read Only		
Z_axis - EOT_Trip_11/EOT_TRIP - 19(MAM), 5(MSO)			
Z_axis - Error_State_10/Error - 3(MAS), 7(MSF), 9(MAFR)			
Z_axis - Initialize/Init - 7(MAFR)			
Z_axis - MoveZ_State_4_5/MoveZ - 12(MAM), 12(MAM), 15(MAS), 16(MAS), 9(MSO)			
Z_axis - Ready_State_1/Ready - 2(MSO)			
Z_axis - UnServo_9/UnServo_All - 10(MAFR), 4(MSF)			
Z_axis.PhysicalAxisFault	0	BOOL	
Z_axis.PhysicalAxisFault - MoveZ_State_4_5/MoveZ - 10(XIC)			
Z_axis.ActualPosition	-1.2237	REAL	
Z_axis.ActualPosition - MainProgram/MainRoutine - 6(CMP)			
Z_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 1(GEQ)			
Z_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 13(CMP), 13(CMP)			
Z_axis.ActualVelocity	0.049999997	REAL	
Z_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT)			
Z_AXIS_STAT		MOTION_INSTRUCTION[32]	Initialize
External Access:	Read Only		
Z_AXIS_STAT[0]		MOTION_INSTRUCTION	
*Z_AXIS_STAT[0] - Initialize/Init - *7(MAFR)*			
Z_AXIS_STAT[0].DN	1	BOOL	
Z_AXIS_STAT[0].DN - Initialize/Init - 8(XIC)			

General

Type:	Ladder Diagram	Class:	Standard
In Program:	Initialize	Number of Rungs:	10

General

Using code from Sarah Johnson

Class:	Standard

Configuration

Main:	PID_Tension	Inhibit program:	No
Fault:	None		

Monitor

Scan Times(Execution Time)

Max:	107 us	Last:	7 us

0

Delay allows time for the tension to build up just after the tension motor is turned on

TENSION_ON_SWITCH

TON
Timer tension_on_switch_delay_on_start
Preset 1000
Accum 0
EN
DN

OSR
Storage Bit oneshotsb[0]
Output Bit tension_on_oneshot
SB
OB

OSF
Storage Bit oneshotsb[1]
Output Bit tension_off_oneshot
SB
OB

1

Avoids spurious "wire breaks" due to electrical blips, e.g. at spring contacts

wire_break_proxy

TON
Timer wire_break_debounce
Preset 20
Accum 1
EN
DN

2

TENSION_ON_SWITCH
Safety_Tripped_S
wire_break_proxy
tension_on_switch_delay_on_start.TT
wire_break_debounce.DN

Enable_tension_motor
<Local:3:O.Pt03.Data>

3

wait for a bit to check whether the wire is broken

tension_on_switch_delay_on_start.TT
TENSION_CONTROL_OK

wire_break_proxy
wire_break_switch_delay_on_start.DN
wire_break_debounce.DN

TENSION_CONTROL_OK

TON
Timer wire_break_switch_delay_on_start
Preset 1000
Accum 0
EN
DN

4

PID_LOOP_TIMER.DN

TON
Timer PID_LOOP_TIMER
Preset 3
Accum 1
EN
DN

5

wire tension from
load cell (N)

CPT
Dest
Expression 2.26*tension_tag-0.503*tension_tag*tension_tag+0.0694*tension_tag*tension_tag*tension_tag-0.00314*tension_tag*tension_tag*

tension
2.08072481e-003
ension_tag*tension_tag

6

CMP
Expression tension_tag<=1

wire tension from
load cell (N)

CPT
Dest tension
2.08072481e-003
Expression tension_tag
<Local:4:I.Ch01.Data>

The electrocraft output is -0.5 - 1.5A, the PLC CV is 0 - 10V. These are all parameters

7

PID_LOOP_TIMER.DN
pid_loop_timer_bypass

TENSION_CONTROL_OK

MOV
Source tension_setpoint
5.5
Dest winding_head_pid.SP
0.0

TENSION_CONTROL_OK

wire tension from load cell (N)

MOV
Source 10
Dest tension
2.08072481e-003

MOV
Source 0
Dest winding_head_pid.SP
0.0

PID
PID winding_head_pid ...
Process Variable tension
Tieback 0
Control Variable tension_motor_cv
PID Master Loop 0
Inhold Bit 0
Inhold Value 0
Setpoint 0.0
Process Variable 10.0
Output % 0.0

TENSION_CONTROL_OK

TENSION_ON_SWITCH

tension_on_switch_delay_on_start.TT

LES
Source A tension_motor_cv
2.5
Source B neutral_cv
2.5

MOV
Source neutral_cv
2.5
Dest tension_motor_cv
2.5

constant_cv_out

MOV
Source SetPoint_Override
6.0
Dest tension_motor_cv
2.5

Tension_Servo_CMD

MOV
Source tension_motor_cv
2.5
Dest cv_to_electrocraft
<Local:5:O.Ch00.Data>
2.5

8

CPT
Dest tension_motor_difference
-2.4978735
Expression tension-tension_motor_cv

Calculate the electrocraft command current output from the control voltage

9

CPT
Dest current_command
0.0
Expression cv_to_electrocraft*(current_command_high-current_command_low)/pid_cv_high_limit+current_command_low

The control voltage which corresponds to 0 current output at the electrocraft

10

CPT
Dest neutral_cv
2.5
Expression -current_command_low/((current_command_high-current_command_low)/(pid_cv_high_limit-pid_cv_low_limit))

Check if the tension has been within a small tolerance of the setpoint for long enough (checked before moving to ensure the tension system responds to motion-initiated disturbances starting from a steady state)

11

MOV
Source tension_stable_time
1
Dest tension_stable_timer.PRE
1

CMP
Expression ABS(tension-tension_setpoint)<tension_stable_tolerance

TON
Timer tension_stable_timer
Preset 1
Accum 0
EN
DN

The maximum tension is the load-cell readout at max compression. We ignore transient spikes less than 100ms

12

wire tension from load cell (N)

GRT
Source A tension
2.12644879e-003
Source B max_tolerable_tension
10.0

TENSION_ON_SWITCH

Beacon - Sounder
Local:3:O.Pt15.Data

TON
Timer overtension_timer
Preset 10
Accum 0
EN
DN

13

overtension_timer.DN

MORE_STATS[2]

CPT
Dest ERROR_CODE
0
Expression 8002

CPT
Dest NEXTSTATE
1
Expression 10

14

tension_on_switch_delay_on_start.DN
wire_break_debounce.DN

NEQ
Source A ERROR_CODE
0
Source B 8002

CPT
Dest NEXTSTATE
1
Expression 10

CPT
Dest ERROR_CODE
0
Expression 8001

15

TENSION_ON_SWITCH

OSF
Storage Bit oneshotsb[0]
Output Bit oneshotob[0]
SB
OB

16

oneshotob[0]

EQU
Source A ERROR_CODE
0
Source B 8002

EQU
Source A ERROR_CODE
0
Source B 8001

CPT
Dest ERROR_CODE
0
Expression 0

(End)

Name	Value	Data Type	Scope
constant_cv_out	0	BOOL	PID_Tension_Servo
Constant	No		
External Access:	Read/Write		
constant_cv_out - PID_Tension_Servo/PID_Tension - 7(XIC)			
current_command	0.0	REAL	DUNEW2PLC1
External Access:	Read/Write		
*current_command - PID_Tension_Servo/PID_Tension - *9(CPT)*			
current_command_high	1.5	REAL	DUNEW2PLC1
Constant	Yes		
External Access:	Read/Write		
current_command_high - PID_Tension_Servo/PID_Tension - 10(CPT), 9(CPT)			
current_command_low	-0.5	REAL	DUNEW2PLC1
Constant	Yes		
External Access:	Read/Write		
current_command_low - PID_Tension_Servo/PID_Tension - 10(CPT), 9(CPT)			
cv_to_electrocraft	2.5	REAL	PID_Tension_Servo
Tension_Servo_CMD			
AliasFor:	Local:5:O.Ch00.Data(C)		
Base Tag:	Local:5:O.Ch00.Data(C)		
Constant	No		
External Access:	Read/Write		
*cv_to_electrocraft - PID_Tension_Servo/PID_Tension - *7(MOV), 9(CPT)*			
Enable_tension_motor	0	BOOL	PID_Tension_Servo
AliasFor:	Local:3:O.Pt03.Data(C)		
Base Tag:	Local:3:O.Pt03.Data(C)		
Constant	No		
External Access:	Read/Write		
*Enable_tension_motor - PID_Tension_Servo/PID_Tension - *2(OTE)*			
ERROR_CODE	0	DINT	DUNEW2PLC1
External Access:	Read Only		
*ERROR_CODE - EOT_Trip_11/EOT_TRIP - *22(CPT)*			
*ERROR_CODE - Error_State_10/Error - *12(CPT)*			
*ERROR_CODE - Initialize/Init - *4(CPT)*			
*ERROR_CODE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *25(CPT), *33(CPT), *36(CPT), *51(CPT)*			
ERROR_CODE - MainProgram/MainRoutine - 36(NEQ)			
*ERROR_CODE - MoveXY_State_2_3/MoveXY - *1(CPT), *1(CPT), *15(CPT), *2(CPT), *2(CPT)*			
*ERROR_CODE - MoveZ_State_4_5/MoveZ - *10(CPT), *13(CPT), *20(CPT), *4(CPT), *4(CPT), *7(CPT)*			
*ERROR_CODE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT), *16(CPT), 14(NEQ), 16(EQU), 16(EQU)*			
*ERROR_CODE - Ready_State_1/Ready - *3(MOV)*			
Local:3:O		AB:5000_DO16:O:0	DUNEW2PLC1
External Access:	Read/Write		
Local:3:O.Pt01.Data	0	BOOL	
Latching signal output			
*Local:3:O.Pt01.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - *44(OTE)*			
Local:3:O.Pt02.Data	0	BOOL	
*Local:3:O.Pt02.Data - Latch_UnLatch_State_6_7_8/Latch_Driver - *22(OTE), *35(OTE), 36(XIC)*			
Local:3:O.Pt03.Data	0	BOOL	
*Enable_tension_motor - PID_Tension_Servo/PID_Tension - *2(OTE)*			
Local:3:O.Pt11.Data	0	BOOL	
Reset PB - Light			
*Local:3:O.Pt11.Data - MainProgram/MainRoutine - *35(OTE)*			
Local:3:O.Pt12.Data	0	BOOL	
Beacon - Green			
*Local:3:O.Pt12.Data - MainProgram/MainRoutine - *35(OTE)*			
Local:3:O.Pt13.Data	0	BOOL	
Beacon - Amber			

Local:3:O (Continued)
*Local:3:O.Pt13.Data - MainProgram/MainRoutine - *36(OTE)*

Local:3:O.Pt14.Data	1	BOOL	
Beacon - Red			
*Local:3:O.Pt14.Data - MainProgram/MainRoutine - *38(OTE)*			
Local:3:O.Pt15.Data	0	BOOL	
Beacon - Sounder			
*Local:3:O.Pt15.Data - MainProgram/MainRoutine - *37(OTE)*			
*Local:3:O.Pt15.Data - PID_Tension_Servo/PID_Tension - *12(OTE)*			
max_tolerable_tension	10.0	REAL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
max_tolerable_tension - PID_Tension_Servo/PID_Tension - 12(GRT)			
MORE_STATS		BOOL[32]	DUNEW2PLC1
External Access:	Read Only		
MORE_STATS[2]	0	BOOL	
*MORE_STATS[2] - PID_Tension_Servo/PID_Tension - *13(OTE)*			
neutral_cv	2.5	REAL	DUNEW2PLC1
External Access:	Read/Write		
*neutral_cv - PID_Tension_Servo/PID_Tension - *10(CPT), 7(LES), 7(MOV)*			
NEXTSTATE	1	DINT	DUNEW2PLC1
External Access:	Read Only		
*NEXTSTATE - EOT_Trip_11/EOT_TRIP - *22(CPT)*			
*NEXTSTATE - Error_State_10/Error - *12(CPT)*			
*NEXTSTATE - Initialize/Init - *4(CPT), *9(CPT)*			
*NEXTSTATE - Latch_UnLatch_State_6_7_8/Latch_Driver - *10(CPT), *13(CPT), *33(CPT), *43(CPT), *51(CPT)*			
NEXTSTATE - MainProgram/MainRoutine - 42(CMP), 43(CMP), 44(CMP), 45(CMP), 46(CMP), 47(CMP), 48(CMP), 49(CMP), 50(CMP), 51(CMP), 52(CMP), 53(CMP)			
*NEXTSTATE - MoveXY_State_2_3/MoveXY - *0(CPT), *1(CPT), *1(CPT), *18(CPT), *2(CPT), *21(CPT)*			
*NEXTSTATE - MoveZ_State_4_5/MoveZ - *10(CPT), *19(CPT), *20(CPT), *4(CPT), *5(MOV), *7(CPT)*			
*NEXTSTATE - PID_Tension_Servo/PID_Tension - *13(CPT), *14(CPT)*			
*NEXTSTATE - Ready_State_1/Ready - *11(CPT), *13(CPT), *15(CPT), *17(CPT), *19(CPT), *5(CPT), *7(CPT), *9(CPT)*			
*NEXTSTATE - UnServo_9/UnServo_All - *11(CPT)*			
oneshotob		BOOL[32]	PID_Tension_Servo
External Access:	Read/Write		
oneshotob[0]	0	BOOL	
*oneshotob[0] - PID_Tension_Servo/PID_Tension - *15(OSF), 16(XIC)*			
oneshotsb		BOOL[32]	PID_Tension_Servo
External Access:	Read/Write		
oneshotsb[0]	0	BOOL	
*oneshotsb[0] - PID_Tension_Servo/PID_Tension - *0(OSR), *15(OSF)*			
oneshotsb[1]	0	BOOL	
*oneshotsb[1] - PID_Tension_Servo/PID_Tension - *0(OSF)*			
overtension_timer		TIMER	DUNEW2PLC1
External Access:	Read/Write		
*overtension_timer - PID_Tension_Servo/PID_Tension - *12(TON)*			
overtension_timer.DN	0	BOOL	
overtension_timer.DN - PID_Tension_Servo/PID_Tension - 13(XIC)			
pid_cv_high_limit	10.0	REAL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
pid_cv_high_limit - PID_Tension_Servo/PID_Tension - 10(CPT), 9(CPT)			
pid_cv_low_limit	0.0	REAL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		

pid_cv_low_limit (Continued)
pid_cv_low_limit - PID_Tension_Servo/PID_Tension - 10(CPT)

PID_LOOP_TIMER TIMER DUNEW2PLC1
External Access: Read/Write
*PID_LOOP_TIMER - PID_Tension_Servo/PID_Tension - *4(TON)*
PID_LOOP_TIMER.DN 0 BOOL
PID_LOOP_TIMER.DN - PID_Tension_Servo/PID_Tension - 4(XIO), 7(XIC)

pid_loop_timer_bypass 0 BOOL DUNEW2PLC1
Constant No
External Access: Read/Write
pid_loop_timer_bypass - PID_Tension_Servo/PID_Tension - 7(XIC)

Safety_Tripped_S 0 BOOL DUNEW2PLC1
External Access: Read/Write
Safety_Tripped_S - MainProgram/MainRoutine - 34(XIO), 35(XIC)
Safety_Tripped_S - PID_Tension_Servo/PID_Tension - 2(XIO)
*Safety_Tripped_S - Safety /Safety - *3(OTL), *5(OTU), 4(XIO)*

SetPoint_Override 6.0 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
SetPoint_Override - PID_Tension_Servo/PID_Tension - 7(MOV)

tension 10.0 REAL DUNEW2PLC1
wire tension from load cell (N)
External Access: Read/Write
tension - MainProgram/MainRoutine - 20(GRT)
tension - MoveXY_State_2_3/xy_speed_regulator - 5(CPT)
*tension - PID_Tension_Servo/PID_Tension - *5(CPT), *6(CPT), *7(MOV), 11(CMP), 12(GRT), 7(PID), 8(CPT)*

TENSION_CONTROL_OK 0 BOOL DUNEW2PLC1
External Access: Read/Write
TENSION_CONTROL_OK - MoveXY_State_2_3/MoveXY - 1(XIO), 12(XIC), 12(XIC), 12(XIO)
TENSION_CONTROL_OK - MoveZ_State_4_5/MoveZ - 6(XIO)
*TENSION_CONTROL_OK - PID_Tension_Servo/PID_Tension - *3(OTE), 3(XIC), 7(XIC), 7(XIO), 7(XIO)*

tension_motor_cv 2.5 REAL DUNEW2PLC1
External Access: Read/Write
*tension_motor_cv - PID_Tension_Servo/PID_Tension - *7(MOV), *7(MOV), *7(PID), 7(LES), 7(MOV), 8(CPT)*

tension_motor_difference -2.4978669 REAL PID_Tension_Servo
External Access: Read/Write
*tension_motor_difference - PID_Tension_Servo/PID_Tension - *8(CPT)*

tension_off_oneshot 0 BOOL DUNEW2PLC1
External Access: Read/Write
*tension_off_oneshot - PID_Tension_Servo/PID_Tension - *0(OSF)*

tension_on_oneshot 0 BOOL DUNEW2PLC1
External Access: Read/Write
*tension_on_oneshot - PID_Tension_Servo/PID_Tension - *0(OSR)*

TENSION_ON_SWITCH 0 BOOL DUNEW2PLC1
External Access: Read/Write
*TENSION_ON_SWITCH - MainProgram/MainRoutine - *12(OTE)*
TENSION_ON_SWITCH - PID_Tension_Servo/PID_Tension - 0(XIC), 12(XIC), 15(XIC), 2(XIC), 7(XIO)

tension_on_switch_delay_on_start TIMER DUNEW2PLC1
External Access: Read/Write
*tension_on_switch_delay_on_start - PID_Tension_Servo/PID_Tension - *0(TON)*
tension_on_switch_delay_on_start.TT
0 BOOL

tension_on_switch_delay_on_start (Continued)
tension_on_switch_delay_on_start.TT - PID_Tension_Servo/PID_Tension - 2(XIC), 3(XIC), 7(XIC)
tension_on_switch_delay_on_start.DN
0 BOOL
tension_on_switch_delay_on_start.DN - PID_Tension_Servo/PID_Tension - 14(XIC)

tension_setpoint 5.5 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
tension_setpoint - PID_Tension_Servo/PID_Tension - 11(CMP), 7(MOV)

tension_stable_time 1 DINT PID_Tension_Servo
Constant No
External Access: Read/Write
tension_stable_time - PID_Tension_Servo/PID_Tension - 11(MOV)

tension_stable_timer TIMER DUNEW2PLC1
External Access: Read/Write
*tension_stable_timer - PID_Tension_Servo/PID_Tension - *11(TON)*
tension_stable_timer.PRE 1 DINT
*tension_stable_timer.PRE - PID_Tension_Servo/PID_Tension - *11(MOV)*
tension_stable_timer.DN 0 BOOL
tension_stable_timer.DN - MoveXY_State_2_3/MoveXY - 1(XIC)
tension_stable_timer.DN - MoveZ_State_4_5/MoveZ - 6(XIC)

tension_stable_tolerance 0.1 REAL PID_Tension_Servo
Constant No
External Access: Read/Write
tension_stable_tolerance - PID_Tension_Servo/PID_Tension - 11(CMP)

tension_tag 2.07500951e-003 REAL DUNEW2PLC1
wire tension (N)
AliasFor: Local:4:I.Ch01.Data
Base Tag: Local:4:I.Ch01.Data
Constant No
External Access: Read/Write
tension_tag - PID_Tension_Servo/PID_Tension - 5(CPT), 6(CMP), 6(CPT)

winding_head_pid PID DUNEW2PLC1
External Access: Read/Write
*winding_head_pid - PID_Tension_Servo/PID_Tension - *7(PID)*
winding_head_pid.SP 0.0 REAL
*winding_head_pid.SP - PID_Tension_Servo/PID_Tension - *7(MOV), *7(MOV)*

wire_break_debounce TIMER PID_Tension_Servo
External Access: Read/Write
*wire_break_debounce - PID_Tension_Servo/PID_Tension - *1(TON)*
wire_break_debounce.DN 0 BOOL
wire_break_debounce.DN - PID_Tension_Servo/PID_Tension - 14(XIC), 2(XIO), 3(XIO)

wire_break_proxy 0 BOOL DUNEW2PLC1
External Access: Read/Write
*wire_break_proxy - MainProgram/MainRoutine - *20(OTE)*
wire_break_proxy - PID_Tension_Servo/PID_Tension - 1(XIO), 2(XIC), 3(XIC)

wire_break_switch_delay_on_start TIMER PID_Tension_Servo
External Access: Read/Write
*wire_break_switch_delay_on_start - PID_Tension_Servo/PID_Tension - *3(TON)*
wire_break_switch_delay_on_start.DN
0 BOOL
wire_break_switch_delay_on_start.DN - PID_Tension_Servo/PID_Tension - 3(XIO)

General

Type:	Ladder Diagram	Class:	Standard
In Program:	PID_Tension_Servo	Number of Rungs:	17

General

Class:	Standard

Configuration

Main:	queue_line_arc	Inhibit program:	No
Fault:	None		

Monitor

Scan Times(Execution Time)

Max:	112 us	Last:	11 us

0

EQU
Source A QueueCtl.POS[0]
Source B 0

QueueEmpty

1

GEQ
Source A QueueCtl.POS[0]
Source B 32

QueueFull

2

MOV
Source QueueCtl.POS[0]
Dest QueueCount[0]

3

NEQ
Source A IncomingSegReqID[44]
Source B LastIncomingSegReqID[44]

EnqueueReq
L

4

EnqueueReq

motion segments for move queue, lines or arcs segment valid flag
IncomingSeg.Valid

motion segments for move queue, lines or arcs 1=line, 2=arc
EQU
Source A IncomingSeg.SegType[1]
Source B 1

motion segments for move queue, lines or arcs >0
GRT
Source A IncomingSeg.Speed[437.4951]
Source B 0.0

motion segments for move queue, lines or arcs >0
GRT
Source A IncomingSeg.Accel[2000.0]
Source B 0.0

motion segments for move queue, lines or arcs >0
GRT
Source A IncomingSeg.Decel[2000.0]
Source B 0.0

motion segments for move queue, lines or arcs 0 to 6
GEQ
Source A IncomingSeg.TermType[4]
Source B 0

motion segments for move queue, lines or arcs 0 to 6
LEQ
Source A IncomingSeg.TermType[4]
Source B 6

SegValidLine

5

EnqueueReq — IncomingSeg.Valid (motion segments for move queue, lines or arcs segment valid flag)

EQU (motion segments for move queue, lines or arcs 1=line, 2=arc)
Source A IncomingSeg.SegType 1
Source B 2

GRT (motion segments for move queue, lines or arcs >0)
Source A IncomingSeg.Speed 437.4951
Source B 0.0

GRT (motion segments for move queue, lines or arcs >0)
Source A IncomingSeg.Accel 2000.0
Source B 0.0

GRT (motion segments for move queue, lines or arcs >0)
Source A IncomingSeg.Decel 2000.0
Source B 0.0

GEQ (motion segments for move queue, lines or arcs 0 to 6)
Source A IncomingSeg.TermType 4
Source B 0

LEQ (motion segments for move queue, lines or arcs 0 to 6)
Source A IncomingSeg.TermType 4
Source B 6

GEQ (motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum)
Source A IncomingSeg.CircleType 1
Source B 0

LEQ (motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum)
Source A IncomingSeg.CircleType 1
Source B 3

GEQ (motion segments for move queue, lines or arcs 0..3 per mmcm enum)
Source A IncomingSeg.Direction 1
Source B 0

LEQ (motion segments for move queue, lines or arcs 0..3 per mmcm enum)
Source A IncomingSeg.Direction 1
Source B 3

SegValidArc

6

SegValidLine
SegValidArc

SegValid

7

EnqueueReq — SegValid — QueueFull

FFL
Source IncomingSeg
FIFO SegQueue[0]
Control QueueCtl
Length 32
Position 0
EN
DN
EM

8
EnqueueReq SegValid QueueFull
MOV
Source IncomingSeg.Seq
143
Dest IncomingSegAck
142

9
EnqueueReq SegValid QueueFull
MOV
Source IncomingSegReqID
44
Dest LastIncomingSegReqID
44

10
EnqueueReq
EnqueueReq U

11
MoveA.ER
MOV
Source 3
Dest FaultCode
0
MoveA.ER U

12
MoveB.ER
MOV
Source 4
Dest FaultCode
0
MoveB.ER U

13
QueueFault
MoveA.ER
MoveB.ER
MotionFault

14
CheckCurSeg
motion segments for move queue, lines or arcs segment valid flag
CurSeg.Valid
motion segments for move queue, lines or arcs Sequence/ack id
GRT
Source A CurSeg.Seq
0
Source B 0
motion segments for move queue, lines or arcs 1=line, 2=arc
EQU
Source A CurSeg.SegType
1
Source B 1
motion segments for move queue, lines or arcs 1=line, 2=arc
EQU
Source A CurSeg.SegType
1
Source B 2
PrepCurMove
L

15
CheckCurSeg
motion segments for move queue, lines or arcs segment valid flag
CurSeg.Valid
MOV
Source 1
Dest FaultCode
0

16
CheckCurSeg
motion segments for move queue, lines or arcs segment valid flag
CurSeg.Valid
QueueFault
L

17
CheckCurSeg
motion segments for move queue, lines or arcs segment valid flag
CurSeg.Valid
motion segments for move queue, lines or arcs Sequence/ack id
LEQ
Source A CurSeg.Seq
0
Source B 0
MOV
Source 6
Dest FaultCode
0

18
CheckCurSeg
motion segments for move queue, lines or arcs segment valid flag
CurSeg.Valid
motion segments for move queue, lines or arcs Sequence/ack id
LEQ
Source A CurSeg.Seq
0
Source B 0
QueueFault
L

19
CheckCurSeg
motion segments for move queue, lines or arcs segment valid flag
CurSeg.Valid
motion segments for move queue, lines or arcs Sequence/ack id
GRT
Source A CurSeg.Seq
0
Source B 0
motion segments for move queue, lines or arcs 1=line, 2=arc
LES
Source A CurSeg.SegType
1
Source B 1
motion segments for move queue, lines or arcs 1=line, 2=arc
GRT
Source A CurSeg.SegType
1
Source B 2
MOV
Source 7
Dest FaultCode
0

20

CheckCurSeg — motion segments for move queue, lines or arcs segment valid flag: CurSeg.Valid

GRT — motion segments for move queue, lines or arcs Sequence/ack id
Source A: CurSeg.Seq
0
Source B: 0

LES — motion segments for move queue, lines or arcs 1=line, 2=arc
Source A: CurSeg.SegType
1
Source B: 1

GRT — motion segments for move queue, lines or arcs 1=line, 2=arc
Source A: CurSeg.SegType
1
Source B: 2

QueueFault (L)

21

CheckCurSeg — CheckCurSeg (U)

22

CheckNextSeg — motion segments for move queue, lines or arcs segment valid flag: NextSeg.Valid

GRT — motion segments for move queue, lines or arcs Sequence/ack id
Source A: NextSeg.Seq
0
Source B: 0

EQU — motion segments for move queue, lines or arcs 1=line, 2=arc
Source A: NextSeg.SegType
1
Source B: 1

EQU — motion segments for move queue, lines or arcs 1=line, 2=arc
Source A: NextSeg.SegType
1
Source B: 2

PrepNextMove (L)

23

CheckNextSeg — motion segments for move queue, lines or arcs segment valid flag: NextSeg.Valid (/)

MOV
Source: 2
Dest: FaultCode
0

24

CheckNextSeg — motion segments for move queue, lines or arcs segment valid flag: NextSeg.Valid (/) — QueueFault (L)

25

CheckNextSeg

motion segments for move queue, lines or arcs segment valid flag
NextSeg.Valid

motion segments for move queue, lines or arcs Sequence/ack id
LEQ
Source A NextSeg.Seq
0
Source B 0

MOV
Source 5
Dest FaultCode
0

26

CheckNextSeg

motion segments for move queue, lines or arcs segment valid flag
NextSeg.Valid

motion segments for move queue, lines or arcs Sequence/ack id
LEQ
Source A NextSeg.Seq
0
Source B 0

QueueFault
L

27

CheckNextSeg

motion segments for move queue, lines or arcs segment valid flag
NextSeg.Valid

motion segments for move queue, lines or arcs Sequence/ack id
GRT
Source A NextSeg.Seq
0
Source B 0

motion segments for move queue, lines or arcs 1=line, 2=arc
LES
Source A NextSeg.SegType
1
Source B 1

motion segments for move queue, lines or arcs 1=line, 2=arc
GRT
Source A NextSeg.SegType
1
Source B 2

MOV
Source 8
Dest FaultCode
0

28
CheckNextSeg
motion segments for move queue, lines or arcs segment valid flag
NextSeg.Valid
motion segments for move queue, lines or arcs Sequence/ack id
GRT
Source A NextSeg.Seq
0
Source B 0
motion segments for move queue, lines or arcs 1=line, 2=arc
LES
Source A NextSeg.SegType
1
Source B 1
motion segments for move queue, lines or arcs 1=line, 2=arc
GRT
Source A NextSeg.SegType
1
Source B 2
QueueFault
L

29
CheckNextSeg
CheckNextSeg
U

30
StartQueuedPath
CurIssued
QueueFault
GEQ
Source A QueueCtl.POS
0
Source B 2
StartCurONS
ONS
LoadCurReq
L

31
LoadCurReq
StartQueuedPath
U

32
LoadCurReq
FFU
FIFO SegQueue[0]
Dest CurSeg
Control QueueCtl
Length 32
Position 0
EU
DN
EM

33
LoadCurReq
CheckCurSeg
L

34
LoadCurReq
LoadCurReq
U

35
PrepCurMove
1=A current/B next
0=b current/A next
UseAasCurrent
COP
Source CurSeg.XY[0]
Dest CmdA_XY[0]
Length 2

36
PrepCurMove
1=A current/B next
0=b current/A next
UseAasCurrent
MOV
Source CurSeg.Speed
437.4951
Dest CmdA_Speed
437.4951

37
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent
MOV
Source CurSeg.Accel 2000.0
Dest CmdA_Accel 2000.0

38
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent
MOV
Source CurSeg.Decel 2000.0
Dest CmdA_Decel 2000.0

39
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent
MOV
Source CurSeg.JerkAccel 20000.0
Dest CmdA_JerkAccel 20000.0

40
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent
MOV
Source CurSeg.JerkDecel 20000.0
Dest CmdA_JerkDecel 20000.0

41
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent
MOV
Source CurSeg.TermType 4
Dest CmdA_TermType 4

42
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent
MOV
Source CurSeg.SegType 1
Dest CmdA_SegType 1

43
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent
MOV
Source CurSeg.CircleType 1
Dest CmdA_CircleType 1

44
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent
COP
Source CurSeg.ViaCenter[0]
Dest CmdA_ViaCenter[0]
Length 2

45
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent

MOV
Source CurSeg.Direction
1
Dest CmdA_Direction
1

46
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent

MOV
Source CurSeg.Seq
0
Dest ActiveSeq
0

47
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent

COP
Source CurSeg.XY[0]
Dest CmdB_XY[0]
Length 2

48
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent

MOV
Source CurSeg.Speed
437.4951
Dest CmdB_Speed
437.4951

49
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent

MOV
Source CurSeg.Accel
2000.0
Dest CmdB_Accel
2000.0

50
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent

MOV
Source CurSeg.Decel
2000.0
Dest CmdB_Decel
2000.0

51
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent

MOV
Source CurSeg.JerkAccel
20000.0
Dest CmdB_JerkAccel
20000.0

52
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent

MOV
Source CurSeg.JerkDecel
20000.0
Dest CmdB_JerkDecel
20000.0

53
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent /
MOV
Source CurSeg.TermType 4
Dest CmdB_TermType 4

54
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent /
MOV
Source CurSeg.SegType 1
Dest CmdB_SegType 1

55
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent /
MOV
Source CurSeg.CircleType 1
Dest CmdB_CircleType 1

56
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent /
COP
Source CurSeg.ViaCenter[0]
Dest CmdB_ViaCenter[0]
Length 2

57
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent /
MOV
Source CurSeg.Direction 1
Dest CmdB_Direction 1

58
1=A current/B next
0=b current/A next
PrepCurMove UseAasCurrent /
MOV
Source CurSeg.Seq 0
Dest ActiveSeq 0

59
PrepCurMove
IssueCurPulse
L

60
PrepCurMove
PrepCurMove
U

61

IssueCurPulse

1=A current/B next
0=b current/A next
UseAasCurrent

EQU
Source A CmdA_SegType
1
Source B 1

MCLM
Coordinate System X_Y ...
Motion Control MoveA
Move Type 0
Position CmdA_XY ...
X_axis ??
Y_axis ??
Speed CmdA_Speed
437.4951
Speed Units Units per sec
Accel Rate CmdA_Accel
2000.0
Accel Units Units per sec2
Decel Rate CmdA_Decel
2000.0
Decel Units Units per sec2
Profile S-Curve
Accel Jerk CmdA_JerkAccel
20000.0
Decel Jerk CmdA_JerkDecel
20000.0
Jerk Units Units per sec3
Termination Type CmdA_TermType
4
Merge Disabled
Merge Speed Programmed
Command Tolerance CmdTolerance
10
Lock Position 0
Lock Direction None
Event Distance 0
Calculated Data 0

EN
DN
ER
IP
AC
PC

62

1=A current/B next
0=b current/A next

IssueCurPulse — UseAasCurrent

EQU
Source A CmdA_SegType
1
Source B 2

MCCM
Coordinate System X_Y ...
Motion Control MoveA
Move Type 0
Position CmdA_XY[0] ...
X_axis 5990.0
Y_axis 1250.0
Circle Type CmdA_CircleType
1
Via/Center/Radius CmdA_ViaCenter[0]
Direction CmdA_Direction
1
Speed CmdA_Speed
437.4951
Speed Units Units per sec
Accel Rate CmdA_Accel
2000.0
Accel Units Units per sec2
Decel Rate CmdA_Decel
2000.0
Decel Units Units per sec2
Profile S-Curve
Accel Jerk CmdA_JerkAccel
20000.0
Decel Jerk CmdA_JerkDecel
20000.0
Jerk Units Units per sec3
Termination Type CmdA_TermType
4
Merge Disabled
Merge Speed Programmed
Command Tolerance CmdTolerance
10
Lock Position 0
Lock Direction None
Event Distance 0
Calculated Data 0

EN
DN
ER
IP
AC
PC

63

IssueCurPulse

1=A current/B next
0=b current/A next
UseAasCurrent

EQU
Source A CmdB_SegType
1
Source B 1

MCLM
Coordinate System X_Y ...
Motion Control MoveB
Move Type 0
Position CmdB_XY ...
X_axis ??
Y_axis ??
Speed CmdB_Speed
437.4951
Speed Units Units per sec
Accel Rate CmdB_Accel
2000.0
Accel Units Units per sec2
Decel Rate CmdB_Decel
2000.0
Decel Units Units per sec2
Profile S-Curve
Accel Jerk CmdB_JerkAccel
20000.0
Decel Jerk CmdB_JerkDecel
20000.0
Jerk Units Units per sec3
Termination Type CmdB_TermType
4
Merge Disabled
Merge Speed Programmed
Command Tolerance CmdTolerance
10
Lock Position 0
Lock Direction None
Event Distance 0
Calculated Data 0

EN
DN
ER
IP
AC
PC

64

IssueCurPulse — 1=A current/B next 0=b current/A next: UseAasCurrent (/)

EQU
Source A: CmdB_SegType 1
Source B: 2

MCCM
Coordinate System: X_Y
Motion Control: MoveB
Move Type: 0
Position: CmdB_XY[0]
X_axis: 5972.74
Y_axis: 1347.2893
Circle Type: CmdB_CircleType 1
Via/Center/Radius: CmdB_ViaCenter[0]
Direction: CmdB_Direction 1
Speed: CmdB_Speed 437.4951
Speed Units: Units per sec
Accel Rate: CmdB_Accel 2000.0
Accel Units: Units per sec2
Decel Rate: CmdB_Decel 2000.0
Decel Units: Units per sec2
Profile: S-Curve
Accel Jerk: CmdB_JerkAccel 20000.0
Decel Jerk: CmdB_JerkDecel 20000.0
Jerk Units: Units per sec3
Termination Type: CmdB_TermType 4
Merge: Disabled
Merge Speed: Programmed
Command Tolerance: CmdTolerance 10
Lock Position: 0
Lock Direction: None
Event Distance: 0
Calculated Data: 0

EN, DN, ER, IP, AC, PC

65

IssueCurPulse — CurIssued (L)

66

IssueCurPulse — IssueCurPulse (U)

67

CurIssued — UseAasCurrent (1=A current/B next 0=b current/A next) — MoveA.IP — X_Y.MovePendingStatus (/) — NextIssued (/) — QueueEmpty (/) — QueueFault (/)

StartNextA_ONS [ONS] — LoadNextReq (L)

68
1=A current/B next
0=b current/A next
CurIssued UseAasCurrent MoveB.IP X_Y.MovePendingStatus NextIssued QueueEmpty QueueFault
StartNextB_ONS ONS
LoadNextReq L

69
LoadNextReq
FFU
FIFO SegQueue[0]
Dest NextSeg
Control QueueCtl
Length 32
Position 0
EU DN EM

70
LoadNextReq
CheckNextSeg L

71
LoadNextReq
LoadNextReq U

72
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent
COP
Source NextSeg.XY[0]
Dest CmdB_XY[0]
Length 2

73
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent
MOV
Source NextSeg.Speed
437.4951
Dest CmdB_Speed
437.4951

74
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent
MOV
Source NextSeg.Accel
2000.0
Dest CmdB_Accel
2000.0

75
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent
MOV
Source NextSeg.Decel
2000.0
Dest CmdB_Decel
2000.0

76
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent
MOV
Source NextSeg.JerkAccel
20000.0
Dest CmdB_JerkAccel
20000.0

77
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent

MOV
Source NextSeg.JerkDecel 20000.0
Dest CmdB_JerkDecel 20000.0

78
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent

MOV
Source NextSeg.TermType 4
Dest CmdB_TermType 4

79
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent

MOV
Source NextSeg.SegType 1
Dest CmdB_SegType 1

80
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent

MOV
Source NextSeg.CircleType 1
Dest CmdB_CircleType 1

81
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent

COP
Source NextSeg.ViaCenter[0]
Dest CmdB_ViaCenter[0]
Length 2

82
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent

MOV
Source NextSeg.Direction 1
Dest CmdB_Direction 1

83
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent

MOV
Source NextSeg.Seq 0
Dest PendingSeq 0

84
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent /

COP
Source NextSeg.XY[0]
Dest CmdA_XY[0]
Length 2

85
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent
MOV
Source NextSeg.Speed 437.4951
Dest CmdA_Speed 437.4951

86
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent
MOV
Source NextSeg.Accel 2000.0
Dest CmdA_Accel 2000.0

87
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent
MOV
Source NextSeg.Decel 2000.0
Dest CmdA_Decel 2000.0

88
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent
MOV
Source NextSeg.JerkAccel 20000.0
Dest CmdA_JerkAccel 20000.0

89
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent
MOV
Source NextSeg.JerkDecel 20000.0
Dest CmdA_JerkDecel 20000.0

90
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent
MOV
Source NextSeg.TermType 4
Dest CmdA_TermType 4

91
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent
MOV
Source NextSeg.SegType 1
Dest CmdA_SegType 1

92
1=A current/B next
0=b current/A next
PrepNextMove UseAasCurrent
MOV
Source NextSeg.CircleType 1
Dest CmdA_CircleType 1

93
PrepNextMove
1=A current/B next
0=b current/A next
UseAasCurrent
COP
Source NextSeg.ViaCenter[0]
Dest CmdA_ViaCenter[0]
Length 2

94
PrepNextMove
1=A current/B next
0=b current/A next
UseAasCurrent
MOV
Source NextSeg.Direction
1
Dest CmdA_Direction
1

95
PrepNextMove
1=A current/B next
0=b current/A next
UseAasCurrent
MOV
Source NextSeg.Seq
0
Dest PendingSeq
0

96
PrepNextMove
IssueNextPulse
L

97
PrepNextMove
PrepNextMove
U

98
IssueNextPulse
1=A current/B next
0=b current/A next
UseAasCurrent
EQU
Source A CmdB_SegType
1
Source B 1
MCLM
Coordinate System X_Y
Motion Control MoveB
Move Type 0
Position CmdB_XY
X_axis ??
Y_axis ??
Speed CmdB_Speed
437.4951
Speed Units Units per sec
Accel Rate CmdB_Accel
2000.0
Accel Units Units per sec2
Decel Rate CmdB_Decel
2000.0
Decel Units Units per sec2
Profile S-Curve
Accel Jerk CmdB_JerkAccel
20000.0
Decel Jerk CmdB_JerkDecel
20000.0
Jerk Units Units per sec3
Termination Type CmdB_TermType
4
Merge Disabled
Merge Speed Programmed
Command Tolerance CmdTolerance
10
Lock Position 0
Lock Direction None
Event Distance 0
Calculated Data 0
EN
DN
ER
IP
AC
PC

99

IssueNextPulse

1=A current/B next
0=b current/A next
UseAasCurrent

EQU
Source A CmdB_SegType
1
Source B 2

MCCM
Coordinate System X_Y ...
Motion Control MoveB
Move Type 0

Position CmdB_XY[0] ...
X_axis 5972.74
Y_axis 1347.2893
Circle Type CmdB_CircleType
1
Via/Center/Radius CmdB_ViaCenter[0]
Direction CmdB_Direction
1
Speed CmdB_Speed
437.4951
Speed Units Units per sec
Accel Rate CmdB_Accel
2000.0
Accel Units Units per sec2
Decel Rate CmdB_Decel
2000.0
Decel Units Units per sec2
Profile S-Curve
Accel Jerk CmdB_JerkAccel
20000.0
Decel Jerk CmdB_JerkDecel
20000.0
Jerk Units Units per sec3
Termination Type CmdB_TermType
4
Merge Disabled
Merge Speed Programmed
Command Tolerance CmdTolerance
10
Lock Position 0

Lock Direction None
Event Distance 0
Calculated Data 0

EN
DN
ER
IP
AC
PC

100

IssueNextPulse

1=A current/B next
0=b current/A next
UseAasCurrent

EQU
Source A CmdA_SegType
1
Source B 1

MCLM
Coordinate System X_Y ...
Motion Control MoveA
Move Type 0

Position CmdA_XY ...
X_axis ??
Y_axis ??
Speed CmdA_Speed
437.4951
Speed Units Units per sec
Accel Rate CmdA_Accel
2000.0
Accel Units Units per sec2
Decel Rate CmdA_Decel
2000.0
Decel Units Units per sec2
Profile S-Curve
Accel Jerk CmdA_JerkAccel
20000.0
Decel Jerk CmdA_JerkDecel
20000.0
Jerk Units Units per sec3
Termination Type CmdA_TermType
4
Merge Disabled
Merge Speed Programmed
Command Tolerance CmdTolerance
10
Lock Position 0

Lock Direction None
Event Distance 0
Calculated Data 0

EN
DN
ER
IP
AC
PC

101

IssueNextPulse — 1=A current/B next 0=b current/A next UseAasCurrent (/)

EQU
Source A CmdA_SegType
1
Source B 2

MCCM
Coordinate System X_Y ...
Motion Control MoveA
Move Type 0
Position CmdA_XY[0] ...
X_axis 5990.0
Y_axis 1250.0
Circle Type CmdA_CircleType
1
Via/Center/Radius CmdA_ViaCenter[0]
Direction CmdA_Direction
1
Speed CmdA_Speed
437.4951
Speed Units Units per sec
Accel Rate CmdA_Accel
2000.0
Accel Units Units per sec2
Decel Rate CmdA_Decel
2000.0
Decel Units Units per sec2
Profile S-Curve
Accel Jerk CmdA_JerkAccel
20000.0
Decel Jerk CmdA_JerkDecel
20000.0
Jerk Units Units per sec3
Termination Type CmdA_TermType
4
Merge Disabled
Merge Speed Programmed
Command Tolerance CmdTolerance
10
Lock Position 0
Lock Direction None
Event Distance 0
Calculated Data 0

EN DN ER IP AC PC

102

IssueNextPulse — NextIssued (L)

103

IssueNextPulse — IssueNextPulse (U)

104

CurIssued NextIssued 1=A current/B next 0=b current/A next UseAasCurrent X_Y.MovePendingStatus (/) MoveB.IP RotateONS_AtoB (ONS) RotateMoves (L)

105

CurIssued NextIssued 1=A current/B next 0=b current/A next UseAasCurrent (/) X_Y.MovePendingStatus (/) MoveA.IP RotateONS_BtoA (ONS) RotateMoves (L)

106
RotateMoves
motion segments for move queue, lines or arcs
COP
Source NextSeg
Dest CurSeg
Length 1

107
RotateMoves
1=A current/B next
0=b current/A next
UseAasCurrent
FlipToB
L

108
RotateMoves
1=A current/B next
0=b current/A next
UseAasCurrent
/
FlipToA
L

109
FlipToB
1=A current/B next
0=b current/A next
UseAasCurrent
U

110
FlipToA
1=A current/B next
0=b current/A next
UseAasCurrent
L

111
RotateMoves
NextIssued
U

112
RotateMoves
MOV
Source PendingSeq
0
Dest ActiveSeq
0

113
RotateMoves
MOV
Source 0
Dest PendingSeq
0

114
RotateMoves
RotateMoves
U

115
FlipToA
FlipToA
U

116
FlipToB
FlipToB
U

117
CurIssued
NextIssued
/
1=A current/B next
0=b current/A next
UseAasCurrent
X_Y.MovePendingStatus
/
MoveA.PC
DoneONS_A
ONS
CurIssued
U

118
CurIssued
NextIssued
/
1=A current/B next
0=b current/A next
UseAasCurrent
/
X_Y.MovePendingStatus
/
MoveB.PC
DoneONS_B
ONS
CurIssued
U

119
AbortQueue
QueueCtl
RES

120 AbortQueue — CurIssued (U)

121 AbortQueue — NextIssued (U)

122 AbortQueue — LoadCurReq (U)

123 AbortQueue — LoadNextReq (U)

124 AbortQueue — PrepCurMove (U)

125 AbortQueue — PrepNextMove (U)

126 AbortQueue — IssueCurPulse (U)

127 AbortQueue — IssueNextPulse (U)

128 AbortQueue — EnqueueReq (U)

129 AbortQueue — RotateMoves (U)

130 AbortQueue — FlipToA (U)

131 AbortQueue — FlipToB (U)

132 AbortQueue — UseAasCurrent (L)
1=A current/B next
0=b current/A next

133 AbortQueue — StartQueuedPath (U)

134 AbortQueue — MOV
Source 0
Dest FaultCode
0

135 AbortQueue — QueueFault (U)

136 AbortQueue — CheckCurSeg (U)

137 AbortQueue — CheckNextSeg (U)

138
AbortQueue
motion segments for move queue, lines or arcs segment valid flag
CurSeg.Valid
U

139
AbortQueue
motion segments for move queue, lines or arcs Sequence/ack id
MOV
Source 0
Dest CurSeg.Seq
0

140
AbortQueue
motion segments for move queue, lines or arcs segment valid flag
NextSeg.Valid
U

141
AbortQueue
motion segments for move queue, lines or arcs Sequence/ack id
MOV
Source 0
Dest NextSeg.Seq
0

142
AbortQueue
MOV
Source IncomingSegReqID
44
Dest LastIncomingSegReqID
44

143
AbortQueue
MOV
Source 0
Dest ActiveSeq
0

144
AbortQueue
MOV
Source 0
Dest PendingSeq
0

145
AbortQueue
motion segments for move queue, lines or arcs
FLL
Source 0
Dest SegQueue[0]
Length 32

(End)

Name	Value	Data Type	Scope
AbortQueue	0	BOOL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
AbortQueue - motionQueue/queue_line_arc - 119(XIC), 120(XIC), 121(XIC), 122(XIC), 123(XIC), 124(XIC), 125(XIC), 126(XIC), 127(XIC), 128(XIC), 129(XIC), 130(XIC), 131(XIC), 132(XIC), 133(XIC), 134(XIC), 135(XIC), 136(XIC), 137(XIC), 138(XIC), 139(XIC), 140(XIC), 141(XIC), 142(XIC), 143(XIC), 144(XIC), 145(XIC)			
ActiveSeq	0	DINT	DUNEW2PLC1
External Access:	Read/Write		
*ActiveSeq - motionQueue/queue_line_arc - *112(MOV), *143(MOV), *46(MOV), *58(MOV)*			
CheckCurSeg	0	BOOL	motionQueue
External Access:	Read/Write		
*CheckCurSeg - motionQueue/queue_line_arc - *136(OTU), *21(OTU), *33(OTL), 14(XIC), 15(XIC), 16(XIC), 17(XIC), 18(XIC), 19(XIC), 20(XIC), 21(XIC)*			
CheckNextSeg	0	BOOL	motionQueue
External Access:	Read/Write		
*CheckNextSeg - motionQueue/queue_line_arc - *137(OTU), *29(OTU), *70(OTL), 22(XIC), 23(XIC), 24(XIC), 25(XIC), 26(XIC), 27(XIC), 28(XIC), 29(XIC)*			
CmdA_Accel	2000.0	REAL	motionQueue
External Access:	Read/Write		
*CmdA_Accel - motionQueue/queue_line_arc - *37(MOV), *86(MOV), 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM)*			
CmdA_CircleType	1	DINT	motionQueue
External Access:	Read/Write		
*CmdA_CircleType - motionQueue/queue_line_arc - *43(MOV), *92(MOV), 101(MCCM), 62(MCCM)*			
CmdA_Decel	2000.0	REAL	motionQueue
External Access:	Read/Write		
*CmdA_Decel - motionQueue/queue_line_arc - *38(MOV), *87(MOV), 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM)*			
CmdA_Direction	1	DINT	motionQueue
External Access:	Read/Write		
*CmdA_Direction - motionQueue/queue_line_arc - *45(MOV), *94(MOV), 101(MCCM), 62(MCCM)*			
CmdA_JerkAccel	20000.0	REAL	motionQueue
External Access:	Read/Write		
*CmdA_JerkAccel - motionQueue/queue_line_arc - *39(MOV), *88(MOV), 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM)*			
CmdA_JerkDecel	20000.0	REAL	motionQueue
External Access:	Read/Write		
*CmdA_JerkDecel - motionQueue/queue_line_arc - *40(MOV), *89(MOV), 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM)*			
CmdA_SegType	1	DINT	motionQueue
External Access:	Read/Write		
*CmdA_SegType - motionQueue/queue_line_arc - *42(MOV), *91(MOV), 100(EQU), 101(EQU), 61(EQU), 62(EQU)*			
CmdA_Speed	437.4951	REAL	motionQueue
External Access:	Read/Write		
*CmdA_Speed - motionQueue/queue_line_arc - *36(MOV), *85(MOV), 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM)*			
CmdA_TermType	4	SINT	motionQueue
External Access:	Read/Write		
*CmdA_TermType - motionQueue/queue_line_arc - *41(MOV), *90(MOV), 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM)*			
CmdA_ViaCenter		REAL[2]	motionQueue
External Access:	Read/Write		
*CmdA_ViaCenter - motionQueue/queue_line_arc - *44(COP), *93(COP)*			
CmdA_ViaCenter[0]	0.0	REAL	
CmdA_ViaCenter[0] - motionQueue/queue_line_arc - 101(MCCM), 62(MCCM)			

CmdA_XY REAL[2] motionQueue
External Access: Read/Write
*CmdA_XY - motionQueue/queue_line_arc - *35(COP), *84(COP), 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM)*

CmdB_Accel 2000.0 REAL motionQueue
External Access: Read/Write
*CmdB_Accel - motionQueue/queue_line_arc - *49(MOV), *74(MOV), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)*

CmdB_CircleType 1 DINT motionQueue
External Access: Read/Write
*CmdB_CircleType - motionQueue/queue_line_arc - *55(MOV), *80(MOV), 64(MCCM), 99(MCCM)*

CmdB_Decel 2000.0 REAL motionQueue
External Access: Read/Write
*CmdB_Decel - motionQueue/queue_line_arc - *50(MOV), *75(MOV), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)*

CmdB_Direction 1 DINT motionQueue
External Access: Read/Write
*CmdB_Direction - motionQueue/queue_line_arc - *57(MOV), *82(MOV), 64(MCCM), 99(MCCM)*

CmdB_JerkAccel 20000.0 REAL motionQueue
External Access: Read/Write
*CmdB_JerkAccel - motionQueue/queue_line_arc - *51(MOV), *76(MOV), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)*

CmdB_JerkDecel 20000.0 REAL motionQueue
External Access: Read/Write
*CmdB_JerkDecel - motionQueue/queue_line_arc - *52(MOV), *77(MOV), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)*

CmdB_SegType 1 DINT motionQueue
External Access: Read/Write
*CmdB_SegType - motionQueue/queue_line_arc - *54(MOV), *79(MOV), 63(EQU), 64(EQU), 98(EQU), 99(EQU)*

CmdB_Speed 437.4951 REAL motionQueue
External Access: Read/Write
*CmdB_Speed - motionQueue/queue_line_arc - *48(MOV), *73(MOV), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)*

CmdB_TermType 4 SINT motionQueue
External Access: Read/Write
*CmdB_TermType - motionQueue/queue_line_arc - *53(MOV), *78(MOV), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)*

CmdB_ViaCenter REAL[2] motionQueue
External Access: Read/Write
*CmdB_ViaCenter - motionQueue/queue_line_arc - *56(COP), *81(COP)*
CmdB_ViaCenter[0] 0.0 REAL
CmdB_ViaCenter[0] - motionQueue/queue_line_arc - 64(MCCM), 99(MCCM)

CmdB_XY REAL[2] motionQueue
External Access: Read/Write
*CmdB_XY - motionQueue/queue_line_arc - *47(COP), *72(COP), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)*

CmdTolerance 10 DINT motionQueue
Constant No
External Access: Read/Write
CmdTolerance - motionQueue/queue_line_arc - 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)

CurIssued 0 BOOL DUNEW2PLC1
External Access: Read/Write
*CurIssued - motionQueue/queue_line_arc - *117(OTU), *118(OTU), *120(OTU), *65(OTL), 104(XIC), 105(XIC), 117(XIC), 118(XIC), 30(XIO), 67(XIC), 68(XIC)*

CurSeg MotionSeg motionQueue
motion segments for move queue, lines or arcs
External Access: Read/Write

CurSeg (Continued)
*CurSeg - motionQueue/queue_line_arc - *106(COP), *32(FFU)*

CurSeg.Valid	0	BOOL	
motion segments for move queue, lines or arcs segment valid flag			
*CurSeg.Valid - motionQueue/queue_line_arc - *138(OTU), 14(XIC), 15(XIO), 16(XIO), 17(XIC), 18(XIC), 19(XIC), 20(XIC)*			
CurSeg.SegType	1	SINT	
motion segments for move queue, lines or arcs 1=line, 2=arc			
CurSeg.SegType - motionQueue/queue_line_arc - 14(EQU), 14(EQU), 19(GRT), 19(LES), 20(GRT), 20(LES), 42(MOV), 54(MOV)			
CurSeg.XY		REAL	
motion segments for move queue, lines or arcs end xy			
CurSeg.XY - motionQueue/queue_line_arc - 35(COP), 47(COP)			
CurSeg.XY[0]	5990.0	REAL	
motion segments for move queue, lines or arcs end xy			
CurSeg.XY[1]	1250.0	REAL	
motion segments for move queue, lines or arcs end xy			
CurSeg.Speed	437.4951	REAL	
motion segments for move queue, lines or arcs >0			
CurSeg.Speed - motionQueue/queue_line_arc - 36(MOV), 48(MOV)			
CurSeg.Accel	2000.0	REAL	
motion segments for move queue, lines or arcs >0			
CurSeg.Accel - motionQueue/queue_line_arc - 37(MOV), 49(MOV)			
CurSeg.Decel	2000.0	REAL	
motion segments for move queue, lines or arcs >0			
CurSeg.Decel - motionQueue/queue_line_arc - 38(MOV), 50(MOV)			
CurSeg.JerkAccel	20000.0	REAL	
motion segments for move queue, lines or arcs			
CurSeg.JerkAccel - motionQueue/queue_line_arc - 39(MOV), 51(MOV)			
CurSeg.JerkDecel	20000.0	REAL	
motion segments for move queue, lines or arcs			
CurSeg.JerkDecel - motionQueue/queue_line_arc - 40(MOV), 52(MOV)			
CurSeg.TermType	4	SINT	
motion segments for move queue, lines or arcs 0 to 6			
CurSeg.TermType - motionQueue/queue_line_arc - 41(MOV), 53(MOV)			
CurSeg.Seq	0	DINT	
motion segments for move queue, lines or arcs Sequence/ack id			
*CurSeg.Seq - motionQueue/queue_line_arc - *139(MOV), 14(GRT), 17(LEQ), 18(LEQ), 19(GRT), 20(GRT), 46(MOV), 58(MOV)*			
CurSeg.CircleType	1	DINT	
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum			
CurSeg.CircleType - motionQueue/queue_line_arc - 43(MOV), 55(MOV)			
CurSeg.ViaCenter		REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
CurSeg.ViaCenter - motionQueue/queue_line_arc - 44(COP), 56(COP)			
CurSeg.ViaCenter[0]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
CurSeg.ViaCenter[1]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
CurSeg.Direction	1	DINT	
motion segments for move queue, lines or arcs 0..3 per mmcm enum			
CurSeg.Direction - motionQueue/queue_line_arc - 45(MOV), 57(MOV)			

DoneONS_A	0	BOOL	motionQueue
External Access:	Read/Write		
*DoneONS_A - motionQueue/queue_line_arc - *117(ONS)*			

DoneONS_B	0	BOOL	motionQueue
External Access:	Read/Write		
*DoneONS_B - motionQueue/queue_line_arc - *118(ONS)*			

EnqueueReq	0	BOOL	motionQueue
External Access:	Read/Write		
*EnqueueReq - motionQueue/queue_line_arc - *10(OTU), *128(OTU), *3(OTL), 10(XIC), 4(XIC), 5(XIC), 7(XIC), 8(XIC), 9(XIC)*			

FaultCode	0	DINT	motionQueue
External Access:	Read/Write		

FaultCode (Continued)
*FaultCode - motionQueue/queue_line_arc - *11(MOV), *12(MOV), *134(MOV), *15(MOV), *17(MOV), *19(MOV), *23(MOV), *25(MOV), *27(MOV)*

FlipToA	0	BOOL	motionQueue
External Access:	Read/Write		
*FlipToA - motionQueue/queue_line_arc - *108(OTL), *115(OTU), *130(OTU), 110(XIC), 115(XIC)*			
FlipToB	0	BOOL	motionQueue
External Access:	Read/Write		
*FlipToB - motionQueue/queue_line_arc - *107(OTL), *116(OTU), *131(OTU), 109(XIC), 116(XIC)*			
IncomingSeg		MotionSeg	DUNEW2PLC1
motion segments for move queue, lines or arcs			
Constant	No		
External Access:	Read/Write		
IncomingSeg - motionQueue/queue_line_arc - 7(FFL)			
IncomingSeg.Valid	1	BOOL	
motion segments for move queue, lines or arcs segment valid flag			
IncomingSeg.Valid - motionQueue/queue_line_arc - 4(XIC), 5(XIC)			
IncomingSeg.SegType	1	SINT	
motion segments for move queue, lines or arcs 1=line, 2=arc			
IncomingSeg.SegType - motionQueue/queue_line_arc - 4(EQU), 5(EQU)			
IncomingSeg.XY		REAL	
motion segments for move queue, lines or arcs end xy			
IncomingSeg.XY[0]	5672.515	REAL	
motion segments for move queue, lines or arcs end xy			
IncomingSeg.XY[1]	444.68442	REAL	
motion segments for move queue, lines or arcs end xy			
IncomingSeg.Speed	437.4951	REAL	
motion segments for move queue, lines or arcs >0			
IncomingSeg.Speed - motionQueue/queue_line_arc - 4(GRT), 5(GRT)			
IncomingSeg.Accel	2000.0	REAL	
motion segments for move queue, lines or arcs >0			
IncomingSeg.Accel - motionQueue/queue_line_arc - 4(GRT), 5(GRT)			
IncomingSeg.Decel	2000.0	REAL	
motion segments for move queue, lines or arcs >0			
IncomingSeg.Decel - motionQueue/queue_line_arc - 4(GRT), 5(GRT)			
IncomingSeg.JerkAccel	20000.0	REAL	
motion segments for move queue, lines or arcs			
IncomingSeg.JerkDecel	20000.0	REAL	
motion segments for move queue, lines or arcs			
IncomingSeg.TermType	4	SINT	
motion segments for move queue, lines or arcs 0 to 6			
IncomingSeg.TermType - motionQueue/queue_line_arc - 4(GEQ), 4(LEQ), 5(GEQ), 5(LEQ)			
IncomingSeg.Seq	143	DINT	
motion segments for move queue, lines or arcs Sequence/ack id			
IncomingSeg.Seq - motionQueue/queue_line_arc - 8(MOV)			
IncomingSeg.CircleType	1	DINT	
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum			
IncomingSeg.CircleType - motionQueue/queue_line_arc - 5(GEQ), 5(LEQ)			
IncomingSeg.ViaCenter		REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
IncomingSeg.ViaCenter[0]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
IncomingSeg.ViaCenter[1]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
IncomingSeg.Direction	1	DINT	
motion segments for move queue, lines or arcs 0..3 per mmcm enum			
IncomingSeg.Direction - motionQueue/queue_line_arc - 5(GEQ), 5(LEQ)			
IncomingSegAck	142	DINT	DUNEW2PLC1
External Access:	Read/Write		
*IncomingSegAck - motionQueue/queue_line_arc - *8(MOV)*			

IncomingSegReqID 44 DINT DUNEW2PLC1
Constant No
External Access: Read/Write
IncomingSegReqID - motionQueue/queue_line_arc - 142(MOV), 3(NEQ), 9(MOV)

IssueCurPulse 0 BOOL motionQueue
External Access: Read/Write
*IssueCurPulse - motionQueue/queue_line_arc - *126(OTU), *59(OTL), *66(OTU), 61(XIC), 62(XIC), 63(XIC), 64(XIC), 65(XIC), 66(XIC)*

IssueNextPulse 0 BOOL motionQueue
External Access: Read/Write
*IssueNextPulse - motionQueue/queue_line_arc - *103(OTU), *127(OTU), *96(OTL), 100(XIC), 101(XIC), 102(XIC), 103(XIC), 98(XIC), 99(XIC)*

LastIncomingSegReqID 44 DINT DUNEW2PLC1
External Access: Read/Write
*LastIncomingSegReqID - motionQueue/queue_line_arc - *142(MOV), *9(MOV), 3(NEQ)*

LoadCurReq 0 BOOL motionQueue
External Access: Read/Write
*LoadCurReq - motionQueue/queue_line_arc - *122(OTU), *30(OTL), *34(OTU), 31(XIC), 32(XIC), 33(XIC), 34(XIC)*

LoadNextReq 0 BOOL motionQueue
External Access: Read/Write
*LoadNextReq - motionQueue/queue_line_arc - *123(OTU), *67(OTL), *68(OTL), *71(OTU), 69(XIC), 70(XIC), 71(XIC)*

MotionFault 0 BOOL DUNEW2PLC1
External Access: Read/Write
*MotionFault - motionQueue/queue_line_arc - *13(OTE)*

MoveA MOTION_INSTRUCTION motionQueue
External Access: Read/Write
*MoveA - motionQueue/queue_line_arc - *100(MCLM), *101(MCCM), *61(MCLM), *62(MCCM)*
MoveA.ER 0 BOOL
*MoveA.ER - motionQueue/queue_line_arc - *11(OTU), 11(XIC), 13(XIC)*
MoveA.PC 0 BOOL
MoveA.PC - motionQueue/queue_line_arc - 117(XIC)
MoveA.IP 0 BOOL
MoveA.IP - motionQueue/queue_line_arc - 105(XIC), 67(XIC)

MoveB MOTION_INSTRUCTION motionQueue
External Access: Read/Write
*MoveB - motionQueue/queue_line_arc - *63(MCLM), *64(MCCM), *98(MCLM), *99(MCCM)*
MoveB.ER 0 BOOL
*MoveB.ER - motionQueue/queue_line_arc - *12(OTU), 12(XIC), 13(XIC)*
MoveB.PC 0 BOOL
MoveB.PC - motionQueue/queue_line_arc - 118(XIC)
MoveB.IP 0 BOOL
MoveB.IP - motionQueue/queue_line_arc - 104(XIC), 68(XIC)

NextIssued 0 BOOL DUNEW2PLC1
External Access: Read/Write
*NextIssued - motionQueue/queue_line_arc - *102(OTL), *111(OTU), *121(OTU), 104(XIC), 105(XIC), 117(XIO), 118(XIO), 67(XIO), 68(XIO)*

NextSeg MotionSeg motionQueue
motion segments for move queue, lines or arcs
External Access: Read/Write
*NextSeg - motionQueue/queue_line_arc - *69(FFU), 106(COP)*
NextSeg.Valid 0 BOOL
motion segments for move queue, lines or arcs segment valid flag
*NextSeg.Valid - motionQueue/queue_line_arc - *140(OTU), 22(XIC), 23(XIO), 24(XIO), 25(XIC), 26(XIC), 27(XIC), 28(XIC)*
NextSeg.SegType 1 SINT
motion segments for move queue, lines or arcs 1=line, 2=arc
NextSeg.SegType - motionQueue/queue_line_arc - 22(EQU), 22(EQU), 27(GRT), 27(LES), 28(GRT), 28(LES), 79(MOV), 91(MOV)
NextSeg.XY REAL

NextSeg (Continued)
motion segments for move queue, lines or arcs end xy
NextSeg.XY - motionQueue/queue_line_arc - 72(COP), 84(COP)

NextSeg.XY[0]	5972.74	REAL	
motion segments for move queue, lines or arcs end xy			
NextSeg.XY[1]	1347.2893	REAL	
motion segments for move queue, lines or arcs end xy			
NextSeg.Speed	437.4951	REAL	
motion segments for move queue, lines or arcs >0			
NextSeg.Speed - motionQueue/queue_line_arc - 73(MOV), 85(MOV)			
NextSeg.Accel	2000.0	REAL	
motion segments for move queue, lines or arcs >0			
NextSeg.Accel - motionQueue/queue_line_arc - 74(MOV), 86(MOV)			
NextSeg.Decel	2000.0	REAL	
motion segments for move queue, lines or arcs >0			
NextSeg.Decel - motionQueue/queue_line_arc - 75(MOV), 87(MOV)			
NextSeg.JerkAccel	20000.0	REAL	
motion segments for move queue, lines or arcs			
NextSeg.JerkAccel - motionQueue/queue_line_arc - 76(MOV), 88(MOV)			
NextSeg.JerkDecel	20000.0	REAL	
motion segments for move queue, lines or arcs			
NextSeg.JerkDecel - motionQueue/queue_line_arc - 77(MOV), 89(MOV)			
NextSeg.TermType	4	SINT	
motion segments for move queue, lines or arcs 0 to 6			
NextSeg.TermType - motionQueue/queue_line_arc - 78(MOV), 90(MOV)			
NextSeg.Seq	0	DINT	
motion segments for move queue, lines or arcs Sequence/ack id			
*NextSeg.Seq - motionQueue/queue_line_arc - *141(MOV), 22(GRT), 25(LEQ), 26(LEQ), 27(GRT), 28(GRT), 83(MOV), 95(MOV)*			
NextSeg.CircleType	1	DINT	
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum			
NextSeg.CircleType - motionQueue/queue_line_arc - 80(MOV), 92(MOV)			
NextSeg.ViaCenter		REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
NextSeg.ViaCenter - motionQueue/queue_line_arc - 81(COP), 93(COP)			
NextSeg.ViaCenter[0]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
NextSeg.ViaCenter[1]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
NextSeg.Direction	1	DINT	
motion segments for move queue, lines or arcs 0..3 per mmcm enum			
NextSeg.Direction - motionQueue/queue_line_arc - 82(MOV), 94(MOV)			

PendingSeq	0	DINT	DUNEW2PLC1
External Access:	Read/Write		
*PendingSeq - motionQueue/queue_line_arc - *113(MOV), *144(MOV), *83(MOV), *95(MOV), 112(MOV)*			
PrepCurMove	0	BOOL	motionQueue
External Access:	Read/Write		
*PrepCurMove - motionQueue/queue_line_arc - *124(OTU), *14(OTL), *60(OTU), 35(XIC), 36(XIC), 37(XIC), 38(XIC), 39(XIC), 40(XIC), 41(XIC), 42(XIC), 43(XIC), 44(XIC), 45(XIC), 46(XIC), 47(XIC), 48(XIC), 49(XIC), 50(XIC), 51(XIC), 52(XIC), 53(XIC), 54(XIC), 55(XIC), 56(XIC), 57(XIC), 58(XIC), 59(XIC), 60(XIC)*			
PrepNextMove	0	BOOL	motionQueue
External Access:	Read/Write		
*PrepNextMove - motionQueue/queue_line_arc - *125(OTU), *22(OTL), *97(OTU), 72(XIC), 73(XIC), 74(XIC), 75(XIC), 76(XIC), 77(XIC), 78(XIC), 79(XIC), 80(XIC), 81(XIC), 82(XIC), 83(XIC), 84(XIC), 85(XIC), 86(XIC), 87(XIC), 88(XIC), 89(XIC), 90(XIC), 91(XIC), 92(XIC), 93(XIC), 94(XIC), 95(XIC), 96(XIC), 97(XIC)*			
QueueCount	0	DINT	DUNEW2PLC1
External Access:	Read/Write		
*QueueCount - motionQueue/queue_line_arc - *2(MOV)*			
QueueCtl		CONTROL	motionQueue
External Access:	Read/Write		

QueueCtl (Continued)
*QueueCtl - motionQueue/queue_line_arc - *119(RES), *32(FFU), *69(FFU), *7(FFL)*

QueueCtl.POS	0	DINT	
QueueCtl.POS - motionQueue/queue_line_arc - 0(EQU), 1(GEQ), 2(MOV), 30(GEQ)			
QueueEmpty	1	BOOL	motionQueue
External Access:	Read/Write		
*QueueEmpty - motionQueue/queue_line_arc - *0(OTE), 67(XIO), 68(XIO)*			
QueueFault	0	BOOL	DUNEW2PLC1
External Access:	Read/Write		
*QueueFault - motionQueue/queue_line_arc - *135(OTU), *16(OTL), *18(OTL), *20(OTL), *24(OTL), *26(OTL), *28(OTL), 13(XIC), 30(XIO), 67(XIO), 68(XIO)*			
QueueFull	0	BOOL	DUNEW2PLC1
External Access:	Read/Write		
*QueueFull - motionQueue/queue_line_arc - *1(OTE), 7(XIO), 8(XIO), 9(XIO)*			
RotateMoves	0	BOOL	motionQueue
External Access:	Read/Write		
*RotateMoves - motionQueue/queue_line_arc - *104(OTL), *105(OTL), *114(OTU), *129(OTU), 106(XIC), 107(XIC), 108(XIC), 111(XIC), 112(XIC), 113(XIC), 114(XIC)*			
RotateONS_AtoB	0	BOOL	motionQueue
External Access:	Read/Write		
*RotateONS_AtoB - motionQueue/queue_line_arc - *104(ONS)*			
RotateONS_BtoA	0	BOOL	motionQueue
External Access:	Read/Write		
*RotateONS_BtoA - motionQueue/queue_line_arc - *105(ONS)*			
SegQueue		MotionSeg[32]	motionQueue
motion segments for move queue, lines or arcs			
External Access:	Read/Write		
*SegQueue - motionQueue/queue_line_arc - *145(FLL), *32(FFU), *69(FFU), *7(FFL)*			
SegQueue[0]		MotionSeg	
motion segments for move queue, lines or arcs			
SegQueue[0].Valid	0	BOOL	
motion segments for move queue, lines or arcs segment valid flag			
SegQueue[0].SegType	0	SINT	
motion segments for move queue, lines or arcs 1=line, 2=arc			
SegQueue[0].XY		REAL	
motion segments for move queue, lines or arcs end xy			
SegQueue[0].XY[0]	0.0	REAL	
motion segments for move queue, lines or arcs end xy			
SegQueue[0].XY[1]	0.0	REAL	
motion segments for move queue, lines or arcs end xy			
SegQueue[0].Speed	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[0].Accel	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[0].Decel	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[0].JerkAccel	0.0	REAL	
motion segments for move queue, lines or arcs			
SegQueue[0].JerkDecel	0.0	REAL	
motion segments for move queue, lines or arcs			
SegQueue[0].TermType	0	SINT	
motion segments for move queue, lines or arcs 0 to 6			
SegQueue[0].Seq	0	DINT	
motion segments for move queue, lines or arcs Sequence/ack id			
SegQueue[0].CircleType	0	DINT	
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum			
SegQueue[0].ViaCenter		REAL	

SegQueue (Continued)

motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[0].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[0].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[0].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[1]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[1].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[1].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[1].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[1].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[1].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[1].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[1].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[1].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[1].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[1].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[1].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[1].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[1].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[1].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[1].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[1].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[1].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[2]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[2].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[2].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[2].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[2].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[2].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[2].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[2].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[2].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[2].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		

SegQueue (Continued)

SegQueue[2].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[2].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[2].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[2].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[2].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[2].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[2].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[2].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[3]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[3].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[3].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[3].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[3].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[3].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[3].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[3].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[3].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[3].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[3].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[3].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[3].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[3].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[3].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[3].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[3].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[3].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[4]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[4].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[4].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[4].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[4].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[4].XY[1]	0.0	REAL

SegQueue (Continued)

motion segments for move queue, lines or arcs end xy		
SegQueue[4].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[4].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[4].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[4].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[4].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[4].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[4].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[4].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[4].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[4].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[4].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[4].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[5]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[5].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[5].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[5].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[5].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[5].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[5].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[5].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[5].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[5].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[5].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[5].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[5].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[5].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[5].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[5].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[5].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[5].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[6]		MotionSeg
motion segments for move queue, lines or arcs		

SegQueue (Continued)

SegQueue[6].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[6].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[6].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[6].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[6].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[6].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[6].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[6].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[6].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[6].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[6].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[6].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[6].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[6].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[6].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[6].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[6].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[7]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[7].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[7].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[7].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[7].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[7].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[7].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[7].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[7].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[7].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[7].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[7].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[7].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[7].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[7].ViaCenter		REAL

SegQueue (Continued)

motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[7].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[7].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[7].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[8]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[8].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[8].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[8].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[8].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[8].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[8].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[8].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[8].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[8].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[8].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[8].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[8].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[8].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[8].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[8].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[8].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[8].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[9]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[9].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[9].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[9].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[9].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[9].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[9].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[9].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[9].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[9].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		

SegQueue (Continued)

SegQueue[9].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[9].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[9].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[9].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[9].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[9].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[9].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[9].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[10]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[10].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[10].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[10].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[10].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[10].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[10].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[10].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[10].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[10].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[10].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[10].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[10].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[10].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[10].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[10].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[10].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[10].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[11]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[11].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[11].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[11].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[11].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[11].XY[1]	0.0	REAL

SegQueue (Continued)

motion segments for move queue, lines or arcs end xy		
SegQueue[11].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[11].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[11].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[11].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[11].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[11].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[11].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[11].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[11].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[11].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[11].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[11].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[12]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[12].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[12].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[12].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[12].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[12].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[12].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[12].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[12].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[12].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[12].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[12].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[12].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[12].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[12].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[12].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[12].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[12].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[13]		MotionSeg
motion segments for move queue, lines or arcs		

SegQueue (Continued)

SegQueue[13].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[13].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[13].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[13].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[13].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[13].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[13].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[13].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[13].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[13].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[13].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[13].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[13].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[13].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[13].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[13].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[13].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[14]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[14].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[14].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[14].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[14].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[14].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[14].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[14].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[14].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[14].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[14].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[14].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[14].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[14].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[14].ViaCenter		REAL

SegQueue (Continued)

motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[14].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[14].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[14].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[15]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[15].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[15].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[15].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[15].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[15].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[15].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[15].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[15].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[15].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[15].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[15].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[15].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[15].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[15].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[15].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[15].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[15].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[16]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[16].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[16].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[16].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[16].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[16].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[16].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[16].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[16].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[16].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		

SegQueue (Continued)

SegQueue[16].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[16].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[16].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[16].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[16].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[16].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[16].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[16].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[17]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[17].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[17].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[17].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[17].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[17].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[17].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[17].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[17].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[17].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[17].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[17].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[17].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[17].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[17].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[17].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[17].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[17].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[18]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[18].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[18].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[18].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[18].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[18].XY[1]	0.0	REAL

SegQueue (Continued)

motion segments for move queue, lines or arcs end xy		
SegQueue[18].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[18].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[18].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[18].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[18].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[18].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[18].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[18].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[18].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[18].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[18].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[18].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[19]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[19].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[19].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[19].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[19].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[19].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[19].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[19].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[19].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[19].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[19].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[19].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[19].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[19].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[19].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[19].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[19].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[19].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[20]		MotionSeg
motion segments for move queue, lines or arcs		

SegQueue (Continued)

SegQueue[20].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[20].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[20].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[20].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[20].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[20].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[20].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[20].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[20].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[20].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[20].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[20].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[20].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[20].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[20].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[20].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[20].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[21]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[21].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[21].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[21].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[21].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[21].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[21].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[21].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[21].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[21].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[21].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[21].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[21].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[21].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[21].ViaCenter		REAL

SegQueue (Continued)

motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[21].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[21].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[21].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[22]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[22].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[22].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[22].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[22].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[22].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[22].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[22].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[22].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[22].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[22].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[22].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[22].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[22].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[22].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[22].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[22].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[22].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[23]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[23].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[23].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[23].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[23].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[23].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[23].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[23].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[23].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[23].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		

SegQueue (Continued)

SegQueue[23].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[23].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[23].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[23].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[23].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[23].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[23].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[23].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[24]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[24].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[24].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[24].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[24].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[24].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[24].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[24].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[24].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[24].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[24].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[24].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[24].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[24].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[24].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[24].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[24].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[24].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[25]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[25].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[25].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[25].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[25].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[25].XY[1]	0.0	REAL

SegQueue (Continued)

motion segments for move queue, lines or arcs end xy		
SegQueue[25].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[25].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[25].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[25].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[25].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[25].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[25].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[25].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[25].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[25].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[25].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[25].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[26]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[26].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[26].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[26].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[26].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[26].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[26].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[26].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[26].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[26].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[26].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[26].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[26].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[26].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[26].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[26].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[26].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[26].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[27]		MotionSeg
motion segments for move queue, lines or arcs		

SegQueue (Continued)

SegQueue[27].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[27].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[27].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[27].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[27].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[27].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[27].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[27].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[27].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[27].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[27].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[27].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[27].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[27].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[27].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[27].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[27].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[28]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[28].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[28].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[28].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[28].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[28].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[28].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[28].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[28].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[28].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[28].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[28].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[28].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[28].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[28].ViaCenter		REAL

SegQueue (Continued)

motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[28].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[28].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[28].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[29]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[29].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[29].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[29].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[29].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[29].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[29].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[29].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[29].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[29].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[29].JerkDecel	0.0	REAL
motion segments for move queue, lines or arcs		
SegQueue[29].TermType	0	SINT
motion segments for move queue, lines or arcs 0 to 6		
SegQueue[29].Seq	0	DINT
motion segments for move queue, lines or arcs Sequence/ack id		
SegQueue[29].CircleType	0	DINT
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum		
SegQueue[29].ViaCenter		REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[29].ViaCenter[0]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[29].ViaCenter[1]	0.0	REAL
motion segments for move queue, lines or arcs [0]=x [1]=y		
SegQueue[29].Direction	0	DINT
motion segments for move queue, lines or arcs 0..3 per mmcm enum		
SegQueue[30]		MotionSeg
motion segments for move queue, lines or arcs		
SegQueue[30].Valid	0	BOOL
motion segments for move queue, lines or arcs segment valid flag		
SegQueue[30].SegType	0	SINT
motion segments for move queue, lines or arcs 1=line, 2=arc		
SegQueue[30].XY		REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[30].XY[0]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[30].XY[1]	0.0	REAL
motion segments for move queue, lines or arcs end xy		
SegQueue[30].Speed	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[30].Accel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[30].Decel	0.0	REAL
motion segments for move queue, lines or arcs >0		
SegQueue[30].JerkAccel	0.0	REAL
motion segments for move queue, lines or arcs		

SegQueue (Continued)

SegQueue[30].JerkDecel	0.0	REAL	
motion segments for move queue, lines or arcs			
SegQueue[30].TermType	0	SINT	
motion segments for move queue, lines or arcs 0 to 6			
SegQueue[30].Seq	0	DINT	
motion segments for move queue, lines or arcs Sequence/ack id			
SegQueue[30].CircleType	0	DINT	
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum			
SegQueue[30].ViaCenter		REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
SegQueue[30].ViaCenter[0]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
SegQueue[30].ViaCenter[1]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
SegQueue[30].Direction	0	DINT	
motion segments for move queue, lines or arcs 0..3 per mmcm enum			
SegQueue[31]		MotionSeg	
motion segments for move queue, lines or arcs			
SegQueue[31].Valid	0	BOOL	
motion segments for move queue, lines or arcs segment valid flag			
SegQueue[31].SegType	0	SINT	
motion segments for move queue, lines or arcs 1=line, 2=arc			
SegQueue[31].XY		REAL	
motion segments for move queue, lines or arcs end xy			
SegQueue[31].XY[0]	0.0	REAL	
motion segments for move queue, lines or arcs end xy			
SegQueue[31].XY[1]	0.0	REAL	
motion segments for move queue, lines or arcs end xy			
SegQueue[31].Speed	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[31].Accel	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[31].Decel	0.0	REAL	
motion segments for move queue, lines or arcs >0			
SegQueue[31].JerkAccel	0.0	REAL	
motion segments for move queue, lines or arcs			
SegQueue[31].JerkDecel	0.0	REAL	
motion segments for move queue, lines or arcs			
SegQueue[31].TermType	0	SINT	
motion segments for move queue, lines or arcs 0 to 6			
SegQueue[31].Seq	0	DINT	
motion segments for move queue, lines or arcs Sequence/ack id			
SegQueue[31].CircleType	0	DINT	
motion segments for move queue, lines or arcs 0,1,2,3 per MCCM enum			
SegQueue[31].ViaCenter		REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
SegQueue[31].ViaCenter[0]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
SegQueue[31].ViaCenter[1]	0.0	REAL	
motion segments for move queue, lines or arcs [0]=x [1]=y			
SegQueue[31].Direction	0	DINT	
motion segments for move queue, lines or arcs 0..3 per mmcm enum			

SegValid	0	BOOL	motionQueue
External Access:	Read/Write		
*SegValid - motionQueue/queue_line_arc - *6(OTE), 7(XIC), 8(XIC), 9(XIC)*			

SegValidArc	0	BOOL	motionQueue
External Access:	Read/Write		
*SegValidArc - motionQueue/queue_line_arc - *5(OTE), 6(XIC)*			

SegValidLine	0	BOOL	motionQueue
External Access:	Read/Write		

SegValidLine (Continued)
*SegValidLine - motionQueue/queue_line_arc - *4(OTE), 6(XIC)*

StartCurONS 0 BOOL motionQueue
External Access: Read/Write
*StartCurONS - motionQueue/queue_line_arc - *30(ONS)*

StartNextA_ONS 0 BOOL motionQueue
External Access: Read/Write
*StartNextA_ONS - motionQueue/queue_line_arc - *67(ONS)*

StartNextB_ONS 0 BOOL motionQueue
External Access: Read/Write
*StartNextB_ONS - motionQueue/queue_line_arc - *68(ONS)*

StartQueuedPath 0 BOOL DUNEW2PLC1
External Access: Read/Write
*StartQueuedPath - motionQueue/queue_line_arc - *133(OTU), *31(OTU), 30(XIC)*

UseAasCurrent 1 BOOL motionQueue
1=A current/B next 0=b current/A next
External Access: Read/Write
*UseAasCurrent - motionQueue/queue_line_arc - *109(OTU), *110(OTL), *132(OTL), 100(XIO), 101(XIO), 104(XIC), 105(XIO), 107(XIC), 108(XIO), 117(XIC), 118(XIO), 35(XIC), 36(XIC), 37(XIC), 38(XIC), 39(XIC), 40(XIC), 41(XIC), 42(XIC), 43(XIC), 44(XIC), 45(XIC), 46(XIC), 47(XIO), 48(XIO), 49(XIO), 50(XIO), 51(XIO), 52(XIO), 53(XIO), 54(XIO), 55(XIO), 56(XIO), 57(XIO), 58(XIO), 61(XIC), 62(XIC), 63(XIO), 64(XIO), 67(XIC), 68(XIO), 72(XIC), 73(XIC), 74(XIC), 75(XIC), 76(XIC), 77(XIC), 78(XIC), 79(XIC), 80(XIC), 81(XIC), 82(XIC), 83(XIC), 84(XIO), 85(XIO), 86(XIO), 87(XIO), 88(XIO), 89(XIO), 90(XIO), 91(XIO), 92(XIO), 93(XIO), 94(XIO), 95(XIO), 98(XIC), 99(XIC)*

X_Y COORDINATE_SYSTEM DUNEW2PLC1
External Access: Read/Write
X_Y - EOT_Trip_11/EOT_TRIP - 0(MCS)
X_Y - Error_State_10/Error - 2(MCS)
X_Y - motionQueue/queue_line_arc - 100(MCLM), 101(MCCM), 61(MCLM), 62(MCCM), 63(MCLM), 64(MCCM), 98(MCLM), 99(MCCM)
X_Y - MoveXY_State_2_3/MoveXY - 12(MCLM), 14(MCCD), 19(MCS), 22(MCLM)
X_Y - MoveXY_State_2_3/xy_speed_regulator - 0(MCLM)
X_Y.MovePendingStatus 0 BOOL
X_Y.MovePendingStatus - motionQueue/queue_line_arc - 104(XIO), 105(XIO), 117(XIO), 118(XIO), 67(XIO), 68(XIO)
X_Y.PhysicalAxisFault 0 BOOL
X_Y.PhysicalAxisFault - MoveXY_State_2_3/MoveXY - 2(XIC)

General

Type:	Ladder Diagram	Class:	Standard
In Program:	motionQueue	Number of Rungs:	146

General

Class:	Safety

Configuration

Type:	Periodic	Priority:	10
Period:	20.000 ms	Watchdog:	20.000 ms

Program Schedule

Scheduled

Safety

Unscheduled

Monitor

Scan Times(Elapsed Time)

Max:	0.577000 ms	Last:	0.472000 ms

Interval Times(Elapsed Times Between Triggers)

Max:	20.094999 ms	Min:	19.903999 ms

General

Class:	Safety

Configuration

Main:	Safety	Inhibit program:	No
Fault:	None		

Monitor

Scan Times(Execution Time)

Max:	49 us	Last:	30 us

0
EStop Rack
DUNEW2PLC2:1:I.Pt02Data
EStop Pullcord 1
DUNEW2PLC2:1:I.Pt03Data
EStop Pullcord 2
DUNEW2PLC2:1:I.Pt04Data
Safety_System_Healthy

1
Run Key
DUNEW2PLC2:1:I.Pt05Data
LC OSS1
DUNEW2PLC2:1:I.Pt00Data
LC OSS2
DUNEW2PLC2:1:I.Pt01Data
X_Axis_SLS.Request

2
Run Key
DUNEW2PLC2:1:I.Pt05Data
MORE_STATS_S[0]

3
Safety_System_Healthy
Safety_Tripped_S
L

4
Safety_Tripped_S
DUNEW2SERVO1:SO.STOOutput1
DUNEW2SERVO1:SO.STOOutput2
DUNEW2SERVO2:SO.STOOutput1

5
resetPB_S
Safety_System_Healthy
Safety_Tripped_S
U
X_Axis_SFX.Reset
X_Axis_SLS.Reset

6

DUNEW2PLC2:1:I.Pt06Data — DUNEW2SERVO1:SI.PrimaryFeedbackValid1

SFX	
Safety Control	X_Axis_SFX
Time Unit	Seconds
Position Scaling	512.0
Feedback Resolution	512
Unwind	512
Home Position	0.0
Feedback Position	DUNEW2SERVO1:SI.FeedbackPosition1 -13
Feedback Velocity	DUNEW2SERVO1:SI.FeedbackVelocity1 0.0
Feedback Valid	DUNEW2SERVO1:SI.PrimaryFeedbackValid1 1
Connection Faulted	DUNEW2SERVO1:SI.ConnectionFaulted 0
Home Trigger	X_Axis_SFX.HomeTrigger 0
Reset	X_Axis_SFX.Reset 0
Safe Feedback Homed	DUNEW2SERVO1:SO.SFHomed1
SFX Fault	DUNEW2SERVO1:SO.SFXFault1
Actual Position	0.9746094
Actual Cycles	-1
Actual Speed	0.0
Fault Type	1
Diagnostic Code	0

O1 FP SFH

7

DUNEW2PLC2:1:I.Pt06Data

SLS	
Safety Control	X_Axis_SLS
Restart Type	AUTOMATIC
Cold Start Type	AUTOMATIC
Check Delay	3000
Active Limit	X_Axis_SLS.ActiveLimit 10.0
Feedback SFX	X_Axis_SFX
Request	X_Axis_SLS.Request 0
Reset	X_Axis_SLS.Reset 0
SLS Active	DUNEW2SERVO1:SO.SLSActive1 0
SLS Limit	DUNEW2SERVO1:SO.SLSLimit1 0
SLS Fault	DUNEW2SERVO1:SO.SLSFault1 0
Fault Type	1
Diagnostic Code	0

O1 RR FP

8

DUNEW2SERVO1:SO.SLSActive1 — MORE_STATS_S[1]

(End)

Name	Value	Data Type	Scope
DUNEW2PLC2:1:I		AB:1734_IB8S_Safety2:I:0	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
DUNEW2PLC2:1:I.Pt00Data	1	BOOL	
LC OSS1			
DUNEW2PLC2:1:I.Pt00Data - MainProgram/MainRoutine - 24(XIC)			
DUNEW2PLC2:1:I.Pt00Data - Safety /Safety - 1(XIO)			
DUNEW2PLC2:1:I.Pt01Data	1	BOOL	
LC OSS2			
DUNEW2PLC2:1:I.Pt01Data - MainProgram/MainRoutine - 24(XIC)			
DUNEW2PLC2:1:I.Pt01Data - Safety /Safety - 1(XIO)			
DUNEW2PLC2:1:I.Pt02Data	1	BOOL	
EStop Rack			
DUNEW2PLC2:1:I.Pt02Data - MainProgram/MainRoutine - 1(XIO)			
DUNEW2PLC2:1:I.Pt02Data - Safety /Safety - 0(XIC)			
DUNEW2PLC2:1:I.Pt03Data	1	BOOL	
EStop Pullcord 1			
DUNEW2PLC2:1:I.Pt03Data - MainProgram/MainRoutine - 1(XIO)			
DUNEW2PLC2:1:I.Pt03Data - Safety /Safety - 0(XIC)			
DUNEW2PLC2:1:I.Pt04Data	1	BOOL	
EStop Pullcord 2			
DUNEW2PLC2:1:I.Pt04Data - MainProgram/MainRoutine - 1(XIO)			
DUNEW2PLC2:1:I.Pt04Data - Safety /Safety - 0(XIC)			
DUNEW2PLC2:1:I.Pt05Data	1	BOOL	
Run Key			
DUNEW2PLC2:1:I.Pt05Data - Safety /Safety - 1(XIO), 2(XIC)			
DUNEW2PLC2:1:I.Pt06Data	0	BOOL	
DUNEW2PLC2:1:I.Pt06Data - MainProgram/MainRoutine - 58(XIO)			
DUNEW2PLC2:1:I.Pt06Data - Safety /Safety - 6(XIO), 7(XIO)			
DUNEW2SERVO1:SI		AB:CIP_Motion_Device_Safety2:SI:0	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
DUNEW2SERVO1:SI.ConnectionFaulted	0	BOOL	
DUNEW2SERVO1:SI.ConnectionFaulted - Safety /Safety - 6(SFX)			
DUNEW2SERVO1:SI.FeedbackPosition1	-13	DINT	
DUNEW2SERVO1:SI.FeedbackPosition1 - Safety /Safety - 6(SFX)			
DUNEW2SERVO1:SI.FeedbackVelocity1	0.0	REAL	
DUNEW2SERVO1:SI.FeedbackVelocity1 - Safety /Safety - 6(SFX)			
DUNEW2SERVO1:SI.PrimaryFeedbackValid1	1	BOOL	
DUNEW2SERVO1:SI.PrimaryFeedbackValid1 - Safety /Safety - 6(SFX), 6(XIC)			
DUNEW2SERVO1:SO		AB:CIP_Motion_Device_Safety2:SO:0	DUNEW2PLC1
External Access:	Read/Write		
DUNEW2SERVO1:SO.SLSActive1	0	BOOL	
*DUNEW2SERVO1:SO.SLSActive1 - Safety /Safety - *7(SLS), 8(XIC)*			
DUNEW2SERVO1:SO.SLSLimit1	0	BOOL	
*DUNEW2SERVO1:SO.SLSLimit1 - Safety /Safety - *7(SLS)*			
DUNEW2SERVO1:SO.SFHomed1	0	BOOL	
*DUNEW2SERVO1:SO.SFHomed1 - Safety /Safety - *6(SFX)*			
DUNEW2SERVO1:SO.SFXFault1	0	BOOL	
*DUNEW2SERVO1:SO.SFXFault1 - Safety /Safety - *6(SFX)*			
DUNEW2SERVO1:SO.SLSFault1	0	BOOL	
*DUNEW2SERVO1:SO.SLSFault1 - Safety /Safety - *7(SLS)*			
DUNEW2SERVO1:SO.STOOutput1	1	BOOL	
*DUNEW2SERVO1:SO.STOOutput1 - Safety /Safety - *4(OTE)*			

DUNEW2SERVO1:SO (Continued)

DUNEW2SERVO1:SO.STOOutput2			
	1	BOOL	
*DUNEW2SERVO1:SO.STOOutput2 - Safety /Safety - *4(OTE)*			
DUNEW2SERVO2:SO		AB:CIP_Motion_Device_Safety2:SO:0	DUNEW2PLC1
External Access:	Read/Write		
DUNEW2SERVO2:SO.STOOutput1			
	1	BOOL	
*DUNEW2SERVO2:SO.STOOutput1 - Safety /Safety - *4(OTE)*			
MORE_STATS_S		BOOL[32]	DUNEW2PLC1
Mapped With:	MORE_STATS		
External Access:	Read/Write		
MORE_STATS_S[0]	1	BOOL	
*MORE_STATS_S[0] - Safety /Safety - *2(OTE)*			
MORE_STATS_S[1]	0	BOOL	
*MORE_STATS_S[1] - Safety /Safety - *8(OTE)*			
resetPB_S	0	BOOL	DUNEW2PLC1
Constant	No		
Mapped With:	ResetPB		
External Access:	Read/Write		
resetPB_S - Safety /Safety - 5(XIC)			
*ResetPB - MainProgram/MainRoutine - *54(OTE)*			
Safety_System_Healthy	1	BOOL	DUNEW2PLC1
Mapped With:	Run_Alarm		
External Access:	Read/Write		
*Safety_System_Healthy - Safety /Safety - *0(OTE), 3(XIO), 5(XIC)*			
Safety_Tripped_S	0	BOOL	DUNEW2PLC1
External Access:	Read/Write		
Safety_Tripped_S - MainProgram/MainRoutine - 34(XIO), 35(XIC)			
Safety_Tripped_S - PID_Tension_Servo/PID_Tension - 2(XIO)			
*Safety_Tripped_S - Safety /Safety - *3(OTL), *5(OTU), 4(XIO)*			
X_Axis_SFX		SAFETY_FEEDBACK_INTERFACE	DUNEW2PLC1
External Access:	Read/Write		
*X_Axis_SFX - Safety /Safety - *6(SFX), *7(SLS)*			
X_Axis_SFX.HomeTrigger	0	BOOL	
X_Axis_SFX.HomeTrigger - Safety /Safety - 6(SFX)			
X_Axis_SFX.Reset	0	BOOL	
*X_Axis_SFX.Reset - Safety /Safety - *5(OTE), 6(SFX)*			
X_Axis_SLS		SAFELY_LIMITED_SPEED	DUNEW2PLC1
External Access:	Read/Write		
*X_Axis_SLS - Safety /Safety - *7(SLS)*			
X_Axis_SLS.Request	0	BOOL	
*X_Axis_SLS.Request - Safety /Safety - *1(OTE), 7(SLS)*			
X_Axis_SLS.Reset	0	BOOL	
*X_Axis_SLS.Reset - Safety /Safety - *5(OTE), 7(SLS)*			
X_Axis_SLS.ActiveLimit	10.0	REAL	
X_Axis_SLS.ActiveLimit - Safety /Safety - 7(SLS)			

General

Type:	Ladder Diagram	Class:	Safety
In Program:	Safety	Number of Rungs:	9

General

Controls to capture data from the Cognex camera.

Class: Standard

Configuration

Main: CameraControl Inhibit program: No

Fault: None

Monitor

Scan Times(Execution Time)

Max: Last:

0
CAM_F_EN
DUNE2CAM1:O.Control.TriggerEnable

1
CAM_F_TRIGGER
OSR
Storage Bit triggerOneShotSB
Output Bit triggerOneShotOB
SB
OB

2
CAM_F_TRIGGER triggerOneShotSB TriggerResetTimer2.TT /
TON
Timer TriggerResetTimer1
Preset 100
Accum 5
EN
DN

3
CAM_F_TRIGGER triggerOneShotSB TriggerResetTimer1.TT /
TON
Timer TriggerResetTimer2
Preset 100
Accum 0
EN
DN

4
TriggerResetTimer1.TT
LightRing.TT
DUNE2CAM1:O.Control.Trigger

5
TriggerResetTimer1.TT
LightRing.TT
MAR
Axis Y_axis ...
Motion Control RegistrationData[1]
Trigger Condition Negative_Edge
Windowed Registration Disabled
Min. Position 0
Max. Position 0
Input Number 2
EN
DN
ER
IP
PC

6
TriggerResetTimer1.TT
LightRing.TT
MAR
Axis X_axis ...
Motion Control RegistrationData[0]
Trigger Condition Negative_Edge
Windowed Registration Disabled
Min. Position 0
Max. Position 0
Input Number 2
EN
DN
ER
IP
PC

7
RegistrationData[0].PC
CPT
Dest X_axis_Cam_F_Position
-5.1799655
Expression X_axis.Registration2NegativeEdgePosition

8
RegistrationData[1].PC
CPT
Dest Y_axis_Cam_F_Position
12.123628
Expression Y_axis.Registration2NegativeEdgePosition

9
RegistrationData[0].PC

FFL
Source X_axis_Cam_F_Position
FIFO X_axis_Cam_F_Pos[0]
Control FIFO_CTRL[0]
Length 10
Position 0
EN DN EM

10
RegistrationData[1].PC

FFL
Source Y_axis_Cam_F_Position
FIFO Y_axis_Cam_F_Pos[0]
Control FIFO_CTRL[1]
Length 10
Position 0
EN DN EM

11
DUNE2CAM1:I.Status.InspectionCompleted

OSR
Storage Bit ossb[3]
Output Bit osob[3]
SB OB

12
RegistrationData[1].PC

FFL
Source DUNE2CAM1:I.InspectionResults[0]
FIFO Cam_F_InspectionRes_0[0]
Control FIFO_CTRL[2]
Length 10
Position 0
EN DN EM

13
RegistrationData[1].PC

FFL
Source DUNE2CAM1:I.InspectionResults[1]
FIFO Cam_F_InspectionRes_1[0]
Control FIFO_CTRL[3]
Length 10
Position 0
EN DN EM

14
RegistrationData[1].PC

FFL
Source DUNE2CAM1:I.InspectionResults[2]
FIFO Cam_F_InspectionRes_2[0]
Control FIFO_CTRL[4]
Length 10
Position 0
EN DN EM

15
RegistrationData[1].PC

FFL
Source DUNE2CAM1:I.InspectionResults[3]
FIFO Cam_F_InspectionRes_3[0]
Control FIFO_CTRL[5]
Length 10
Position 0
EN DN EM

16
FIFO_CTRL[5].DN
READ_FIFOS
L

17
READ_FIFOS

FFU
FIFO X_axis_Cam_F_Pos[0]
Dest FIFO_Data[0]
Control FIFO_CTRL[0]
Length 10
Position 0
EU DN EM

18 FIFO_CTRL[0].EU

FFU
FIFO Y_axis_Cam_F_Pos[0]
Dest FIFO_Data[1]
Control FIFO_CTRL[1]
Length 10
Position 0
EU DN EM

19 FIFO_CTRL[1].EU

FFU
FIFO Cam_F_InspectionRes_0[0]
Dest FIFO_Data[2]
Control FIFO_CTRL[2]
Length 10
Position 0
EU DN EM

20 FIFO_CTRL[2].EU

FFU
FIFO Cam_F_InspectionRes_1[0]
Dest FIFO_Data[3]
Control FIFO_CTRL[3]
Length 10
Position 0
EU DN EM

21 FIFO_CTRL[3].EU

FFU
FIFO Cam_F_InspectionRes_2[0]
Dest FIFO_Data[4]
Control FIFO_CTRL[4]
Length 10
Position 0
EU DN EM

22 FIFO_CTRL[4].EU

FFU
FIFO Cam_F_InspectionRes_3[0]
Dest FIFO_Data[5]
Control FIFO_CTRL[5]
Length 10
Position 0
EU DN EM

23 FIFO_CTRL[5].EU

READ_FIFOS
U

24 EN_POS_TRIGGERS

OSR
Storage Bit ossb[1]
Output Bit osob[1]
SB OB

25 osob[1]

CPT
Dest x_pos
564.005
Expression X_axis.ActualPosition

26 osob[1]

CPT
Dest y_pos
2465.8398
Expression Y_axis.ActualPosition

27 osob[1]

Pos_trigger
L

28

EN_POS_TRIGGERS Pos_trigger

CMP Expression X_DELTA<0 — CMP Expression X_axis.ActualPosition<(x_pos+X_DELTA)

CMP Expression X_DELTA>0 — CMP Expression X_axis.ActualPosition>(x_pos+X_DELTA)

CMP Expression Y_DELTA<0 — CMP Expression Y_axis.ActualPosition<(y_pos+Y_DELTA)

CMP Expression Y_DELTA>0 — CMP Expression Y_axis.ActualPosition>(y_pos+Y_DELTA)

foundIt (L)

29

EN_POS_TRIGGERS foundIt | Manual_Trig

OSR
Storage Bit ossb[0]
Output Bit osob[0]
(SB) (OB)

Pos_trigger (U)

30

EN_POS_TRIGGERS osob[0]

CPT
Dest x_pos_next
564.005
Expression x_pos+X_DELTA

31

EN_POS_TRIGGERS osob[0]

CPT
Dest y_pos_next
2465.8398
Expression y_pos+Y_DELTA

32

EN_POS_TRIGGERS ossb[0]

TON
Timer LightRing
Preset 100
Accum 0
(EN) (DN)

33

EN_POS_TRIGGERS LightRing.DN

foundIt (U)

34

EN_POS_TRIGGERS LightRing.DN

CMP Expression x_pos_next<>x_pos

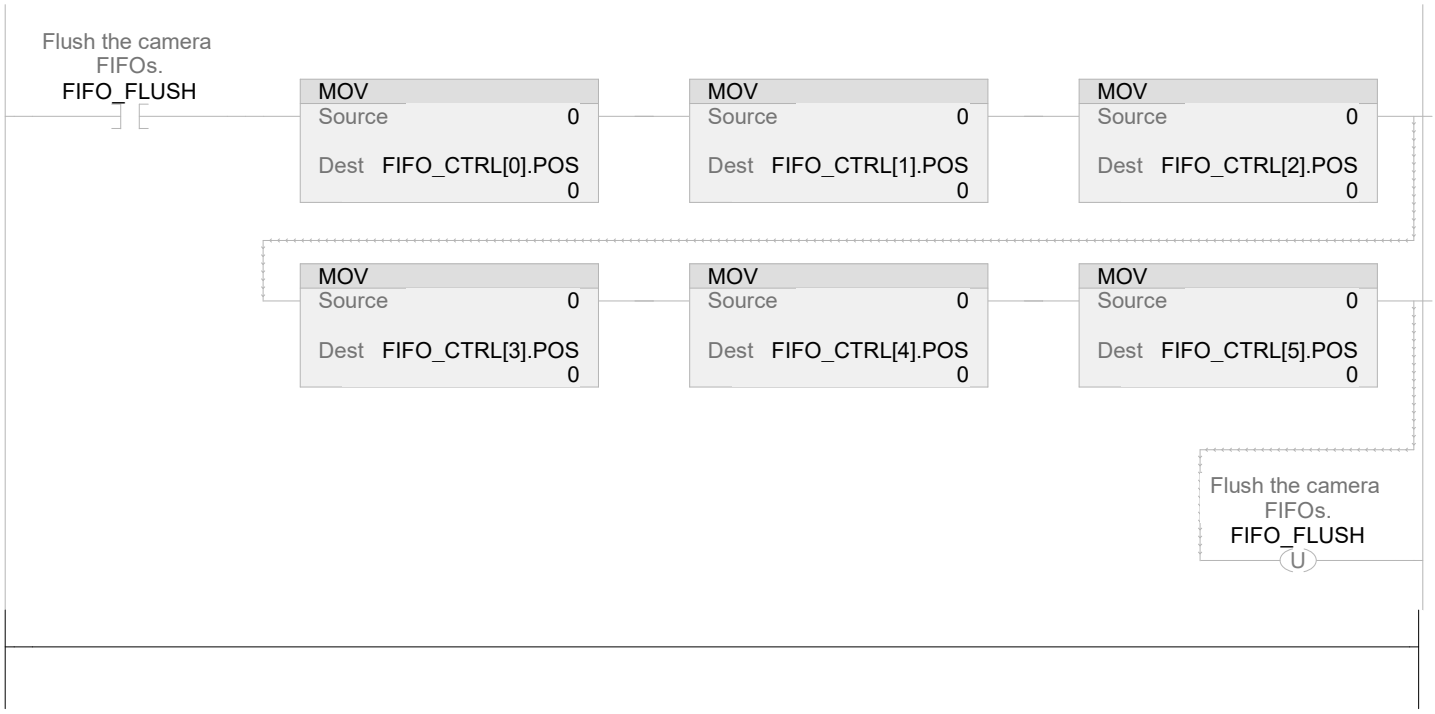
CMP Expression y_pos_next<>y_pos

CPT
Dest x_pos
564.005
Expression x_pos_next

CPT
Dest y_pos
2465.8398
Expression y_pos_next

Pos_trigger (L)

35


Flush the camera FIFOs.
FIFO_FLUSH
MOV
Source 0
Dest FIFO_CTRL[0].POS
0
MOV
Source 0
Dest FIFO_CTRL[1].POS
0
MOV
Source 0
Dest FIFO_CTRL[2].POS
0
MOV
Source 0
Dest FIFO_CTRL[3].POS
0
MOV
Source 0
Dest FIFO_CTRL[4].POS
0
MOV
Source 0
Dest FIFO_CTRL[5].POS
0
Flush the camera FIFOs.
FIFO_FLUSH
U

(End)

Name	Value	Data Type	Scope
CAM_F_EN	1	BOOL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
CAM_F_EN - Camera/CameraControl - 0(XIC)			
Cam_F_InspectionRes_0		REAL[32]	Camera
External Access:	Read Only		
*Cam_F_InspectionRes_0 - Camera/CameraControl - *12(FFL), *19(FFU)*			
Cam_F_InspectionRes_1		REAL[10]	Camera
External Access:	Read Only		
*Cam_F_InspectionRes_1 - Camera/CameraControl - *13(FFL), *20(FFU)*			
Cam_F_InspectionRes_2		REAL[10]	Camera
External Access:	Read Only		
*Cam_F_InspectionRes_2 - Camera/CameraControl - *14(FFL), *21(FFU)*			
Cam_F_InspectionRes_3		REAL[10]	Camera
External Access:	Read Only		
*Cam_F_InspectionRes_3 - Camera/CameraControl - *15(FFL), *22(FFU)*			
CAM_F_TRIGGER	1	BOOL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
CAM_F_TRIGGER - Camera/CameraControl - 1(XIC), 2(XIC), 3(XIC)			
DUNE2CAM1:I		CC:InSight11_DINT8:I:0	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
DUNE2CAM1:I.Status.InspectionCompleted	1	BOOL	
DUNE2CAM1:I.Status.InspectionCompleted - Camera/CameraControl - 11(XIC)			
DUNE2CAM1:I.InspectionResults[0]	0	DINT	
DUNE2CAM1:I.InspectionResults[0] - Camera/CameraControl - 12(FFL)			
DUNE2CAM1:I.InspectionResults[1]	0	DINT	
DUNE2CAM1:I.InspectionResults[1] - Camera/CameraControl - 13(FFL)			
DUNE2CAM1:I.InspectionResults[2]	0	DINT	
DUNE2CAM1:I.InspectionResults[2] - Camera/CameraControl - 14(FFL)			
DUNE2CAM1:I.InspectionResults[3]	0	DINT	
DUNE2CAM1:I.InspectionResults[3] - Camera/CameraControl - 15(FFL)			
DUNE2CAM1:O		CC:InSight11_DINT8:O:0	DUNEW2PLC1
External Access:	Read/Write		
DUNE2CAM1:O.Control.TriggerEnable	1	BOOL	
*DUNE2CAM1:O.Control.TriggerEnable - Camera/CameraControl - *0(OTE)*			
DUNE2CAM1:O.Control.Trigger	1	BOOL	
*DUNE2CAM1:O.Control.Trigger - Camera/CameraControl - *4(OTE)*			
EN_POS_TRIGGERS	0	BOOL	DUNEW2PLC1
Constant	No		
External Access:	Read/Write		
EN_POS_TRIGGERS - Camera/CameraControl - 24(XIC), 28(XIC), 29(XIC), 30(XIC), 31(XIC), 32(XIC), 33(XIC), 34(XIC)			
FIFO_CTRL		CONTROL[6]	Camera
External Access:	Read Only		
FIFO_CTRL[0]		CONTROL	
*FIFO_CTRL[0] - Camera/CameraControl - *17(FFU), *9(FFL)*			
FIFO_CTRL[0].POS	0	DINT	

FIFO_CTRL (Continued)
*FIFO_CTRL[0].POS - Camera/CameraControl - *35(MOV)*

FIFO_CTRL[0].EU	0	BOOL	
FIFO_CTRL[0].EU - Camera/CameraControl - 18(XIC)			
FIFO_CTRL[1]		CONTROL	
*FIFO_CTRL[1] - Camera/CameraControl - *10(FFL), *18(FFU)*			
FIFO_CTRL[1].POS	0	DINT	
*FIFO_CTRL[1].POS - Camera/CameraControl - *35(MOV)*			
FIFO_CTRL[1].EU	0	BOOL	
FIFO_CTRL[1].EU - Camera/CameraControl - 19(XIC)			
FIFO_CTRL[2]		CONTROL	
*FIFO_CTRL[2] - Camera/CameraControl - *12(FFL), *19(FFU)*			
FIFO_CTRL[2].POS	0	DINT	
*FIFO_CTRL[2].POS - Camera/CameraControl - *35(MOV)*			
FIFO_CTRL[2].EU	0	BOOL	
FIFO_CTRL[2].EU - Camera/CameraControl - 20(XIC)			
FIFO_CTRL[3]		CONTROL	
*FIFO_CTRL[3] - Camera/CameraControl - *13(FFL), *20(FFU)*			
FIFO_CTRL[3].POS	0	DINT	
*FIFO_CTRL[3].POS - Camera/CameraControl - *35(MOV)*			
FIFO_CTRL[3].EU	0	BOOL	
FIFO_CTRL[3].EU - Camera/CameraControl - 21(XIC)			
FIFO_CTRL[4]		CONTROL	
*FIFO_CTRL[4] - Camera/CameraControl - *14(FFL), *21(FFU)*			
FIFO_CTRL[4].POS	0	DINT	
*FIFO_CTRL[4].POS - Camera/CameraControl - *35(MOV)*			
FIFO_CTRL[4].EU	0	BOOL	
FIFO_CTRL[4].EU - Camera/CameraControl - 22(XIC)			
FIFO_CTRL[5]		CONTROL	
*FIFO_CTRL[5] - Camera/CameraControl - *15(FFL), *22(FFU)*			
FIFO_CTRL[5].POS	0	DINT	
*FIFO_CTRL[5].POS - Camera/CameraControl - *35(MOV)*			
FIFO_CTRL[5].EU	0	BOOL	
FIFO_CTRL[5].EU - Camera/CameraControl - 23(XIC)			
FIFO_CTRL[5].DN	0	BOOL	
FIFO_CTRL[5].DN - Camera/CameraControl - 16(XIC)			

FIFO_Data		REAL[6]	DUNEW2PLC1
External Access:	Read Only		
FIFO_Data[0]	0.0	REAL	
*FIFO_Data[0] - Camera/CameraControl - *17(FFU)*			
FIFO_Data[1]	0.0	REAL	
*FIFO_Data[1] - Camera/CameraControl - *18(FFU)*			
FIFO_Data[2]	0.0	REAL	
*FIFO_Data[2] - Camera/CameraControl - *19(FFU)*			
FIFO_Data[3]	0.0	REAL	
*FIFO_Data[3] - Camera/CameraControl - *20(FFU)*			
FIFO_Data[4]	0.0	REAL	
*FIFO_Data[4] - Camera/CameraControl - *21(FFU)*			
FIFO_Data[5]	0.0	REAL	
*FIFO_Data[5] - Camera/CameraControl - *22(FFU)*			

FIFO_FLUSH	0	BOOL	DUNEW2PLC1
Flush the camera FIFOs.			
External Access:	Read/Write		
*FIFO_FLUSH - Camera/CameraControl - *35(OTU), 35(XIC)*			

foundIt	0	BOOL	Camera
External Access:	Read Only		
*foundIt - Camera/CameraControl - *28(OTL), *33(OTU), 29(XIC)*			

LightRing		TIMER	Camera
External Access:	Read Only		
*LightRing - Camera/CameraControl - *32(TON)*			

LightRing (Continued)

LightRing.TT	0	BOOL	
LightRing.TT - Camera/CameraControl - 4(XIC), 5(XIC), 6(XIC)			
LightRing.DN	0	BOOL	
LightRing.DN - Camera/CameraControl - 33(XIC), 34(XIC)			
Manual_Trig	0	BOOL	Camera
Constant	No		
External Access:	Read Only		
Manual_Trig - Camera/CameraControl - 29(XIC)			
osob		BOOL[32]	Camera
External Access:	Read Only		
osob[0]	0	BOOL	
*osob[0] - Camera/CameraControl - *29(OSR), 30(XIC), 31(XIC)*			
osob[1]	0	BOOL	
*osob[1] - Camera/CameraControl - *24(OSR), 25(XIC), 26(XIC), 27(XIC)*			
osob[3]	0	BOOL	
*osob[3] - Camera/CameraControl - *11(OSR)*			
ossb		BOOL[32]	Camera
External Access:	Read Only		
ossb[0]	0	BOOL	
*ossb[0] - Camera/CameraControl - *29(OSR), 32(XIC)*			
ossb[1]	0	BOOL	
*ossb[1] - Camera/CameraControl - *24(OSR)*			
ossb[3]	1	BOOL	
*ossb[3] - Camera/CameraControl - *11(OSR)*			
Pos_trigger	1	BOOL	Camera
External Access:	Read Only		
*Pos_trigger - Camera/CameraControl - *27(OTL), *29(OTU), *34(OTL), 28(XIC)*			
READ_FIFOS	0	BOOL	DUNEW2PLC1
External Access:	Read/Write		
*READ_FIFOS - Camera/CameraControl - *16(OTL), *23(OTU), 17(XIC)*			
RegistrationData		MOTION_INSTRUCTION[8]	Camera
External Access:	Read Only		
RegistrationData[0]		MOTION_INSTRUCTION	
*RegistrationData[0] - Camera/CameraControl - *6(MAR)*			
RegistrationData[0].PC	0	BOOL	
RegistrationData[0].PC - Camera/CameraControl - 7(XIC), 9(XIC)			
RegistrationData[1]		MOTION_INSTRUCTION	
*RegistrationData[1] - Camera/CameraControl - *5(MAR)*			
RegistrationData[1].PC	0	BOOL	
RegistrationData[1].PC - Camera/CameraControl - 10(XIC), 12(XIC), 13(XIC), 14(XIC), 15(XIC), 8(XIC)			
triggerOneShotOB	0	BOOL	Camera
External Access:	Read Only		
*triggerOneShotOB - Camera/CameraControl - *1(OSR)*			
triggerOneShotSB	1	BOOL	Camera
External Access:	Read Only		
*triggerOneShotSB - Camera/CameraControl - *1(OSR), 2(XIC), 3(XIC)*			
TriggerResetTimer1		TIMER	Camera
External Access:	Read Only		
*TriggerResetTimer1 - Camera/CameraControl - *2(TON)*			
TriggerResetTimer1.TT	1	BOOL	
TriggerResetTimer1.TT - Camera/CameraControl - 3(XIO), 4(XIC), 5(XIC), 6(XIC)			
TriggerResetTimer2		TIMER	Camera
External Access:	Read Only		

TriggerResetTimer2 (Continued)
*TriggerResetTimer2 - Camera/CameraControl - *3(TON)*
TriggerResetTimer2.TT 0 BOOL
TriggerResetTimer2.TT - Camera/CameraControl - 2(XIO)

X_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
X_axis - Camera/CameraControl - 6(MAR)
X_axis - EOT_Trip_11/EOT_TRIP - 10(MAM), 5(MSO), 7(MAM)
X_axis - Error_State_10/Error - 11(MAFR), 5(MSF)
X_axis - MoveXY_State_2_3/MoveXY - 3(MSO)
X_axis - Ready_State_1/Ready - 2(MSO)
X_axis - UnServo_9/UnServo_All - 2(MSF), 6(MAFR)
X_axis.ActualPosition -898.1537 REAL
X_axis.ActualPosition - Camera/CameraControl - 25(CPT), 28(CMP), 28(CMP)
X_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)
X_axis.ActualPosition - MoveZ_State_4_5/MoveZ - 2(LIM), 2(LIM)
X_axis.ActualVelocity 0.0 REAL
X_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)
X_axis.Registration2NegativeEdgePosition
0.0 REAL
X_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 7(CPT)
X_axis.CommandAcceleration 0.0 REAL
X_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)
X_axis.SafeTorqueOffInhibit 0 BOOL
X_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)
X_axis.SLSActiveStatus 0 BOOL
X_axis.SLSActiveStatus - MainProgram/MainRoutine - 36(XIC)

X_axis_Cam_F_Pos REAL[10] Camera
External Access: Read Only
*X_axis_Cam_F_Pos - Camera/CameraControl - *17(FFU), *9(FFL)*

X_axis_Cam_F_Position -5.1799655 REAL Camera
External Access: Read Only
*X_axis_Cam_F_Position - Camera/CameraControl - *7(CPT), 9(FFL)*

X_DELTA 0.0 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
X_DELTA - Camera/CameraControl - 28(CMP), 28(CMP), 28(CMP), 28(CMP), 30(CPT)

x_pos 564.005 REAL Camera
External Access: Read Only
*x_pos - Camera/CameraControl - *25(CPT), *34(CPT), 28(CMP), 28(CMP), 30(CPT), 34(CMP)*

x_pos_next 564.005 REAL Camera
External Access: Read/Write
*x_pos_next - Camera/CameraControl - *30(CPT), 34(CMP), 34(CPT)*

Y_axis AXIS_CIP_DRIVE DUNEW2PLC1
External Access: Read Only
Y_axis - Camera/CameraControl - 5(MAR)
Y_axis - EOT_Trip_11/EOT_TRIP - 13(MAM), 16(MAM), 5(MSO)
Y_axis - Error_State_10/Error - 10(MAFR), 6(MSF)
Y_axis - MoveXY_State_2_3/MoveXY - 3(MSO)
Y_axis - Ready_State_1/Ready - 2(MSO)
Y_axis - UnServo_9/UnServo_All - 3(MSF), 8(MAFR)
Y_axis.ActualPosition -15.0942 REAL
Y_axis.ActualPosition - Camera/CameraControl - 26(CPT), 28(CMP), 28(CMP)
Y_axis.ActualPosition - MainProgram/MainRoutine - 31(LIM), 32(LIM), 33(LIM)
Y_axis.ActualPosition - MoveXY_State_2_3/MoveXY - 15(CMP), 15(CMP), 6(MOV), 8(CPT)
Y_axis.ActualVelocity 0.0 REAL
Y_axis.ActualVelocity - MainProgram/MainRoutine - 55(CPT), 56(CPT), 57(CPT)

Y_axis (Continued)

Y_axis.Registration2NegativeEdgePosition 0.0 REAL
Y_axis.Registration2NegativeEdgePosition - Camera/CameraControl - 8(CPT)

Y_axis.CommandAcceleration 0.0 REAL
Y_axis.CommandAcceleration - MainProgram/MainRoutine - 57(CPT)

Y_axis.SafeTorqueOffInhibit 0 BOOL
Y_axis.SafeTorqueOffInhibit - MoveXY_State_2_3/MoveXY - 2(XIC)

Y_axis_Cam_F_Pos REAL[10] Camera
External Access: Read Only
*Y_axis_Cam_F_Pos - Camera/CameraControl - *10(FFL), *18(FFU)*

Y_axis_Cam_F_Position 12.123628 REAL Camera
External Access: Read Only
*Y_axis_Cam_F_Position - Camera/CameraControl - *8(CPT), 10(FFL)*

Y_DELTA 5.75 REAL DUNEW2PLC1
Constant No
External Access: Read/Write
Y_DELTA - Camera/CameraControl - 28(CMP), 28(CMP), 28(CMP), 28(CMP), 31(CPT)

y_pos 2465.8398 REAL Camera
External Access: Read/Write
*y_pos - Camera/CameraControl - *26(CPT), *34(CPT), 28(CMP), 28(CMP), 31(CPT), 34(CMP)*

y_pos_next 2465.8398 REAL Camera
External Access: Read/Write
*y_pos_next - Camera/CameraControl - *31(CPT), 34(CMP), 34(CPT)*

General

Type:	Ladder Diagram	Class:	Standard
In Program:	Camera	Number of Rungs:	36

Signature Listing

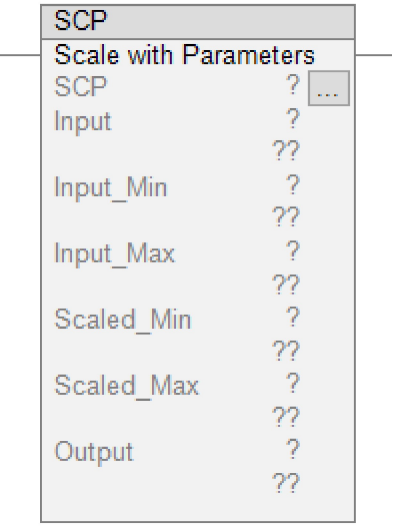
SCP v1.0
CVKI

Scale with Parameters

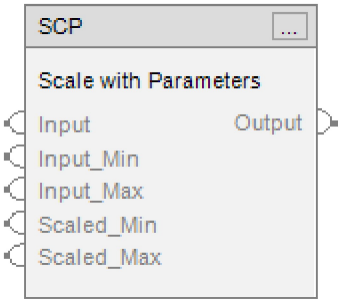
Class: Standard

Available Languages

Relay Ladder



Function Block



Structured Text
SCP(Input, Input_Min, Input_Max, Scaled_Min, Scaled_Max, Output);

Parameters

Required	Name	Data Type	Usage	Description
X	SCP	SCP	InOut	Scale with Parameters
	EnableIn	BOOL	Input	
	EnableOut	BOOL	Output	
X	Input	REAL	Input	Value to be Scaled
X	Input_Min	REAL	Input	Minimum Value for the Input
X	Input_Max	REAL	Input	Maximum Value for the Input
X	Scaled_Min	REAL	Input	Minimum Scaling Value
X	Scaled_Max	REAL	Input	Maximum Scaling Value
X	Output	REAL	Output	Scaled Value [Slope*Input + Offset]

Extended Description

Rockwell Software RSLogix 500 Description:
The Input value is scaled to a range determined by creating a linear relationship between input min and max values and scaled min and max values. The scaled result is returned to the address indicated by the output parameter.

Execution

Condition	Description
EnableIn is true	

Revision v1.0 Notes

Name	Default	Data Type	Scope
Input	0.0	REAL	SCP
Value to be Scaled			
Usage:	Input Parameter		
Required:	Yes		
Visible:	Yes		
External Access:	Read/Write		
Input - SCP/Logic - 2(CPT)			
Input_Max	0.0	REAL	SCP
Maximum Value for the Input			
Usage:	Input Parameter		
Required:	Yes		
Visible:	Yes		
External Access:	Read/Write		
Input_Max - SCP/Logic - 0(CPT)			
Input_Min	0.0	REAL	SCP
Minimum Value for the Input			
Usage:	Input Parameter		
Required:	Yes		
Visible:	Yes		
External Access:	Read/Write		
Input_Min - SCP/Logic - 0(CPT), 1(CPT)			
Output	0.0	REAL	SCP
Scaled Value [Slope*Input + Offset]			
Usage:	Output Parameter		
Required:	Yes		
Visible:	Yes		
External Access:	Read Only		
*Output - SCP/Logic - *2(CPT)*			
Scaled_Max	0.0	REAL	SCP
Maximum Scaling Value			
Usage:	Input Parameter		
Required:	Yes		
Visible:	Yes		
External Access:	Read/Write		
Scaled_Max - SCP/Logic - 0(CPT)			
Scaled_Min	0.0	REAL	SCP
Minimum Scaling Value			
Usage:	Input Parameter		
Required:	Yes		
Visible:	Yes		
External Access:	Read/Write		
Scaled_Min - SCP/Logic - 0(CPT), 1(CPT)			

Name	Default	Data Type	Scope
offset	0.0	REAL	SCP
Offset Value [Scaled Min - (Input Min * Slope)]			
Usage:	Local Tag		
External Access:	Read/Write		
*offset - SCP/Logic - *1(CPT), 2(CPT)*			
slope	0.0	REAL	SCP
Slope Value [(Scaled Max - Scaled Min)/(Input Max - Input Min)]			
Usage:	Local Tag		
External Access:	Read/Write		
*slope - SCP/Logic - *0(CPT), 1(CPT), 2(CPT)*			

0

Slope Value [(Scaled Max - Scaled Min)/(Input Max - Input Min)]

CPT	
Dest	slope
	0.0
Expression	(Scaled_Max-Scaled_Min)/(Input_Max-Input_Min)

1

Offset Value [Scaled Min - (Input Min * Slope)]

CPT	
Dest	offset
	0.0
Expression	Scaled_Min-(Input_Min*slope)

2

Scaled Value [Slope*Input + Offset]

CPT	
Dest	Output
	0.0
Expression	(slope*Input)+offset

(End)

Data type Name: MotionSeg

Description:
motion segments for move queue, lines or arcs

Size: 56 byte(s)

Name	Value	Data Type	Style
Valid		BOOL	Decimal
segment valid flag			
External Access:	Read/Write		
SegType		SINT	Decimal
1=line, 2=arc			
External Access:	Read/Write		
XY		REAL[2]	Float
end xy			
External Access:	Read/Write		
Speed		REAL	Float
>0			
External Access:	Read/Write		
Accel		REAL	Float
>0			
External Access:	Read/Write		
Decel		REAL	Float
>0			
External Access:	Read/Write		
JerkAccel		REAL	Float
External Access:	Read/Write		
JerkDecel		REAL	Float
External Access:	Read/Write		
TermType		SINT	Decimal
0 to 6			
External Access:	Read/Write		
Seq		DINT	Decimal
Sequence/ack id			
External Access:	Read/Write		
CircleType		DINT	Decimal
0,1,2,3 per MCCM enum			
External Access:	Read/Write		
ViaCenter		REAL[2]	Float
[0]=x [1]=y			
External Access:	Read/Write		
Direction		DINT	Decimal
0..3 per mmcm enum			
External Access:	Read/Write		

Data type Name: AB:5000_String16_Struct:I:0

Description:

Size: 20 byte(s)

Name	Value	Data Type	Style
LEN		DINT	Decimal
External Access:	Read/Write		
DATA		SINT[16]	ASCII
External Access:	Read/Write		

Data type Name: AB:5000_String32_Struct:I:0

Description:

Size: 36 byte(s)

Name	Value	Data Type	Style
LEN		DINT	Decimal
External Access:	Read/Write		
DATA		SINT[32]	ASCII
External Access:	Read/Write		

Data type Name: STRING

Description:

Size: 88 byte(s)

Name	Value	Data Type	Style
LEN		DINT	Decimal
External Access:	Read/Write		
DATA		SINT[82]	ASCII
External Access:	Read/Write		

Data type Name: STRING_16

Description:

Size: 20 byte(s)

Name	Value	Data Type	Style
LEN		DINT	Decimal
External Access:	Read/Write		
DATA		SINT[16]	ASCII
External Access:	Read/Write		

Data type Name: STRING_32

Description:

Size: 36 byte(s)

Name	Value	Data Type	Style
LEN		DINT	Decimal
External Access:	Read/Write		
DATA		SINT[32]	ASCII
External Access:	Read/Write		

Data type Name: SCP

Description:
Scale with Parameters

Size: 36 byte(s)

Name	Value	Data Type	Style
EnableIn		BOOL	Decimal
Enable Input - System Defined Parameter			
External Access:	Read Only		
EnableOut		BOOL	Decimal
Enable Output - System Defined Parameter			
External Access:	Read Only		
Input		REAL	Float
Value to be Scaled			
External Access:	Read/Write		
Input_Min		REAL	Float
Minimum Value for the Input			
External Access:	Read/Write		
Input_Max		REAL	Float
Maximum Value for the Input			
External Access:	Read/Write		
Scaled_Min		REAL	Float
Minimum Scaling Value			
External Access:	Read/Write		
Scaled_Max		REAL	Float
Maximum Scaling Value			
External Access:	Read/Write		
Output		REAL	Float
Scaled Value [Slope*Input + Offset]			
External Access:	Read Only		

5069 Backplane : Local Modules

Local: [0] 5069-L310ERMS2 DUNEW2PLC1

Type:	5069-L310ERMS2 Compact GuardLogix® 5380 Safety Controller	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	0	Safety Network Number:	4555_0322_E9A5
Electronic Keying:	Disabled	Revision:	33.11
Status:	Running	Inhibit Flag	Off

Local: [1] 5069-IB16/A <no name>

Type:	5069-IB16/A 16 Point 24V DC Input, Sink	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	1	Electronic Keying:	Compatible Module
Revision:	2.14	Status:	Running
Inhibit Flag	Off	Use Unicast:	n/a

Module Defined Configuration Tag	Value	Data Type
Local:1:C		AB:5000_DI16:C:0
.Pt00		AB:5000_DI_Channel:C:0
.Pt01		AB:5000_DI_Channel:C:0
.Pt02		AB:5000_DI_Channel:C:0
.Pt03		AB:5000_DI_Channel:C:0
.Pt04		AB:5000_DI_Channel:C:0
.Pt05		AB:5000_DI_Channel:C:0
.Pt06		AB:5000_DI_Channel:C:0
.Pt07		AB:5000_DI_Channel:C:0
.Pt08		AB:5000_DI_Channel:C:0
.Pt09		AB:5000_DI_Channel:C:0
.Pt10		AB:5000_DI_Channel:C:0
.Pt11		AB:5000_DI_Channel:C:0
.Pt12		AB:5000_DI_Channel:C:0
.Pt13		AB:5000_DI_Channel:C:0
.Pt14		AB:5000_DI_Channel:C:0
.Pt15		AB:5000_DI_Channel:C:0

Local: [2] 5069-IB16/A <no name>

Type:	5069-IB16/A 16 Point 24V DC Input, Sink	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	2	Electronic Keying:	Compatible Module
Revision:	2.14	Status:	Running
Inhibit Flag	Off	Use Unicast:	n/a

Module Defined Configuration Tag	Value	Data Type
Local:2:C		AB:5000_DI16:C:0
.Pt00		AB:5000_DI_Channel:C:0
.Pt01		AB:5000_DI_Channel:C:0
.Pt02		AB:5000_DI_Channel:C:0
.Pt03		AB:5000_DI_Channel:C:0
.Pt04		AB:5000_DI_Channel:C:0
.Pt05		AB:5000_DI_Channel:C:0
.Pt06		AB:5000_DI_Channel:C:0
.Pt07		AB:5000_DI_Channel:C:0
.Pt08		AB:5000_DI_Channel:C:0
.Pt09		AB:5000_DI_Channel:C:0
.Pt10		AB:5000_DI_Channel:C:0
.Pt11		AB:5000_DI_Channel:C:0

.Pt12	AB:5000_DI_Channel:C:0
.Pt13	AB:5000_DI_Channel:C:0
.Pt14	AB:5000_DI_Channel:C:0
.Pt15	AB:5000_DI_Channel:C:0

Local: [3] 5069-OB16/A <no name>

Type:	5069-OB16/A 16 Point 24V DC Output, Source	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	3	Electronic Keying:	Compatible Module
Revision:	2.14	Status:	Running
Inhibit Flag	Off	Use Unicast:	n/a

Module Defined Configuration Tag	Value	Data Type
Local:3:C		AB:5000_DO16_Diag:C:0
.Pt00		AB:5000_DO_Diag_Channel:C:0
.Pt01		AB:5000_DO_Diag_Channel:C:0
.Pt02		AB:5000_DO_Diag_Channel:C:0
.Pt03		AB:5000_DO_Diag_Channel:C:0
.Pt04		AB:5000_DO_Diag_Channel:C:0
.Pt05		AB:5000_DO_Diag_Channel:C:0
.Pt06		AB:5000_DO_Diag_Channel:C:0
.Pt07		AB:5000_DO_Diag_Channel:C:0
.Pt08		AB:5000_DO_Diag_Channel:C:0
.Pt09		AB:5000_DO_Diag_Channel:C:0
.Pt10		AB:5000_DO_Diag_Channel:C:0
.Pt11		AB:5000_DO_Diag_Channel:C:0
.Pt12		AB:5000_DO_Diag_Channel:C:0
.Pt13		AB:5000_DO_Diag_Channel:C:0
.Pt14		AB:5000_DO_Diag_Channel:C:0
.Pt15		AB:5000_DO_Diag_Channel:C:0

Local: [4] 5069-IF8/A <no name>

Type:	5069-IF8/A 8 Channel Voltage/Current Analog Input	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	4	Electronic Keying:	Compatible Module
Revision:	2.11	Status:	Running
Inhibit Flag	Off	Use Unicast:	n/a

Module Defined Configuration Tag	Value	Data Type
Local:4:C		AB:5000_AI8:C:0
.Ch00		AB:5000_AI_Channel:C:0
.Ch01		AB:5000_AI_Channel:C:0
.Ch02		AB:5000_AI_Channel:C:0
.Ch03		AB:5000_AI_Channel:C:0
.Ch04		AB:5000_AI_Channel:C:0
.Ch05		AB:5000_AI_Channel:C:0
.Ch06		AB:5000_AI_Channel:C:0
.Ch07		AB:5000_AI_Channel:C:0

Local: [5] 5069-OF8/A <no name>

Type:	5069-OF8/A 8 Channel Voltage/Current Analog Output	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	5	Electronic Keying:	Compatible Module
Revision:	2.1	Status:	Running

Inhibit Flag	Off	Use Unicast:	n/a

Module Defined Configuration Tag	Value	Data Type
Local:5:C		AB:5000_AO8:C:0
.Ch00		AB:5000_AO_Channel:C:0
.Ch01		AB:5000_AO_Channel:C:0
.Ch02		AB:5000_AO_Channel:C:0
.Ch03		AB:5000_AO_Channel:C:0
.Ch04		AB:5000_AO_Channel:C:0
.Ch05		AB:5000_AO_Channel:C:0
.Ch06		AB:5000_AO_Channel:C:0
.Ch07		AB:5000_AO_Channel:C:0

Local: [6] 5069-IB16/A <no name>

Type:	5069-IB16/A 16 Point 24V DC Input, Sink	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	6	Electronic Keying:	Compatible Module
Revision:	2.14	Status:	Running
Inhibit Flag	Off	Use Unicast:	n/a

Module Defined Configuration Tag	Value	Data Type
Local:6:C		AB:5000_DI16:C:0
.Pt00		AB:5000_DI_Channel:C:0
.Pt01		AB:5000_DI_Channel:C:0
.Pt02		AB:5000_DI_Channel:C:0
.Pt03		AB:5000_DI_Channel:C:0
.Pt04		AB:5000_DI_Channel:C:0
.Pt05		AB:5000_DI_Channel:C:0
.Pt06		AB:5000_DI_Channel:C:0
.Pt07		AB:5000_DI_Channel:C:0
.Pt08		AB:5000_DI_Channel:C:0
.Pt09		AB:5000_DI_Channel:C:0
.Pt10		AB:5000_DI_Channel:C:0
.Pt11		AB:5000_DI_Channel:C:0
.Pt12		AB:5000_DI_Channel:C:0
.Pt13		AB:5000_DI_Channel:C:0
.Pt14		AB:5000_DI_Channel:C:0
.Pt15		AB:5000_DI_Channel:C:0

Local: [7] 5069-OB16/A <no name>

Type:	5069-OB16/A 16 Point 24V DC Output, Source	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	7	Electronic Keying:	Compatible Module
Revision:	2.14	Status:	Running
Inhibit Flag	Off	Use Unicast:	n/a

Module Defined Configuration Tag	Value	Data Type
Local:7:C		AB:5000_DO16_Diag:C:0
.Pt00		AB:5000_DO_Diag_Channel:C:0
.Pt01		AB:5000_DO_Diag_Channel:C:0
.Pt02		AB:5000_DO_Diag_Channel:C:0
.Pt03		AB:5000_DO_Diag_Channel:C:0
.Pt04		AB:5000_DO_Diag_Channel:C:0
.Pt05		AB:5000_DO_Diag_Channel:C:0
.Pt06		AB:5000_DO_Diag_Channel:C:0

.Pt07	AB:5000_DO_Diag_Channel:C:0
.Pt08	AB:5000_DO_Diag_Channel:C:0
.Pt09	AB:5000_DO_Diag_Channel:C:0
.Pt10	AB:5000_DO_Diag_Channel:C:0
.Pt11	AB:5000_DO_Diag_Channel:C:0
.Pt12	AB:5000_DO_Diag_Channel:C:0
.Pt13	AB:5000_DO_Diag_Channel:C:0
.Pt14	AB:5000_DO_Diag_Channel:C:0
.Pt15	AB:5000_DO_Diag_Channel:C:0

A1/A2, Ethernet : Local Modules

1783-BMS10CGP DUNEW2NETSW

Type:	1783-BMS10CGP Stratix 5700 10 Port Managed Switch, Gigabit Uplinks, Full FW, PTP	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Electronic Keying:	Compatible Module	Revision:	10.1
Status:	Running	Inhibit Flag	Off
RPI:	1000 ms	Use Unicast:	Yes

1734-AENT/C DUNEW2PLC2

Type:	1734-AENT/C 1734 Ethernet Adapter, Twisted-Pair Media	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	0	Electronic Keying:	Disabled
Revision:	6.12	Status:	Running
Inhibit Flag	Off		

2198-P031 DUNEW2AMP

Type:	2198-P031 Kinetix 5700 Bus Power Supply, 10A, 195-264VAC, 324-528VAC	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Electronic Keying:	Exact Match	Revision:	11.3
Status:	Running	Inhibit Flag	Off
Motion Connection Update Period:	2.0 ms		

2198-D006-ERS4 DUNEW2SERVO1

Type:	2198-D006-ERS4 Kinetix 5700, 2x2.5A, Dual-Axis Inverter, Networked Advanced Safety	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Safety Network Number:	448F_0298_455C	Electronic Keying:	Compatible Module
Revision:	11.3	Status:	Running
Inhibit Flag	Off	Motion Connection Update Period:	2.0 ms
RPI:	20 ms	Input Type:	Unicast
Input Trigger:	Application		
RPI:	10 ms	Input Type:	Unicast
Input Trigger:	Application		

2198-D006-ERS4 DUNEW2SERVO2

Type:	2198-D006-ERS4 Kinetix 5700, 2x2.5A, Dual-Axis Inverter, Networked Advanced Safety	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Safety Network Number:	448F_0298_455C	Electronic Keying:	Compatible Module
Revision:	11.2	Status:	Running
Inhibit Flag	Off	Motion Connection Update Period:	2.0 ms
RPI:	20 ms	Input Type:	Unicast
Input Trigger:	Application		
RPI:	10 ms	Input Type:	Unicast
Input Trigger:	Application		

In-Sight 8000 Series DUNE2CAM1

Type:	In-Sight 8000 Series Vision System	Parent:	Local
Vendor:	Cognex Corporation	Vendor ID:	678
Electronic Keying:	Compatible Module	Revision:	11.1
Status:	Running	Inhibit Flag	Off
RPI:	60 ms	Use Unicast:	Yes

PointIO 3 Slot Chassis : DUNEW2PLC2

Path: Local: [0] DUNEW2PLC2

[1] 1734-IB8S/B <no name>

Type:	1734-IB8S/B 8 Point 24V DC Sink Input	Parent:	DUNEW2PLC2
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	1	Safety Network Number:	44AC_0304_FFB7
Electronic Keying:	Compatible Module	Revision:	2.1
Status:	Running	Inhibit Flag	Off
Use Unicast:	Yes		

[2] 1734-OB8S/B <no name>

Type:	1734-OB8S/B 8 Point 24V DC Source Output	Parent:	DUNEW2PLC2
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	2	Safety Network Number:	44AC_0304_FFB7
Electronic Keying:	Compatible Module	Revision:	2.1
Status:	Running	Inhibit Flag	Off
Use Unicast:	Yes		

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